OECD Economic Surveys

Portugal

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OECD ECONOMIC SURVEYS

2002-2003

Portugal

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BASIC STATISTICS OF PORTUGAL

THE LAND

Area (thousand sq. km)	91.9
Major cities, resident population in thousands (2001)	
Greater Lisbon	1 893
Greater Porto	1 261

THE PEOPLE

Population (average in 2001)	10 299
Number of inhabitants per sq. km	112
Net natural increase 2000-2001 (thousands)	8
Net increase 2000-2001 (thousands)	65
Civilian labour force (2001, thousands)	5 167
Civilian employment (2001, thousands)	4 989
As a percentage of total	
Agriculture	12.6
Industry (including construction)	34.4
Services	52.3

PRODUCTION

Gross domestic product, 2001 (million of euros)	122 978
Gross domestic product per head, 2001 (euros)	11 941
Gross fixed capital formation, 2001	
Per cent of GDP	28.1
Gross domestic product at factor cost by origin (2001, per cent of total)	
Agriculture	3.6
Industry (including construction)	28.5
Services	67.9

THE GOVERNMENT

Public consumption, 2001, per cent of GDP	20.7
Public investment, 2001, per cent of GDP	4.1
(Per cent of total investment)	14.5
General government current revenue 2001, per cent of GDP	37.8
Composition of Parliament, 17 March 2002 (Number of seats)	
Popular Democrat/Social Democrat (PPD/PSD)	105
Socialist (PS)	96
Social Center Democrat-Popular party (CDS-PP)	14
Communist	10
Left Block (BE)	3
Ecologist	2
Total	230

FOREIGN TRADE

Exports of goods and services 2001, per cent of GDP	31.0
Main exports as a percentage of commodities exports, 2001 (SITC)	
Machinery and equipment	19.1
Clothing and shoes	17.4
Basic and semi-finished materials	17.3
Road vehicles and transport equipment	16.9
Mineral and metal products	9.2
Other manufactured products	20.1
Imports of goods and services 2001, per cent of GDP	40.7
Main imports as a percentage of commodities imports, 2001 (SITC)	
Chemical and energetic products	22.2
Machinery and equipment	21.9
Road vehicles and transport equipment	14.8
Food, beverages and tobacco	12.2
Basic and semi-finished materials	10.8
Other manufactured products	18.1

THE CURRENCY

Monetary unit : Euro	
Currency unit per $, averge of daily figures:	
Year 2002	1.0611
December 2002	0.9817

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

•

The economic situation and policies of Portugal were reviewed by the Committee on 2 December 2002. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 13 January 2003.

•

The Secretariat's draft report was prepared for the Committee by Bénédicte Larre, Stéphanie Guichard, Chiara Bronchi, Sonia Cabral and Boris Cournède under the supervision of Nicholas Vanston.

•

The previous Survey of Portugal was issued in April 2001.

Assessment and recommendations

Growth has slowed…

A 5-year long economic boom ended in 2001, but not before major imbalances had built up. Falling real and nominal interest rates encouraged the private sector to incur high debt levels, and blunted the government's incentive to rein back the growth of primary spending. The public sector debt-to-GDP ratio stopped falling, and prolonged high demand pressure widened the inflation differential with the euro area. Private demand started to weaken in 2001, but GDP growth was sustained that year by buoyant public spending on both investment and consumption, especially wages and salaries. By early 2002, it became evident that the 2001 budget deficit was going to exceed the Stability and Growth Pact 3 per cent limit by a substantial margin, and that public debt was on the rise. Retrenchment has had to begin in the public sector, and is still ongoing in the private sector. Domestic demand is expected to have stagnated in 2002, and there are no fundamental reasons for expecting it to stage a strong recovery soon. Hence the challenge for policy over the next few years will be the adjustment to weakened domestic demand in an international environment which has yet to regain strength, and with no demand-side help from fiscal policy. The adjustment will be eased, and gains in real incomes increased, to the extent that the supply side of the economy functions more efficiently, with resources moving to more productive uses.

… but inflation remains high

The slowdown in real growth to the EU average in 2001 was not accompanied by a similar convergence in inflation rates. On the contrary, the difference in inflation rates (as measured by the harmonised consumer price, HICP) with the rest of the euro area widened from about half a percentage point in 2000 to nearly 2 percentage points in 2001. The

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increase in the gap on core inflation was much less marked, although the gap itself was high, at 1¾ points. It has widened further in 2002, to 2¾ percentage points by October. It is disturbing that core inflation continued to accelerate in 2002 despite the opening of a significant output gap. The underlying reason for this poor inflation performance is clear. With Portuguese labour markets still relatively tight, compensation per employee continued to outpace that in the euro area by a substantial margin even in the slow-growth years of 2001 and 2002, while labour productivity growth fell to euro area rates, or below.

Recovery will largely depend on external demand

Given the tight fiscal policies that will have to be pursued in the short term, tighter monetary conditions consequent on the appreciation of the euro, depressed consumer and business confidence, and still-high private sector indebtedness, it is unlikely that GDP growth in 2003 or 2004 will return to anything like the rates of the late 1990s. In the near term, the most dynamic source of demand is expected to be foreign, when Portugal's main trading partners in Europe recover from the 2002 slowdown. It is expected that Portuguese firms will continue to attempt to preserve foreign market shares as in 2001 and 2002, given poor sales prospects domestically and despite weakened competitiveness. OECD projections suggest that net exports may add about ¾ of a percentage point to GDP growth in 2003, and almost half a point the following year. Nevertheless, GDP may expand by only 1½ per cent in 2003 and 2½ per cent in 2004. Wage settlements are expected to start reflecting weaker activity. Employment is projected to keep on growing, albeit at a slowing rate, and unemployment may rise to above 5 per cent. With the output gap remaining substantial, unit labour costs decelerating significantly, and import price increases subdued, both core and actual HICP inflation rates are expected to fall to under 2½ per cent by 2004. The gap with the euro area would narrow to around half a point on this hypothesis.

Risks are mainly on the downside

There are several risks to the projections, mostly on the downside. The economic outlook in major trading partners looks more optimistic than in Portugal, but clear signs of an upswing in Europe are not yet apparent. A delayed pick-up

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in foreign demand is most unlikely to be offset by a spontaneous recovery in domestic demand. Indeed, weak foreign demand would probably encourage households to further raise their precautionary savings, and business to postpone investment plans. Wage bargains need to reflect the poor short-term outlook and deteriorated international competitiveness;. otherwise an external demand led recovery will become problematical. It is especially important that public sector wage increases be considerably lower than in recent years, not only as a signal to the private sector, but also to contain public spending pressures. Both public-sector and private-sector wage settlements should take into account the evolution of wages in other euro-area countries, and be based on forward-looking inflation expectations.

Public finances have moved into very large deficit

Containing wage pressures is all the more important at present, and for the next few years, because of the very large deterioration of public finances, and the unfavourable cyclical outlook. The original budget proposal for 2002, prepared in October 2001, set a deficit target of 1.1 per cent of GDP, revised to 1.8 per cent in December. However, it became obvious early in 2002 that the 2001 outturn was going to be considerably worse than targeted, throwing doubt on the achievability of the 2002 objectives. Indeed, by mid-2002, the 2001 outturn was calculated to be 4.1 per cent of GDP, because of both substantial primary spending overruns – especially on wages and salaries – in the lead-up to the elections, and shortfalls in tax revenue, reflecting less success than expected in efforts to combat tax evasion, and the effects of slowing activity. Changes in accounting methodology also had an unfavourable impact. The primary balance swung into deficit for the first time in almost 20 years. The new government, which had previously promised tax cuts to improve the functioning of the economy, approved a supplementary budget in May 2002 to limit the damage by recourse to both tax hikes and spending cuts. These were aimed at containing the 2002 deficit to 2.8 per cent of GDP. The OECD projection published in *Economic Outlook* 72 of December 2002 is for a 3.4 per cent deficit. In cyclically-adjusted terms, the correction from 2001 would nevertheless be substantial, around 1¾ percentage points of potential GDP. Official estimates, taking into account more recent

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one-off measures, put the nominal deficit at just below 3 per cent of GDP in 2002.

Emergency measures were taken in 2002 to lower the deficit

The emergency measures implemented, or announced, as of May 2002 include a 2 percentage point hike of the standard VAT rate (to 19 per cent), a suppression or restructuring of public agencies with overlapping functions, a partial hiring freeze accompanied by enhanced labour mobility in the administration, a freeze on renewal of individual and fixed-term contracts, measures to prevent local governments building up debt, and a 2 per cent limit on the growth of expenditures of Autonomous Funds and Services except for health, EU co-financed investment, and base salaries. These measures seem to be proving effective in reducing the deficit despite weak activity. Some of them – for example placing tighter limits on local government spending and the potential for more flexible management of human resources in public administration – are desirable structural reforms in their own right, or could be the basis of such reforms. Cutbacks in infrastructure spending though, are undesirable from a longer-term view, while hiring freezes give distorted incentives to younger, better-educated aspirants to permanent posts in the Administration, and also to older, possibly low-productivity, public employees. Hiring freezes also greatly complicate human resource management in all but the very short term. It is important that such temporary measures be phased out as soon as possible and be replaced by deeper structural reforms.

Further corrective measures were announced at the end of the year

When it became clear that the 2001 budget outturn had exceeded the 3 per cent threshold, the European Commission proposed to the Council in October 2002 that Portugal be subject to the excessive deficit procedure. The Council adopted the same decision on 5 November 2002. Following this decision the government announced corrective measures. The budget for 2003, presented to Parliament in October, foresees a primary surplus of 0.8 per cent of GDP and a minor reduction in the actual deficit, to 2.4 per cent of GDP. Corrected for cyclical influences, this would represent a tightening of about 0.8 percentage point of GDP on a structural basis, according to OECD estimates. The main features are reduced spending on administration, compulsory mobility

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within the administration (which provoked a wave of strikes), and limits on spending increases in some social areas. Infrastructure spending is to be freed up to some extent.

Reducing the deficit is the top priority for 2003

Judging the appropriateness of the 2003 budget is not easy. A significant structural tightening in the face of very weak domestic demand – and uncertain foreign demand – is an unusually pro-cyclical stance, but it is nevertheless necessary in the particular circumstances facing Portugal. This stance is a direct consequence of the fiscal errors and misjudgements of the past, which led to strongly procyclical movements in the opposite direction. These have to be unwound, and it would be unwise to delay taking measures until the economy has started growing strongly again. Indeed, should the short-term outlook prove even weaker than expected, it would be appropriate to ensure that the nominal deficit target for 2003 be met nevertheless. Any windfall gains should be used to pay down debt.

Achieving budget balance has been postponed, though consolidation continues

The Stability Programme 2003-06 targets a steady reduction of the structural deficit, by a ½ per cent point of GDP each year, which is in line with the recent proposal of the European Commission. Under the Programme's assumptions, the structural deficit would reach near balance in 2006, at about ½ per cent of GDP. The Stability Programme also envisages a reduction of the debt-to-GDP ratio from its estimated 2002 level of a little less than 60 per cent, to 52.7 per cent by end-2006. Such a reduction would imply that primary surpluses rise gradually to 2½ per cent of GDP in 2006. Achieving the pace of consolidation foreseen between now and 2006 is by no means an easy target. It will require better control of public spending in health, education and social security, as well as better allocation of this spending, as stressed in the 2003-06 Stability Programme. If the measures already taken or envisaged are effective in containing spending, then the Programme foresees a gradual reduction in the corporate income tax rate (by a cumulative 10 percentage points), to improve the country's international competitiveness. The tax cut will have to be postponed if difficulties are encountered in meeting the budget targets. Even when existing imbalances have been unwound, further reforms

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will be needed to assure longer-term fiscal sustainability. There will be growing pressures from public spending on pensions as the baby-boomer generation starts moving into retirement as from the middle of the decade. Even to maintain balance in these circumstances will require ongoing measures to reduce the growth of spending and more effective tax collection.

Budget process reforms should be intensified

The imbalances and inefficiencies in the Portuguese public sector impede better economic performance. Administrative control, the cost-effectiveness of service provision, and incentives for efficient resource allocation all need to be improved. The authorities have recently recognised the need to modernise the budget process, and intend to introduce multi-annual spending programmes and a spending ceiling in the first year of such programmes. They have also taken action to strengthen financial coordination between all entities of the general government. These are steps in the right direction but further action could be envisaged. Following the experience of other OECD countries, firm multi-year expenditure ceilings based on prudent macroeconomic assumptions would provide useful hard budget constraints to prevent overspending and undesirable discretionary mid-year cuts in budget allocations. Multi-year spending ceilings should be set and followed through so as to give them the necessary credibility. Ongoing efforts to develop accruals-based accounting in line with ESA95 norms should be accelerated and completed. In this context, the implementation of the accrual-based *Plano Oficial de Contas Publicas* (public accounting system) should no longer be delayed. This would give better information on costs and may therefore improve cost effectiveness. Finally, the internal audit system should be reinforced and frequent external auditing should be established.

Local governments need both more freedom and more responsibility

Decentralisation can help to make government more accountable and improve allocation of the economy's resources by placing decision-making closer to the ultimate users and payers of public services at the regional and local levels. But progress towards greater devolution in Portugal has not gone hand in hand with stronger fiscal accountability.

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Local governments currently depend on grants that do not provide adequate incentives to contain spending or to allocate it optimally. Local authorities need to operate under strict budget constraints, but also need more discretion in how to operate within such constraints in order to make fiscal responsibility operational. This entails making local policy more transparent and accountable before the local electorate. Local public administration and public good delivery should also improve. Local governments could be given more freedom to organise local expenditure programmes, avoiding overlapping responsibilities with the central government, and to design the appropriate fees and charging structures. Meanwhile, the devolution of tax and spending power needs to be supported by better internal and external control mechanisms and improved information flows between levels of government. The new restriction on indebtedness of local governments was introduced by the Budgetary Stability Law of 2002 and subsequently applied in the 2003 Budget Law. It would be wise to change the financing rules set in theLaw for local finances accordingly, making it long-lasting. Sanctions have been introduced, and should be made effective, when spending ceilings are not met or when financial information is not made available. The strict enforcement of a no bailout rule would be instrumental in this regard.

Administration needs more flexibility in management and an outcome-oriented approach

To provide public services in a more cost-efficient way, public sector reform should focus on moving further towards a budget and management approach geared towards outcomes. Activity-based funding schemes, similar to that for pilot hospitals, should be extended to other public services, but at the same time price signals should be used more frequently to avoid excessive demand and hence public spending. To enhance public sector efficiency, more flexible personnel management practices are also required. The use of fixed-term or individual employment contracts could allow public managers more flexibility in attracting the best suited employees and to allocate resources according to needs. The recent reforms intended to increase labour mobility within public administration notably the new Law on labour mobility and a public "employment pool" as of 2003 are steps in the right direction. These measures are key for

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reallocating human resources in a more effective way, especially as there seem to be an excessive number of public employees relative to the services that they deliver, and this is putting chronic pressure on public spending. The performance evaluation system should be revised to reflect more accurately individual effort and productivity. Finally, public bodies should be given more freedom to set remuneration levels, with the possibility to use efficiency gains to reward good performance.

Delivery of public services can be improved via market-oriented solutions

More competition between public and private sector providers could also induce a more cost-effective supply of public services. In this context, equal funding of public and private providers becomes instrumental. For example, some OECD countries have allowed parents to choose the school they consider best while letting public funding follow the students as a type of voucher. Thereby, public and private schools can compete. More user choice may give better incentives to the providers of compulsory education in Portugal to improve outcomes, which have recently been shown as being well below average in the OECD PISA study despite high government outlays. The decision to make an annual assessment of all schools, and publish the results, is an important innovation. However, equity considerations need be taken into account and the expansion of user choices supplemented by measures to avoid social segregation. Furthermore, geographic restrictions on user choices could be removed to let citizens use the services of other municipalities. Finally, increased recourse to outsourcing and competitive tendering would contribute to a better delivery of services.

User charges could lead to better resource allocation

Increases in user charges should be considered as a means to reduce excessive demand for some public services. Higher user charges could make households more cost aware and help them for instance to limit excessive pharmaceutical consumption. It is important, however, that any increase in co-payments should not conflict with equity objectives. This can be achieved through increased differentiation of co-payments according to income levels or, alternatively, tax credits for drugs could be subject to a ceiling equivalent to the price of generics.

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Market oriented solutions will also aid sustainable development

A greater use of charging and economic instruments in general would help to maintain a sustainable use of natural resources. The economy needs to end situations in which free access and subsidisation leads to depletion of natural resources. The use of water is one particular area that merits close consideration by the authorities. Household face charges for the water they use, though not generally fully in line with the supply costs. The farming sector, however, benefits from irrigation projects that have been subsidised both nationally and supranationally. A system in which existing water extraction permits could be traded and in which irrigation water was priced according to its cost would ensure that scarce water was used in the highest value uses. However, price signals in this area are affected by limits on international trade emanating from the EU Agricultural Policy. In the area of fishing, free access to existing stocks has been ended in the context of the European Fisheries Policy by the introduction of quotas. Current instruments to manage stocks have played a useful role. A more efficient industry would emerge by allowing a market for the transfer of quota rights. In addition, subsidies remain high and need to be reduced.

Air quality is good, and could be improved further

Air pollution has not been a major problem in Portuguese cities, as a result of the prevailing weather conditions, with the exception of fine particles that have localised impacts. Consequently, although in the past there was some effort to contain emissions, now there is need to make substantial reductions in order to comply with European agreements. Existing compliance plans place an emphasis on the administrative allocation of pollution quotas. This is unlikely to generate the lowest cost solution and the authorities should consider a cap and trade regime in accordance with European and international market-based mechanisms, as is the case for climate change agreements. Pollution from fine particles poses different problems. Most of it comes from road vehicles, notably those using diesel fuel. Despite this, taxes on diesel fuel are lower than those on gasoline. In addition, high taxes on new cars tend to prolong the life of the existing fleet, so postponing the gains from emission standards that have fallen over time, and should preferably be replaced with a tax regime that encourages the scrapping

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of older, more polluting, vehicles. In addition, the government should adopt measures oriented towards a further reduction of emission limits on diesel vehicles and the introduction of very low sulphur fuel.

Structural reforms can help convergence to EU-average living standards

Reforms are also desirable in other areas of the economy, so that once recovery is under way, it can gather strength and allow Portuguese living standards to resume convergence with those in existing EU member countries. The new government has defined a broadly-based programme to strengthen competitiveness and efficiency, which closely follows the policies for growth identified in the OECD "Growth Project". The government's Productivity and Growth Programme is designed to promote investment, exports, competition, deregulation and innovation. The analysis in the "Growth Project" suggested that potential growth rates in Portugal could be boosted by raising the level of educational attainment of the workforce, improving mobility in labour markets, raising the technological base, and assuring more competitive product markets, issues which are taken up below.

Educational and training standards and outcomes need improvement

The announced reforms in the delivery of compulsory education, described above, are likely to both improve educational outcomes and lower its cost. The reforms should be accompanied by a stronger emphasis on vocational training, to ease the school-to-work transition. New rules permitting employment of unqualified youths on condition that training is provided by the employer may help, but it would be better if the schools themselves were better able to provide school-leavers with qualifications useful to employers. The high private returns to tertiary education in Portugal suggest that increasing the supply of graduates would add significantly to productivity. Several OECD countries have targeted support for higher education through fees, loans and grants schemes that include provisions for the less well-off. Portuguese authorities could envisage similar measures and at the same time increase tuition fees and eliminate education tax credits that are regressive. Besides reducing public costs in post-compulsory education, such a system can reduce inequality without compromising efficiency goals. Since the diffusion of larger numbers of higher-educated

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people in the labour force is bound to be a gradual process, these reforms should be accompanied by the training and re-training of existing workers. The government's aim to provide employment-related training to all youths that are registered in employment centres and a renewed emphasis on life-long training are welcome. If successful, these incentives would increase the productivity and earnings capacity of those who might otherwise exit the labour force.

Labour markets could function better…

Portuguese labour market performance has several positive aspects. Unemployment rates are lower than the OECD average, among the lowest in the EU, and with no trend increase. Employment has been growing briskly even during economic downturns, and analysis suggests that wage flexibility is high across the private sector and when faced with shocks. At the same time, there is a high level of job protection, and high private costs to geographic mobility, because of housing market restrictions. Both factors hinder the necessary restructuring of the economy as patterns of demand and technology change, and may help explain the continuing high importance of traditional labour-intensive industries. Measures that increase geographic mobility, and which enable employers to allocate human resources more flexibly would result in higher average productivity and real wages, and possibly narrower wage differentials also. Fixed-term contracts and other atypical forms of work, that provide comparatively low employment protection, are an important channel into the labour market and they have developed rapidly. However, they can only be a partial solution, as there is the risk of creating more precarious positions for those groups of workers that experience the lowest attachment to the labour market. The relaxation of general job-protection legislation, included in the recent Labour Code reform proposal, would encourage employers to create more jobs of a permanent nature.

… and this would assist innovation and use of better technology

Greater geographic and within-firm job mobility combined with training and life-long training would also help the economy to raise its technological level, as new techniques require new skills and new working practices. The continuing importance of traditional low-skill labour intensive industries suggests that technological diffusion leaves

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much to be desired, and spending on R&D in Portugal is comparatively low, though there has been significant improvement in the last decade. Indicators of the penetration of the Internet show it to be still lagging behind other EU countries. Successful innovation often requires the creation of new firms, and recent reforms have been introduced to simplify administrative procedures and shorten delays in starting up new enterprises. New firms also need to be able to hire new labour, which at the macro level requires that existing non-viable low-productivity firms are able to go out of business speedily and at low cost, releasing their resources to more productive uses. Bankruptcy procedures are being reformed but it is also necessary to reduce legal fees and fiscal charges related to mergers, restructuring and divestiture.

Implementation of competition policy needs to be strengthened

Stronger competition in product markets is needed to give incentives to raise productivity. In this regard, an independent Competition Authority responsible for competition policy implementation has been recently created. Its aim is to make sure that competition rules are respected as regards efficient markets, resources allocation and consumers interests. This will be reinforced, for example by assigning terms for the Competition Authority members longer than the government's term of office, and imposing strict time constraints for issuing decisions. Of paramount importance is giving the Competition Authority adequate numbers of qualified staff to ensure that it can monitor product market developments, and also giving it powers to implement decisions in a timely fashion. A new competition law will be proposed soon that will go some way to dealing with these concerns. Competition would be further strengthened if remaining privatisations were carried out in ways which ensured the creation of competitive conditions in the privatised sectors. The structure and regulation of the electricity industry provides little incentive to reduce costs and pass lower costs to consumers, and prices remain high by EU standards. At present, strong commercial links remain between the former publicly-owned generator and the regulated transmission grid, and the independent generators have few incentives to supply electricity on a competitive basis. Competition from imports is low but will increase

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gradually as the transmission infrastructure for the Iberian electricity market develops in the period up to 2006.

In summary

The size of the imbalances that have accumulated in recent years have weakened the outlook for domestic demand growth, and also left Portugal with little choice but to pursue budgetary rigour for some time. Recovery will have to wait on that in European trading partner countries, but Portuguese competitiveness has been damaged by comparatively high inflation rates. Wage settlements will need to take this into account. The recovery is therefore likely to be gradual, and a return to the fast growth rates of the late 1990s may have to wait for some years. The authorities have rightly stressed the need for fiscal consolidation during this period, especially as a much improved budgetary position will be needed to cope with ageing pressures over the longer term. Return to budget balance will be speeded, and fiscal sustainability thereafter will be assisted, by measures that contain the growth of primary spending, especially on wages and salaries in public administration. Such measures will be more effective if they take place in the context of a medium-term process that includes hard budget constraints and greater recourse to market-oriented solutions in the delivery of public services. Such reforms would help avert future fiscal shocks that lower growth in the short term. Macroeconomic performance would be enhanced in the longer term by a range of structural reforms in labour and product markets, such as proposed in the Productivity and Growth Programme. Potential growth could be raised by educational and training policies that raise average competence levels in both school leavers and existing employees. Also, university graduates command high salary premiums, and the beneficiaries should be required to participate financially to a greater extent in their studies. Other growth-enhancing reforms include raising the mobility of employees within and between firms and encouraging the use of better technologies and management methods. More competition, especially in network industries, will also raise productivity levels and put downward pressure on prices. Portugal has great potential for catch-up with other euro-area countries, as demonstrated in the past. Reforms are required across a broad front to release this potential.

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I. Macroeconomic developments and prospects

Overview

An extended period of real convergence in living standards towards the euro area average came to a halt in 2000 and the Portuguese economy continued to slow down in 2002 (Figure 1). The downturn reflected the international weakening of activity and, most of all, the adjustment process of private demand that had started mid-2000. These developments took place against a background of sizeable macroeconomic imbalances generated in the late 1990s, including a large external deficit and high household and corporate indebtedness. In 2001, these factors outweighed the positive stimulus associated with an expansionary fiscal policy and the relatively favourable monetary conditions that characterised the Portuguese economy. In 2002, the more restrictive policy stance, in particular the tight fiscal measures adopted, intensified the slowdown of domestic demand. The recovery of the Portuguese economy is projected to lag the euro area and will be heavily dependent on external demand, as domestic demand is expected to remain subdued for some time.

Inflation continued to exceed the euro area average in 2002, reflecting relatively stronger cost pressures in Portugal. With low unemployment, wage increases (including in the public sector) have remained above increases in the euro area and the differential has not been compensated by higher productivity growth. In 2002, the slowdown of economic activity helped to moderate wage growth, but the core inflation differential *vis-à-vis* the euro area still remains above 2½ percentage points, damaging external cost competitiveness.

Portugal's cycle in the European context

Economic growth in Portugal was particularly strong during the five years to 2000, averaging 4 per cent and exceeding the euro area average by 1½ percentage points (Table 1). This good performance reflected a much stronger growth of domestic demand than in the euro area, as would be expected in a catching-up economy. However the imbalances that built up during this period triggered the

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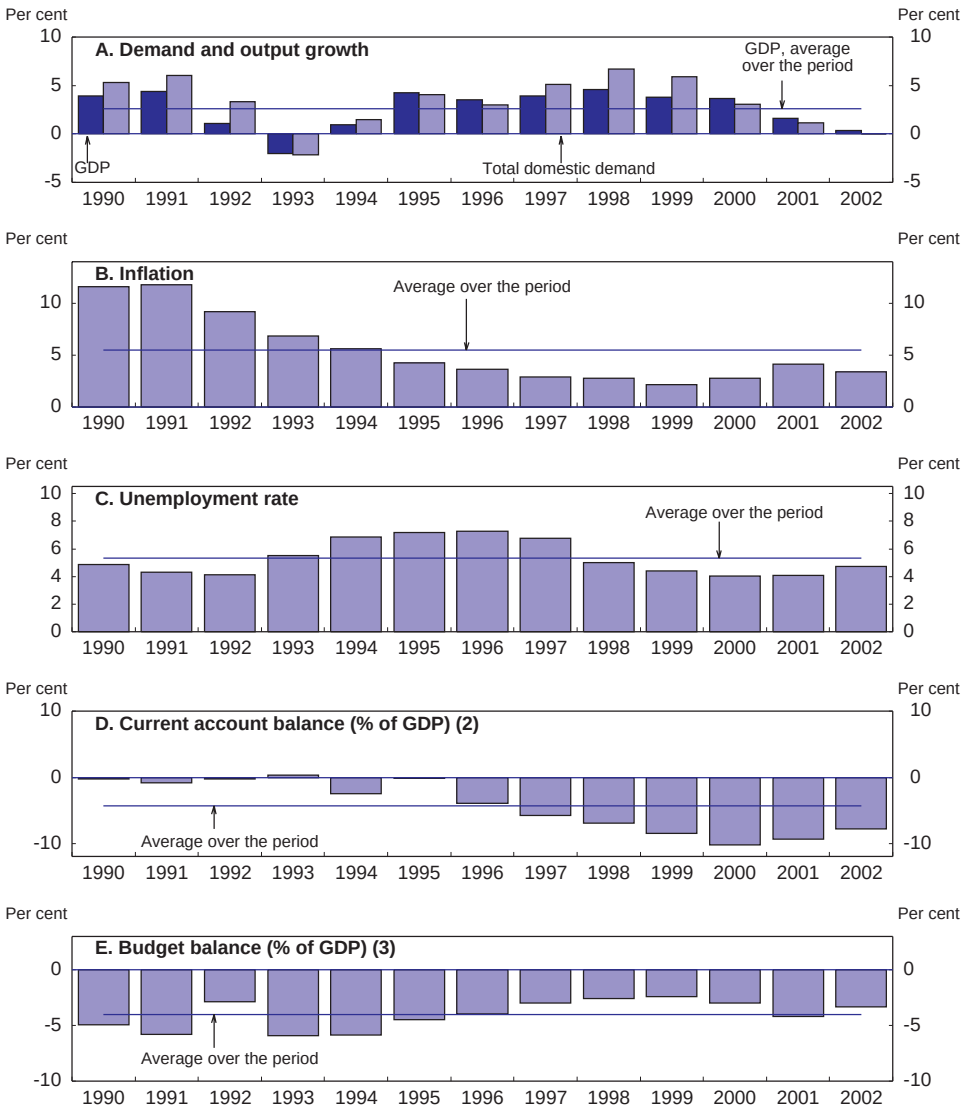
Figure 1. **Macroeconomic performance at a glance**[1]



1. OECD estimates for 2002.
2. Break in the series, as from 1996 the current account balance excludes capital transfers, amounting to around 2 per cent of GDP.
3. Includes the sale of UMTS licences in 2000. Breaks in the series in 1995 and 2000.

Source: OECD.

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Table 1. **Portugal and the euro area: key indicators**

	Portugal			Euro area		
	Average 1995-2000	2001	2002[1]	Average 1995-2000	2001	2002[1]
Real GDP[2]	4.0	1.6	0.4	2.5	1.5	0.8
Total domestic demand[2]	4.7	1.1	0.0	2.5	1.0	0.3
Employment rate[3]	65.6	68.7	68.9	59.7	62.4	62.5
Unemployment rate	5.8	4.1	4.7	10.0	8.0	8.3
Private consumption deflator[2]	3.1	4.2	3.4	2.0	2.4	2.2
Overall budget deficit (% of GDP)	–3.2	–4.2	–3.4	–2.6	–1.5	–2.2
Cyclically-adjusted budget deficit (% of potential GDP)	–3.7	–4.6	–2.9	–2.2	–1.5	–1.6
Current account balance (% of GDP)	–5.9	–9.4	–7.9	0.8	0.1	0.9

1. OECD estimates.
2. Growth rate.
3. Total employment as a percentage of population, aged 15-64 years.
Source: OECD.

current slowdown and are likely to continue to affect growth prospects adversely for several years.

Economic developments, especially since 1997, were strongly conditioned by Portugal's participation in the single currency from the very start. The sizeable interest rate cuts implied by the necessary convergence of interest rates to the lower levels of other euro area countries were interpreted by economic agents as largely irreversible, as they were associated with a new macroeconomic regime characterised by price stability. The reduction of interest rates translated into a significant easing of liquidity constraints for households and corporations, increasing their wealth and their capacity for taking on more debt. Corporations seized the opportunity to expand and modernise their productive capacity. The counterpart to the rise in the indebtedness of households was increased access to housing and durable consumer goods, leading to significant improvements in living standards. Fiscal policy did not counter the stimulus to activity from falling interest rates, failing to match the consolidation efforts of other smaller euro area countries (see Chapter II). The very strong growth of domestic demand, public and private, led to a widening of the external deficit, as inflationary pressures were partly damped by competition from strongly growing imports.

By mid-2000, the high debt levels of private agents induced them to adjust borrowing downward. Expenditures slowed and saving levels raised. This desirable adjustment of the growth of domestic demand to a more sustainable, though still brisk, pace was compounded by a series of shocks, most of them

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Table 2. **Demand and output: recent trends**[1]

	1998 current prices	1999	2000	2001	2002
	Billions of euros	Percentage change			
Demand and output (volume)[2]					
Private consumption	62.8	5.1	2.6	1.2	0.8
Government consumption	19.1	5.6	4.0	2.8	1.1
Gross fixed capital formation	27.1	6.4	4.4	0.0	–2.5
Final domestic demand	109.0	5.5	3.3	1.2	–0.0
Change in stockbuilding[3]	0.8	0.5	–0.3	0.0	–0.0
Total domestic demand	**109.9**	**5.9**	**3.1**	**1.1**	**0.0**
Exports of goods and services	31.1	2.9	8.0	1.4	1.1
Imports of goods and services	40.1	8.5	5.4	0.3	0.0
Change in foreign balance[3]	–8.9	–2.6	0.3	0.4	0.4
GDP at market price	**101.0**	**3.8**	**3.7**	**1.6**	**0.4**
Memorandum item					
GDP deflator	..	3.1	3.2	4.7	3.7
Private consumption deflator	..	2.1	2.8	4.2	3.4
Household saving rate[4]	..	8.5	10.1	11.0	11.2

1. OECD estimates for 2002.
2. At 1995 constant prices.
3. Expressed as a percentage of GDP in the previous year.
4. As a percentage of disposable income.

Source: Ministry of Finance; OECD.

common to all OECD countries. These included the surge in oil prices in 1999-2000, the bursting of the ICT bubble, the ensuing slump in stock markets, and the 11 September events. Whereas the beginning of the European slowdown in 2000, the Portuguese economy had appeared more resilient than others, the situation was progressively reversed. In 2002, as private domestic demand and export growth further weakened and the fiscal stimulus was withdrawn, the Portuguese growth was among the weakest of the euro area, below ½ per cent (Table 2). The Portuguese economy is now in the group of smaller euro-area countries with mediocre GDP growth, like Austria, Belgium and the Netherlands, while much stronger economic performance is observed in Greece, Ireland and Spain.

Economic slowdown deepened in 2002

Household demand is cooling off…

Private consumption remained weak in 2002, growing by less than 1 per cent, after the sharp slowdown recorded in 2001 (Figure 2). In addition to temporary factors,[1] the slowdown of consumer expenditure reflects a lower growth of real disposable income.[2] It also reflects a continued increase in the saving ratio: consumers

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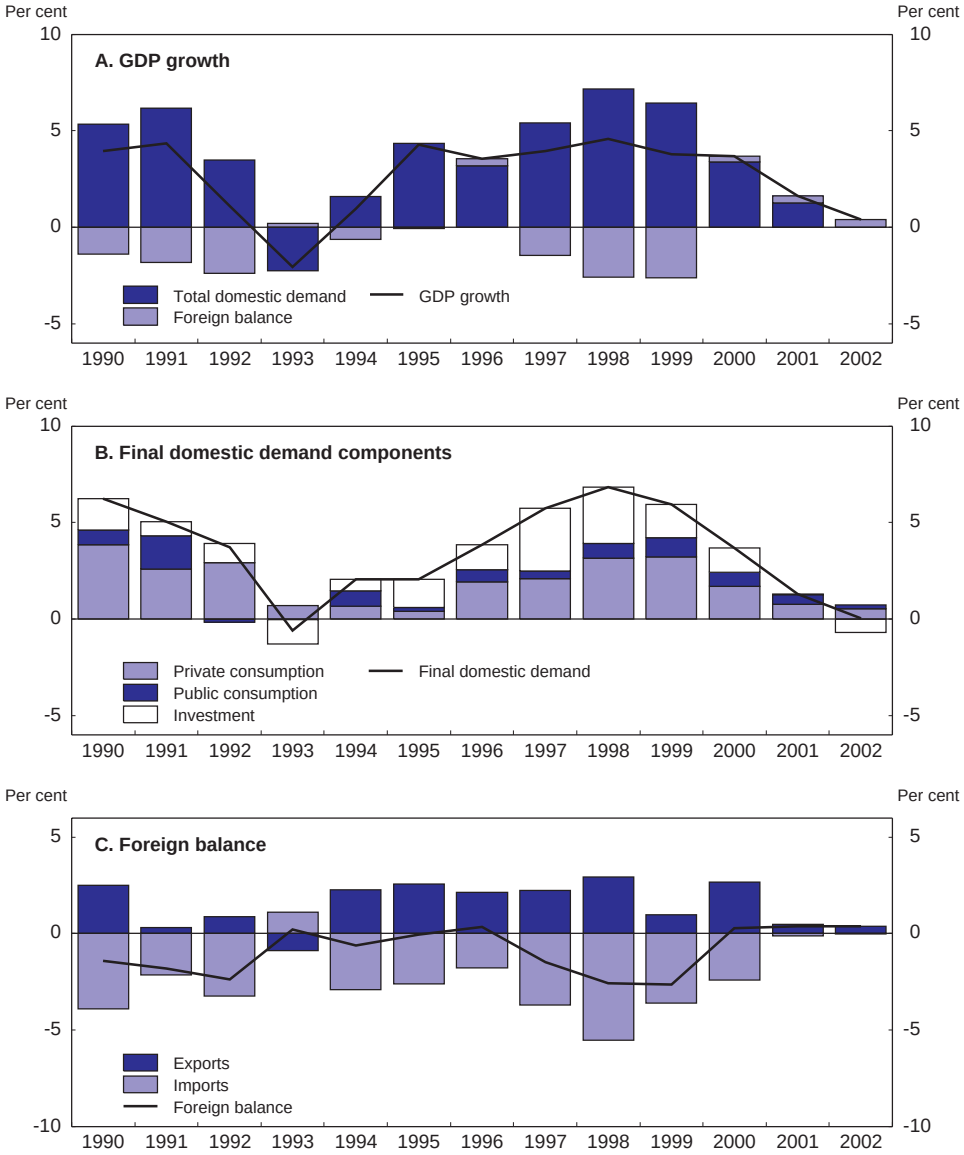
Figure 2. **Contribution to real GDP growth**[1]

Percentage points



1. Based on ESA95; data prior to 1995 have been adjusted for the statistical breaks. 2002 data are OECD estimates.
Source: OECD.

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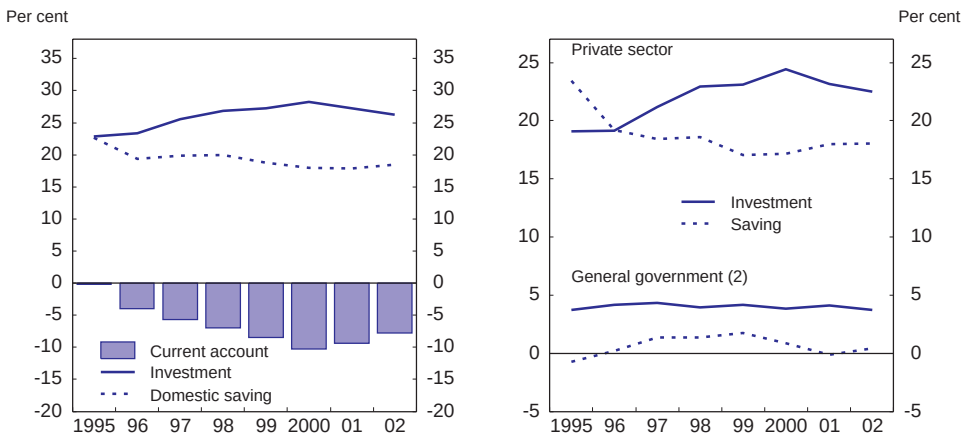
tried to stabilise indebtedness at a level consistent with expectations for future earnings growth while uncertainty about future economic developments increased and consumer confidence fell to very low levels.[3] In addition, the current economic situation also led to a more cautious attitude in households' investment decisions, which resulted in a correction in residential construction after several years of strong growth.

... and both corporate and public investment fell

The high investment growth rates of the 1990s fell sharply at the beginning of the new century. The investment ratio reached a peak at 28½ per cent of GDP in 2000 (Figure 3), with foreign savings increasing to over 10 per cent of GDP. Subsequently, the adjustment process started, leading to some increase in private savings and to a decline in the investment ratio. In 2002, gross fixed capital formation declined by 2.5 per cent, after stagnating in 2001. The outlook for investment became more and more unfavourable: business confidence indicators turned sharply down, order books (internal and external) came down and the rate of capacity utilisation was on a downward trend. The deceleration of private investment was reinforced by massive public investment cuts, following the 2001 boom.

Figure 3. **Aggregate saving and investment**[1]
Per cent of GDP



1. Data are based on the national accounts for 1995-2001; 2002 data are OECD estimates.
2. Government data are affected by a change in methodology and recording practices in 2000.
Source: OECD.

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Labour market slack is increasing…

Until 2001, labour market developments were characterised by a rise in the participation rate, a strong growth of employment, in particular dependent employment, and low unemployment rates (Table 3). The participation rate is estimated to have recorded a further rise in 2002, reaching close to 75 per cent, a relatively high rate in the OECD context. This increase continued to be largely explained by demographic factors[4] and the upward trend of female participation. In the last two years, inflows of economically active migrants also contributed to the rise in the participation rate. As in the rest of the euro area, employment lagged in the cycle, with the services sector again contributing to total employment growth. However, employment growth may have expanded by only around

Table 3. **Labour-market indicators**

Percentages

	1998	1999	2000	2001	2002[1] Q2	2002 Q3
	Percentage changes from a year earlier					
Total labour force	0.6	1.2	1.3	1.7	1.5	1.6
Employment, total	2.5	1.8	1.7	1.6	0.9	0.5
Male	1.7	0.9	1.6	1.5	0.5	0.0
Female	3.4	3.0	1.8	1.9	1.3	1.0
Agriculture	–2.6	–4.9	1.4	2.0	–3.8	–1.9
Industry, including construction	1.2	–0.0	1.5	–0.2	1.1	–0.1
Services	4.8	4.7	2.2	2.8	1.9	1.4
Dependent employment	3.9	3.3	2.5	1.7	1.5	1.1
	Per cent					
Participation rate, total[2]	70.1	70.6	71.1	71.8	n.a.	n.a.
Male	78.0	78.6	78.8	79.4	n.a.	n.a.
Female	61.9	62.7	63.7	64.6	n.a.	n.a.
Unemployment rate, total[3]	5.0	4.4	4.0	4.1	4.5	5.1
Male	4.0	3.9	3.2	3.2	3.8	4.2
Female	6.3	5.1	5.0	5.1	5.3	6.2
Youth (15-24)	10.2	8.7	8.7	9.3	10.4	12.0
Long-term unemployment[4]	42.5	37.6	40.4	36.8	38.7	38.0
Job vacancies/total labour force	0.26	0.27	0.30	0.24	0.18	0.19
Fixed-term contracts[5]	17.51	19.04	20.41	20.65	21.8	21.7

Note: Data for employment refer to quarterly labour force surveys. New survey sample as of 1998; it represents persons aged 15 and over and refers to the whole country.

1. As of the second quarter 2002 the *Employment Survey* incorporates new estimates of population calculated in census 2001. The revised series, available since the second quarter 2001, are used to calculate the 2002 figures.
2. As a percentage of population aged 15-64 years.
3. As a percentage of the labour force of the group or age group.
4. 12 months and over as a percentage of total unemployment.
5. As a percentage of total contracts for wage-earners.

Source: Ministry of Social Security and Labour; INE.

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Table 4. **Wages and labour costs**
Percentage changes

	1998	1999	2000	2001	2002[1]
Contractual wages[2]	3.2	3.3	3.3	3.9	3.7
Minimum wage	3.9	4.1	4.1	5.0	4.1
Compensation per employee					
Total economy[4]	4.9	4.8	5.6	5.6	3.9
Business sector	3.3	4.2	4.9	6.1	4.0
Unit labour costs					
Total economy[4]	3.7	2.7	3.5	5.4	4.2
Business sector	0.8	2.0	2.7	5.9	4.4
Memorandum items:					
CPI harmonised, Portugal	2.2	2.2	2.8	4.4	4.1[3]
Euro area:					
CPI harmonised	1.2	1.1	2.4	2.5	2.3[3]
Compensation per employee, business sector	0.9	1.2	2.3	2.7	2.9
Unit labour costs, business sector	–0.1	0.5	1.0	2.7	2.5

1. OECD estimates.
2. All sectors excluding general government. Ten-months' average for 2002.
3. October 2002 (year-on-year rate of change).
4. Estimates by Ministry of Social Security and Labour.

Source: Ministry of Finance; Ministry of Social Security and Labour; INE; OECD.

0.5 per cent in 2002. The unemployment rate stopped declining in 2001 and rose in 2002, reaching 5.1 per cent by end-September.

Wage pressures eased somewhat in 2002 (Table 4). Nominal wage increases implicit in collective agreements for the private sector decelerated only slightly in 2002, reaching 3.7 per cent on average.[5] But the increase in compensation of employees is likely to come down more rapidly, as the drift between contractual and effective wages tends to vary in line with fluctuations in economic activity. Given the emergence of slack in the labour market, wage drift might be below ½ percentage point in 2002, compared with 1-2 percentage points in previous years.

... but inflation remained high

The inflation differential with the euro area widened in 2001. All euro area countries were affected by the lagged effects of the rise in international commodity prices aggravated by the euro depreciation. In Portugal, the strong increase of some fresh food prices and the acceleration of wages amplified the rise of inflation, measured by the annual rate of change of the Harmonised Index of Consumer Prices (HICP), to 4.4 per cent (Figure 4). In 2002, the consumer inflation rate

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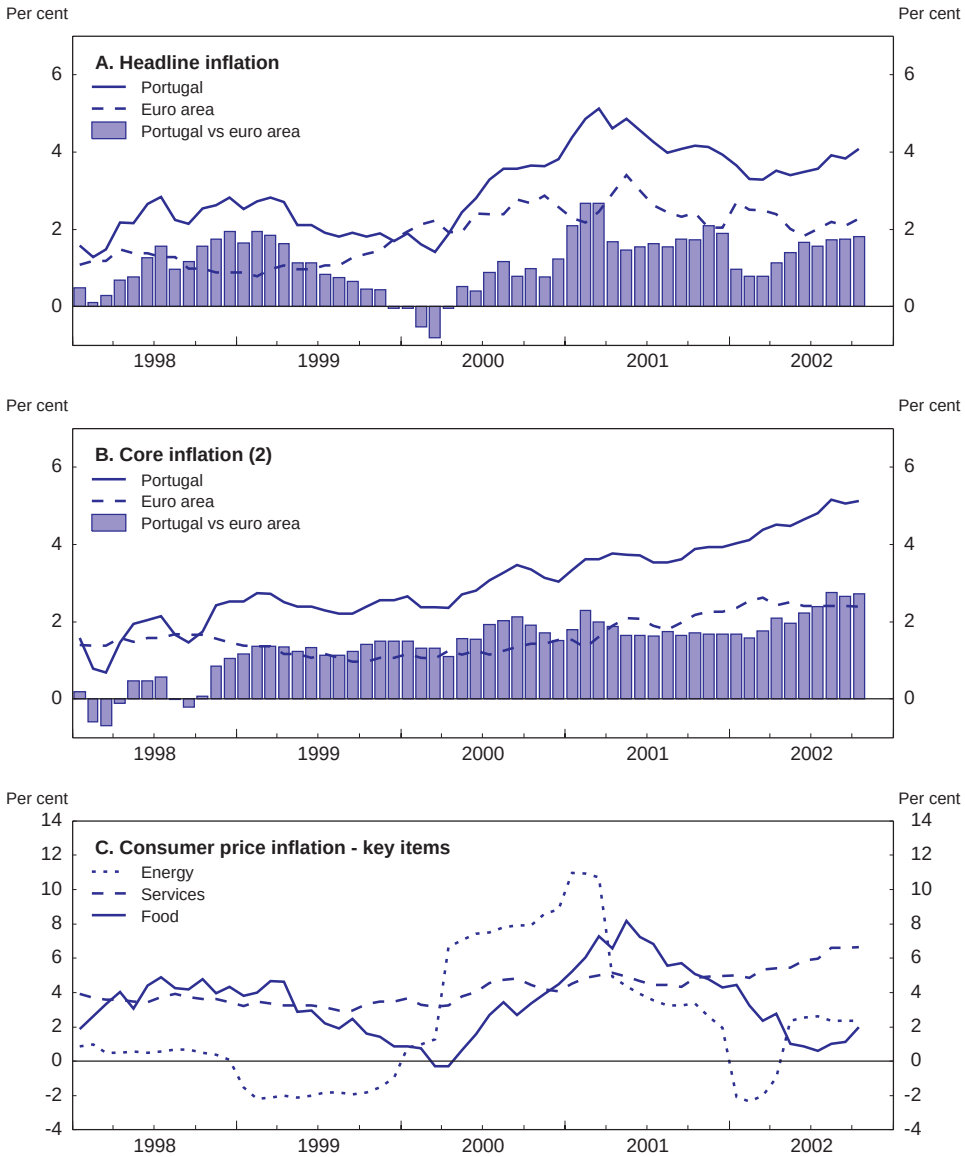
Figure 4. **Inflation developments**[1]



1. Year-on-year percentage changes of the harmonised index of consumer prices (HICP).
2. HICP excluding energy, food, alcohol and tobacco.

Source: EUROSTAT; OECD.

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declined to 41 per cent,[6] reflecting: *i*) the reduction of wage pressures as the economy weakened; *ii*) relatively low import prices; and *iii*) the fall of unprocessed food prices from the abnormal levels observed in 2001. The two percentage points rise in the VAT rate led to a mechanical effect from June 2002 onwards[7] estimated at about ½ percentage points as part of the impact was absorbed by a reduction in profit margins, given the weak demand environment. The headline inflation differential with the euro area fell slightly.

By contrast, core inflation continued to rise steadily, reaching 5.1 per cent in autumn 2002,[8] more than 2½ percentage points above the euro area average. The increase in core inflation in a period of slack is worrying and cannot be fully explained by temporary effects associated with the VAT increase. The prices of services, which are particularly sensitive to cost developments, showed an especially strong acceleration in 2002. High inflation in some service sectors probably reflects insufficient product-market competition. In addition, the introduction of the euro may have played a role in the rapid price rises observed in "hotels, cafés and restaurants", in particular.

A very high external debtor position

Export performance improved, despite the erosion of competitiveness

With low unemployment and high wage increases, cost pressures have kept inflation well above the euro area average and above what can be explained by Balassa-Samuelson effects.[9] On balance, given weak productivity growth, the increase in wages has resulted in rising labour costs that have been passed on to producer prices. In 2002, Portugal's cost competitiveness deteriorated for the sixth consecutive year (Figure 5). Nevertheless, and against a deteriorating external environment, Portuguese exporters virtually stabilised market shares in 2001 and are expected to gain some in 2002, interrupting the trend decline observed over earlier years, a cumulative loss of around 10 per cent over 1997-2000. This recent gain in market share reflects in part the redirection of sales to external markets as domestic demand weakened, and profit margins were squeezed. Lower costs based on wage moderation and a higher productivity growth are needed to reverse the steady deterioration of cost competitiveness, especially at a moment where the pick-up of economic activity is likely to be driven largely by on export growth.

Although potentially vulnerable to competitiveness developments, the recent gains in market share may also reflect changes in Portugal's trade specialisation (Table 5). Revealed comparative advantage indicators show that since the early 1990s Portugal's trade has moved away from traditional labour-intensive goods, which tend grow less than average, in favour of machinery and equipment.

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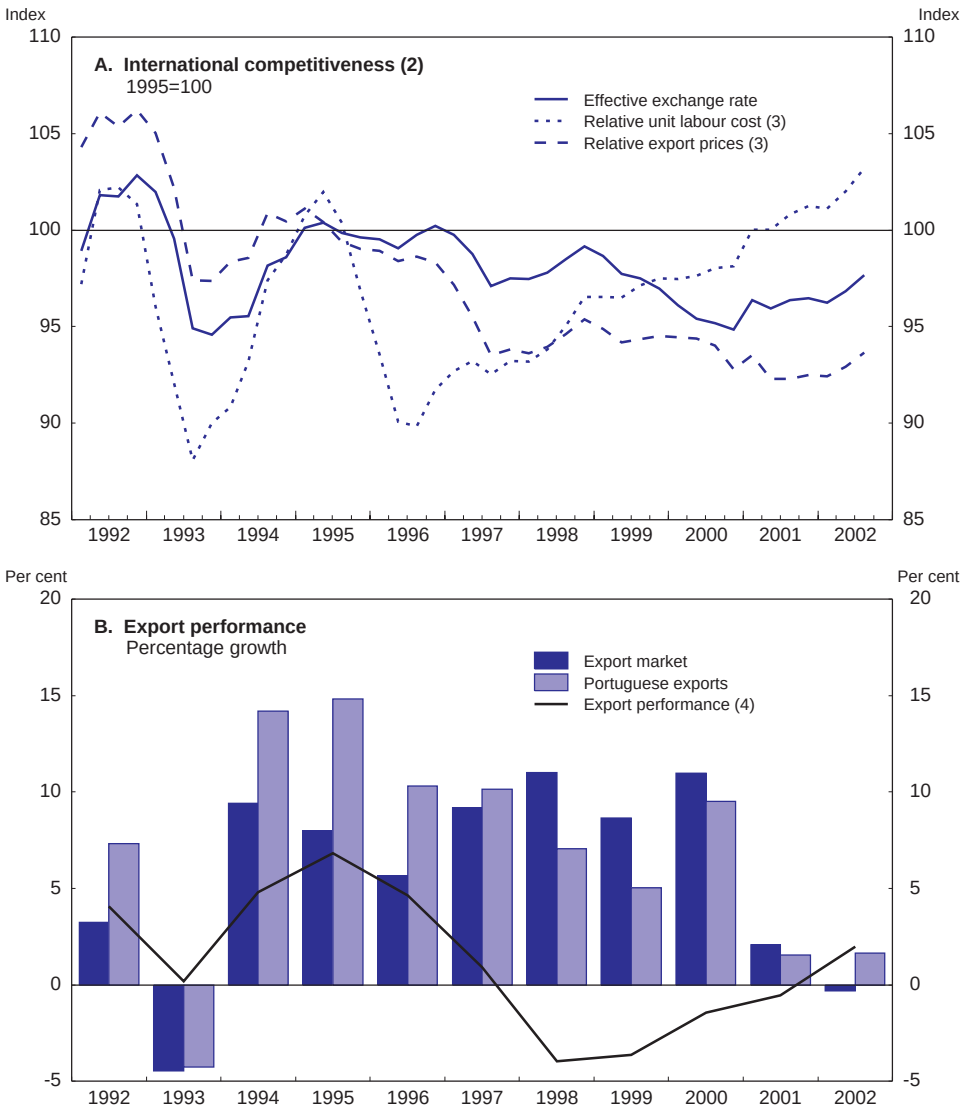
Figure 5. **Competitiveness indicators and export performance**[1]



1. 2002 data are OECD estimates.
2. Indices in common currency. An increase indicates a loss in competitiveness.
3. Manufacturing.
4. Volume index of Portugal's manufactured exports divided by volume index of Portugal's export market.

Source: OECD.

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Table 5. **Change in the trade specialisation**

SITC-3 code	Main comparative advantages	RCA[1]	Cumulative share of exports	Share of exports[2]	RCA[1]	Cumulative share of exports	Share of exports[2]
		1993			2000		
84	Articles of apparel and clothing accessories	17.48	20.33	20.33	9.16	11.60	11.60
85	Footwear	8.77	29.93	9.60	5.23	17.67	6.07
63	Cork and wood manufactures, excluding furniture	4.15	34.47	4.54	3.60	21.86	4.19
66	Non-metallic mineral manufactures, n.e.s.	3.24	39.25	4.78	1.84	25.20	3.34
11	Beverages	2.45	42.35	3.10	1.41	27.45	2.25
25	Pulp and waste paper	2.24	44.69	2.35	2.12	29.73	2.28
65	Textile yarn and related products	1.97	52.50	7.80	2.29	36.64	6.91
77	Electrical machinery, apparatus and appliances, n.e.s.	1.71	59.69	7.20	3.83	46.73	10.09
28	Metalliferous ores and metal scrap	1.45	61.26	1.57	0.60	47.46	0.72
64	Paper and paper manuvactures	0.59	63.70	2.44	0.99	50.37	2.91
79	Other transport equipment	0.57	64.77	1.07	–0.49	51.56	1.19
82	Furniture and parts thereof	0.55	66.00	1.23	0.60	52.98	1.42
69	Manufactures of metals, n.e.s.	0.44	68.38	2.37	0.46	55.67	2.69
71	Power generating machinery and equipment	–0.01	69.88	1.51	–0.92	56.55	0.88
76	Telecommunication and sound recording apparatus	–0.39	72.70	2.81	0.55	60.71	4.17
51	Organic chemicals	–0.70	73.77	1.07	–0.16	61.94	1.23
05	Vegetables and fruits	–0.74	74.81	1.04	–0.46	62.98	1.03
03	Fish, crustaceans, molluscs and preparations thereof	–1.27	76.11	1.30	–1.02	64.16	1.18
74	Other industrial machinery and parts	–1.60	78.17	2.06	–0.28	67.39	3.24
89	Miscellaneous manufactured articles, n.e.s.	–1.94	80.03	1.87	–1.81	69.09	1.70
33	Petroleum, petroleum products and related materials	–4.15	83.41	3.37	–5.95	71.36	2.27
78	Road vehicles	–8.94	88.75	5.35	–0.56	84.87	13.51

n.e.s. = Not elsewhere specified.

1. RCA: Revealed comparative advantage indicator (Xi/X – Mi/M)*100, where Xi(Mi) and X(M) denote respectively Portuguese exports (imports) of product i, and total Portuguese exports (imports).
2. As a percentage of total exports in 1993 and 2000, respectively.

Source: OECD, *Annual Foreign Trade Statistics.*

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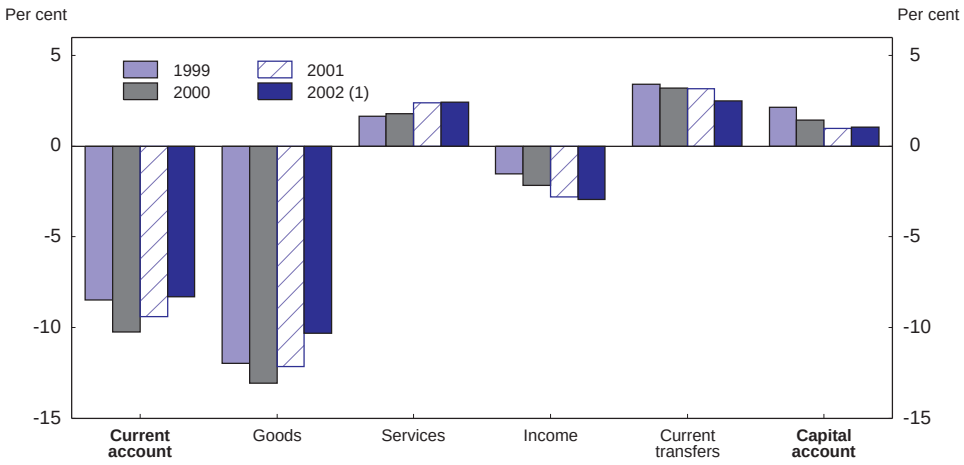
Portuguese exporters of more traditional products lost market share in recent years, revealing difficulty in competing with producers from countries with lower labour costs. There has been a shift in recent years in the structure of exports towards more technology and capital intensive products. The increased specialisation in machinery and transport equipment goods resulted to a large extent from foreign direct investment projects. But this driving force should attenuate, as, since 1998, foreign direct investment in Portugal has been more oriented to non-tradable sectors.

The current account deficit narrowed, but still remains high

From 1996 to 2000, the current account deficit widened to a peak of around 10 per cent of GDP in 2000 (Figure 6). This deterioration resulted mainly from a widening of the trade deficit reflecting the growth differential between domestic and foreign demand and market share losses. Nevertheless the trade deficit declined in 2001 and has continued to narrow in 2002 due to more favourable terms of trade and weaker imports. With domestic demand slackening, the external current account deficit declined to around 8 per cent of GDP in 2002.[10]

Figure 6. **Current and capital account breakdown**

Balances, per cent of GDP



1. January-September annualised.
Source: Bank of Portugal.

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Financial inflows reflected mainly banks' external borrowing

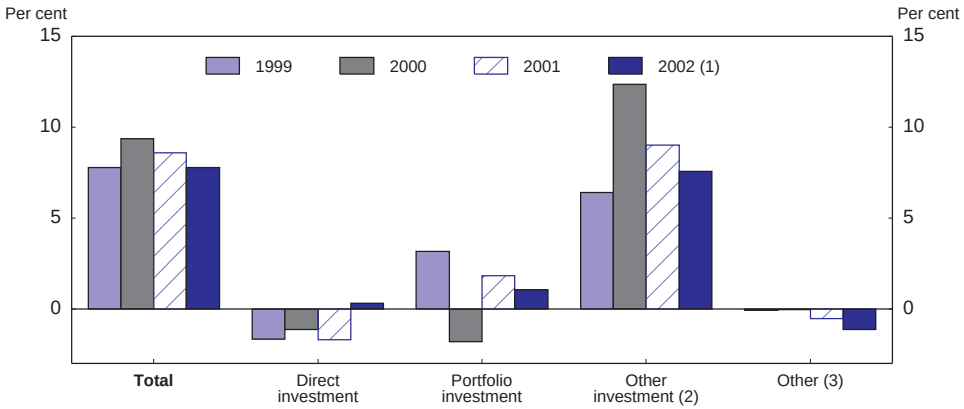
Resident banks' external borrowing has continued to largely finance the external deficit, in line with the trend observed since 1996 (Figure 7). The financing structure of the banking system has changed in recent years, because of a reduced growth of domestic deposits combined with strong growth of internal credit. As customers' deposits did not mirror the rapid credit expansion, the underlying financing gap was filled, in a first phase, through recourse to international money markets. Subsequently, the sources of external financing were progressively diversified, as banks regularly placed international bond issues (subordinated and non-subordinated) through foreign subsidiaries. In the past two years, most of the rise in banks' net external liabilities corresponded to the recording of transfers of funds stemming from the issue of medium and long-term debt securities in international markets by foreign branches of resident banks.

Direct investment inflows remained strong, but their nature has changed

Since 2000, there has been a significant increase in both inflows and outflows of direct investment (Figure 8). This reflects the rise in cross-shareholdings between Portuguese holding companies, mainly in the services and network industries sectors, and foreign companies. Hence, while FDI inflows are still high

Figure 7. **Financial account breakdown**

Balances, per cent of GDP



1. January-September annualised.
2. Includes operations on loans, currency and deposits of the resident banking sector.
3. Includes reserve assets and financial derivatives.

Source: Bank of Portugal.

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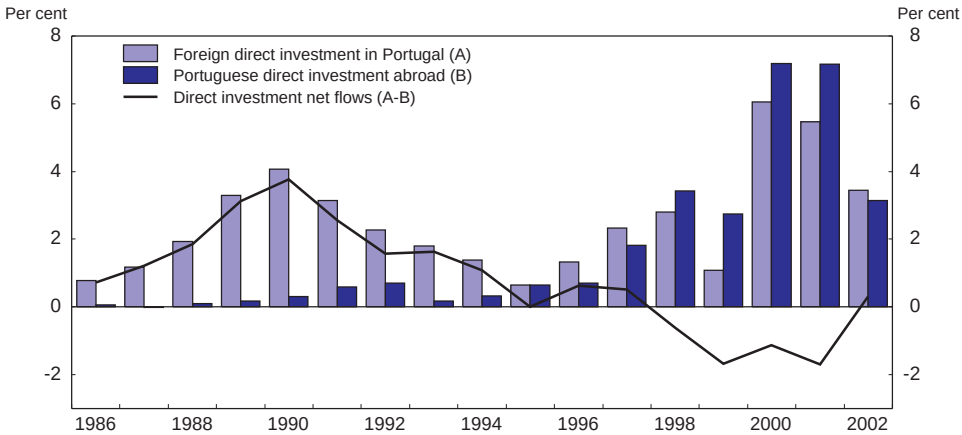
Figure 8. **Direct investment flows**[1]
Balances, per cent of GDP



1. As from 1996, foreign direct investment flows include reinvested earnings and, therefore, are not exactly comparable with data from previous years. For 2002, January-September annualised.
Source: Bank of Portugal.

as a share of GDP, they are less closely associated than earlier with export-oriented industrial plants that contributed to the technological improvement and the diversification of the productive structure, and thereby to the broadening and strengthening of the export basis. The ability of Portugal to attract these kinds of projects has diminished in part because of the deepening of EU economic relations with Central and Eastern European countries. On the other hand, the growing internationalisation of Portuguese companies has resulted in a sharp increase of direct investment abroad.

Forces at work and the short-term outlook

Monetary conditions are becoming somewhat more restrictive…

After early 2002, the fall in real short-term interest rates was offset by the appreciation of the real effective exchange rate,[11] leaving monetary conditions broadly stable, and tighter than in the late 1990s (Figure 9).

Banks' nominal lending rates have remained fairly stable since 2001.[12] Nevertheless, the deceleration of credit to the private sector has continued,[13]

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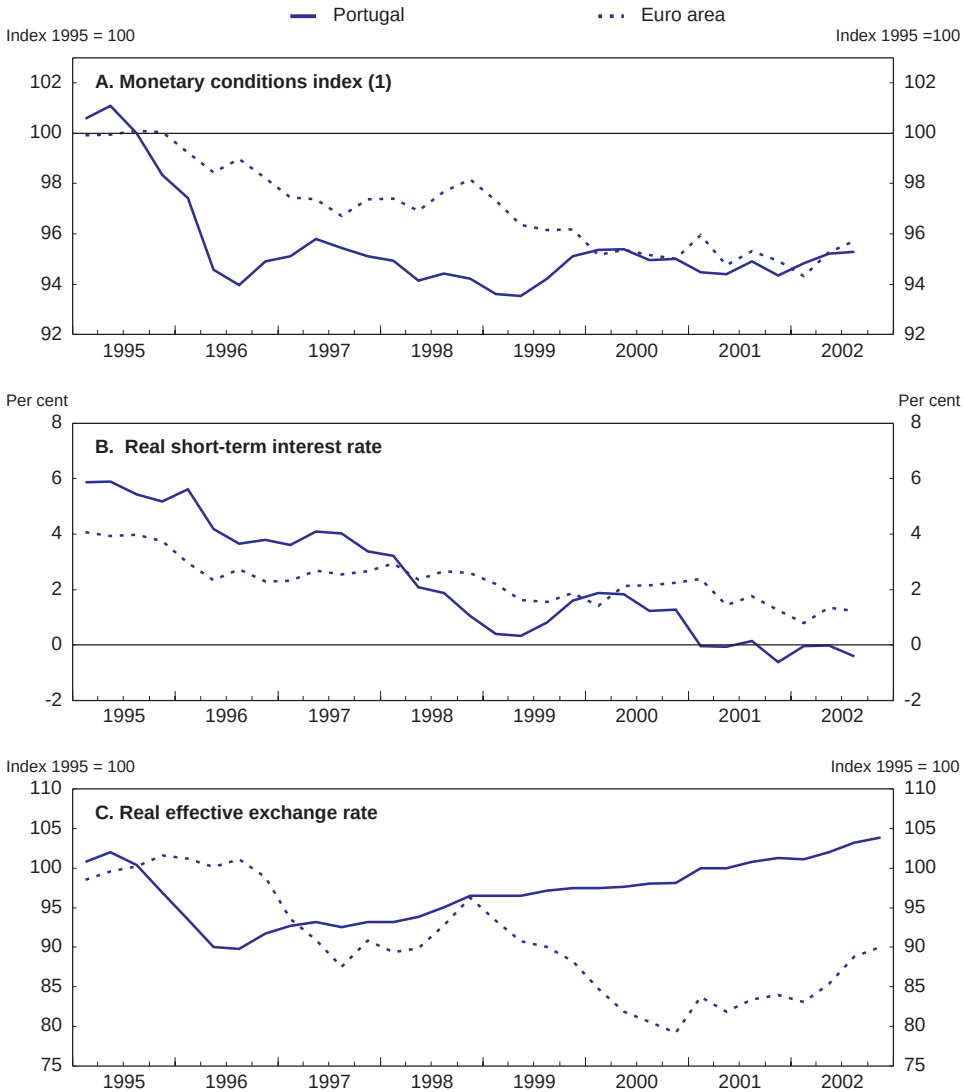
Figure 9. **Monetary conditions**



1. The Monetary Conditions Index is defined as MCI = MCI[t – 1] * (1 + (r – r[t – 1])/100 + (e/e[t – 1] –1)/w), where:
   r is the real short-term interest rate, CPI deflated;
   e the real effective exchange rate, based on unit labour costs in manufacturing;
   1/w the constant weight of the exchange rate relative to the weight of the interest rate (0.40 for Portugal, 0.15 for the euro area).
   A decline of the MCI implies looser monetary conditions.

Source: OECD.

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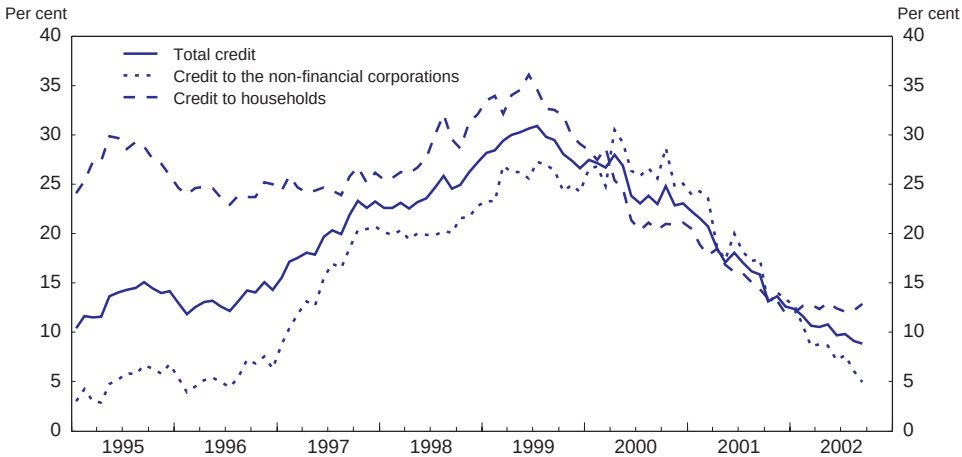
Figure 10. **Credit to the non-financial private sector**[1]
Year-on-year rates of change



1. Adjusted for securitisation.
Source: Bank of Portugal.

reflecting the high indebtedness levels, as well as economic agents' uncertainty concerning future income (Figure 10). Despite this slowdown, the growth of credit aggregates remained higher than the growth of disposable income and nominal GDP, implying a further increase of the indebtedness of households and non-financial corporations, although of much smaller magnitude than in previous years.[14] This indicates that the process of endogenous correction in the private sector has probably not yet been completed.

... and fiscal imbalances are being corrected

In recent years, the stance of fiscal policy has been broadly expansionary,[15] amplifying the stimulus from lower interest rates. In contrast to private expenditure, general government expenditure did not undergo a correction in 2001, and primary current expenditure continued to grow briskly (see Chapter II). The sharp widening of the budget deficit in 2001 led to a series of emergency measures by the new government in 2002, including an increase in the VAT rate and spending cuts. Given the need for further consolidation, the fiscal stance is likely to remain tight in the short term.

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The outlook is for a modest recovery and risks are mainly on the downside

The gradual readjustment process of the Portuguese economy is expected to continue in the short-term. Although showing some recovery after the very low 2002 growth rates, private components of domestic demand should remain subdued in 2003. There is a still need for slower growth of indebtedness. Moreover the economic climate continued to deteriorate for much of 2002 and households' and firms' confidence stood at extremely low levels at end-year (Figure 11). Against this background and the announced fiscal consolidation efforts, foreign demand will need to lead a recovery. Consequently, Portugal is likely to lag behind the rest of the European recovery. The projections discussed below were prepared for the OECD *Economic Outlook* 72, released on 21 November 2002. It was assumed than that the main refinancing rate with euro area would be lowered by ½ percentage point over the following months. The cut was announced by the ECB on 5 December 2002. Growth is expected to pick up only gradually in 2003, driven by stronger exports to the rest of Europe, and reaching around 1½ percent. Although edging up in 2004 when private domestic demand strengthens, GDP growth is likely to remain below 2½ per cent, constrained by severe budget consolidation efforts (Table 6). The output gap is therefore expected to widen further in 2003 and would only stabilise in 2004.

Domestic demand is expected to recover very gradually. Public consumption is likely to remain subdued, in line with the 2003 Budget proposal and the consolidation needs. Private consumption growth is expected to maintain a modest pace in 2003 and pick up somewhat in 2004. Household disposable income growth is expected to remain low, given the expected weak evolution of real wages and employment. Moreover, the household saving ratio may rise further, reflecting the building up of precautionary savings and the use of part of the disposable income to finance the repayments of accumulated debts. In addition, the suspension of subsidised housing loans in 2002 will reinforce the slowdown of house purchases.

In 2003, low growth is projected for gross fixed capital formation, underpinned by some recovery of private investment, which should compensate a further reduction of public investment. The evolution of business investment will continue to be affected by the depressed economic outlook. In the present cyclical position and given the large increase of the capital stock registered in previous years, the recovery is likely to be slow. The still high corporate indebtedness level is also likely to continue to affect business investment growth. The reassessment of major public investment projects is likely to lead to further cutbacks, but the infrastructure projects associated with the European football championship, due to be held in Portugal in 2004, are expected to provide some support to non-residential construction.

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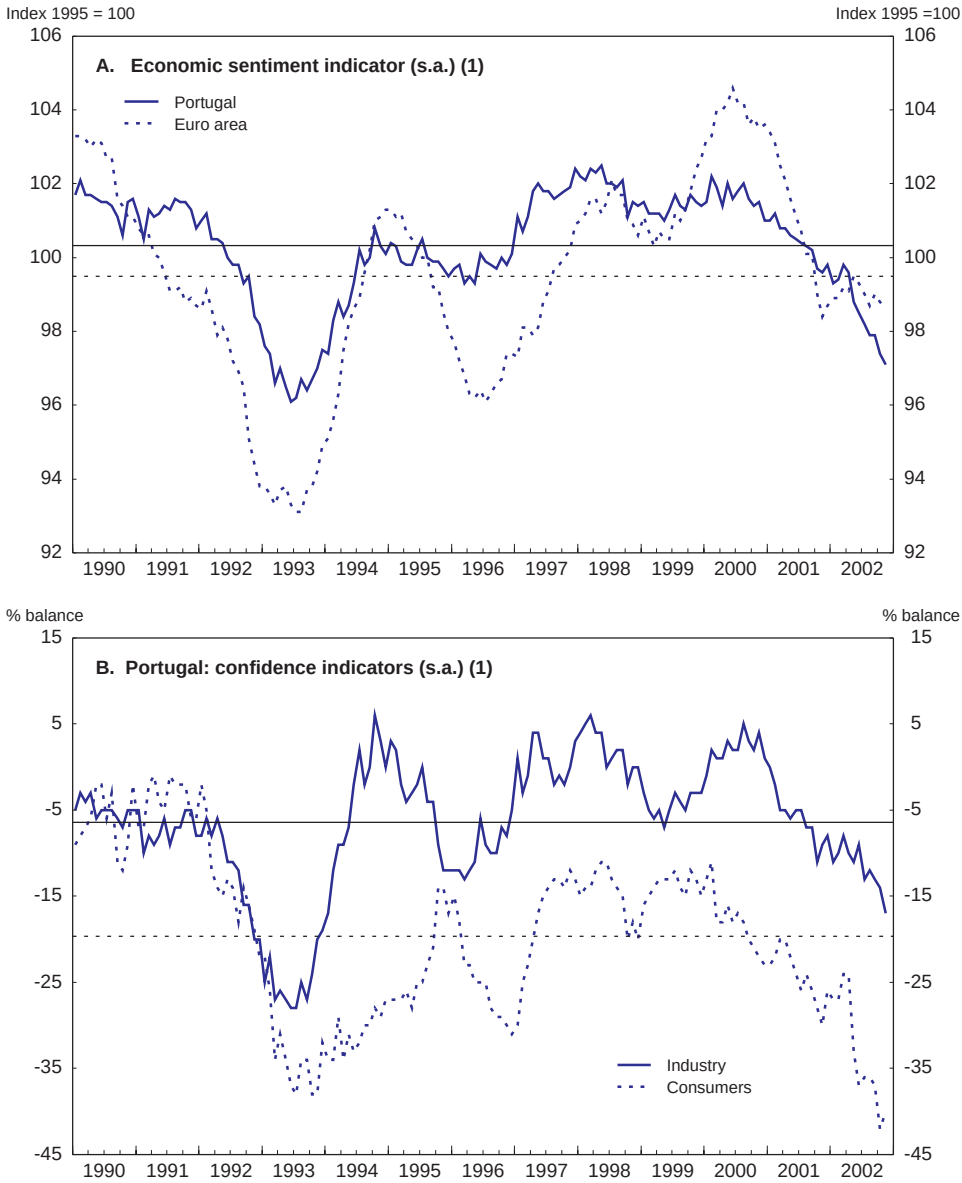
Figure 11. **Economic confidence indicators**



1. The associated horizontal lines show the period average.
Source: European Commission.

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Table 6. **Short-term projections**
Percentage changes

	2001	2002	2003	2004
A. Demand and output				
Private consumption	1.2	0.8	1.0	1.8
Government consumption	2.8	1.1	–0.2	0.0
Gross fixed capital formation	0.0	–2.5	0.3	3.0
Final domestic demand	1.2	0.0	0.6	1.8
Stockbuilding[1]	0.0	0.0	0.0	0.0
Total domestic demand	1.1	0.0	0.6	1.8
Exports of goods and services	1.4	1.1	5.7	8.0
Imports of goods and services	0.3	0.0	2.6	5.7
Net exports of goods and services[1]	0.4	0.4	0.8	0.4
GDP at constant prices	1.6	0.4	1.5	2.3
B. Prices, costs and employment				
GDP deflator	4.7	3.7	2.9	2.6
Private consumption deflator	4.2	3.4	2.8	2.4
Private compensation per employee	6.1	4.0	3.6	3.4
Total employment	1.6	0.6	0.5	1.2
Unemployment rate	4.1	4.7	5.1	5.0
Memorandum items:				
General government balance[2]	–4.2	–3.4	–3.0	–2.4
Cyclically adjusted general government balance[3]	–4.6	–2.9	–2.1	–1.5
Potential output	3.2	2.8	2.6	2.5
Output gap[3]	1.1	–1.3	–2.4	–2.5
Current account[2]	–9.4	–7.8	–6.9	–6.4

1. Changes expressed as a percentage of GDP in the previous period.
2. As a percentage of GDP.
3. As a percentage of potential GDP.

Source: OECD, *Economic Outlook* No. 72.

The main engine of growth seems likely to be exports of goods and services which are expected to pick up in 2003 in line with higher external demand. Despite an acceleration of imports, net exports are expected to contribute positively to GDP growth in the short-term. Given these developments and the projected small gains in the terms of trade, the current account deficit is expected to narrow further in 2003 and 2004.

The deceleration in employment in 2003 is projected to translate only partially into an increase in unemployment, as participation rates stabilise and labour force growth slows. The unemployment rate should rise again in 2003 to levels above its estimated structural rate. Labour market trends are expected to improve in the course of 2004. Inflation should continue to decrease, reflecting the reduction of wage pressures and the sizeable output gap. The high volatility of international oil prices constitutes a risk to the inflation outlook, as the recent

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change of the price setting regime increases the sensitivity of domestic fuel prices.[16]

Risks to the outlook for activity are mainly on the downside, as the Portuguese economy has an increased sensitivity to exogenous economic and financial shocks in this transition period. The timing and scale of the recovery are somewhat uncertain and will largely depend on external developments, most notably on the strength of the recovery in Europe. On the domestic front, the main risks are related to wage developments and to the fiscal stance. If private wage increase do not moderate in the face of weak activity, competitiveness would erode further, putting the export-led recovery at risk. The government's ability to limit public sector pay will also be key, not only because it traditionally serves as a benchmark for private sector wage settlements, but also to help the adjustment in public expenditure and facilitate budgetary consolidation.

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II. Fiscal policy issues

Introduction

A sustained consolidation of public finances is one of the most important challenges facing the authorities. During the latter part of the high growth period, between 1997 and 2000, the pace of fiscal consolidation slowed significantly from previous years, in the face of activity levels that would have permitted an acceleration of the process. The resources derived from the increase in receipts and the fall in interest payments on the public debt were used to increase primary current expenditure instead of reducing the deficit (Figure 12). After 1997, the general government deficit remained virtually unchanged during the following two years on a cyclically adjusted basis (Figure 13). The deficit was then reduced in 2000, though mainly *via* a temporary freeze on investment expenditure during the year[17] and by exceptional proceeds from the sale of mobile phone licences (UMTS) that year. In 2001, the budget forecasts envisaged a slight reduction in the government deficit as a proportion of GDP. In fact, however, growth of primary expenditure was not curbed, while the weakness of economic activity resulted in substantial revenue losses. The general government deficit widened, overshooting the 3 per cent of GDP limit by more than 1 percentage point. The new government formed in April 2002 presented a set of emergency measures designed to bring the deficit back down under this limit. Past experience suggests that it is the fast growth of public spending that explains the lack of progress made. Therefore, if the government deficit is to be steadily reduced on a sustainable basis and brought back to equilibrium, reforms aimed primarily at restraining this buoyancy will have to be implemented. The question of public spending and the attendant recommendations are discussed in detail in Chapter III.

Disappointing results in 2001

The 2001 budget envisaged a reduction in the government deficit by ½ of a percentage point of GDP, to 1.1 per cent of GDP, which implied an appreciable tightening of the fiscal policy stance on a cyclically adjusted basis. Current revenues were projected to increase as a proportion of GDP as a result of tighter tax control. Outlays were scheduled to remain more or less the same as a proportion

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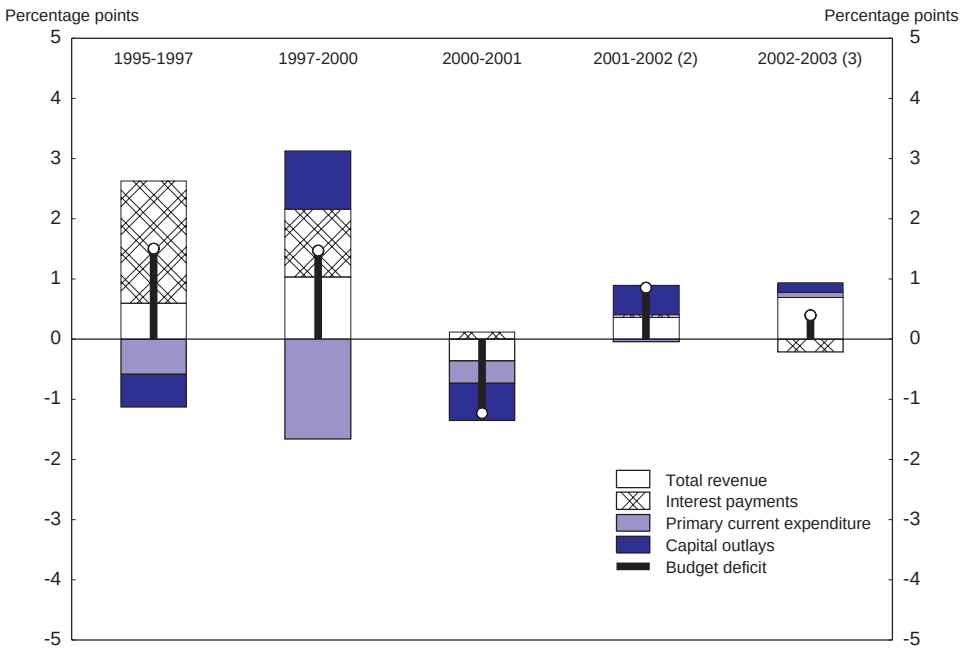
Figure 12. **Contributions to fiscal consolidation 1995-2003**

Cumulative change in budget balance in percentage points of GDP[1]



Note: Data based on former methodology and recording practices for 1995 to 2000 sub-periods. Data based on new methodology and recording practices for sub-periods as from 2000.

1. General government, national accounts basis. A positive change contributes to a deficit reduction.
2. OECD estimates for the 2002 budget outturn.
3. Public accounts basis, 2003 Budget.

Source: Ministry of Finance; OECD.

of GDP, with capital expenditure rising faster than current expenditure. The implementation of the budget ran into considerable difficulties, however: the slowdown in real GDP growth (down to 1.7 per cent, or half the rate envisaged in the budget forecast); mediocre tax receipts, in contrast with the forecasts; and expenditure overruns, especially by the local authorities in a pre-election period. In addition, government capital spending picked up in early 2001, boosted by the start-up of projects cofinanced by the European Union under the heading of the Third Community Support Framework.[18] Restrictive measures had to be adopted in June 2001 in an effort to halt the worsening budget balances but, despite these

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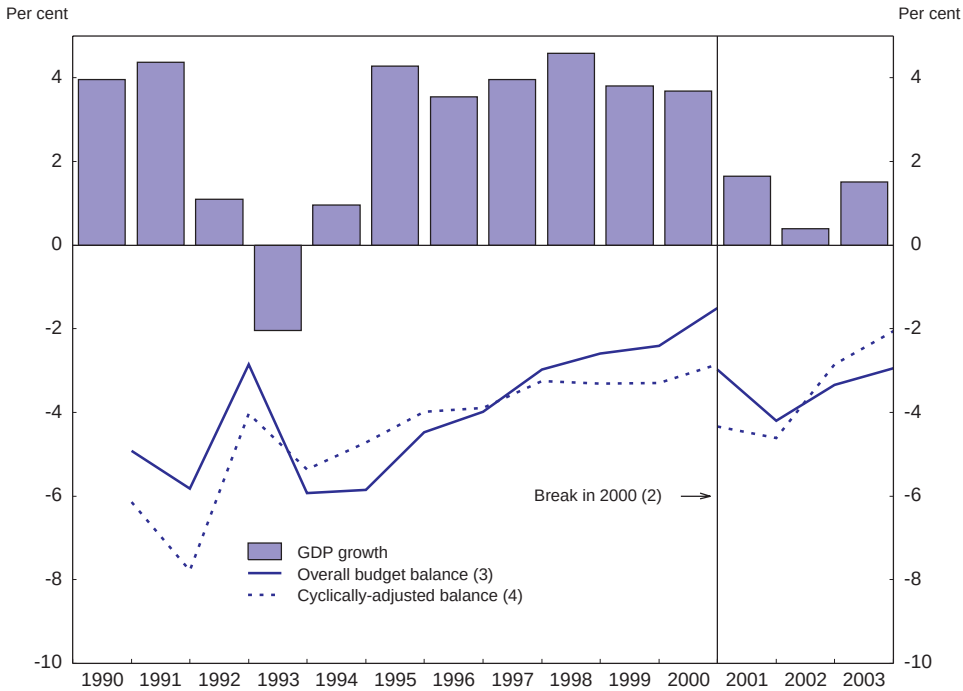
Figure 13. **The fiscal policy stance**[1]



Note: Budget balances are measured at end-year.
1. National accounts basis. 2002 and 2003 are OECD estimates.
2. Break due to changes in methodology and recording practices.
3. As a percentage of GDP.
4. As a percentage of potential output.
Source: OECD.

measures, it became obvious by the end of the year that the general government deficit would be at least twice the budgeted figure.

In April 2002, the European Commission, responsible for overseeing the implementation of the Stability Programmes, presented an estimate of Portugal's 2001 budget deficit (2¾ per cent of GDP, compared with 2¼ per cent in 2000, excluding receipts from the sale of UMTS mobile concessions)[19] which was higher than the estimates made a month earlier in the context of the excessive budget deficit notification. Apart from the impact of further downward revisions to economic growth, other factors were involved, including the authorities' underestimating the shortfall deriving from the reform of direct taxation implemented in 2001 and the mediocre results regarding tax collection. Lastly, the

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upward revision was largely attributable to the local government deficit which proved much bigger than originally estimated, and to the accounting regularisation of capital transfers by the state to public enterprises, which had so far been recorded off-budget.[20] Other regularisation operations that became retroactively applicable in 2002 included, in particular, the change of the recording of taxes on income and wealth, and social contributions from an accrual to a cash basis.

Shortly after coming to power, the new government appointed a Commission to draw up a clear appraisal of the public finance situation. The Commission's report, published in July 2002, indicated a general government deficit of 4.1 per cent of GDP in 2001 (on the "excessive deficit procedure" basis). For the first time since 1983, the primary balance showed a deficit, of –1.1 per cent of GDP. This very substantial upward revision of the deficit reflected more recent data, particularly concerning the autonomous services (the health system) and local governments; changes in the accounting classification of a number of operations: more rigorous recording of tax revenues (henceforth done on a cash basis),[21] recording of the compensatory amounts paid to public enterprises; and recording of a larger proportion of expenditure on a commitments basis (Table 7).

Overall, approximately one-third of the deterioration of the deficit relative to the initial budget stems from methodological changes, but the other two-thirds

Table 7. **Budget balance in 2001**[1]

	Percentage points of GDP in 2001
General government deficit (% of GDP) excluding methodological changes	
March 2002	–2.4
July 2002	–3.1
Methodological and other changes:	
Compensations "due" and unpaid to public transport enterprises	–0.2
Expenses of 2001 to be paid in 2002 budget[2]	–0.3
Neutral recording of EU transfers[3]	+0.1
Taxes and social contributions recorded on cash basis[4]	–0.6
General government deficit (% of GDP) including methodological changes	–4.1

1. Values calculated in accordance with the rules of the Excessive Deficit Procedure. The difference *vis-à-vis* the recording on a National Accounts basis in ESA95 results from considering swap and Forward Rate Agreement (FRA) operations as non-financial operations in the Excessive Deficit Procedure, therefore affecting interest payments and the deficit (€ 89.2 million, *i.e.*, 0.07 per cent of GDP in 2001). The general government deficit obtained on the basis of ESA95 National Accounts is 4.2 per cent of GDP in 2001.
2. Recording expenditure on an annual basis is now done on a wider scale.
3. Recording of EU transfers in the period in which the co-financed spending takes place.
4. Doubtful taxes and social security contributions (due but not received) were previously recorded in revenue; they are now recorded on a cash basis.

Source: Estimates by the Portuguese Committee for the Analysis of Public Accounts released July 2002.

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Table 8. **From the initial 2001 budget to the outturn**[1]

	Initial budget (1)	Estimation (2)	Difference (2 – 1)	*Of which* methodology
Total receipts	44.9	42.1	–2.8	–0.6
Current receipts	42.3	40.3	–2.0	–0.6
Capital receipts	2.6	1.8	–0.8	
Total disbursements	46.0	46.2	0.2	0.4
Current disbursements	39.3	40.4	1.1	0.3
Capital disbursements	6.7	5.8	–0.9	0.1
Overall budget balance	–1.1	–4.1	–3.0	–1.0

1. Values calculated in accordance with the rules of the excessive deficit procedure. The difference *vis-à-vis* the recording on a National Accounts basis in ESA95 results from considering swap and FRA operations as non-financial operations in the excessive deficit procedure, therefore affecting interest payments and the deficit (€ 89.2 million, *i.e.*, 0.07 per cent of GDP in 2001). The general government deficit obtained on the basis of ESA95 National Accounts is 4.2 per cent of GDP in 2001.

Source: Estimates by the Committee for the Analysis of Public Accounts released July 2002 and OECD calculations.

reflect overestimated receipts and underestimated expenditure (Table 8). Whereas the overrun on spending, largely attributable to the wage bill[22] and to spending by local authorities, was contained by means of temporary freezes decided on during the year and mainly affecting investment expenditure, the tax shortfalls due to the slowdown in activity and tax avoidance could not be offset.

On a cyclically adjusted basis, and without taking account of the exceptional UMTS receipts in 2000, the primary balance worsened by ½ of a percentage point between 2000 and 2001. The automatic stabilisers therefore played their role and might have delay the cyclical slowdown, though at the cost of the 3 per cent reference ceiling being overshot. Portugal is not the only European Union country to have experienced difficulties in implementing the Stability Pact. A majority of them did achieve their targets in 2000, when growth was still brisk, but only six were able to do so in 2001. Like Germany and Italy, Portugal failed to achieve its objectives in both 2000 and 2001. The European Commission proposed in February 2002 that Germany and Portugal be given an early warning. As the two countries committed to take measure to contain the deficit below 3 per cent of GDP and balance the accounts by 2004, the Council decided not to follow up the Commission proposal (Box 1). In October 2002, the Commission launched the excessive deficit procedure concerning 2001; Portugal is the first EU member country to be submitted to this procedure. Portugal's situation differs from that of the other small EU countries which, having made very substantial progress in putting their public finances on a sounder structural footing, gave themselves a sufficient margin to allow the automatic stabilisers to take effect in the event of a cyclical slowdown. Having failed to take advantage of the growth years to ensure lasting fiscal consolidation, Portugal saw its cyclically-based gains evaporate with

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Box 1. Early warning and excessive deficit procedure

Twice a year, member countries present their government accounts. If the Commission judges that there is likely to be an excessive deficit or believes that the commitments entered into with regard to the deficit or the debt are not being respected, it draws up a report which triggers the procedure. The Member State may then respond. Next, on the basis of a recommendation addressed by the Commission to the Council, the latter may decide either to pursue or not pursue the procedure (by qualified majority vote), depending on whether it considers the deficit to be "exceptional", *i.e.* resulting from an unusual event outside the control of the Member State concerned or from a severe economic downturn, or "excessive". If the deficit is deemed to be "exceptional", the procedure is halted. If the deficit is considered "excessive", a series of mechanisms are put into operation, deadlines are set for correcting the deficit and sanctions may be imposed. The country may be required to set up a non-interest-bearing deposit with the Commission, and this will be converted into a fine if the excessive deficit has not been corrected after two years. The amount of the fine may not exceed ½ per cent of GDP.

In late January 2002, the European Commission proposed to the Council that Germany and Portugal should receive an early warning so as to avoid triggering the excessive deficit procedure. The Council decided, without voting, to cancel the early warning procedure in light of commitments by Germany and Portugal to take action to keep the deficit below the 3 per cent limit and to balance their accounts before 2004.

In the light of the revisions to the 2001 outturn, and judging that cyclical factors did not explain everything, the Commission decided in mid-October 2002 that Portugal was running an excessive deficit in 2001. Council adopted a similar decision on 5 November 2002. Following the Commission's recommendations, the Council asked Portugal to remedy its excessive deficit situation as rapidly as possible. The Council established a deadline of 31 December 2002 for the Portuguese government to adopt the all the necessary measures needed to end the excessive deficit situation. The Programme for Stability and Growth presented at end-December 2002 includes a serie of measures to be adopted in 2003-06, which are designed to contain expenditures on a durable basis. The Council will issue an opinion on this programme by end February 2003.

the economic slowdown. As a result, it was obliged to implement distinctly restrictive measures in 2002, just when the economy was experiencing virtual stagnation.

The 2002 budget (and supplementary budget): implementation difficulties

The 2002 budget, drawn up in October 2001, envisaged the general government deficit being cut to 1.3 per cent of GDP (*i.e.* 0.4 per cent of GDP down on the estimates made at the time for 2001). It was based on a real GDP growth

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assumption of 1¾-2¼ per cent, which was somewhat optimistic, given the uncertainty surrounding the international environment following the events of 11 September.[23] Social expenditure and investment were given priority. The measures taken to combat tax evasion and avoidance were supposed to widen the tax base, thereby offsetting the effect on revenues of the rate cuts decided on as part of the 2000 tax reform (cut in the rate of corporation tax and revision of tax schedules). As in 2001, it soon became apparent that the budget forecasts as a whole were unrealistic.

In order to cope with the rapid deterioration in public finances in early 2002, the new government passed a supplementary budget in May 2002 (Box 2) and took executive measures, the object of which was to keep the general

Box 2. The 2002 Supplementary Budget

In May 2002 the incoming government approved a supplementary budget for 2002 that should slow public expenditure growth over the next two years

Revision of the number of public agencies and institutes. As a preliminary measure, 41 government agencies with similar functions that create a duplication of services will be suppressed, 31 merged and 19 restructured. In a second phase, cost-benefit analysis will apply to the remaining departments and agencies. In a third phase, a restructuring will take place, leading to a reduction in the number of civil servants and a reallocation of resources.

Measures to reduce the wage bill

- The renewal of individual contracts and fixed term contracts is frozen. Any new hiring will henceforth require special permission by the Ministry of Finance. This is expected to translate into a reduction of staff numbers.
- Each ministry will assess its needs in terms of human resources. Redundant civil servants will be listed in a special list and can be called by other ministries that are understaffed. This measure is key for the re-allocation of redundant employees from the extinction and merging of public agencies and institutes.
- Careers restructuring is frozen until further notice.

Restrictions on the debt of local authorities. Loans that would result in an increase of the annual liquid debt of local authorities are forbidden with the exception of the loans destined to the Special Recommendation Programme and to projects co-financed by the Community Support Framework III.

An *annual limit of 2 per cent is imposed on the expenditure growth of Autonomous Funds and Services*, with the exception of National Health Service departments, the investments co-financed by the EU and of basic salaries.

Finally, *subsidies to new home loans* are ended, starting in October 2002, with most of the impact being felt in 2004.

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government deficit under 3 per cent. The supplementary budget goes further than the implementation of the previous two years' safeguard clauses.

On the expenditure side, while the supplementary budget contains measures that will have an immediate but temporary effect, it also contains other measures that should have a substantial impact on future spending. Amongst the measures that can only be applied temporarily, there is the freeze on recruitment and the deferral of certain investment projects.[24] It is in fact desirable that these emergency measures be terminated as soon as the fiscal situation permits, given Portugal's need for investment in infrastructure and recruitment of young staff in government. Other measures, on the other hand, that ought to have an in-depth impact include the shutting down of government agencies and the introduction of civil servant mobility within government departments. Other reforms approved after the supplementary budget have similar aims (see Chapter III for more details). In the hospital sector, one-third of hospitals are to be converted into public corporations. The reform of public administration which is also under way ought to make budget processes more transparent. Lastly, related reforms could make it easier to control expenditure. Central government, for example, which has traditionally had little influence over local authorities, now has a new control instrument at its disposal, with the ceiling on local authority borrowing incorporated in the supplementary budget and confirmed by the more recently approved Law of Budgetary Stability.[25] This package of measures should contribute to a sustainable improvement in the medium-term structural balance.

The authorities were aware that the measures taken on the expenditure side would not suffice to bring the deficit down to less than 3 per cent of GDP in 2002, and have raised the standard VAT rate, from 17 to 19 per cent, close to the EU average, but well above the rate applied in Spain. Also, tax controls were to be stepped up and the recovery of arrears accelerated. Last, the tax reductions announced in the electoral programme have been put off until a later date.

It would seem, however, that this package of measures might not suffice to bring the budget deficit down below 3 per cent of GDP in 2002, since the measures contained in the supplementary budget only have an impact on the second part of the year. Furthermore, there is a great deal of inertia in the wage bill and a sizeable reduction looks difficult to achieve, though the measures taken could slow down the momentum observed in recent years. Estimates of the State budget execution from January till October show no slippage in expenditure growth control. Above all, demand has been very weak. According to official estimates, the 2 point rise in the rate of VAT ought to have just about offset the shortfall in tax receipts resulting from the slowdown in growth. But according to the latest OECD estimates, domestic demand

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has almost stagnated this year in real terms, whereas in the supplementary budget it was still expected to grow by between 0.5 and 1.1 per cent. For indirect taxation alone, this will probably represent an additional shortfall of some 0.1 per cent of GDP, and for current receipts overall, a shortfall ranging between 0.2 and 0.4 per cent of GDP. Additional corrective measures will probably be needed to contain the deficit. The government could proceed with a number of *ad hoc* measures (such as asset sales) at the end of the year to ensure that the budget deficit remains below the 3 per cent limit. Since no specific measure of this sort had been announced by end-October, the OECD based its forecasts[26] on the assumption that the target would be overshot, with a general government deficit above 3 per cent of GDP in 2002. As part of the excessive deficit procedure, the Commission expressed similar worries in mid-October regarding the possibility of bringing the deficit down under 3 per cent in 2002. On November 5th, the Council confirmed end-year 2002 as the deadline for ending the excessive deficit situation (see Box 1). Some new expenditure control measures were announced soon thereafter, but it is too early to assess their potential impact. Official estimates taking into account more recent one-off measures (sale of the fixed telephone network, highway tolls and a tax amnesty) put the deficit just below 3 per cent of GDP in 2002.

The 2003 budget: a restrictive stance

The draft 2003 budget submitted to Parliament in early October is designed to reduce the general government deficit to 2.4 per cent of GDP, *i.e.* approximately ½ of a point less than the corresponding deficit for 2002 (Table 9). It is planned to cut general government operating outlays and to limit the growth of grants to the social security sub-sectors to 8 per cent (a rate of growth nevertheless higher than that of nominal GDP). Expenditure on public works is spared, in particular investment financed by EU transfers. The budget estimate is based on the assumption that there will be no past debt to regularise.[27] Also, local authority borrowing is still capped, but certain strategic projects could be authorised. According to OECD estimates on a cyclically-adjusted basis, the budget deficit is set to fall by 0.8 percentage point of GDP, implying a restrictive fiscal policy stance in 2003.[28]

The budgetary objective of a ½ percentage point of GDP cut in the deficit for 2003 would seem to be achievable, bearing in mind the measures taken to improve budgetary processes (2001 Framework Law) and the in-depth reforms launched in 2002, especially in the civil service, which should begin to produce results. Substantial savings therefore appear possible. However, other components of expenditure will continue to exert upward pressure (pension systems, for example, see Chapter IV). In view of the need to reduce the deficit in a fairly short

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Table 9. **General government financial accounts**[1]
Per cent of GDP

	2000[2]	2001[2]	2002[3]	2003 Budget
Current receipts	**40.9**	**40.2**	**41.7**	**41.8**
Direct taxes	10.4	9.8	9.8	9.2
Social security contributions	11.8	12.0	12.1	12.0
Indirect taxes	14.5	14.4	14.9	14.9
Other current receipts	4.2	4.0	4.9	5.7
Current disbursements	**40.2**	**40.1**	**41.4**	**41.4**
Intermediate consumption	4.4	4.1	4.2	4.0
Social security	14.1	14.2	14.6	14.6
Interest on public debt	3.2	3.1	3.1	3.2
Other current disbursements	18.5	18.7	19.5	19.5
Current balance	**0.7**	**0.1**	**0.3**	**0.4**
Capital receipts	**1.5**	**1.9**	**2.5**	**2.4**
Capital outlays	**5.1**	**6.1**	**5.7**	**5.2**
Gross investment	3.9	4.1	3.1	3.7
Capital transfers	1.2	2.0	2.0	1.5
Overall budget balance	**–2.9**	**–4.1**	**–2.8**	**–2.4**
Memorandum items:				
Primary balance[4]	0.3	–1.1	0.2	0.8
Primary current spending[4]	37.0	37.0	38.3	38.2

1. National accounts basis, new methodology; in accordance with excessive deficit procedure rules for 2001.
2. Estimate.
3. Provisional data.
4. Excluding "interest on public debt".

Source: Ministry of Finance.

space of time, the government has again had to postpone the reductions in tax rates; the reduction of the corporate income tax is now scheduled to be implemented gradually as from 2004.

Medium-term budget strategy

The Stability Programme

The December 2002 Stability Programme targets a 0.4 per cent general government budget deficit for 2006 (Figure 14). The structural deficit will be reduced by ½ percentage point of GDP every year from 2003 to 2006 to reach around ½ per cent of GDP in 2006. This consolidation path is in line with the European Commission's proposal of September 2002: following the worsening public finance situation

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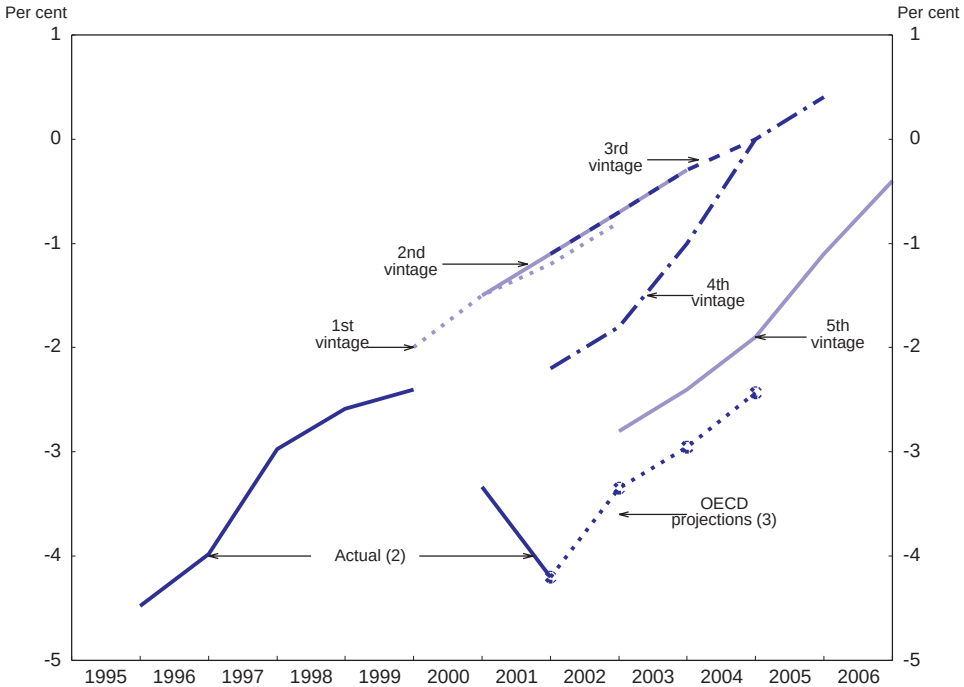
Figure 14. **Fiscal targets**[1]
General government balance as a per cent of GDP



1. The various vintages of the Stability and Growth Programmes were released over the following periods: 1st: December 1998; 2nd: February 2000; 3rd: January 2001; 4th: December 2001; 5th: December 2002.
2. Excluding UMTS in 2000.
3. According to the assumptions made in *Economic Outlook* No. 72, December 2002.

Source: Eurostat; OECD.

in a number of EU countries, including three of the main ones, the Commission proposed that the terms of the Stability Pact be adjusted in order to give EU countries a certain amount of room for manoeuvre. In return, the countries have to commit to reduce their structural deficits by approximately ½ of a percentage point of GDP per year. The reduction of Portugal's deficit will result from structural reforms recently implemented or announced in the areas of public administration management, social security as well as education and health.[29] If the measures are effective in containing spending, then the Programme foresees a gradual reduction in the corporate income tax rate from 30 per cent at present to 20 per cent by 2006. If on the other hand, difficulties are encountered in meeting the budget targets, the corporate income tax cut will have to be postponed further.

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Public debt trends and management

The debt/GDP ratio has stopped declining

After falling steadily for five years, Portugal's gross public debt grew by 2 percentage points in 2001, reaching 55.5 per cent of GDP at end-December according to recent official estimates. Apart from the rise in the general government borrowing requirement, this increase is a reflection of major regularisation operations – about 1½ percentage points of GDP – mainly reflecting, as in previous years, the state absorbing the debt of the health sector. Moreover, the privatisation receipts earmarked for debt redemption fell by half between 2000 and 2001 (to 0.3 per cent of GDP). For 2002, 0.5 per cent of GDP of privatisation receipts are to be used to reduce public debt. However, further regularisation operations in the health sector and net acquisitions of financial assets should add to the borrowing requirements, pushing up the debt ratio.[30] According to OECD estimates, overall gross public debt would increase to almost 60 per cent of GDP.

In order to bring public debt gradually down to around 52.7 per cent of GDP, as targeted in the Stability Programme, it would be necessary to generate primary surpluses averaging 2 per cent of GDP from 2003 to 2006, unless there are pleasant surprises, such as larger privatisation receipts, brisker growth or lower interest rates than anticipated (Figure 15).

Debt management

Since the mid-90s the government has been pursuing a policy aimed at minimising the cost and risks associated with public debt. In the new context of monetary union, the object of this strategy is to increase the attractiveness of Portuguese bonds *vis-à-vis* other European sovereign bonds and, in particular, to increase their liquidity. It would in fact appear that the interest rate spread between Portuguese and German Treasury bonds with the same maturity stems in large part from a liquidity premium. The Portuguese authorities have therefore continued to reduce the range of debt instruments available, concentrating on 5 to 10-year, fixed-rate Treasury bonds denominated in euros. The latter accounted for almost two-thirds of public debt in May 2002, compared with less than 40 per cent in 1998 (Figure 16). With financing policy focused on liquidity, debt management has resorted more strongly to the derivatives markets in order to adjust the duration of primary debt portfolio to cost and risk minimisation goals. Therefore, the number of derivative transactions increased fivefold in 2001 so that at the year-end the derivatives portfolio included outstanding contracts with a nominal value of € 10.8 billion[31]. Lastly, the government also continued to modernise the infrastructure of the primary and secondary debt markets with the object of improving efficiency and transparency and thus attracting foreign investors.

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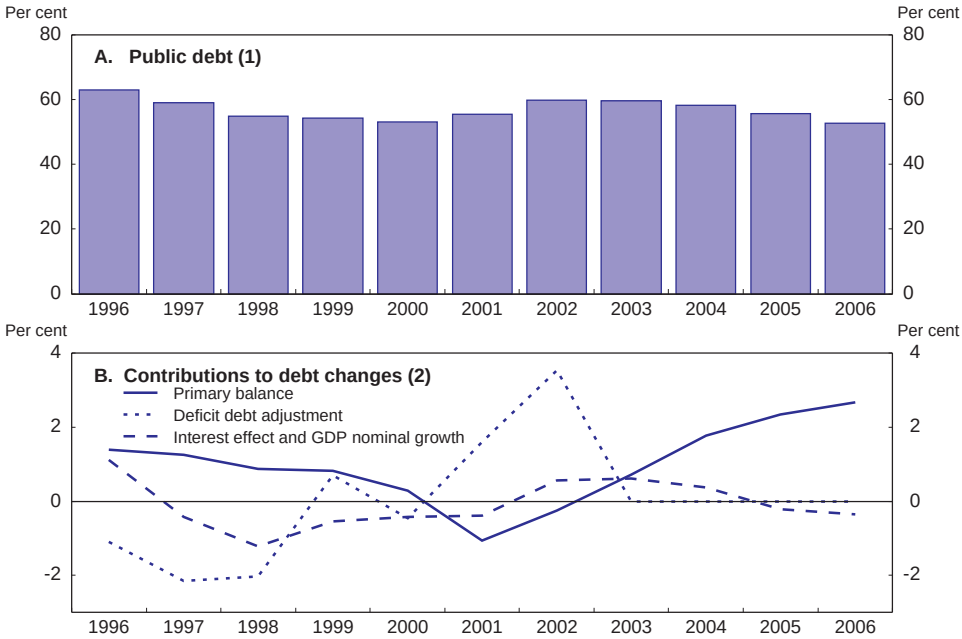
Figure 15. **Public debt scenario**
Per cent of GDP



1. According to the Stability and Growth Programme, December 2002, the debt/GDP ratio will be 52.6 per cent in 2006.
2. The primary surplus was determined from the following basic formula for debt dynamics:
   (D/Y)[t] – (D/Y)[t – 1] = (P/Y)[t] + (r[t] – g[t])/ (1 + g[t]) * (D/Y)[t – 1] + O[t] where:
   (P/Y)[t] = government primary balance per cent of GDP at time t;
   r[t] = nominal interest rate on government debt at time t;
   g[t] = nominal growth rate of the economy at time t;
   O[t] = Other factors (deficit debt adjustment at time t.
   The underlying growth and interest rate assumptions are based for the period 2002-2004 on *Economic Outlook* No. 72, and for 2005-2006 on the Stability and Growth Programme, December 2002.

Source: OECD.

The above strategy was successful in both reducing the cost of public debt and making the budget less sensitive to interest and exchange rate shocks. The implicit interest rate on the debt fell in 2001, reflecting the progress made in public debt management and a reduction in the rate spread *vis-à-vis* Germany (of some 20 basis points) following the fall in the liquidity premium on Portuguese securities. In addition, the *average residual maturity* of Portuguese debt remained broadly stable, around 4.8 years. Recourse to swap operations minimised exposure

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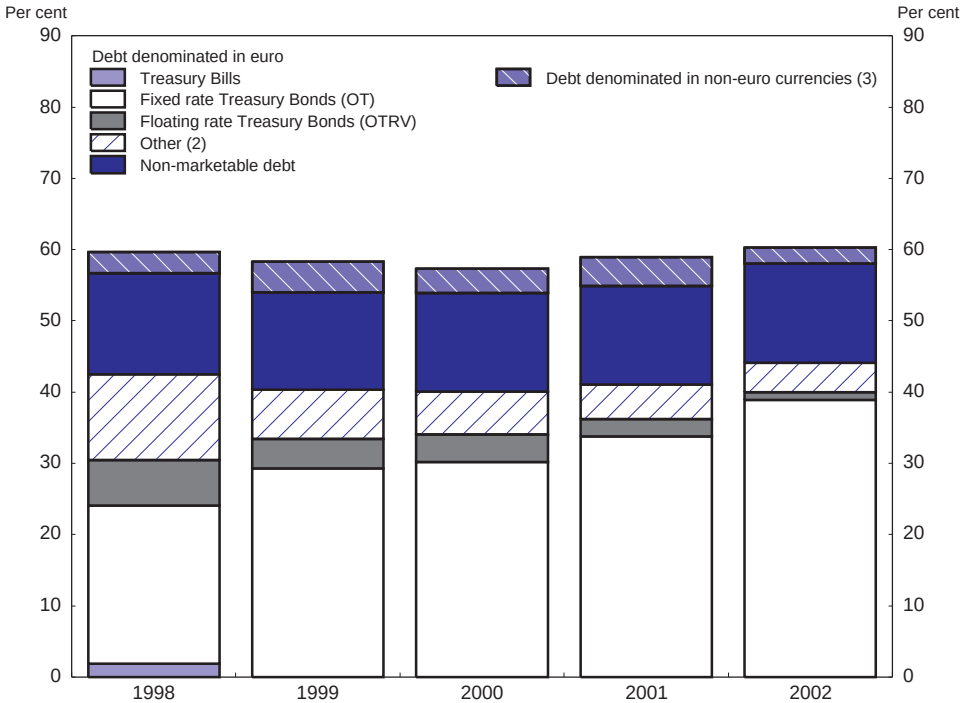
Figure 16. **Trends in the debt structure, 1998-2002**
Stocks outstanding, end of period[1]
Per cent of GDP



1. September data for 2002.
2. Including Euro-Commercial Paper (ECP), Euro and Global bonds, and Medium Term Notes (MTN).
3. Including marketable debt (ECP and MTN) and non-marketable debt.

Source: Portuguese Government Debt Agency.

to the exchange rate risk by increasing euro-denominated debt to 99 per cent of total debt in 2001.

Progress with tax reform

Some of the measures proposed or envisaged in late 2000 have since been approved and implemented, while others were postponed or abandoned. Box 3 summarises all the tax policy initiatives, indicating which ones have actually come into force.

A number of additional measures introduced in 2002 in the framework of the supplementary budget are designed to help combat tax avoidance and evasion:

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Box 3. Tax policy initiatives

The 2001 Budget Law

The 2001 budget marks the first stage of the reform and included the following provisions:

- An increase in family tax benefits with the main purpose of improving family protection and fostering savings and housing.
- Changes in the income schedule with a reduction of the tax rate for the lowest brackets.

Both were approved and implemented.

The December 2000 Tax Reform Act

Personal income tax (IRS)

The following measures were approved and implemented:

- In order to simplify the system, incomes derived from business activities and independent personal services are now assessed under a single income category and subject to the same assessment regime. Agricultural income is excluded from taxation when the yearly gross amount does not exceed five times the national minimum wage.
- Independent workers and business activities with annual sales up to 149 639.37 euros and/or other income up to 99 759.58 euros are subject to a simplified IRS tax regime – a minimum taxable income – based on coefficients fixed by law. A floor equivalent to the annual minimum wage, in the case of sales, or of 50 per cent of that amount in the case of other income, applies. The new regime applies automatically unless people opt out.
- Tax auditing is facilitated by the possibility of assessing taxable income against an income presumption deduced from wealth indications: indirect evaluation in case of non-submission of the tax return or where the declared income and lifestyle appear incompatible (*i.e.* declared income is less than 50 per cent of deemed income, which is estimated with reference to purchased assets such as property, cars, boats, planes). In these cases the taxpayer is allowed to produce substantiated evidence.

The new regime for capital gains is not being introduced.

Corporate income tax (IRC)

The principal elements of the corporation tax reform were approved and implemented:

- The corporate income tax rate was reduced to 30 per cent as of 2002.
- A simplified tax regime for the determination of taxable profits under IRC similar to the one that applies to individuals will be available as an option for small businesses with yearly turnover not exceeding 149 639.37 euros. Here the simplified regime is the exception.

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Box 3. **Tax policy initiatives** (*cont.*)

– Introduction of a number of rules and methods for determining prices for tax purposes in inter-company transactions (transfer pricing).

– Introduction of a number of anti-avoidance rules concerning transactions with non-resident entities subject to "privileged tax regimes".

– Change from a consolidated tax regime to an integrated tax regime as far as the method of tax assessment for company groups is concerned: taxable profits will be the sum of net profits (or losses) realised by each company within the same group.

– Creation of an Investment Tax Credit (those previously in force having ceased) together with a legislative authorisation in order to encourage investments in R&D.

The proposed regime for re-invested capital gains of holding companies (SGPS) has been abandoned.

Free Zone of Madeira (and the Azores)

The following measures were approved and implemented:

– Greater precision and accuracy as far as it concerns the definition of *non-residents* as well as operations carried out by entities operating in Madeira Free Zone with non-residents. Certification of eligibility as non-resident is required.

The proposed gradual introduction of taxation of financial activities *via* a moderate rate has been abandoned. New licences to financial activities in the Free Zone are not granted anymore.

Bank secrecy – all measures listed below were approved

– Access to any information protected by professional secrecy or bank secrecy will still require legal authorisation. The novelty is that there will be no need for authorisation and no suspensive measure will take place in case of the taxpayer's disagreement if the taxpayer benefits from tax incentives or he/she refuses to present documents supporting accounting records. However, the taxpayer will be entitled to *judicial appeal* with *suspensive effect* when the tax administration requires lifting bank secrecy as a consequence of one of the following:

- it is not possible to substantiate and have a direct appraisal of taxable income;
- income subject to IRS is incongruent with income standards;
- there is circumstantial evidence of malicious fraud (*dolus*), namely fake invoices ;
- there is a need to give proof of application of public grants or aids.

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Box 3. **Tax policy initiatives** (*cont.*)

Announced measures which were not introduced

Taxation of immovable property

Elimination of the municipal tax on transfer of properties (sisa); elimination of the gift and inheritance tax. Re-examination of the local tax on real estate (CA). These measures have not been introduced.

Energy and vehicle taxation

Assessment of measures concerning the energy sector in view of harmonisation with the EU directives. Replacement of the present tax on motor vehicles (IA) (effectively levied on the buyers of new vehicles) with two new taxes: a "registration tax" (levied at purchase – similar to old IA but with lower rates) and an annual "circulation tax" (levied on old and new vehicles). These measures have not been introduced.

consumers are encouraged to ask for bills – for repairs for example – because they will, within certain limits, be able to enjoy tax rebates on VAT. Tax rebates will be awarded only to taxpayers who do not have tax and social security contributions arrears. Mechanisms are to be brought in to prevent residents and non-residents from evading tax on interest earnings.[32] Also, the taxation of oil products was changed and excise taxes are not anymore systematically adjusted to offset pre-tax price variations. The standard rate of VAT was raised by 2 points. It is planned to reduce the rate once fiscal consolidation has been achieved. Tax incentives in favour of saving for housing and pension funds are maintained.

Amongst the initiatives not approved, more will need to be done to boost local government revenue. In particular, national registers of property values (land and buildings) need to be updated. Even if the receipts to be expected from a reform of property tax are small, the main objective is to improve the equity of the system. However, a smooth transition to the new system of taxation of property is desirable in order to avoid disruptive effects. The vehicle tax also needs to be looked at again because it does not satisfy environmental considerations (see last part of Chapter IV).

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III. The effectiveness of public expenditure in Portugal

Introduction and overview of issues

Over the past decade, OECD *Economic Surveys* have made frequent reference to problems of public expenditure control in Portugal. General government spending relative to GDP is still lower than the EU average. But a persistent reliance on contingency measures to meet planned spending objectives – often associated with emergency supplementary budgets – has been symptomatic of fundamental weaknesses in the budget process. Recent *Surveys* have identified unplanned shifts in the composition of public spending as a result of such short-term expedients, which have usually been based on intra-year spending freezes falling disproportionately on capital outlays. The overshooting of the general government deficit in 2001, and the need to create a sustainable budget balance path, has made it all the more critical to establish a viable medium-term profile for spending. This, in turn, requires substantial improvements to public expenditure management systems, so as to provide the authorities with the tools they need for effective planning and control.

Against this background, the chapter begins with an examination of the trends in aggregate spending and the evolution of public spending patterns. It then discusses the main budget-management factors behind the questionable record of Portuguese fiscal control and the recent measures taken to correct some of the evident defects in the budget process. This is followed by an assessment of the major efficiency aspects of public service provision: deficiencies in human resource management, lack of competition in the production of public goods in areas where it has proved effective in other countries, and difficulties between central and local governments' co-ordination. The subsequent section focuses on the outcomes achieved, for the costs involved, in major sectors: health, education and social spending. The policy challenges and recommendations arising from this review are then identified.

Pressures on public spending

Primary current spending has been rising rapidly

At 42 per cent of GDP, Portugal's government spending ratio was below the EU average in 2001 (Figure 17, panel A), though when compared with economies

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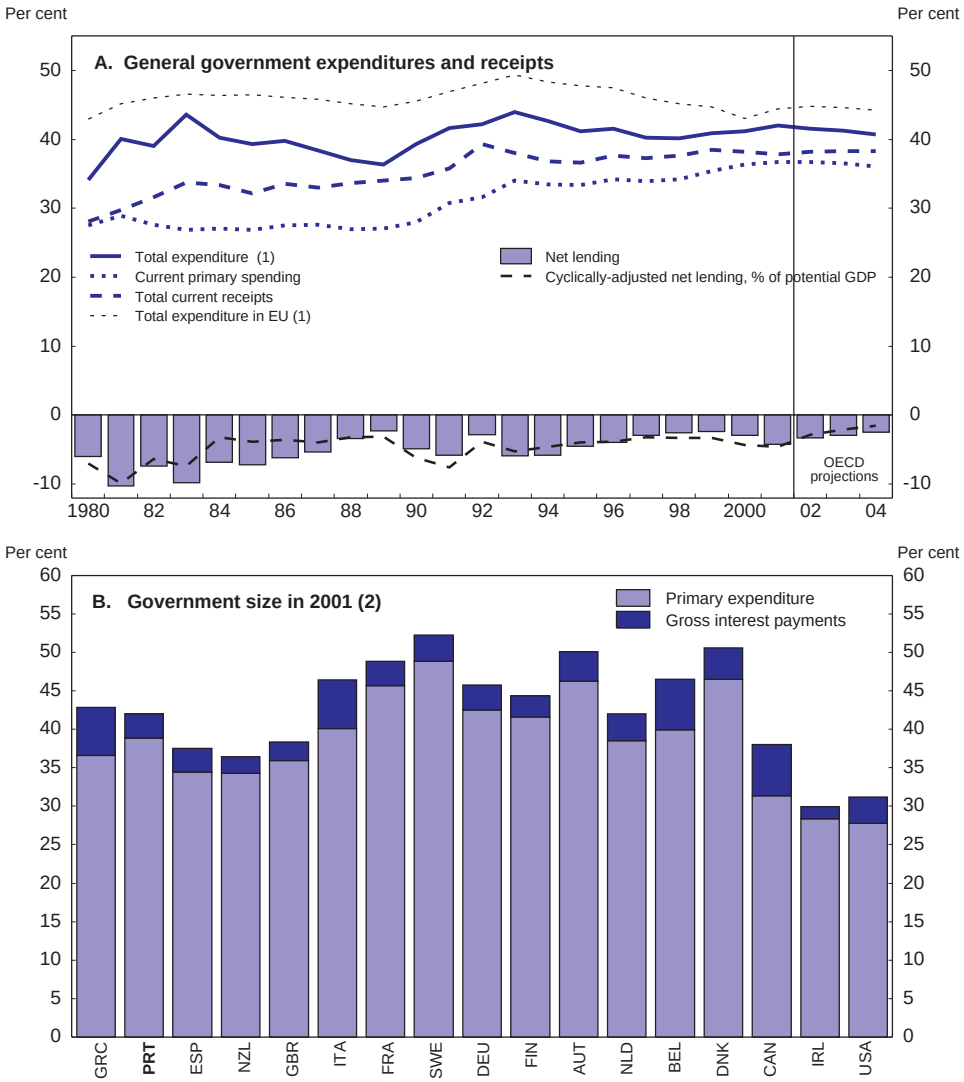
Figure 17. **Government spending**

Per cent of GDP



1. Total spending is defined as current outlays plus net capital outlays. In 2000 one off revenues from the sale of mobile telephone licenses are recorded as negative capital outlays, amounting to 0.3 per cent of GDP.
2. Share of total government expenditure in GDP. Countries are ranked according to GDP per capita at constant prices and in constant PPPs, from the lowest level to the highest.

Source: OECD, *Economic Outlook* No. 72, December 2002.

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of similar income levels (*e.g.* Spain and New Zealand) it appears to be a relatively high-spending country (panel B).[33] Primary current spending (current goods and services plus transfers) is particularly high, at over 36.8 per cent of GDP. Furthermore, Portugal is exceptional among EU members as to the extent of the rise in primary spending during the latter half of the 1990s (equivalent to 3 percentage points of GDP). The fast growth of tax receipts and the decline in interest payments allowed Portugal to reduce the deficit and meet the criteria for participating in the third phase of the European Monetary Union. Nevertheless, the pace of deficit reduction slowed considerably after 1997, despite the strong economic expansion. Over the period 1997-2000 the gains from higher revenues and lower interest payments – equivalent to accumulative 4.4 percentage points of GDP over three years – were mostly offset by higher primary current spending.

The rise in government primary consumption (Figure 18) has been the result of both cost and input factors. Over the past six or so years the ratio of the wage bill to GDP has increased significantly, reaching over 15 per cent, nearly half the primary current spending. The upward trend in payroll expenditures is explained by the steady increase in both public sector employment and real wages, as well as growing pressures from the civil servants' social security system (see section below):

- Public sector employment increased at an average of 3.3 per cent per year between 1995 and 2001. This was almost twice the growth rate of total economy-wide employment, so that in 2000 the share of government in total employment rose above the OECD and EU averages. Most new entrants joined the public administration and the Ministries of Health or Education, which have experienced a persistent increase in staff (see below). In recent years, the creation of several autonomous services and institutes has also contributed to the hiring expansion, and legal obstacles to redeploying public servants also help explain part of the trend.

- More importantly, the average compensation of public sector employees increased by over 20 per cent in real terms in the same period, *vis-à-vis* 14 per cent in the private sector. This was mostly the result of an upgrading of salary levels in the public administration and automatic advancements based on seniority that have increased average compensation at rates above those recorded in the private sector. These reclassification exercises increased the government wage bill by an average of 3½ per cent per year. It is not easy to demonstrate that these increases were more than necessary to bring public sector remuneration into line with that in the private sector. One pointer is that public sector employees rarely leave the public employment to take up jobs elsewhere.

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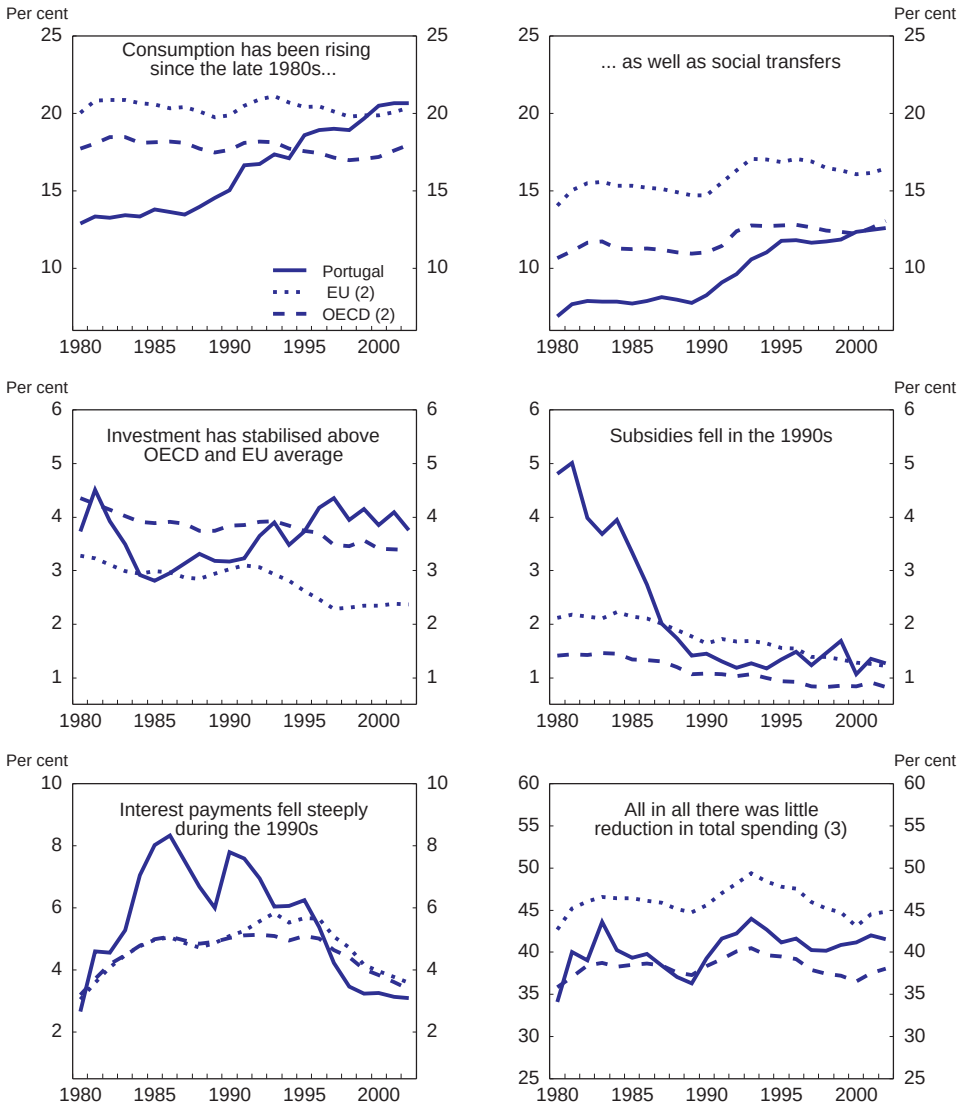
Figure 18. **General government outlays by economic category**[1]

Per cent of GDP



1. Based on ESA 95. OECD estimates for EU and OECD in 2002.
2. Weighted average of available countries.
3. Total spending is defined as current outlays plus net capital outlays.

Source: OECD, *Economic Outlook* No. 72, December 2002.

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Looking ahead, some structural factors will continue to put pressure on payroll expenditures. It will be difficult to revise downward automatic advancements based on seniority and salary upgrading that apply to the wide public sector. More importantly, while reforms are being undertaken in the pension system of the private sector, there has been only limited reform attempts to relieve the pressure on the already fragile *Caixa Geral de Aposentações* (CGA), the pension system of civil servants (Box 4). Measures have been taken to address the long-term

Box 4. Pension system for civil servants (Caixa Geral de Aposentações, CGA)

The CGA system is designed as a PAYG defined-benefit scheme, receiving contributions from subscribers and employers and distributing these as pensions. The State contributes with the amount necessary to balance the CGA, which is not actuarially balanced, making it particularly vulnerable to the effects of population ageing and the ensuing jump in internal dependency ratios. Moreover, the average pension has still to reflect the recent rapid jump in public sector wages.

The system has already been facing significant strains related to ageing. Between 1990 and 2001, the number of pensioners more than doubled, whereas the number of active contributors increased by an estimated 17 per cent. By 2000, 20 per cent of state payroll expenditures was being spent on civil servants' pensions, and direct subsidies to the system had reached 2 per cent of GDP. The disbursements of the CGA are expected to rise rapidly in the coming decades. This is due both to the fact that civil servant pensions are indexed to civil service salaries and to the growing number of pensioners: 40 per cent of civil servants are due to retire in less than ten years. The size of the CGA will rise and general government transfers to CGA will increase more rapidly than civil servant contributions in order to meet commitments to those retiring in years to come.[1]

The State Budget for 2003 has modified the CGA in order to control future expenditure growth. As of 1st January 2003 the relevant wage to determine the pension are net of CGA contributions. Moreover, those civil servants who have reached full pension entitlements but are still below the statutory minimum retirement age (60), can retire earlier but with a lower pension. Up to the end of 2002, a CGA subscriber with full pension rights can retire earlier with the right to a full pension. As of the beginning of 2003, pension benefits are reduced by 4.5 per cent per year of anticipation. Though these measures are in the right direction, they will have only a small financial impact in the short-term and are insufficient to restore financial equilibrium of the CGA.[2]

1. All civil servants, excluding employees of public enterprises contribute to the civil servants' social security system.
2. It has been estimated that a large number of CGA subscribers with full pension entitlements will apply before the end of 2002.

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sustainability of the system, including making benefits less generous for those joining the civil service after 1993 in order to meet commitments to those retiring in years to come.[34] These are expected to bear fruit in the longer-run. In the meantime, the burden on taxpayers should be alleviated by whatever means are available, by raising contributions and reducing the growth of benefits.

Additionally, the demand for social services nearer to European standards has brought increasing expenditure in health, education and social protection (in 1999 the minimum-guaranteed income was introduced), raising the collective share of such programmes to just under 60 per cent of total spending by the end of the 1990s (Figure 19, Panel B). As the Portuguese population ages, pressures on public spending will intensify, as in other OECD countries. OECD simulations suggest that spending on health care could rise by a further 3 percentage points of GDP by 2030 as a result of population ageing alone (OECD, 2001a). Furthermore, the price of health services in Portugal remains below those in other EU countries. As it is likely that non-tradeable prices will rise relative to these of tradeable, *via* the Balassa-Samuelson effect, health service prices are likely to rise faster than the general price level. These sectors thus pose special problems, which are discussed below.

Public investment has been boosted by EU *membership despite spending freezes*

In the second half of the 1990s Portuguese public investment stabilised at a level above the OECD and EU averages, mainly as a consequence of EU co-financing (Figure 18). Since joining the EU in 1986, Portugal has received, on average, 3 per cent of GDP in structural transfers per year in the context of successive Community Support Frameworks (CSF), which include both structural and cohesion funding. These transfers, equivalent to 15 per cent of public spending on goods and services, have played an important role in the modernisation and restructuring of the Portuguese economy, aiding convergence to EU average productivity levels. They have had a significant impact on infrastructure development. The density of motorways doubled between 1994 and 2000 and a significant proportion of railway lines were electrified (see Chapter IV). They have been instrumental in pushing public investment from 2.8 per cent in 1985 to 4.1 per cent of GDP in 2001. According to recent estimates this expansion has "crowded in" private investments as well as employment in Portugal, with a positive impact on output growth.[35]

Despite the observed relatively high level of capital spending, however, problems of planning and execution may have impeded Portuguese convergence to productivity levels in more advanced EU countries. Results have not been very visible in the domain of innovation, research and development, while the *ad hoc* temporary spending freezes used to meet the Stability and Growth Pact (SGP) targets over the past few years have resulted in disproportionate cuts and deferrals to capital outlays not related to structural funds. Moreover, the large share of capital transfers that serves to finance the chronic deficits of certain public enterprises

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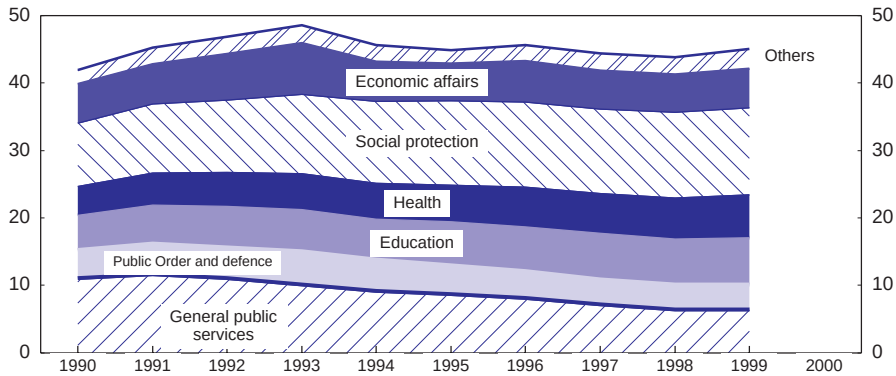
Figure 19. **Main components of general government outlays**[1]
As a percentage of GDP

A. Portugal



B. International comparison, 1999[2]
% of GDP

	General public services	Defence, public order and safety	Education	Health	Social protection	Economic affairs	Others	Total outlay
Austria	8.2	2.5	6.1	8.1	21.5	5.0	2.4	53.8
Belgium	10.4	2.8	6.2	6.3	17.5	4.8	2.1	50.2
Denmark	4.6	2.6	8.0	5.3	24.0	4.4	2.5	56.0
France (1993)	4.6	4.2	6.0	10.8	19.5	4.7	6.7	56.6
Germany	6.5	2.9	4.4	6.2	22.0	4.3	2.5	48.8
Finland	6.5	3.1	6.6	5.9	22.4	5.2	2.3	52.1
Greece	10.4	4.2	4.1	3.9	19.3	0.2	1.1	43.4
Ireland	5.9	2.2	4.2	5.7	7.1	7.3	2.4	34.8
Italy	9.9	3.2	5.1	5.8	18.2	4.0	2.6	48.8
Netherlands	8.9	3.1	4.8	4.0	18.0	5.1	3.3	47.1
Portugal	**6.4**	**3.7**	**6.9**	**6.4**	**13.0**	**5.8**	**2.9**	**45.1**
Spain (1995)	1.8	3.8	4.9	5.5	15.2	6.2	9.9	47.2
Sweden	10.1	3.8	6.8	6.3	25.3	4.8	3.2	60.3
United Kingdom (1995)	1.9	5.5	5.4	5.8	16.4	3.3	6.0	44.3
EU unweigthed average	6.9	3.4	5.7	6.2	18.5	4.7	3.6	49.2

1. Based on ESA 95.
2. For France, Spain and the United Kingdom, due to lack of information, data are presented on the basis of SNA 1968.
Source: OECD, *National Accounts*.

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(around 0.2 per cent of GDP) waste resources that could go into more productive public investments.

The third *Community Support Framework* (CSF III) will provide the equivalent of an annual 2.7 per cent of GDP between 2000 and 2006, focusing particularly on the development of information and communication technology (ICT). Investment projects co-financed by CSF III funds are expected to represent about 20 per cent of total investment.[36] However, there is no guarantee that CSF funds will be available for Portugal at these levels after 2006, implying even greater pressure on public finances thereafter, if infrastructure spending is to continue at rates that will help faster convergence to EU norms.[37]

Addressing problems in the budget management process

Spending control has relied on stopgap measures…

Apart from issues related to economic development, where Portugal has been catching up with its European partners, many spending pressures are shared – to differing degrees – with other OECD economies. However, in Portugal's case budgetary pressures have also been a function of inadequate spending control, since primary spending overruns have been the norm in recent years, often by a margin exceeding 3 percentage points.[38] Containing the deficit in the face of such overruns has meant a heavy reliance on budgetary safeguard clauses in order to meet fiscal targets. These allow wide discretion in implementing mid-year cuts in budget allocations. The Executive (in practice, the Ministry of Finance) uses its discretion as to whether a freeze is justified in order to meet the deficit target. It also decides which expenditures will be frozen, both within and among spending categories.[39] Typically, a freeze is seen as a pre-emptive action. It may or may not turn into a definite cut, depending on overall budgetary developments (on both the revenue and spending sides of the budget).[40] However, their continuous use may lead to inefficiencies and imbalances in the composition of public spending. Such chronic overruns are also symptomatic of grave deficiencies in the preparation of the original budgets in the first place, as well as in their implementation.

… due to flaws in the budget management process…

During the 1990s, several reform measures were introduced to try to improve the efficiency and accountability of public administration, to enhance internal control and to modernise accounting systems. However, such measures were limited in scope and insufficient to prevent recurrent spending control problems, culminating in a renewed effort at comprehensive reform (OECD 1999 and 2001a). In 2001, under pressure to keep medium-term current spending on target, the government launched a comprehensive reform strategy for reviewing the budgetary

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process and spending control mechanisms. A Task Force within the Ministry of Finance, the so-called ECORDEP (*Estrutura de Coordenação da Reforma da Despesa Pública*) was created to identify major areas for reform. Its report, the *Relatório de Coordenação para a Reforma da Despesa Pública*, was issued in September 2001.

In this report, which was intended to guide the reform of public spending, the Task Force identifies several shortcomings in the budgetary process and control mechanisms:

- *There are no explicit budgetary rules for spending control*. The Stability and Growth Pact targets a downward trend for the Portuguese deficit in order to reach budget balance over a time horizon of several years, but does not set rules for budgetary discipline.
- *Macroeconomic assumptions are not thoroughly examined and debated either by independent experts or Parliament*. The macroeconomic assumptions included in the annual draft budget documents are often based on an overly optimistic macroeconomic scenario that does not take the business cycle into consideration, and which is not thoroughly debated by Parliament or independent experts.
- *The accounting system is mainly on a cash basis*.[41]
- *Control and evaluation are inadequate*. The Ministry of Finance initiates the budgetary process and negotiates with line ministries and independent agencies for their respective programmes, before presenting the draft budget to Parliament for approval. The law defines spending procedures and spending limits at each level of the hierarchy. Unspent resources cannot be carried forward, encouraging line managers to spend all of their annual resources while surpluses in one department or ministry cannot be carried forward and/or cannot be re-allocated to activities elsewhere. There is an absence of agency or department-specific performance evaluation, with no internal financial control in some areas, and an overlapping of competencies, in others.
- *Co-ordination between central and local government is inadequate*. Local governments depend on unconditional central government transfers (see below) but they do have some autonomy in spending. The central government budget law ensures that current spending cannot exceed current revenues, and sets a ceiling on annual debt servicing, but the Local Finances Law does not impose a limit on the annual net borrowing for investment spending. It is possible for local governments to run large deficits, especially when they are faced with political pressure and the ceiling on debt-servicing is weakened by low interest rates (for example in 2001, ahead of the December municipal elections). In any case there are no sanctions for overshooting the ceiling on debt servicing while the monitoring of local government finances is hampered by the lack of timely budget statistics.

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... which have brought efforts at administrative reform

In view of these problems, in August 2001, the Parliament passed a new framework law for the State Budget, together with a set of measures aimed at the better administration of spending (Box 5). The new legislation sets out general principles for a rationalisation of budgetary and financial decisions of the different branches of the administration, as well as the basis for financial co-ordination among different layers of government that should help to improve budget co-ordination and to monitor sub-central government indebtedness. As endorsed by the new administration, the centrepiece of the new law is the introduction of a two-step budget procedure. At the first stage, in the spring, the government establishes a multi-annual spending program that includes an expenditure ceiling for the first year; in the second stage, taking place in the summer, the government prepares the annual budget bill, which has to be presented to Parliament on the first of October. This will make the Government's budget process more transparent.

However, the budget and the Parliamentary appropriation are not subject to pre-approved spending ceilings over the whole period covered by the expenditure program. Expenditure ceilings for the remaining three years are only indicative and can be modified depending on the evolution of macroeconomic and fiscal variables. The ceiling on the annual level of indebtedness of local authorities is accompanied by a sanction that reduces state transfers if the ceiling is not met, but there is no rule preventing the central government from bailing-out sub-central governments. Nor are the local authorities obliged to provide the necessary information on time to allow monitoring and auditing of local finances. Some EU countries have gone as far as approving a Law on the internal stability of the budget to ensure that all levels of government are committed to comply with the Stability and Growth Pact.[42] Though a major step in the right direction, the new framework law is thus unlikely to curtail all spending overruns in future, and it is also important to ensure that the new rules are fully implemented in practice.

In July 2002, the Parliament approved the Stability Law that complements the 2001 Framework Law. This law strengthens the financial co-ordination between all levels of the general government, setting up three principles: the first principle, budgetary stability, requires that the general government fiscal position be in balance or in surplus on the basis of National Accounts (ESA-95); the second principle, the solidarity principle, imposes that all sectors of the general government contribute to budget stability; the third principle relates to transparency and implies that all levels of government must provide financial information about their budget implementation. Moreover, the Budgetary Stability Law allows the government to define, in the State budget, measures to achieve budget stability (Box 5).

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Box 5. 2001 Framework Law for the State Budget and the 2002 Budgetary Stability Law

The 2001 Framework Law for the State Budget marks the first step of the reform of budgetary procedures. The Budgetary Stability Law, approved in July 2002 and integrated in the Framework Law for the State Budget, adds new provisions that are instrumental to reach budgetary stability.

The 2001 Framework Law for the State Budget includes the following provisions:

Multi-annual financial programming

- Multi-annual financial planning is to be implemented, starting with the budgetary procedures for the 2003 budget law.
- Such planning should cover four years, in line with the EU Stability and Growth Pact and include an explicit and credible target for current primary expenditure.
- The State Budget Law must include multi-annual estimates of assets and liabilities of the central administration, regional and local administrations. The social security fund, and autonomous funds and services should also present multi-annual estimates in their budgets.

Two-step budget procedure

- In May the Government must present the pre-budget to the Parliament for discussion. The pre-budget must indicate the orientation of public expenditure within a medium-term macroeconomic scenario and must set expenditure ceilings for the first year and some expenditure ceiling indication for the following years. The Government also must submit an indicative spending ceiling for local governments. The first stage of the process concludes with the adoption by the Parliament of the spending ceiling.
- Stage two entails the preparation of the annual budget bill by the Government and the voting of appropriation by the Parliament.

Activity-based budgeting (ABB)

- The state budget must be structured by programme. Structuring by activities should help to determine more precisely, *via* management indicators, the links between outcomes and human and financial resources. The inclusion of sunset clauses and "learning by doing" procedures is one of the main motivations for implementing the ABB.

Strengthening audits and controls mechanisms

- In order to strengthen audit and control mechanisms, each ministry will have an internal financial controller, responsible for controlling the budget execution and producing consolidated accounts. The budgetary execution will also be subject to external audits at regular intervals of 6-8 years.

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Box 5. **2001 Framework Law for the State Budget and the 2002 Budgetary Stability Law** (*cont.*)

Better financial co-ordination within the public sector

The 2002 Budgetary Stability Law includes the following provisions:

- The budget of the Autonomous Funds and Services must be in balance or in surplus. All Autonomous Funds and Services, with the exception of those that are required by the Constitution, or are integrated in the National Health Service or are supervision bodies, lose financial autonomy if their own resources do not cover at least two thirds of their expenditure (with receipts other than transfers from the central government).
- The Ministry of Finance can require other sub-sectors of general government to provide financial data on budget outturns during the year, on a regular and timely basis
- If sub-central governments or autonomous funds exceed the limits on annual debt increase set in the State Budget or if the financial information is not provided in time, State transfers can be reduced or suspended.

Improving the cost effectiveness of public spending

The spending pressures which make the adoption of new management and control measures so essential are not unique to Portugal. But they do emanate in part from the institutional characteristics of the Portuguese system which have determined the cost and efficiency with which public services are provided. Three such features stand out: the effectiveness of human resource management; the respective roles of the government and private sectors in public service provision; and the incentive structures implied by fiscal-federal relations.

Inefficiencies in human resource management

Although the size of the public sector labour force is now close to the OECD average, as noted above, Portugal is characterised by both a comparatively high expenditure on wages and salaries and by a large wage premium in favour of the public sector (Figure 20), the wage gap between the public and private sectors being the highest in the EU.[43] Considering identical workers' characteristics, a Portuguese woman working in the general government receives an hourly wage premium 64.6 per cent higher than that earned by a woman working in the private sector, while the wage premium for men corresponds to 37.6 per cent (Table 10).[44] Furthermore, government compensation conditions become even more favourable when wages are corrected for non-wage benefits (*e.g.* generous pension schemes,

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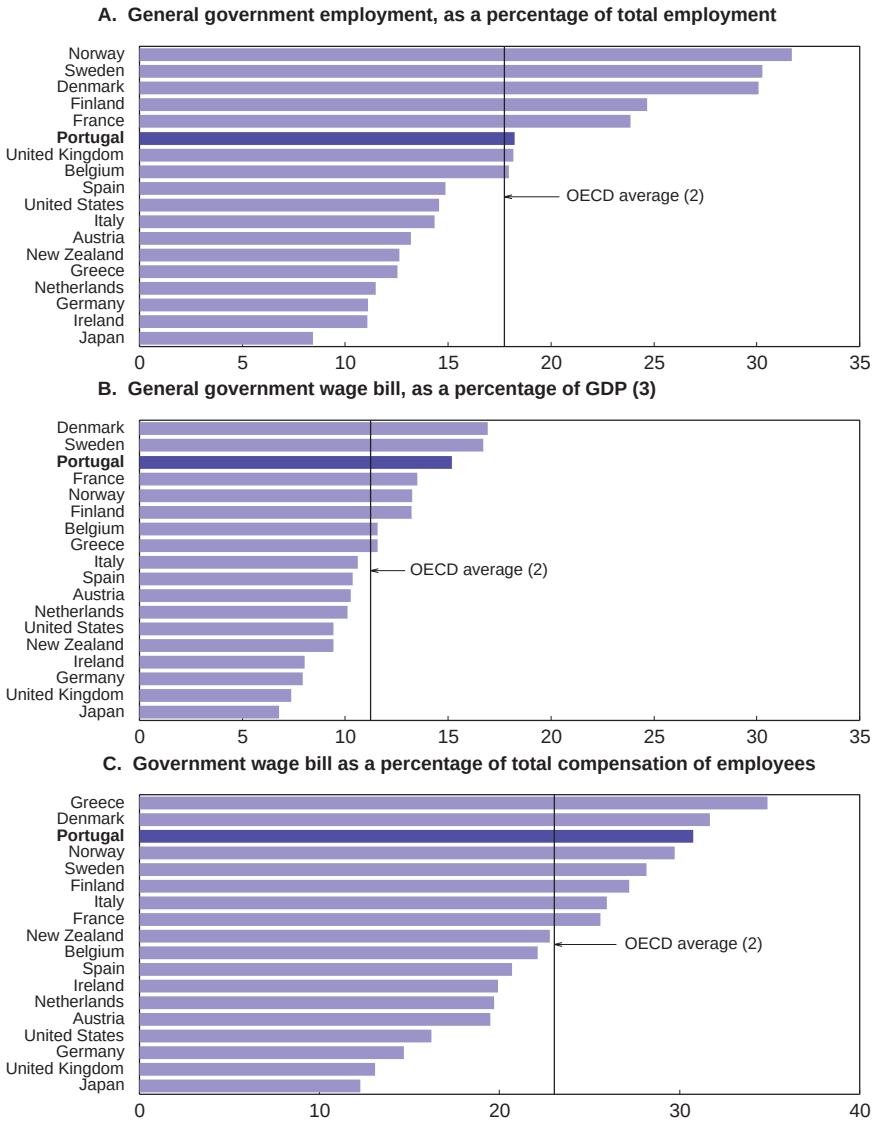
Figure 20. **General government employment and wage bill in OECD countries**[1]

Per cent



1. 2001. Data are OECD estimates. Excluding public enterprises, except for Japan.
2. Unweighted average.
3. Wages and social security contributions.

Source: OECD, *Economic Outlook* No. 72, December 2002.

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Table 10. **Wage differences between the public and private sectors**[1]

Percentage

	Men	Women
Austria	–2.3	11.9
Belgium	3.6	2.9
Denmark	–9.0	7.1
France	7.7	14.7
Germany	0.8	13.7
Greece	10.3	22.0
Netherlands	2.8	5.7
Ireland	32.8	41.0
Italy	10.3	25.8
Luxembourg	31.0	36.5
Portugal	**37.6**	**64.6**
Spain	19.7	34.8
United Kingdom	2.8	10.5
Average of the above countries	11.2	21

1. Estimates are based on panel data for households in the European Union area, harmonised by Eurostat.
Source: Portugal and Centeno (2001).

flexible working hours, less stressful conditions and predominant life-long tenures) making public jobs more attractive. General government workers hardly ever leave the public sector on a voluntary basis and the number of job applications is high whenever new vacancies are advertised in the general government.

New working arrangements were introduced as of 1998, in order to modernise the workforce in the public sector and improve skills and performance.[45] Permanent staff are allowed to opt for a four-day week, subject to certain conditions. Public servants near retirement age are allowed to complete their careers on a part-time basis. The authorities expected that this would permit a partial renewal of the government's workforce without increasing the wage bill. However, in 1999 when the last public administration survey took place, only 2.7 per cent of total public employees had a part-time job, despite 26 per cent of civil servants being over 50 years old. Some efforts have also been made to introduce internal mobility among staff. Employees can compete for vacancies available at both local and central government levels, and can apply for transfers or secondment from one administration to another in the same way as they can move from one department to another.

While public employees are free to move within the public administration, they cannot be required to do so. The legal barriers impeding managers from transferring public servants between different services, levels of government and functions have not been removed. This has brought duplication of tasks in some services, and understaffing in others, and has resulted in a very inefficient deployment of human resources in the public sector. For example, the Ministry of Education has little flexibility in reallocating teachers, who have life-long tenures in a particular school, to

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the schools most in need and away from schools whose pupil base has shrunk. Similar problems have been faced by other Ministries. The low mobility in public employment has made public spending behaviour asymmetric. Public employment has tended to adjust upwards when a new demand emerged in a specific sector, but failed to adjust downwards when demand waned, thus contributing to the steady rise in the number of public employees since 1990 (Table 11). However, in mid-2002, the government proposed new rules to increase staff mobility. Legislation to create an employment "pool" of civil servants was then approved. And the employment "pool" should be in place in early 2003.[46] The government seems also committed to use "hard budget" constraints for many departments (with the exclusion of the Health Sector).

Finally, there is a broad consensus that the lack of incentive schemes in the Portuguese civil service pay system weakens the performance of managers and employees. As in many OECD countries, wages are based on rigid pay scales linked to professional category and seniority. In addition, promotion tends to be based more on seniority than on merit. As a general rule the Portuguese public-sector pay scheme has no provision for individual productivity or performance-related pay systems:[47] civil servants competencies are not clearly defined, a proper performance assessment system is not in place, and hence poor performance is not sanctioned, nor is exceptional performance easily rewarded. Nevertheless, since 1998, performance-related compensation has been introduced, at an experimental level, to improve the quality of the National Health Service (NHS), tax administration and justice.[48] Those in the health service are described below. In tax administration all employees share a team-based productivity premium based on the ability of the department to recover tax arrears and fight tax evasion. Similarly, a group premium for professionals operating in the judiciary system is now granted when procedural delays and the length of court trials are reduced.

Table 11. **Trends in general government employment**
As a percentage of total employment

	1990	1995	2000	2002[1]
Portugal	**14.8**	**17.2**	**18.1**	**18.2**
Belgium	18.9	18.2	18.4	18.6
Ireland	13.7	13.1	11.0	11.6
Greece	12.3	12.2	12.5	12.6
Spain	13.5	15.0	14.4	14.6
EU	18.3	18.8	17.9	18.0
Euro area	15.9	16.5	15.8	15.8
OECD	16.2	17.6	16.9	16.9

1. OECD estimates.
Source: OECD.

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The involvement of private finances is still limited

Contracting out for the provision of publicly funded services and support functions (*e.g.* cleaning and building maintenance) is not systematically used in Portugal. Only 8 per cent of value of public services provided by the public administration and defence comes from inputs purchased from other service sectors (telecommunications, transport, catering, insurance, IT, real estate, and other business service sectors), whereas the corresponding figure for other OECD countries varies between 10 and 20 per cent (Figure 21).[49]

On the other hand, for large infrastructure investment, Portugal has made significant use of public-private partnerships. Beginning with motorways, the provision of public services *via* the application of public-private partnerships has expanded rapidly with 14 deals being concluded in 2000 across the areas of transport, water and energy. Notable examples have been the construction of the Vasco da Gama Bridge in 1998, which was undertaken through a concession contract to build, own, operate and transfer (BOOT) over a period of 33 years, and the motorways construction concession. In the latter case, running costs are paid by the State on a shadow toll basis with no charges for users.

Public-private partnerships can be an effective tool for promoting efficiency and improving the delivery of certain public goods,[50] but there is a lack of systematic evaluation of the performance of public-private partnerships projects in Portugal. From a budgetary point of view, however, some of the privately funded infrastructure projects have entailed large explicit and contingent liabilities for future years, which have not been reflected in the budget (OECD 2001a).[51] Only if efficiency in delivery is enhanced or if payment is shifted to users through charges as part of the private finance arrangement will there be a net benefit. In any case, public-private partnerships initiatives need to be carefully designed to ensure an adequate risk sharing with the private sector and that the costs of setting up and monitoring public-private partnerships are not excessive (Box 6).

An important aspect of ensuring competition for public contracts is to announce calls for tender internationally. Portugal has been active in advertising public procurement internationally. In the second half of the 1990s, around 15 per cent of total public procurement value were advertised internationally each year between 1995 and 2000, which was above the average for EU-countries. More recently, though, other EU countries such as Greece, Ireland and Spain, with which Portugal might typically be compared, have progressively increased the share of public procurement being advertised internationally to almost a quarter of the total (Table 12). There is thus scope in Portugal as well to inject more competition by this means.

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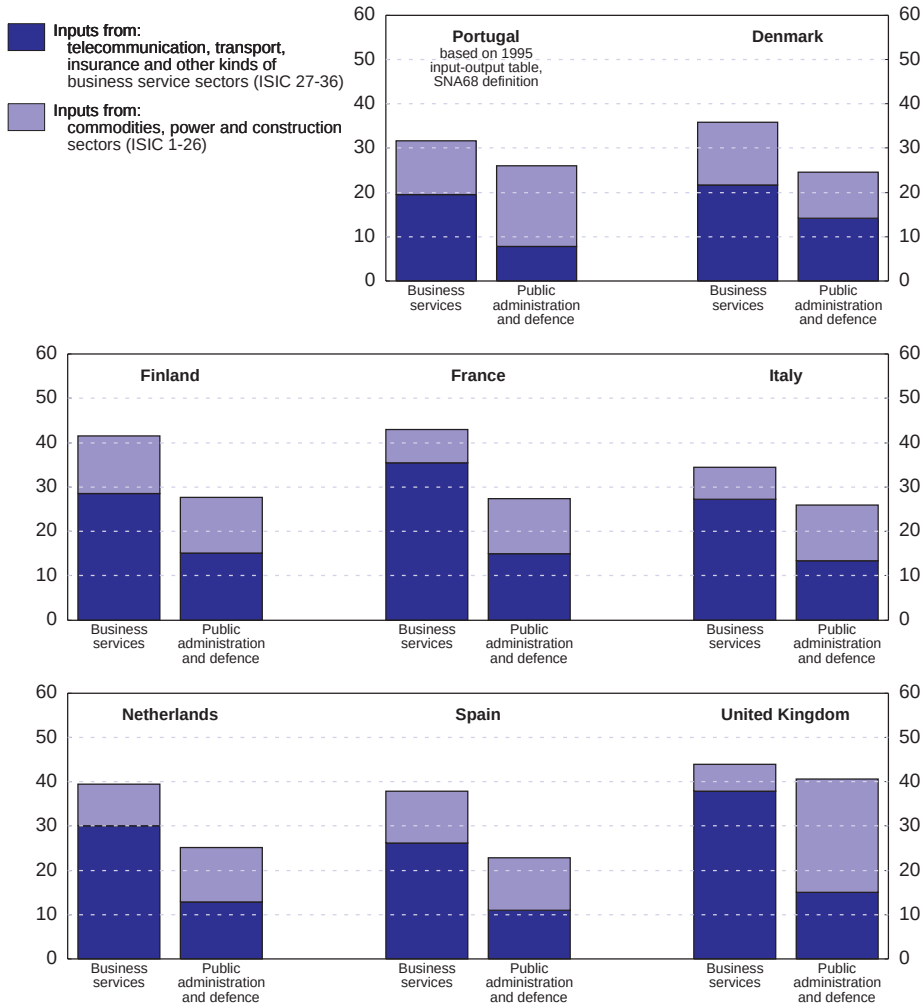
Figure 21. **Sub-contracting**

Inputs purchased from outside the sector as a share of total sector production value, 1995-1998



Note: The business services shown for comparison in the first column include accounting, consultancy, architecture, engineering and legal services (ISIC 36). Public administration includes also agencies for compulsory social security schemes. Purchases from, as well as payments to and among public administration, education, health, care and social and community services have been eliminated, as statistical methods differ across countries.

Source: OECD, STI/EAS based on Input-output tables from national accounts.

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Box 6. Public-private partnerships: considerations to bear in mind

The performance of public-private partnerships is affected by the following considerations:

- Often the Government is a co-owner of the public-private partnerships in order to ensure that any financial gains arising from the partnership are shared with the taxpayer. This set-up may confuse the roles of the Government as a shareholder and regulator, and could establish a *de facto* monopoly position of the incumbent private partner. It also entails a greater exposure of the budget if a private partner fails.
- The empirical evidence suggests that most efficiency gains stem not from the tender as such but rather from the permanent exposure of potential contractors to competition: competition *in* the market is more important than competition *for* the market. It is therefore essential that the process of tendering and contracting be organised in such a way that they reduce the government's dependence on the incumbent franchise or concession holder.
- Keeping transactions simple could reduce the relatively high costs of writing contracts and bidding for tenders.

Table 12. **Share of public procurement being openly advertised internationally in European Union countries**[1]

Value of public procurement which is openly advertised in per cent of total value of public procurement

	1995	1996	1997	1998	1999	2000
Austria	5	8	7	8	7	13
Belgium	7	8	11	14	16	16
Germany	5	6	6	7	5	6
Denmark	16	13	13	13	14	21
Finland	8	9	8	9	10	13
France	5	7	8	11	12	15
Greece	34	37	43	45	38	n.a.
Ireland	11	16	19	16	17	21
Italy	10	10	11	11	13	18
Luxembourg	5	7	9	14	13	12
Netherlands	5	5	6	5	6	11
Portugal	**15**	**18**	**15**	**16**	**15**	**15**
Spain	9	11	11	12	17	25
Sweden	12	12	13	13	14	20
United Kingdom	15	15	17	16	15	22
Total EU	8	9	11	11	11	15

1. Value of public procurement published in the EU official journal as percentage of estimated total value of public procurement.

Source: European Commission, Internal Market Directorate General.

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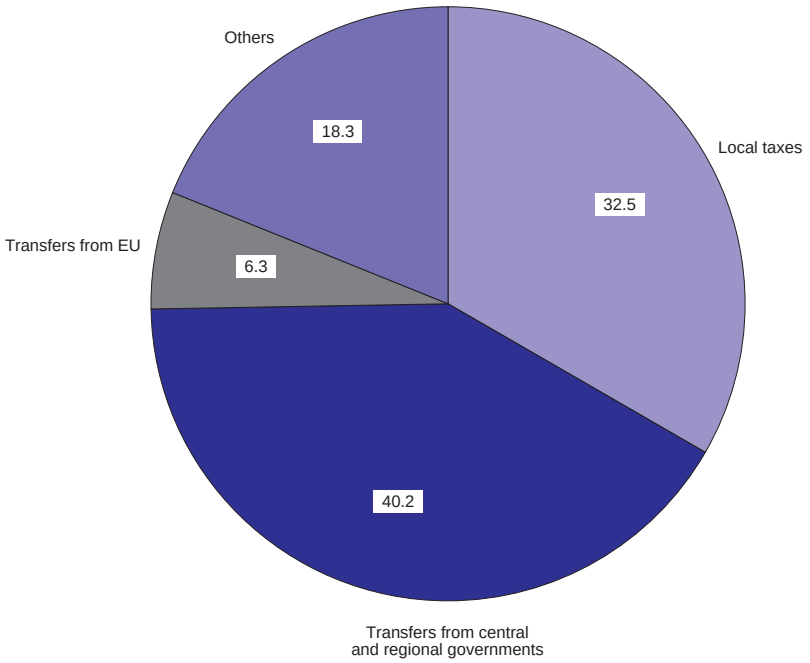
Local government spending incentives are inefficient

The current legislation that defines how local governments can raise funds does not seem to establish an adequate relationship between the decisions to increase expenditure and the responsibility for raising additional revenue (in particular through tax increases). The financial resources of municipalities come largely from the central budget in the form of unconditional grants, while local governments own resources account for a little more than one-third of the local budget (Figure 22 and Annex I). The Local Finance Law, last revised in 1998, strictly governs local funding, leaving very limited margin of manoeuvre to local authorities to affect the amount of their revenues, including local taxes. As a result, local governments do not take the marginal social cost of raising these revenues into

Figure 22. **Local government receipts by source of finance**



Source: In Jorge Correia da Cunha and Patrícia Silva, Bank of Portugal, *Boletim economíco* (March 2002).

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account when deciding how to allocate resources and tend to use additional transfers for further spending even when its social utility is low (Correia da Cunha *et al.*, 2002). This need not matter insofar as the local provision of key areas such as health care, higher education and social services is still tightly regulated by the central government, which explains the low local government share in total spending. However, local governments have spending autonomy in strategic areas such as the staff of public offices and institutes, including pre-primary schools and nurseries, as well as investments in local infrastructure (*e.g.* schools, playgrounds, etc.). In 2000, for example, investment expenditure accounted for 41 per cent of total local spending and for a larger share (45.5 per cent) of total investment of the general government. Local governments have therefore an important influence on the cost and output of a number of spending programmes.

Moreover, the Local Finance Law in Portugal does not provide local governments with adequate incentives to contain spending. Given the limited scope for adjusting other sources of financing, municipalities tend to finance gaps between receipts and expenditure by additional borrowing that is not subject to legal limits. As explained above, the Local Finance Law sets an annual ceiling only on debt servicing of local governments and on short-term loans. These ceilings depend positively on the total amount of funds transferred by the central government to the local authorities and past investment spending.[52] In the period 1999-2001, the decline in interest rates as well as additional transfers induced by the 1998 revision of the law for local finances eased, to a large extent, the limits to borrowing based on the debt service. Table 13 compares over the period 1999 to 2002

Table 13. **Central Government transfers and bank liabilities of local authorities**

Million euros

		1998	1999	2000	2001	2002
Central government transfers according to:						
1998 law for local finances	(A)		1 620.3	1 775.4	2 012.2	2 243.0
Percentage change			19.7[2]	9.6	13.3	11.5
1987 law for local finances[1]	(B)	1 354.1	1 438.8	1 594.3	1 757.8	1 888.8
Percentage change			6.3	10.8	10.3	7.4
Difference	(C) = (A) – (B)		181.5	181.2	254.4	354.3
In per cent of GDP			0.2	0.2	0.2	0.3
Bank liabilities						
Net bank liabilities[3]		570.2	739.5	1 081.3	1 649.8	
In per cent of GDP		0.6	0.7	0.9	1.3	
Changes in net bank liabilities		53.8	169.3	341.9	568.4	
In per cent of GDP		0.1	0.2	0.	0.5	

1. Values are based on the 1998 Central government transfers and on VAT estimates enclosed in the State Budgets of the following years.
2. Growth rate based on the 1998 Central government transfers.
3. Values at December 31.

Source: General Government Budget 1998-2002 and estimates by the Central Bank of Portugal.

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transfers from the central government to local authorities based on the 1998 Law for local finances with the transfers that would have taken place under the previous law. Central transfers to local authorities rose by an annual average of 13.5 per cent, compared with the 8.7 per cent annual average assumed under the 1987 law on local finances. At the same time, the bank liabilities of local authorities rose by an annual average of 1 percentage point of GDP, indicating that the additional transfers induced by the 1998 Law was used for further spending rather than containing local net additional borrowing. The Central Bank of Portugal estimated that this has resulted, *ceteris paribus*, in 0.2-0.3 per cent of GDP increase in the total general government deficit.

So far, local authorities have not worried too much about debt accumulation because, as mentioned above, there have been no sanctions for overshooting debt servicing ceilings. Control over local spending should improve with the new limits on additional borrowing by local governments introduced by the State Budget for 2003.

Assessing public expenditure outcomes in selected areas

Assessing efficiency in public spending on merit goods is a difficult exercise. In spending areas such as education, health and social spending, matching costs to outcomes is most difficult. The three spending areas together accounted for more than 60 per cent of general government outlays in Portugal in 2001. They have risen more rapidly than other spending components over the 1990s and, by international standards, are high. The number of personnel working in education and health, including administrative personnel, increased steadily and by 2000, reached around 20 and 35 per cent (respectively) of total public employment. Despite this heavy weight on public spending, little has been done to assess the costs and benefits of public education and social policy spending, and piecemeal reforms in the health care system have not made much progress in enhancing efficiency in this sector.

Education outcomes do not match costs

Primary and secondary education is costly…

At around 5.7 per cent of GDP, public spending on education is close to the OECD average (Table 14). However, expenditure per student is about 14 per cent higher than the OECD average in secondary education and has been increasing since the 1990s. Unlike in some other OECD countries, private-sector co-payments in tertiary education are negligible. Despite high costs, educational attainment in Portugal remains significantly below international standards[53] (see also Chapter IV), encouraging increasing numbers of middle-class families to send their children to private schools. In addition, there are costs because of

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Table 14. **Spending and unit costs at various levels of education**

1999

	Public spending on education				Total spending on educational institutions	Spending per student (2000)[3]	Unit costs		
	Primary and Secondary[1]	Tertiary	*Of which*: Student grants[2]	All levels of education	All levels of education		Primary	Secondary	Tertiary
	Per cent of GDP						Expenditure per full-time student relative to GDP per capita[4]		
Australia	3.8	1.2	0.4	5.0	5.8	1.7	19	27	46
Austria	4.1	1.7	0.2	6.3	6.3	2.6	26	33	47
Belgium	3.5	1.5	0.2	5.5	5.5	1.8	16	26	39
Denmark	4.8	2.4	0.8	8.1	6.7	2.3	24	28	39
Finland	3.8	2.1	0.3	6.2	5.8	1.8	18	25	35
France	4.2	1.1	0.1	6.0	6.2	2.1	18	31	34
Germany	3.0	1.1	0.1	4.7	5.6	1.6	16	27	42
Greece	2.4	1.1	0.0	3.6	3.9	1.7	14	18	27
Ireland	3.1	1.2	0.2	4.3	4.6	1.0	12	17	37
Italy	3.2	0.8	0.1	4.5	4.8	2.4	22	27	32
Japan	2.7	0.5	0.0	3.5	4.7	2.0	21	24	41
Netherlands	3.1	1.3	0.3	4.8	4.7	n.a.	16	21	46
New Zealand	4.8	1.2	0.3	6.3	n.a.	n.a.	n.a.	n.a.	n.a.
Norway	4.6	2.0	0.6	7.4	6.6	1.7	20	26	43
Portugal	**4.2**	**1.0**	**0.1**	**5.7**	**5.7**	**2.0**	**20**	**30**	28
Spain	3.3	0.9	0.1	4.5	5.3	1.8	19	26	30
Sweden	5.1	2.1	0.6	7.7	6.7	2.1	24	25	61
Switzerland	4.0	1.2	0.0	5.5	5.9	2.2	23	34	63
United Kingdom	3.3	1.1	0.4	4.7	5.2	1.8	16	24	41
United States	3.5	1.4	0.3	5.2	6.5	1.9	20	24	57
Country mean	**3.7**	**1.3**	**0.3**	**5.5**	**5.6**	**1.9**	**19**	**26**	**41**

1. Includes post-secondary non-tertiary education.
2. Scholarships/other grants to households and student loans.
3. Cumulative expenditure on educational institutions up to age 15 in USD, converting using purchasing power parities.
4. Multiplied by 100. Includes both public and private institutions except for Austria, Norway (1998), Italy (1998), Portugal and Switzerland (1998) where data refer to public institutions only and Greece (1998) and the United Kingdom for which they concern public and government-dependent institutions.

Source: OECD, *Education at a Glance* (2002); OECD, *Knowledge and Skills for life – First results from* PISA 2000, Table 3.6.

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pupils abandoning school and students who take one or more years longer to complete upper secondary education.[54]

High outlays per student seem to result from high educational staff salaries: in compulsory education, the staff wage bill represents 90 per cent of total spending on education. Cross-country comparison shows that teachers' relative compensation tends to fall as per capita income levels rise (OECD 2002c). However, this has not been the case for Portugal. Measured in purchasing-power parity terms, teachers' starting salaries are below the OECD average but they increase rapidly above the average with experience (OECD 2002c). Overall, as a multiple of per capita income, salaries are at least 9 per cent above the average for other OECD countries.

Geographical restrictions and the limited responsibility of local authority spending have an impact on school size and performance at the compulsory school level. Pupils who are residents in one municipality are not entitled to go to school in another. This largely stems from the funding system of municipalities, whereby transfers from central government are based on the number of inhabitants, whether or not they use the services provided by their municipality of residence. Moreover, local authorities are responsible only for the maintenance of school buildings and local school transport, and do not bear the high wage costs. Therefore, local communities have no incentive to give up their schools. In 2001-2002 nearly two thirds of schools had fewer than 30 students, and 1 700 schools had fewer than 10 students.[55] Maintaining such small schools may also entail excessive variation in learning outcomes with students in some schools not receiving adequate teaching.[56] However, local communities have few incentives to co-ordinate and merge local schools into larger and less costly units because neither local governments nor their residents directly bear the high costs per student.

… while private returns to tertiary education are high

The public education system heavily subsidises tertiary education in Portugal. Public university fees are very low and deductible from the personal income tax. At the same time wage premia for people who have a university degree are high and seem to have increased during the 1990s (see Chapter IV).[57] This combination implies very high private returns to education.[58] Such heavy subsidisation seems debatable on equity grounds, and because tertiary education tends to be less prone to positive externalities than lower education levels.[59] Moreover, students from a favoured background still account for an important proportion of participants in public tertiary education, despite the increase in the participation of students from disadvantaged backgrounds and with modest income backgrounds that took place in the wake of the expansion of enrolment rates in secondary education.

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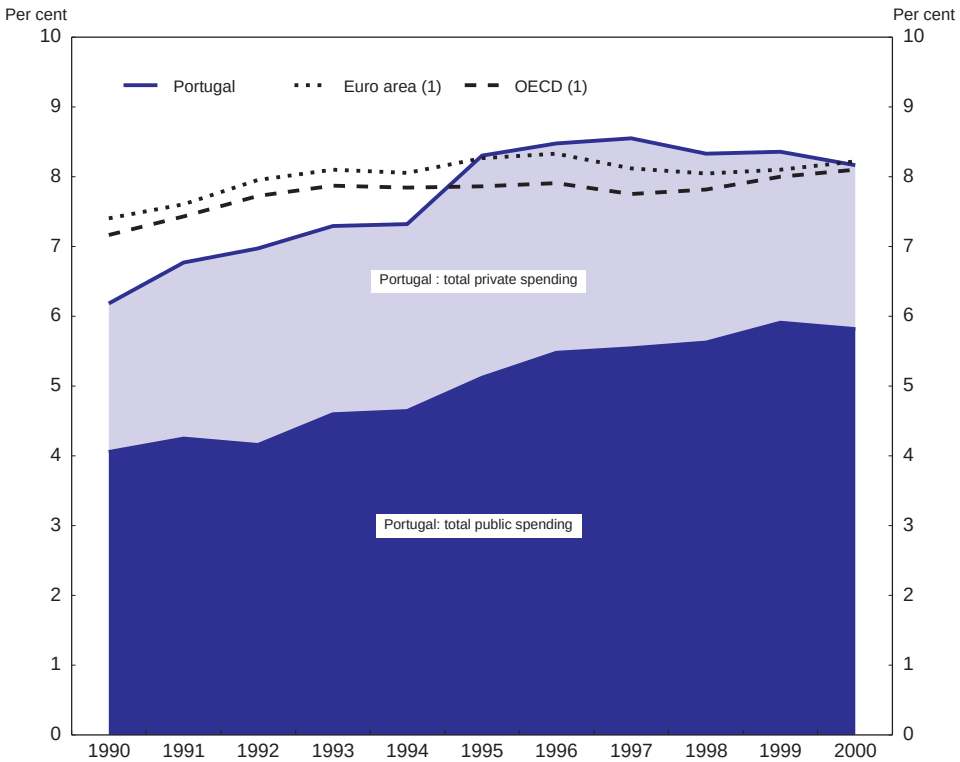
Areas of inefficiency in health care

The public health-care system does not impose the right incentives…

Public health care spending in Portugal accounts for some 6 per cent of GDP, up from 4 per cent a decade ago. This increase in spending, accompanied by a steady growth in private outlays has brought overall health spending above the OECD average in the second half of the 1990s (Figure 23).[60] Although this has been associated with an improvement in health outcomes, several studies indicate that Portugal's health performance is still lagging behind the EU average and that the

Figure 23. **Total expenditure on health care**
Per cent of GDP



1. Mean of available data.
Source: OECD *Health Data 2002*.

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sector displays important inefficiencies.[61] In 1998, the OECD *Economic Survey of Portugal* highlighted the main problems of the public health system and sources of spending pressure. These derive from a multitude of factors, including an inefficient system of doctors' remuneration, lack of management flexibility and accountability, inadequate co-ordination between public health institutions, and insufficient competition in health provision and sale of pharmaceuticals.

… with respect to doctors' remuneration…

The remuneration system for physicians employed by the National Health System (NHS) does not set the right incentives and it is a cause for spending pressures. Doctors' remuneration there is linked to professional category and length of service, irrespective of productivity.[62] About half of NHS doctors work in the private sector to compensate for the lower remuneration in the public sector.[63] The dual employment status of hospital doctors leads to low work intensity and thus to long waiting lists for some non-urgent surgical procedures. It also exacerbates problems of accessibility and encourages people to seek private-sector health care, the cost of which is partly deductible from personal income tax. Since patients face only a low marginal cost of obtaining a "second opinion", the result can be duplication of diagnostic tests and prescriptions. Hospitals are short of nurses, while doctors dedicate increasing hours to private practice, leading to large purchases by hospitals of laboratory services in the private sector and long waiting lists made worse by the lack of public infrastructure for the care of the elderly that tend to crowd hospitals seeking assistance. Some progress has been made in making budget allocations more therapy-based, including the introduction in 1997 of a case-mix system whereby diagnostic related groups (DRG) are used to define in-patient and ambulatory surgery cases. This system, however, affects only 20 per cent of hospitals' initial budget allocation.

… while the budget constraints on hospitals are too soft…

Public Budget allocation in public hospitals is based on historical figures and in practice, public hospitals operate with an open-ended budget, as the NHS directly and passively covers unexpected outlays by supplementary allocations that have become the norm.[64] Administrative penalties for excess spending have been increasingly applied, but remain rare. Managers and administrators have no personal incentives to improve hospital performance. Salaries typically account for 80 per cent of hospitals' current outlays, but health institutions have limited management autonomy, notably in the area of staff policy. The lack of effective budget constraints, combined with administrative hurdles, has contributed to inflate current costs, crowding out investment in capital equipment and new recruits in understaffed areas.

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Box 7. Transforming public hospitals into corporate entities

A new framework Law on Health Services was approved in July 2002. One of the most salient changes proposed is the transformation of the financial relationship between the State and hospitals and the consequent modernisation of hospital management. The government has allocated € 900 million in the 2002 Supplementary Budget (about 0.7 per cent of GDP) and an additional 400 million in the 2003 Budget Law for the transformation of around one third of public hospitals into public enterprises, with an executive board composed of managers. The State will be the sole shareholder, *i.e.* the hospitals will remain in public ownership, but incentives towards efficiency and demand-orientation of service provision should be heightened. The objective is to simulate a market for hospitals where the National Health System will be free to purchase from either the public or private sector. The details of the reform are likely to be based on experience drawn from the pilot programmes put into place during the past six years or so.

The five year old pilot programme in the public hospital of *Nossa Senhora do Rosário* (NSR) in the outskirts of Lisbon shows that it is possible to make efficiency gains by making each director fully responsible for the management of his service (*i.e.* hiring, purchasing of goods and service, time management, etc.). This should help break down the resistance of doctors and nurses in traditional public hospitals to changes to their familiar environment, even if the changes are intended to increase efficiency. At the same time, activity-based costing (ABC) has been introduced. The objective, (which has met with growing support from professional staff in the NSR hospital), is to set up a system in which each patient-based activity (appointments, operations, reception, discharge, medication etc.) has been costed in each service, so that comparisons can be made, benchmarks established and inefficiencies spotted. While the ABC is a long-term programme, some efficiency gains can already be observed. At present, savings from spotting some of the inefficiencies are paying for the running costs of the programme. But higher gains are expected in the future.

Another pilot programme in the public hospital of *São Sebastião* (HSS), might serve as an example of how private employment contracts can co-exist with those of the civil service. In 1998, when the programme started, hospital employees where given the choice of to leaving their civil servant status and entering into individual private contracts where salary was also based on individual performance as assessed by managers. In 2000, approximately 80 per cent of personnel had entered into individual contracts, allowing more flexibility in the organisation of services. According to the 2002 review carried out by the *National Instituto of Administration* (INA), personnel productivity enabled the HSS to reduce the percentage of losses from one year to another, and to keep personnel spending below the average of the other public hospital of same size, in the same region.

The government is also planning the creation of ten new hospitals in the next four years or so. These will have the legal form of public-private partnerships. Ownership will remain in public hands, while the management will be completely private. The government expects to shift part of the risks of the investment to the private sector, while retaining the benefits of a privately-run service. The experience

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Box 7. **Transforming public hospitals into corporate entities** (*cont.*)

in Portugal not yet conclusive because of the few cases of these types of partnerships in the public sector. According to the INA report, the hospital *Fernando da Fonseca* (HFF), one of the few existing examples of public-private partnerships in Portugal, is indeed profitable with a more efficient personal management (that is they employ less people for the same services), but the prices of services that it provides are not competitive.

Non-critical health-care facilities are allocated suboptimally. With the public sector having to compete with high private-sector fees, and hospitals' wage bills and medical fees claiming the bulk of resources, health centres have suffered from under-equipment of capital, and under-endowment of quality labour, especially in poor regions. Such centres are generally responsible for public ambulatory or primary health care. The absence of financial and managerial autonomy, combined with the bureaucratic rules governing health centres, has led to inefficient modes of service. Rural and low-income areas are characterised by a general shortage of health professionals. Health centres unable to meet the demand for ambulatory services have diverted demand to hospitals, thus contributing to the overcrowding of hospital emergency rooms.

Apart from some pilot programmes, there has until now been only modest action to improve the managerial structure of hospitals. In 2002 a new Framework Law on Health was approved, whereby 35 out of 114 public hospitals will be converted into public enterprises (Box 7). The remaining 79 traditional hospitals will not be modified for the time being. Ideally, the transformation into corporate entities should enhance accountability and efficiency. The details of the reform, however, are not known yet. A few pilot programmes, where either public hospitals have incorporated private practices in cost management, or hospitals are privately run, show that costs can be reduced. The government has also planned to build ten new hospitals, which will all be public-private partnerships.

… and pharmaceutical consumption is excessive

Expenditure on pharmaceuticals is a major source of total health care spending (around 26 per cent) and its share out of total health expenditure is the highest in the OECD after Hungary (OECD 2002d) and pharmaceutical sales per inhabitant are among the highest in the OECD. The drug market is characterised by freedom of prescription for doctors (outside the hospital inpatient setting), largely unregulated drug prices, disincentives for the production of generic medicines,[65] and monopolistic power for the pharmacists who sell drugs.

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On the demand side, low private co-participation, deductibility provisions of health costs from the personal income tax and NHS reimbursement for drugs prescribed under private treatment limit consumers' perception of real costs.

Social spending does not have a strong redistributive role

There is very little state provision of community care services in Portugal, including long term care, day centres and social services for the elderly, the chronically ill and other groups with special needs such as the mentally and physically disabled. The family is the first line of social care in Portugal. However, two out of three families receive social benefits other than pensions, and their share in total spending is around the OECD average (Table 15, panel A). Non-pension income transfers do not play much of a redistributive role in Portugal, largely because such a broad section of the population receives them. The share of such benefits in total income per "adult equivalent"[66] is negligible: they are equivalent to one quarter of pension benefits and make up 5 per cent of total income per "adult equivalent", one of the lowest shares in the EU (Table 15, panel B). In the majority of EU member countries, half of non-pension benefits are distributed to the lowest quintile of the income distribution, whereas Portugal pays out less than 40 per cent to this income group, even though Portugal has one of the widest income distributions. After benefits, income disparity remains important with the Gini coefficient being 6 points above the EU average (Table 15, panel C). If the wide range of tax credits and deduction of personal income tax is taken into consideration, the fiscal system probably exacerbates the inequality of income distribution.[67]

Table 15. **Social benefits and their redistributive effects**

A. Share of population receiving transfers

	Pensions	Other social benefits	All social benefits together
Austria	34	69	86
Belgium	28	69	89
Denmark	19	75	85
France	25	62	79
Germany	29	57	78
Greece	39	19	50
Ireland	23	82	90
Italy	40	18	51
Luxembourg	29	66	86
Netherlands	19	66	81
Portugal[1]	**36**	**68**	**89**
Spain	34	33	58
United Kingdom	28	71	85
European Union	30	52	73

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Table 15. **Social benefits and their redistributive effects** (*cont.*)

B. Percentage of income from social benefits in total income per "adult equivalent" in the EU[2]

	All households				
	IBB	P	SBOTP	TI	SBOTP/TI
Austria	10 144	2 600	1 589	14 334	0.11
Belgium	9 250	2 609	1 961	13 820	0.14
Denmark	10 034	1 635	2 338	14 007	0.17
France	9 855	2 337	1 285	13 475	0.10
Germany	10 180	2 818	1 018	14 015	0.07
Greece	6 577	1 626	169	8 372	0.02
Ireland	8 416	1 112	1 406	10 935	0.13
Italy	7 389	2 383	313	10 085	0.03
Luxembourg	15 988	3 965	2 003	21 956	0.09
Netherlands	9 686	2 045	1 651	13 382	0.12
Portugal	**6 058**	**1 234**	**418**	**7 710**	**0.05**
Spain	6 784	1 647	662	9 093	0.07
United Kingdom	10 506	1 677	1 524	13 707	0.11
European Union	9 078	2 203	1 040	12 320	0.08

C. Measures of inequality in income distributions in the EU[2]

	Gini coefficient %		Quintile ratio S80/S20	
	OI	TI	OI	TI
Austria	31	26	5.4	3.9
Belgium	32	28	10.3	4.4
Denmark	30	23	8.0	3.3
France	34	29	7.4	4.4
Germany	30	28	6.8	4.5
Greece	35	34	6.7	6.3
Ireland	39	33	26.8	5.4
Italy	34	33	6.9	5.8
Luxembourg	32	28	6.2	4.1
Netherlands	33	29	11.5	4.5
Portugal	**39**	**37**	**9.1**	**7.1**
Spain	36	33	9.2	5.7
United Kingdom	39	33	16.9	5.5
European Union	35	31	8.4	4.9

Key: IBB = income before benefits; P = pensions; SBOTP = Social benefits other than pensions; TI = total income; OI = official income.

1. The first line of the table reveals that 36 per cent of Portuguese live in households where at least one member draws a pension. For other social benefits, the figure is 68 per cent. For all social benefits together, the figure is smaller than 104 per cent (*i.e.* 36 per cent plus 68 per cent) since some households receive more than one type of social benefit.
2. All amounts quoted in the table are for adult equivalent. This concept is based on household income not on the income of each member of the household taken separately.

Source: Eurostat.

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Assessment and agenda for reform

The foregoing analysis points to major defects in the Portuguese system of public spending planning and control. This is potentially important for economic performance, which depends, *inter alia*, on an economically efficient composition of public expenditure at overall levels that minimise the distorting effects of the taxes needed to finance it. Though Portugal's total spending is somewhat below the EU average, the rapid and unstructured growth of its primary current spending now above the OECD and EU average, is disturbing.

To an extent, some of the above problems were addressed in the 2001 Framework Law of the previous government. As well, the 2002 supplementary budget also included a number of measures with a longer-term impact on the control of spending. In particular, the merging and suppression of public organisms with similar functions providing a duplication of services in the public administration should help to reduce spending and free-up resources. The ban on the growth of the annual local administration debt, unrelated to structural fund spending, should also help to limit future spending *via* a better control of local spending. However, efficient public spending requires a strategic overhaul of the incentive systems built into the budget systems, institutional arrangements and the process of public service provision. An array of recommendations are highlighted in this section and summarised in Box 8, and discussed in greater detail below.

Strengthening monitoring and control of the budget process

The 2001 Framework Law indicates the direction for reforming budget management and control. This needs to be quickly endorsed.

- With pre-budget discussions taking place in May, and a medium-term budget framework, covering four years the budgetary process will be more transparent. However, to enhance fiscal sustainability the budget and the Parliamentary appropriation should be subject to pre-approved spending ceilings over the whole period covered by the expenditure program.
- In order to make budget targets more credible, the assumptions behind the macroeconomic scenario underlying the draft budget should be made public and subjected to scruting by independent forecasters. The budget law should be approved only if a quantification of the budget's short- and long-term effects is carried out beforehand. Furthermore, it is important that budget proposals include a "margin of safety" to allow for unexpected imbalances.
- The ongoing transition towards accrual accounting of public expenditure needs to be stepped up in order to improve transparency and set budget priorities at all levels of government.

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Box 8. A synopsis of recommendations to improve public spending efficiency and effectiveness

Strengthening monitoring and control of the budget process

- The 2001 Framework Law and the 2002 Budgetary Stability Law need to become operational and effective as soon as possible.
- Set aggregate spending ceilings, based on prudent macroeconomic assumptions, over the four year period covered by the expenditure program. This should prevent overspending and undesirable mid-year cuts in budget allocations.
- Consider introducing a quantification of short and long term effects before the budget is approved. This should be done sufficiently in advance to allow the supervision of independent experts.
- Allow managers to transfer efficiency savings between programmes and fiscal years.
- Strengthen internal and external auditing.
- Move to accrual accounting of public expenditure.

Improve matching of resources

- Allow greater fund raising autonomy to enforce local government's spending discipline and increase local accountability.
- The internal budget stability rules endorsed by 2003 State Budget Law should be accompanied by compulsory deadlines for the presentation of local budget outturns and by credible sanctions when spending ceilings are not met. The strict enforcement of a no bailout rule would be instrumental in this regard.
- Consider reassessing the non-pension transfer system and the tax credits for health and education to make the spending system more redistributive.

Increase flexible incentives and strengthen control mechanisms in public administration

- Define public sector objectives in a more outcome-oriented fashion, and devise instruments to evaluate managers' performance. The establishment of an independent body for this purpose and of activity-based funding systems for public entities might be instrumental.
- Rely more extensively on performance-oriented career management schemes for public employees and reconsider the extent of their non-pecuniary benefits (*e.g.* long-life tenure and pension arrangements in particular). Encourage internal mobility of civil servants.
- Rely more on fixed-term contracts, or recourse to outside consultants, to handle issues that could prove temporary. This could allow mangers further flexibility to allocate resources according to needs.

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Box 8. **A synopsis of recommendations to improve public spending efficiency and effectiveness** (*cont.*)

Encourage competition and extend market signals

- Strengthen the user choice, both across public providers and between them and private providers. This would require the development of cost-accounting, in particular for local governments.
- Consider funding compulsory education on the basis of the number of student served and less dependent on the institutional structure.
- Reduce public funding for tertiary education by raising private fees for tuition. At the same time introduce mean-tested mechanisms and loan arrangements to ease access to young people from disadvantaged backgrounds.
- Pursue pilot schemes to introduce efficiency incentives in the provision of public health care. Introduce them more widely if they prove successful.
- Increase co-payments to limit excessive pharmaceutical consumption, paying due regard to equity objectives.
- Pursue efforts to encourage outsourcing and competitive tendering, subject to a careful assessment of benefits and transaction costs.

- Similar to other OECD countries, the carry-over of a certain percentage of the unspent appropriation to the next fiscal year should be envisaged.[68] This would help to avoid "spending sprees" at the end of each fiscal year that normally lead to unplanned and inefficient spending.
- As concerns audit activities, the internal audit system should be reinforced and some form of external auditing should be established.

Better matching of resources could enforce local governments' spending discipline…

Many of the incentive problems derive from the nexus of central-local government relations. Some progress has been made towards greater devolution of power; but local authorities should be given more spending and taxing powers to make fiscal responsibility operational within harder budget constraints. Local government finances currently depend on grants that do not provide adequate incentives to allocate spending efficiently, notably in staffing, health and education. Local governments should be given more freedom to organise local expenditure programmes, avoiding overlapping responsibilities with the central government, and to design the appropriate fees and charging structures. This would make local

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policy more transparent and accountable before the local electorate. Local public administration and public good delivery should also improve. Meanwhile, the restrictions on additional borrowing by local governments proposed by the 2001 Framework Law and endorsed in the State Budget for 2003 should be imposed each year and incorporated in the Local Finance Law. Compulsory deadlines for the compilation of local governments budget statistics should also be introduced. Sanctions have been introduced, when spending ceilings or deadlines are not met; they should be made effective.[69] Strict enforcement of a no bailout rule would be instrumental for effective compliance.

... and improve income distribution

Insofar as progressive tax and non-pension transfers have an income-distributive objective (though at some cost in terms of labour and other incentives) the marginal excess burden of the system could be reduced by simplification and streamlining without significant social cost. A step in this direction came with the ending of subsidies to home loans in October 2002; the elimination of the tax credits for private health and education expenses could also be considered since they tend to be regressive.[70]

Setting outcome objectives and increasing incentives for managers

Empowering and motivating managers to improve performance offers scope for efficiency gains from lower staffing levels and reduced operating expenditures, as well as improved public services. But achieving these gains requires more flexibility in public management, to enable performance assessment that ensures that accountability goes hand-and-hand with greater autonomy. Health services could be made more cost-effective, for example, if hospital managers were given greater discretion and autonomy, including in those public hospitals which are not slated to be run as private enterprises.

Managers' activities should be governed by properly-defined objectives and consistent evaluation. This should be supported by measures that create the appropriate incentives for public employees. In those areas where outcomes – individual or team based – are measurable, public bodies should be able to use their efficiency gains to reward employees associated with those gains. In addition, there should be a general shift towards a gradually weakening of seniority-based wage and promotion elements. Pilot schemes in health, justice and tax administration should be reviewed and extended. More generally, public employees' compensation should more closely reflect economy-wide labour market conditions. This implies a re-assessment of the generosity of civil servants' wage and non-wage compensation to reduce the gap with the private sector and enhance mobility between the public and private sectors.

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In less than ten years, 40 per cent of civil servants are expected to retire, which could represent an opportunity to replace older, low-skilled workers with smaller numbers of better-trained recruits. The more general use of fixed-term employment contracts could allow managers more flexibility in attracting the best suited employees and to allocate resources according to needs. This could help avoid blind hiring freezes in future. In the meantime, labour mobility within public administration should be strongly encouraged. Recent reform measures will help increasing mobility within the administration. These measures, including the creation of the public employment "pool", are key to re-allocate staff for the restructuring of central government, while maintaining some control over the wage bill. Enhanced mobility would help to maintain public service quality across the country. The average quality of education, for example, could be improved across regions if the Ministry of Education had greater possibilities to re-allocate teachers according to educational needs or from schools with a decreasing student/school ratio to schools that are relatively understaffed, while taking account the family circumstances of the teachers involved. In addition, if salary levels were allowed to take local conditions more into account, health provision as well as teaching resources could be improved in needy municipalities.

The structure of public hospitals could be improved and made more cost-effective if managers were given greater authority over procurement and staff policy (and had the training to enable them to do so efficiently). The availability of improved facilities to measure efficiency gains could also be used to reward productivity gains. The activity-based costing analysis, currently applied as a pilot programme in the hospital of *Nossa Senhora do Rosário*, could be instrumental to allow managers to detect areas where resources could be used more effectively and keep cost control. Transforming hospitals into public enterprises, subject to private-sector law and hard-budgeting constraints as planned, could also help to improve performance while containing costs.

Encouraging competition and extending market signals

A number of ways of improving the efficiency and the effectiveness of public outlays have been tried in OECD countries. Possible techniques include user charges, enhanced consumer choice (vouchers), creating internal competition and contracting-out. Some of these market mechanisms have been used in Portugal, but their extension could improve allocative efficiency.

User choice could be strengthened across jurisdictions…

Enhanced competition between alternative providers of public services would help to raise public spending efficiency and responsiveness to users' needs. Stimulating demand-side pressure to improve public sector performance would require removing the existing geographic restrictions on user choice, particularly for

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hospitals and compulsory education, as residents from one municipality are not entitled to use the services offered by another. The "money follows the user" principle could be applied more consistently through co-operation agreements between municipalities. This, though, would require accurate and transparent information on the cost structure of public service providers. To facilitate the development of cost-accounting, guidelines could be provided by the central government.

… and between public and private sectors

Empirical studies indicate that competition between public and private schools can improve learning outcomes in both. A central element in compulsory education spending could be to move towards funding based on the more number of students served and less on whether the institutions concerned are public or private. Some OECD countries have allowed parental choice of school, while letting the public funding follow the students as a type of voucher, thereby allowing public and private schools to compete. At present, existing differences in parental fees encourage social segregation, with public facilities taking a greater share of the children from low-income families and with special needs. To ensure a level playing field for this competition, and to avoid social segregation, Sweden and the Netherlands, for example, provide private schools with the same level of public funding, as public schools. Sweden also prevents private schools from charging tuition fees, while in the Netherlands market competition is sufficient to prevent private schools from charging fees.

An extended use of price signals could reduce excessive demand for costly public provision

There is scope to raise the price individual users pay for publicly-funded services, without raising serious distributional concerns, at least in tertiary education and in some areas of health spending. The high subsidy element involved in public education, together with high rates of return to tertiary education, call for an assessment of the level of cost recovery in public universities. University tuition fees could be raised and student loans could be made available at the same time. Education tax credits should be reconsidered to focus more on lowest income groups. Evidence from some OECD countries (*e.g.* United Kingdom, Australia and New Zealand) shows that participation in tertiary education does not fall with higher tuition fees when accompanied by means-testing mechanisms and loan arrangements that take graduates income into consideration (Blondal, Field and Girouard, 2002). Moreover, besides reducing public costs in post-compulsory education, such a system can reduce inequality without compromising efficiency goals. An increase in tuition fees would reduce the regressive nature of public funding of post-compulsory education, while the greater availability of student loans would act to offset the impact of increased private costs of enrolment. Easier

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access to student loans may be important to young people from disadvantaged backgrounds, improving opportunities for all individuals to develop their potential.

In the health care area, higher co-payments could make households more cost aware and help them to limit excessive pharmaceutical consumption. It is important, however, that any increase in co-payments should not conflict with equity objectives. This can be achieved through increased differentiation of co-payments according to indicators of income levels or general health status. Alternatively, tax credits for drugs could be subject to a ceiling equivalent to the price of generics. Entry restrictions in the pharmacy sector should be eased; the market share of generic treatments should be raised and fixed margins on the sale of pharmaceuticals should be abolished.

Improving and expanding competitive tendering and public-private partnerships

The appropriate use of competitive tendering and contracting can raise efficiency and thereby allow a constant quality of service to be provided at lower costs – especially for technical services like cleaning, where savings typically range from 10 to 30 per cent. The sources of these savings include efficiency gains from economies of scale and specialisation, as well as improved work practices. However, contracting is not free from pitfalls and a careful assessment should be made in each case to weigh the benefits from efficiency gains against transaction costs and to ensure that contracting is made in a way that does not lock the public purchaser in with a single private monopoly supplier. Moreover, public managers need to acquire the capabilities associated with managing competitive tenders and contractual relations.

The recourse to private sources of financing through public-private partnerships will continue with the creation of ten new hospitals. Public-private partnerships are often used as a means both of raising more funds for investment and benefiting from private sector expertise. Moreover, involving private firms as suppliers may make competition more dynamic, as they are more likely than public agencies and private non-profit institutions to expand their activities. However, experience in other countries shows that contracts need to be well designed[71] to insulate the budget from the risk of project failure.

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IV. Structural reform for sustaining high growth

The weak macroeconomic conditions, the urgency of budgetary consolidation and the need for improving the effectiveness of public expenditure which are described in the previous chapters should not divert the attention of policy makers from moving forward in structural reform. The new government, in place since April 2002, has also emphasised the need to advance on a wide front in order to foster growth and productivity. This chapter starts by reviewing Portugal's economic performance. It then focuses on key areas of a comprehensive growth-oriented strategy for Portugal: enhancing human capital and improving labour market adaptability; putting in place conditions conducive to the adoption of new technologies and to the dynamism of entrepreneurial activity; and strengthening competition in product markets. A summary of the OECD's main recommendations for further structural reforms is included at the end (Box 11). The chapter concludes with the discussion of three environmental dimensions of sustainable development: air pollution, water pollution and the sustainable use of natural resources.

Portugal's growth performance in comparison

The Portuguese economy has recorded rapid growth in the past 20 years, allowing some catching up in living standards with the European average.[72] Per capita GDP now stands at three-quarters of the EU average (2001, measured in PPP), about 20 percentage points more than at the time of accession to the EU in 1986. Nonetheless, with the second lowest per capita GDP in the EU, Portugal still has some way to go (Figure 24). The gap in income levels between Portugal and the EU average reflects only to a small extent lower utilisation of labour, as employment rates are relatively high in Portugal. The main cause of the income gap is the lower level of labour productivity.

This lower level of labour productivity raises the question of why productivity has not grown at a faster rate, and, what can policies do to raise it. In standard growth accounting analysis, growth of output is "explained" in terms of growth of inputs of labour and capital and a residual factor interpreted as technological progress, or multifactor productivity. Such analyses do not leave a role for

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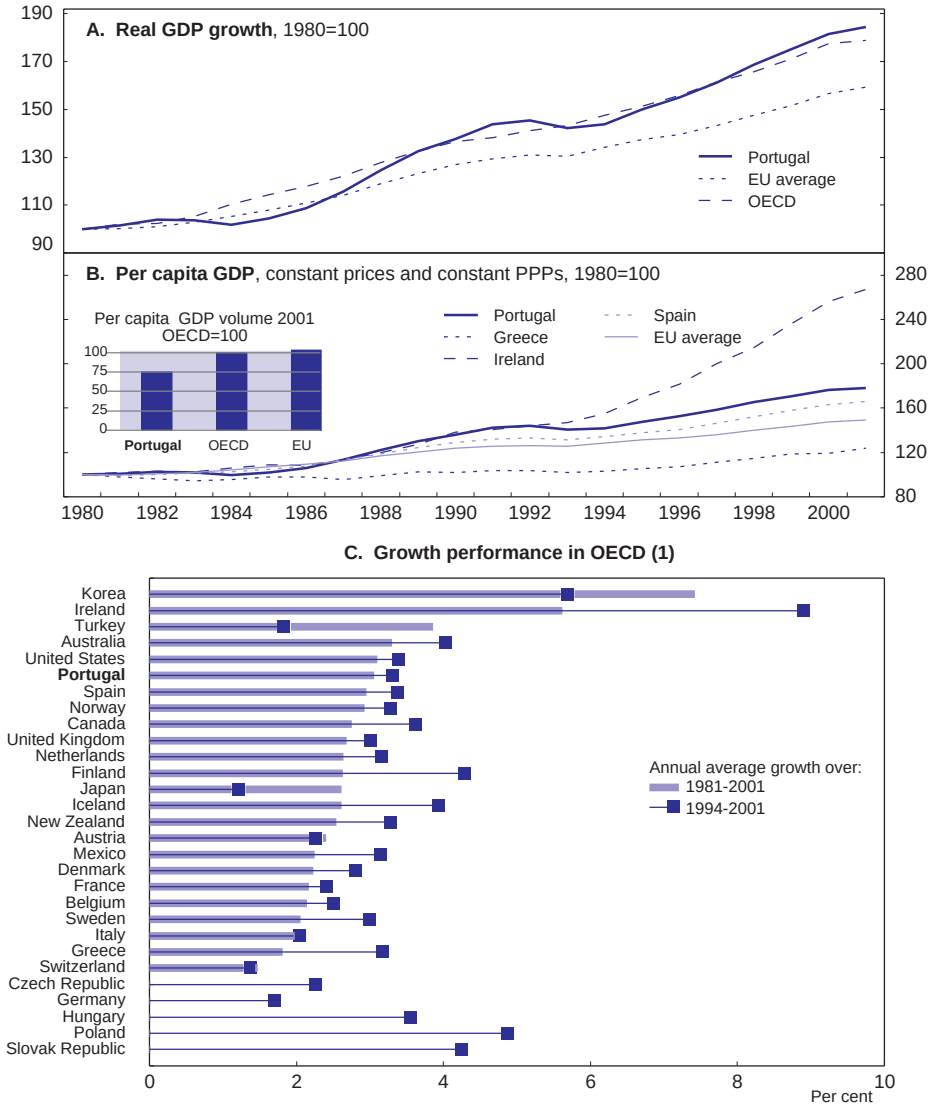
Figure 24. **Portugal's growth performance in comparison**

Measured by the real GDP



1. The series for the Czech Republic, Germany (due to unification), Hungary, Poland and Slovak Republic are incomplete before 1991. These countries are ranked according to alphabetical order.

Source: OECD, *Main Economic Indicators;* OECD, *National Accounts*.

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policy in explaining growth, which by hypothesis depends on exogenous factors such as savings rates and population growth, and the unexplained residual term. An alternative approach is based on a (conditional) convergence model, in which growth of per capita GDP depends on initial conditions (levels of human and physical capital as well as of output) and policy-related variables, that help explain the convergence of per capita GDP growth to an equilibrium value over time. This approach was one adopted in the OECD "Growth Project" (2001), using panel data for 19 OECD countries between 1974 and 1997.[73]

In the particular case of Portugal, that analysis showed that during this period, *i*) actual Portuguese per capita growth was above the OECD average by more than half a percentage point (*i.e.* there was convergence of per capita incomes towards the OECD average); *ii*) the main reason for the fast growth was the low starting level, which boosted growth by 2½ points relative to the OECD average, *ceteris paribus*; *iii*) because the level of human capital was considerably lower than the OECD average, the growth rate was lowered by 1.2 percentage points, again *ceteris paribus*; *iv*) an above-average share of investment in GDP (Figure 25) raised growth by about a half-point; *v*) the separate effects of demographic factors, of trade opening and of the "size of government" were each comparatively small, but cumulatively more than offset the negative impact of macroeconomic instability; and *vi*) a "country-specific effect" lowered growth by 1½ points, *ceteris paribus* (Table 16). This estimated country-specific effect was the largest for all countries in the sample, and its interpretation is that Portuguese growth has been held back in the past by inefficiencies that have wasted resources of labour and capital and prevented them from being used optimally. In this sense, growth of labour productivity was "too slow".

Indeed, during the 1990s, in particular, annual Portuguese labour productivity growth rate was below 2 per cent.[74] If the raw labour productivity data are corrected for increases in the quality of labour, as proxied by the educational attainment of the labour force, hourly labour productivity growth in Portugal, at 1 per cent annually in the 1990s, has lagged behind that in other OECD countries, on average (Figure 26). Although the levels of educational attainment in the Portuguese labour force have risen over past decades, they are still below the OECD average. In the early 1970s, compulsory schooling lasted only 4 years, helping to explain why only 10 per cent of older workers have at least upper secondary educational qualifications (see section below). Subsequent reforms have tripled this ratio for younger workers, and the "Growth Project" analysis estimated that without this progress, the acceleration in per capita income growth would have been less marked in the 1990s.[75] Nevertheless, there is ample room for substantial increases in output via reforms that remove inefficiencies in the way resources are used, allowing the economy to reach its true potential, as well as other reforms that increase the potential rate of growth itself.[76] Such reforms should be directed to making product markets function more efficiently *via* competition-oriented

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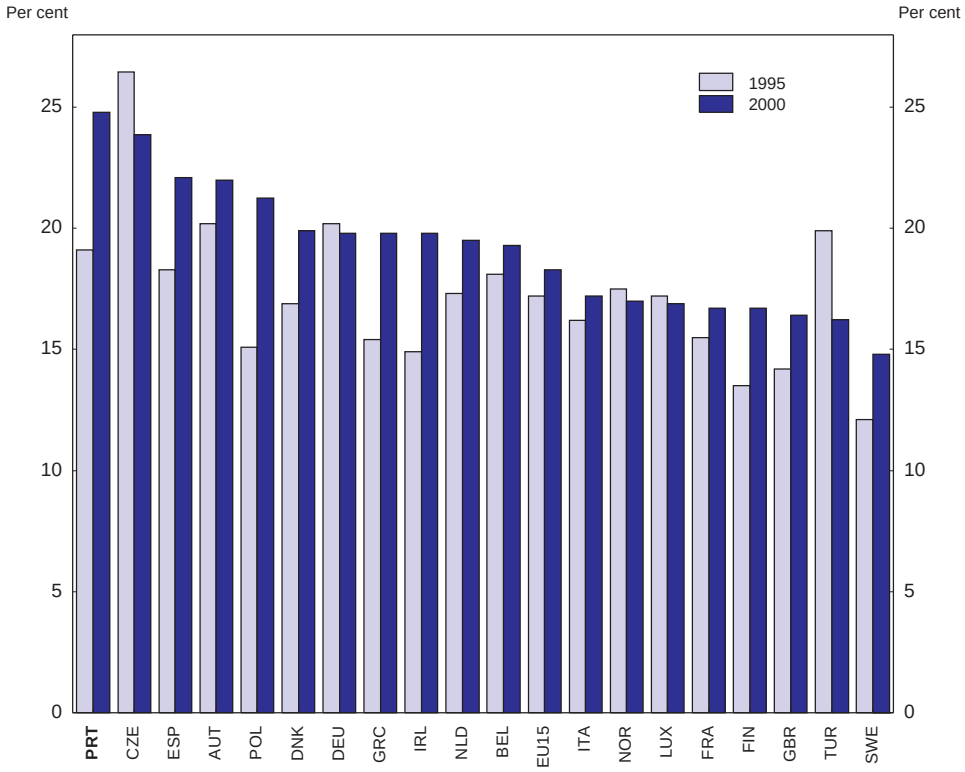
Figure 25. **Private investment share**
Per cent of GDP



Source: European Commission, Structural Indicators.

policies, and reducing the regulatory burden on enterprises which, though improved, compares poorly with other countries (Table 17), and discourages innovation. The results of Scarpetta and Tressel (2002) suggest that an alignment of the Portuguese regulatory stance to "best practice" would narrow the technology gap and increase productivity levels.[77]

The effects of labour market reforms go beyond their direct impact on labour market outcomes because they also enhance innovative activity and hence output growth. The potential growth effects of labour market reforms (like easing employment protection, reducing the administrative extension of collective

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Table 16. **Factors explaining differences in growth rates across OECD countries**[1]

1970s-1990s, annual percentage point growth rates, per capita

	Annual average growth rate	Growth differential	Initial conditions (real GDP/pop)	Investment share	Human capital	Population growth	Variability of inflation (SDinfl)	Government consumption	Trade exposure (Trade exp^{adj})	Residual country specific effect
Ireland	3.02	**1.47**	1.54	–0.18	–0.32	–0.18	0.01	0.09	0.17	0.34
Portugal	**2.15**	**0.60**	**2.56**	**0.58**	**–1.20**	**0.07**	**–0.10**	**0.10**	**0.11**	**–1.52**
United States	1.93	**0.38**	–1.62	–0.34	0.63	–0.09	0.07	0.09	–0.25	1.89
Finland	1.82	**0.27**	0.51	0.05	0.02	0.15	0.00	–0.06	–0.26	–0.14
Italy	1.73	**0.18**	0.22	–0.13	–0.69	0.13	0.02	0.01	0.14	0.48
Norway	1.72	**0.17**	–0.12	–0.05	0.35	0.07	0.03	–0.06	–0.04	–0.01
Denmark	1.69	**0.14**	–0.57	0.28	0.21	0.12	0.02	–0.14	–0.05	0.27
Australia	1.68	**0.13**	–0.37	0.20	0.52	–0.25	0.03	0.01	–0.41	0.40
Belgium	1.66	**0.11**	–0.53	0.02	–0.15	0.20	0.03	–0.05	0.53	0.06
United Kingdom	1.63	**0.08**	0.05	–0.21	0.17	0.15	–0.03	–0.02	0.31	–0.34
Austria	1.57	**0.02**	–0.41	0.07	0.26	0.01	0.05	0.00	0.03	0.01
France	1.35	**–0.20**	–0.59	–0.09	–0.10	0.07	0.07	–0.08	0.05	0.48
Canada	1.32	**–0.23**	–0.90	–0.21	0.62	–0.18	0.04	–0.07	0.14	0.32
Spain	1.28	**–0.27**	0.73	0.04	–1.12	0.00	0.03	0.07	–0.14	0.11
Netherlands	1.26	**–0.29**	–0.47	–0.03	0.25	0.01	0.06	–0.13	0.52	–0.50
Sweden	1.20	**–0.35**	–0.60	–0.10	0.21	0.11	–0.10	–0.17	0.01	0.30
Greece	1.15	**–0.40**	2.00	0.19	–0.56	–0.07	–0.16	0.17	–0.51	–1.48
Switzerland	0.81	**–0.74**	–1.75	0.08	0.59	–0.04	0.00	0.15	0.02	0.21
New Zealand	0.53	**–1.02**	0.34	–0.17	0.31	–0.29	–0.07	0.10	–0.36	–0.87

1. Decomposition of cross-country differences in annualised growth rate (in %) from the average output per capita of 1974-77 to the average output per capita of 1994-97. See Bassanini A. and S. Scarpetta (2001) for details of the analytic approach behind these results.

Source: OECD.

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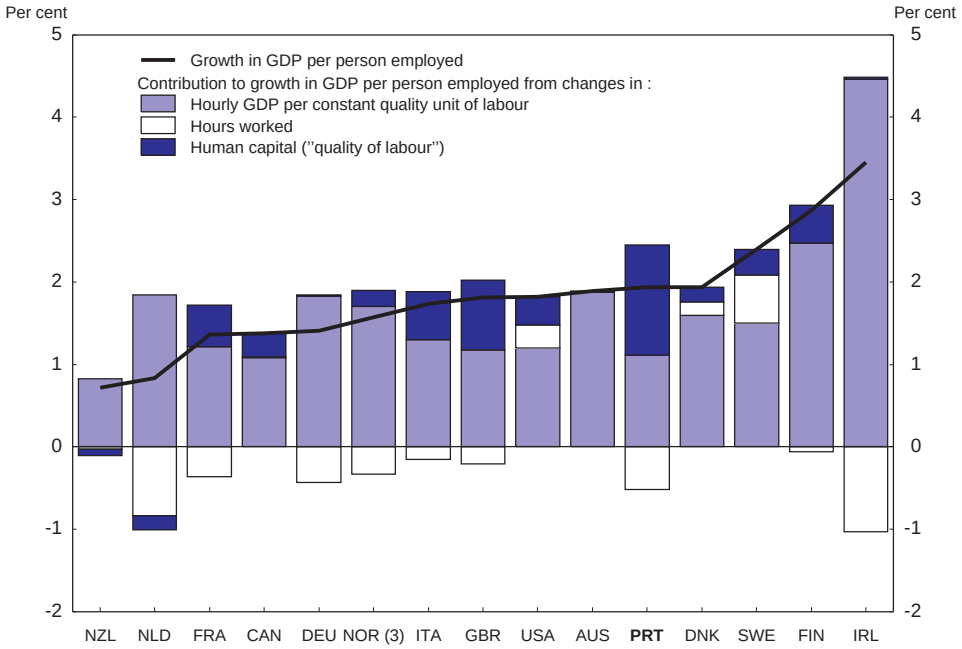
Figure 26. **The role of skills and labour utilisation in labour productivity growth**[1]

Average annual percentage change, 1990-2000[2]



1. This is based on a simple quantitative decomposition: growth in GDP per person employed = hourly GDP per constant quality unit of labour + growth in average hours worked + human capital adjustment.
2. 1991-2000 for Germany, 1990-1999 for Ireland.
3. Mainland only.

Source: OECD.

agreements and lowering tax wedges) are likely to be reinforced since they increase specialisation in R&D intensive industries. In a rapidly changing environment, improvements in MFP are largely the result of innovation by best-practice firms, technological catch up by other firms and reallocation of resources across firms and industries. Bassanini and Scarpetta (2001) found evidence of a clear positive effect of innovation activity on output: a persistent 0.1 percentage point increase in the share of business-sector R&D spending in GDP was estimated to boost the level of GDP *per capita* by 1¼ per cent in the long term. It follows from the above that action on a wide front is required, with continuing efforts to strengthen competition pressures in product markets, to raise human capital, to ease labour

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Table 17. **The regulatory environment: aggregate indicator**[1]
Time series regulatory indicators[2] (scale 0-6 from the least to the most restrictive)

	1978	1982	1988	1993	1998	1998-1978	Percentage change
Greece	5.7	5.7	5.7	5.5	5.1	–0.6	–0.10
Italy	5.8	5.8	5.8	5.3	4.3	–1.5	–0.25
Portugal	**5.9**	**5.9**	**5.4**	**4.9**	**4.1**	**–1.8**	**–0.30**
Ireland	5.7	5.7	5.1	4.8	4.0	–1.7	–0.29
Switzerland	4.5	4.5	4.5	4.4	3.9	–0.6	–0.14
France	6.0	5.9	5.7	4.7	3.9	–2.1	–0.35
Spain	4.7	4.7	4.6	4.2	3.2	–1.5	–0.31
Austria	5.2	5.1	4.5	3.9	3.2	–2.0	–0.39
Belgium	5.5	5.5	5.0	4.3	3.1	–2.4	–0.43
Netherlands	5.3	5.5	5.5	4.1	3.0	–2.4	–0.44
Japan	5.2	5.2	3.9	3.2	2.9	–2.3	–0.44
Denmark	5.6	5.5	5.5	4.0	2.9	–2.7	–0.48
Finland	5.6	5.5	4.8	4.0	2.6	–3.0	–0.53
Norway	5.0	5.0	4.3	3.2	2.5	–2.5	–0.49
Canada	4.2	4.2	2.8	2.6	2.4	–1.9	–0.44
Germany	5.2	5.2	4.7	3.8	2.4	–2.8	–0.54
Sweden	4.5	4.4	4.2	3.5	2.2	–2.3	–0.51
Australia	4.5	4.5	4.2	3.3	1.6	–2.9	–0.65
United States	4.0	3.3	2.5	2.0	1.4	–2.7	–0.66
New Zealand	5.1	5.1	3.6	2.2	1.4	–3.7	–0.73
United Kingdom	4.3	4.2	3.5	1.9	1.0	–3.3	–0.76

1. Countries in descending order of their 1998 results.
2. Simple averages of indicators for 7 industries: gas, electricity, post, telecoms, air transport, railways, road freight. Depending on the industry the following dimensions have been included: barriers to entry, public ownership, market structure, vertical integration, price controls.

Source: Nicoletti *et al.* (2001).

market regulations, to facilitate the re-allocation of labour among firms and among sectors, and to foster innovation and technological progress.

Upgrading skills and human capital

Despite considerable improvements in the last 10 to 15 years, one of the main remaining impediments to higher income levels is the large gap in terms of human capital. The ineffectiveness of the Portuguese education system in imparting relevant skills to all the relevant age groups, notwithstanding substantial allocation of resources, justifies strong policy attention. The upgrading of skills and human capital is key to increase productivity and competitiveness and ensure a successful response to the challenges ahead. At present, Portugal is one of the few OECD countries where job creation in low-skill jobs is still strong. As this is not likely to last, efforts must continue on a wide front to raise the quality of formal education and ensure that all students leaving school have a qualification. In

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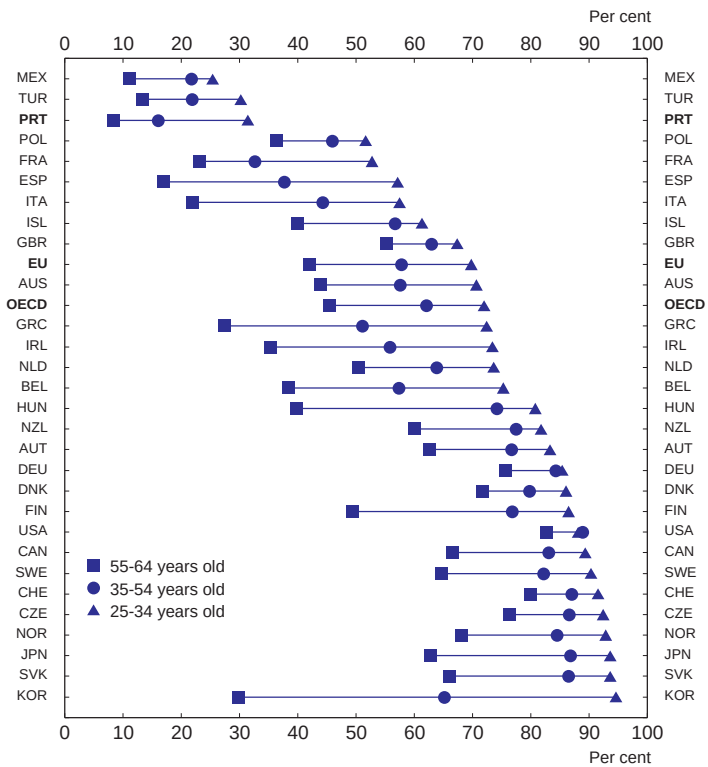
parallel, actions to make the workforce adaptable to new job opportunities will need to be sustained, stressing the importance of life long learning.

Assessing the stock of human capital

Despite improvements in recent years, Portugal is still one of the worst performers in terms of education attainment of the working age population (Figure 27). In 2001, only one fifth of the population aged between 25 and 64 had completed upper secondary education or better (against more than three-fifths on average in the OECD).[78] Even for younger age groups, Portugal remains far below

Figure 27. **Educational attainment of the working-age population**

Population with at least an upper-secondary qualification, 2000[1]



1. Per cent of each age group; 1999 for Austria, Finland, the Netherlands and Norway.
Source: OECD, Labour Market Statistics database.

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average: in 2001 about one third of people aged 25 to 34 had attained at least upper secondary education, compared with around three-quarters in the OECD. The results of the work-oriented tests of the International Adult Literacy Survey clearly show that adult literacy is an area for concern. Portugal is among the lowest ranking countries: more than 40 per cent of the population is at the lowest level in all literacy tests.[79] The Portuguese poor performance essentially reflects results obtained by those who have not benefited from upper secondary education.

The poor performance of the educational system

The share of resources (public and private) that Portugal devotes to education has increased sharply over the past twenty years (see Chapter III).[80] This has financed the increase in enrolment rates and expected duration of studies to the levels prevailing in more advanced countries. However, the Portuguese education system does not seem efficient, comparing the level of spending with education outcomes.[81] The percentage of students graduating from upper secondary school at 18 years is relatively low, because of high failure and dropout rates. In 2001, around 45 per cent of young people aged 14-18 left school with only basic qualifications, the highest proportion in the European Union.[82] In the past decade, the opening of the tertiary education system to the private sector led to an explosion in the number and diversity of courses provided. However, at that level as well there are problems of high failure/repetition and dropout rates.

Based on international comparisons, the performance of Portugal's education system rates poorly. The OECD Programme for International Student Assessment (PISA) shows that the ability of Portuguese youth (15-year olds about to finish compulsory schooling) to use their knowledge and skills in order to meet real-life challenges is significantly below the OECD average.[83] In reading literacy of students, Portugal, ranking 26th out of 32 countries, is below Spain or Ireland and close to Latvia and Russia, countries with significantly lower per capita incomes. Similar results were observed in mathematical and in scientific literacy tests (Portugal ranking 28th). Poor achievements by the cohorts that are currently going through school call for immediate policy attention, because they suggest that skill shortages might remain a problem in Portugal for a long time.

The returns to education are high

The low levels of human capital of the Portuguese workforce make skills particularly scarce and, hence, valuable. Portugal exhibits large economic returns to education[84] (see Box 9). The premium for attending university seems to have increased during the 1990s and is high by international comparison. In addition, recent applied research on wage formation in the Portuguese labour market found that years of education, experience and tenure have positive influence on the individual's wage, with education having the highest impact.[85] The high returns to

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Box 9. Rates of return to education in Portugal, 1999

To obtain information on returns to education in Portugal, the direct derivations of internal rates of return made in Blondal *et al.* (2002) for 10 OECD countries were reproduced for Portugal using data for 1999.* The results confirm the existence of very high returns to education in Portugal, especially from upper secondary to tertiary education. Overall, the estimates indicate that there are strong incentives for the average student to continue studying beyond the compulsory schooling age.

The overall incentives to invest in human capital that are embedded in the labour market benefits, financing and tax arrangements can be summarised in estimates of private internal rates of return. The internal rate is equal to the discount rate that equalises the real costs of education during the period of study to the real gains from education thereafter. In Portugal, these net gains due to human capital investment in tertiary education are estimated to be above 20 per cent (see Table), which is much higher than for the other 10 countries included in the OECD study. The high private internal rates of return that are available to those who successfully complete upper-secondary and, especially, tertiary education programmes suggest that there are strong incentives for the average Portuguese student to engage in post-compulsory education.

Private rates of return[1] to education for men in selected OECD countries

Per cent

	Upper secondary	Tertiary
Portugal (1999)	**11.3**	**23.9**
United Kingdom (2000)	15.1	17.6
United States (1999)	16.4	17.5
France (2000)	13.5	14.1
Netherlands (1997)	7.9	9.7
Canada (2000)	13.6	9.2
Sweden (1999)	6.4	9.1
Japan (1999)	6.8	8.6
Italy (1998)	11.2	8.3
Denmark (1999)	11.3	6.8
Germany (1998)	10.8	6.7
Unweighted average	11.3	12.0

1. Rates of return based on after-tax earnings and the length of studies taking into account the unemployment risk.

* See Annex II for a discussion of methodological issues.

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education may reflect the scarcity of skilled labour supply relative to the demand for skilled workers. The demand for graduates is in any case boosted by the public administration, which employs more than half of the annual flow of university graduates.

Enhancing the stock of human capital

Ambitious targets for upgrading education and workers' skills were set out in the National Action Plan for Employment (NAPE) adopted in 1998 within the framework of the 1997 Luxembourg process in the EU. The NAPE update for 2002 followed closely the guidelines and recommendations drawn up by the European Commission. Among its main priorities, special attention is to be given to improving education and training systems.[86]

How to improve educational achievements

Recent research work suggests that the improvement of schools depends more on better use of resources than on providing additional funding. Indeed, the gap in Portugal's educational outcomes does not seem to reflect inadequate levels of spending. Hence, reforms in the administrative structure of education, and the combination of incentives and competition, are required.

The Portuguese authorities acknowledge the need for a stronger focus on the quality of education. This may involve an improvement in professional management of schools, namely by enhancing the accountability of teachers and school managers. Related guidelines include the provision of more information about results achieved through different initiatives to facilitate evaluation, with a view to strengthening or discontinuing some programmes. The setting of explicit targets on key quantitative measures of performance could also be useful. Since 2001, a yearly assessment of the performance of a sample of pre-primary to upper secondary schools is made and a detailed report is published on the Internet.[87] In July 2002, a new law for the creation of a comprehensive system of evaluation of all non-tertiary schools was approved; it emphasised the following requirements: auto-evaluation of schools as well as external evaluation based on quantitative measures; public disclosure and certification of results; and rewarding of best performers.

There is substantive evidence that participation in pre-school programmes is positively associated with later success in school. Efforts have been made in Portugal over recent years to improve early childhood education, and by 1999, the coverage of the pre-school education system (3-4 year-olds) had reached European averages. In the nine years of basic education, the range of courses offered has also been broadened and modernised, with the objective of reducing dropout rates.[88] The reunification of the basic education levels, instead of the current division in three cycles, could also contribute to the definition of a

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uniform and more modern curriculum and improve student's achievements. Extending compulsory education to the upper-secondary level (12 years, in line with EU standards) should be an important near-term objective. As compulsory education is extended, vocational and technical education will also need to be further developed, as recognised in the 2002 NAPE, to give adequate skills to students entering the labour market.[89] In this context, the recent "nine plus one" years of schooling programme aims at ensuring that those who finish basic schooling with a view to entering the labour market have access to vocational training. Moreover, it is essential to strengthen the general upper-secondary education, making it more demanding to better prepare students to tertiary education programmes. However the need to achieve an appropriate balance between efficiency concerns and a higher enrolment rate in tertiary education requires that the minimum positive mark in upper-secondary graduation, introduced as a condition to access tertiary education institutions, is associated with a strong improvement in the quality of secondary level education.

Skills learned at the tertiary level are critical in providing the basis for technological development, good management and entrepreneurship. In Portugal, the number of universities and courses offered grew strongly in the last decade, as did tertiary enrolment rates. Since 2001, the number of places available is higher than the number of students applying to universities/polytechnics. This fact is explained by demographic factors, which will be reinforced in the future as ageing of the population continues. Currently, some courses already operate with very few students and there has been a progressive reduction of admission requirements, especially in private universities. Given also clear mismatches on the supply-side,[90] it seems essential to re-evaluate the operating and financing conditions of higher education institutions. The Ministry of Higher Education and Science seems committed to increase the quality of tertiary education, while maintaining an even-handed treatment of private and public institutions.[91] The creation of a systematic and independent system of evaluation and certification of universities and courses was one of the measures announced. It seems essential to modernise tertiary education curricula in ways that take into account labour market needs and to eventually close or merge courses with very few students. Additionally, government subsidies should be based on factors other than costs: greater funding should be given to priority areas, and courses/universities that do not meet a quality or desirability test should not be funded at all (see Chapter III for a discussion on the financing of tertiary education).

Facilitating transition from school to work

A characteristic at upper levels of education in Portugal has been the traditional bias in favour of general education programmes.[92] Maintaining a close relationship between schools and the business community is essential to ensure

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that the skills and competencies provided match the needs of the labour market. Exchange of experiences and information among educational, social and economic partners can improve responsibility and understanding. In February 2001, the social partners signed a tripartite agreement with the government on "employment, the labour market, education and training policy", that seeks to achieve changes regarding the training of employees, especially young people. In March 2002, new rules were introduced allowing companies to recruit young people, aged 16 years or above, who have not completed compulsory schooling or do not possess any vocational qualifications, on condition that they are provided with training opportunities.[93]

Significant advances have been made in boosting technical courses and giving greater emphasis to information and communication technology. Since the beginning of 2002, all schools are connected to the Internet. The challenge is now to increase the quality of the equipment and services available and to improve the ratio of pupils per Internet PC.[94] The Basic Skills Diploma in Information Technology, associated with the finishing of the basic education, was created in 2001 and made available to any citizen.[95]

Supporting lifelong learning

Increasing demand for workers who are good at using and interpreting knowledge flexibly can be only partially addressed through curriculum changes in schools and universities, because of the usual lags involved in education. Lifelong learning, outside formal education, provides an opportunity for individuals to repair and/or complement previously received education or training. Although survey data are of doubtful reliability for Portugal, they suggest that the proportion of people aged 25 to 64 participating in some continuing education and training is low by EU standards.[96] The Portuguese system for further education and vocational training is characterised by the dominant role of the state, a separation between government agencies responsible for education on the one hand and training on the other till recently and relatively little development of training within firms.[97] A new arrangement introduced in 2002 concentrates responsibilities for education and vocational training in the Ministry of Education.

Significant measures and reforms have been introduced in recent years to improve the adult learning system.[98] Portugal prepared a strategy for the development of adult education in 1998, focusing in particular on providing greater access to those with low education levels. The creation of a specialised national agency (ANEFA – *Agência Nacional de Educação e Formação de Adultos*) in 2000, now integrated in the Ministry of Education, responds partly to the need to co-ordinate all adult learning policies towards a common objective. Portugal is also in the process of defining a new national system of recognition, validation and certification of prior learning and experience. The social partners have a crucial role to play in this

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area. They can contribute to the identification of educational requirements and to the development of relevant education and training programmes, providing qualifications that are useful and recognised. The 2001 tripartite agreement included measures to consolidate the adult education public system and set specific objectives for firms' provision of training.[99]

Labour market and social policies

Portugal's labour market performance has been impressive, despite strict regulations

In recent years, the overall performance of the labour market in Portugal has been stronger than in its European counterparts (Figure 28). Labour force participation rates as well as employment rates have increased and in 2001 they stood above both the OECD average and those in most EU countries. Overall, unemployment declined to around 4 per cent in 2001. During the last decade, the unemployment rate in Portugal was four percentage points below the EU average and the employment rate around eight percentage points above.[100] Despite the recent deterioration following the economic downturn (with unemployment reaching 5.1 per cent in September 2002), the labour market situation still compares favorably with most other EU countries.

In recent years, youths and long-term unemployed, the categories most at risk, have also seen their situation improve. The rate of youth unemployment fell from over 16 per cent in 1996 to around 9 per cent in 2001. The share of long-term unemployment also declined in Portugal, though remaining above the OECD average.[101] Policy measures were taken in recent years to facilitate employment (or re-employment), in particular measures targeted at youths and long-term unemployed. However, the still-high proportion of long-term unemployment remains a concern, given its implications in terms of depreciation of a worker's human capital and the subsequent loss of motivation for job search.[102]

High wage flexibility but low adaptability of the workforce

The implications of labour demand changes on labour market outcomes depend on the specific institutions that regulate the market and the workforce characteristics, such as the degree of labour mobility and adaptability, the nature of contractual arrangements and wage flexibility.[103] Labour mobility and adaptability are low in Portugal compared with many other industrialised countries. The employment protection legislation (EPL) is rather restrictive, reducing the ability of firms to react quickly to shocks. Even after some easing during the 1990s, EPL in Portugal is among the strictest in the OECD but is under review (see below). In short, the legislation on collective dismissals imposes a long, complex, and costly process on employers.[104] High dismissal costs can lead to a continuation of unproductive employer-worker matches and reduce the scale of job-creation, while

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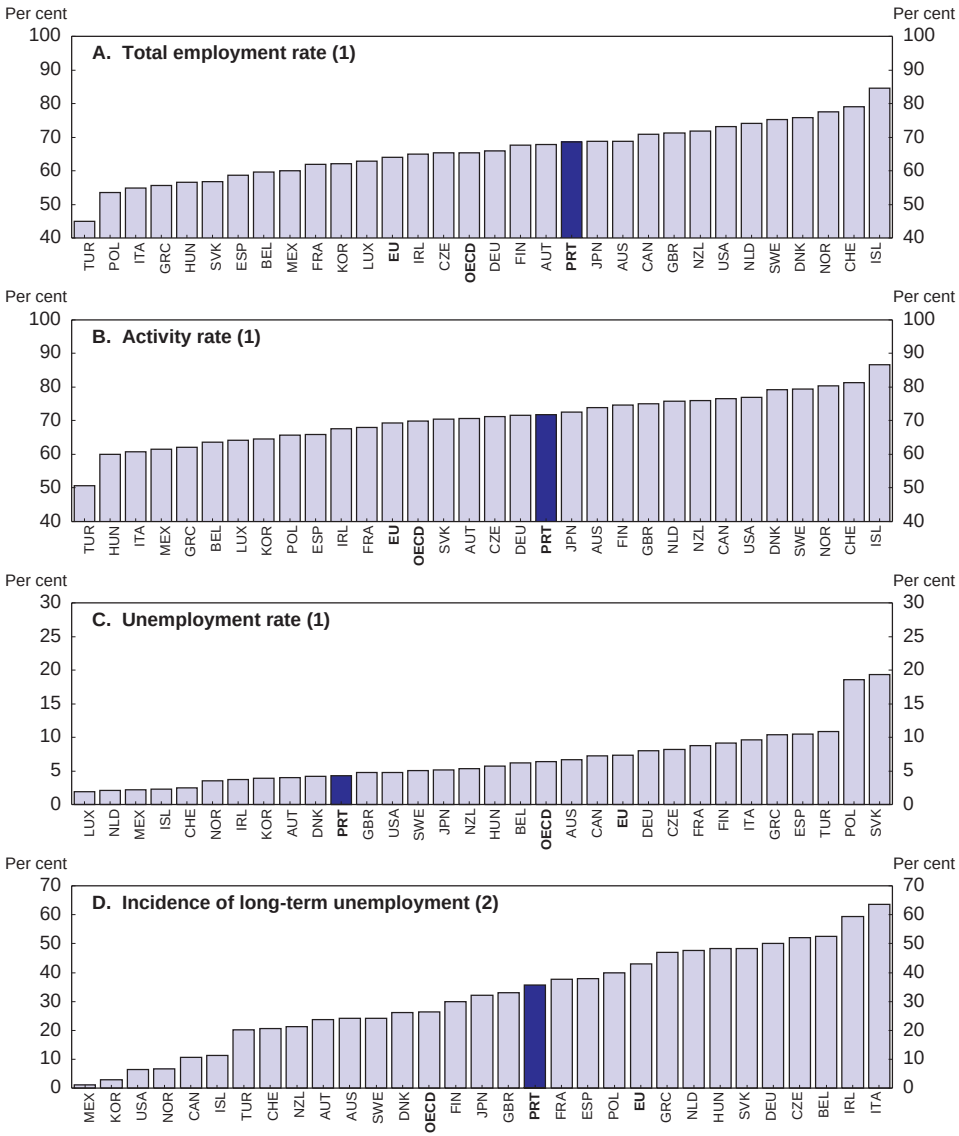
Figure 28. **Labour market indicators: an international comparison, 2001**



1. Ratios refer to persons aged 15 to 64 years who are in employment or in the labour force divided by the working age population, or in unemployment divided by the labour force.
2. Persons who are unemployed for 12 months and over as a percentage of total unemployment.

Source: OECD, *Employment Outlook*, July 2002.

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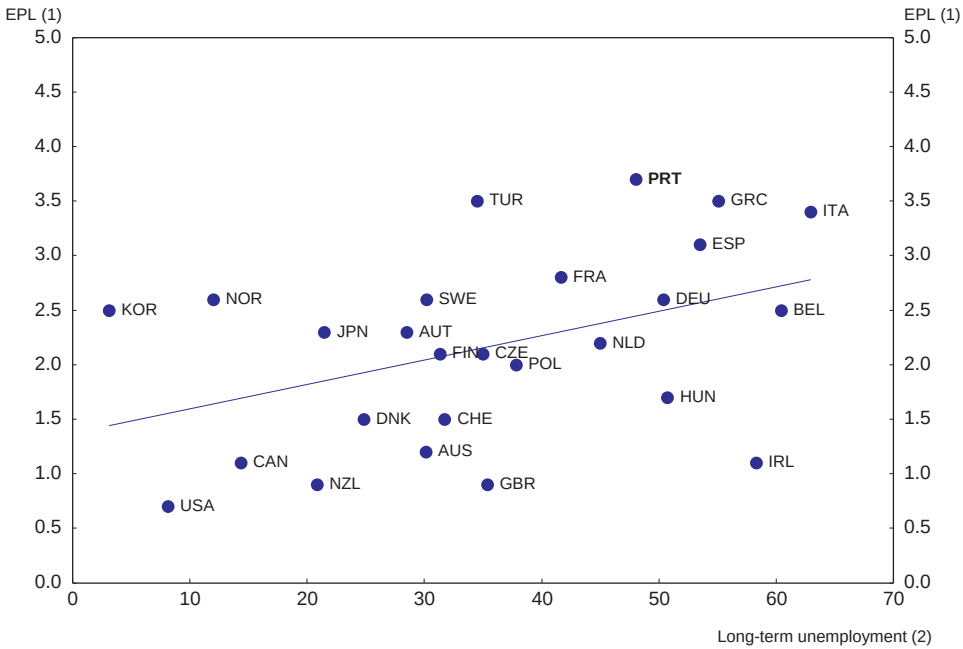
increasing the duration of unemployment[105] (Figure 29). In addition, the low educational level of the labour force hinders its ability to adapt to new industries. Improving the efficacy of the public employment service in terms of job-brokerage would facilitate labour mobility. Furthermore, easing housing market restrictions would also enhance the mobility of workers.

By contrast, several forces have ensured a high degree of flexibility of Portuguese wages,[106] thus helping to offset some of the negative effects of low labour mobility. Wage formation in Portugal is characterised by sectoral wage bargaining with limited co-ordination between bargaining unions, which has tended to weaken their bargaining power.[107] Furthermore, the traditional adjustment of the wage drift to the economic cycle also helps to ensure the flexibility of Portuguese wages. Portuguese

Figure 29. **Employment protection legislation: selected OECD countries**



1. The overall indicator of the strictness of employment protection legislation (EPL) is a weighted average of EPL indicators for regular contracts, temporary contracts and collective dismissals. Each of these indicators can range from 0 to 6, with higher values representing stricter regulation. The calculation of these indicators as well as the weighting scheme are explained in OECD *Employment Outlook*, June 1999 Chapter 2.
2. Average incidence of long-term unemployment (12 months and over) as a percentage of total unemployment over the period 1995-2000.

Source: OECD *Employment Outlook*, June 1999; OECD Labour Market Statistics database.

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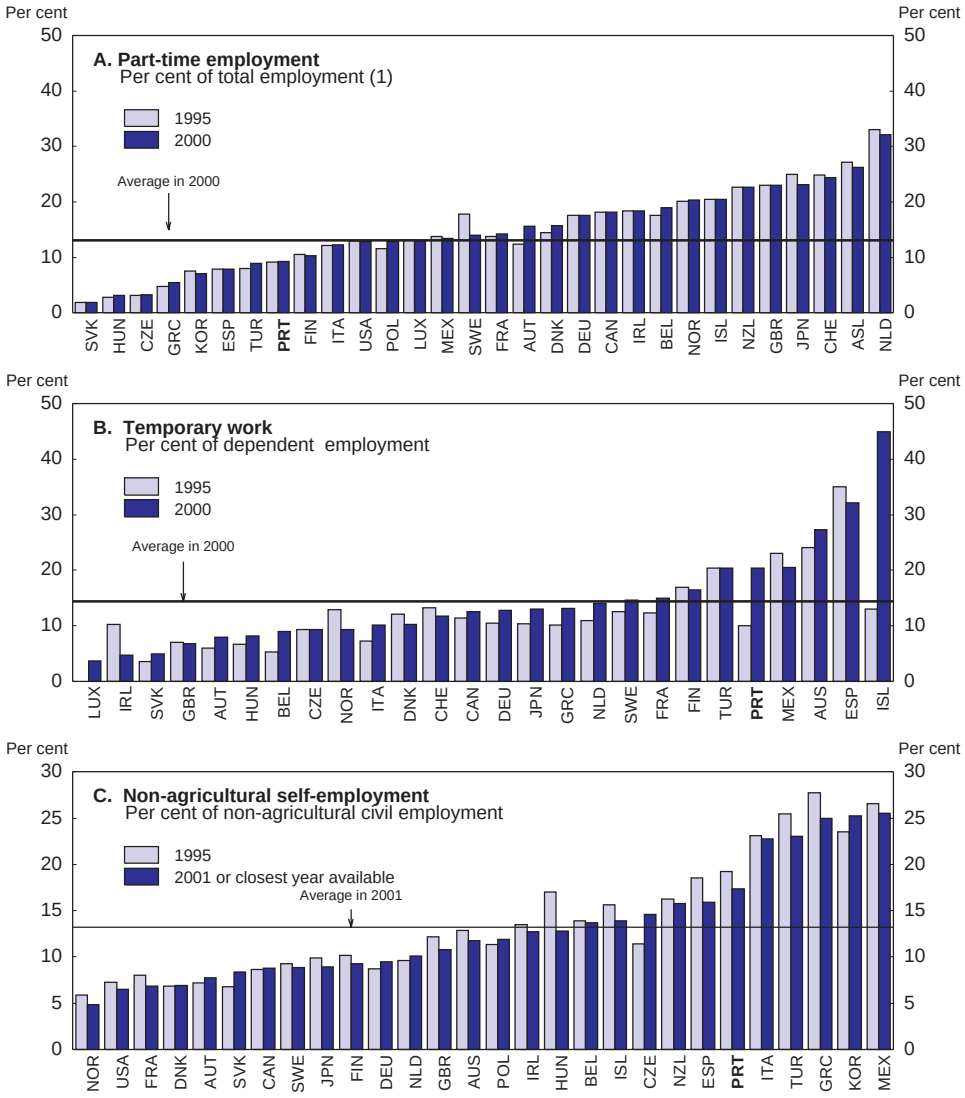
Figure 30. **Types of employment: an international comparison**

Per cent



1. Part-time employment refers to persons who usually work less than 30 hours per week in their main job. Data are expressed as a proportion of total employment, apart from the case of the United States where the data refers to wage and salary workers.

Source: OECD, *Employment Outlook*, July 2002; OECD, *Annual Labour Force Statistics*.

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nominal wages react significantly to changes in the unemployment rate and inflation compared with most other OECD countries.[108] In addition, Portugal has a high inter-industry wage dispersion compared with the European standard. In recent years, wage adjustment has acted as a "second-best" for adjustments in employment/unemployment, as large and flexible wage differentials supported job creation and helped to keep the unemployment rate low.

The share of "atypical" employment contracts has increased

As a consequence of high firing costs for regular workers, fixed-term contracts have gained increasing weight in Portugal, reaching more than 20 per cent of total employment in 2001 (Figure 30).

There is increasing evidence that Portuguese firms use this type of contracting as a device to select and test workers at an early stage of their work life, to ensure a good matching of the job characteristics with the worker's skills.[109] Such a mechanism would not significantly penalise workers' wage income. However, additional evidence also suggests that a substantial proportion of workers apparently does remain in a chronic situation of temporary contracting. In contrast, the Portuguese labour market is characterised by a relatively low incidence of part-time employment.

In Portugal, the level of self-employment accounts for almost 17 per cent of total non-agricultural civil employment, the third highest share in the EU (where the average share is around 12.5 per cent). A large proportion of self-employed work is in fact economically dependent on a single company, with workers drawing their income from a single employer for prolonged periods.[110] Portuguese firms have taken advantage of the possibilities of contracting out for services to individuals with independent worker contracts, which are subject to less rigid employment regulations.[111] Differences in social security contributions according to the form of contract tented to create a bias in favour of self-employment, but contribution rates have been aligned. Still it remains less costly for the employer to hire workers under such contracts, as there are no fringe benefits involved, *i.e.* holiday and Christmas bonuses, maternity leaves or insurance for accidents at work.

Recent changes in employment legislation

Temporary work contracts have been subjected to new regulations…

To adjust labour legislation to new forms of employment, the relevant legal framework has been under revision and updating since 1999. The aim is to facilitate the use of contracts adapted to these new forms and to ensure that workers involved in such arrangements are legally protected. In the recent Seville Council recommendation of 21 June 2002 on the Broad Guidelines of the Economic Policies, one of the main priorities for Portugal was to "modernise the labour market

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institutions, *inter alia* by adapting employment contract regulations taking into account the need for a proper balance between flexibility and security".

New legislation for fixed-term employment contracts came into effect in August 2001, tightening the rules of such contracts in the private sector. One of the objectives of the law was to ensure that workers who, *de facto*, were employed on permanent jobs would benefit from a standard permanent employment contract. As a result, temporary contracts are now covered by a set of strict rules governing their scope and termination conditions. In particular, they may be used only in special situations indicated in the law, corresponding to temporary labour force needs.[112] The maximum legal length of general fixed-term contracts in the private sector is 3 years.[113] Then the legal status of the fixed-term contract is automatically changed into a standard (open ended) contract. A proposal for a new Labour Code was approved by the Council of Ministers at the end of November 2002 and is currently open to public discussion. It includes a proposal to increase the legal maximum period of fixed-term contracts.

Supplementary legislation was adopted in March 2002 to encourage permanent employment by means of financial subsidies to small companies that convert a fixed-term contract, on expiry, into a permanent contract.[114] The impact of rules governing temporary employment should be carefully monitored. Offering incentives to permanent contracts and/or putting restrictions on temporary contracts may impede the response to downturns in activity. In addition, the need for such measures may be an indication that employment protection legislation is an obstacle to hiring on "regular" contracts. If job protection legislation is too restrictive and reduces the number of permanent contracts below what would be expected, it would be preferable to ease the general legal framework rather than to restrict the rules for fixed-term contracts.

… while initiatives have proposed to ease employment protection legislation

Portugal's economic performance would benefit from efforts to ease employment protection legislation (EPL). Low labour mobility, due in part to strong employment protection, may have hindered more efficient resource allocations in Portugal and negatively influenced productivity growth. Scarpetta and Tressel (2002) found that the negative impact of strict EPL on productivity is stronger in countries, like Portugal, where sectoral wage bargaining without co-ordination is predominant. In addition, cross-country estimates suggest that comparatively stringent EPL and benefit regulations may have reduced employment rates in Portugal by about 4.5 percentage points over the period 1982-98.[115] Efforts should be made to allow more flexibility to negotiate conditions of termination on an individual/firm basis, to reduce barriers to collective dismissals and to liberalise procedural barriers to dismissals.[116]

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One of the priorities of the current government is the revision of labour legislation. A new Labour Code (*Código do Trabalho*) was proposed by mid 2002, encountering union opposition. Some of the key points are: an easing of the conditions required for fair dismissals,[117] new measures to fight against absenteeism and fraudulent sick leaves, increasing the geographical/functional mobility of labour and the flexibility of working hours and revising collective agreements procedures. The proposal, if approved by the Parliament, would be a useful step to ease the excessive rigidity of the Portuguese labour legislation.

Active labour market policies

Use of active labour market policies (ALMP) has increased dramatically in recent years, the objective being to get people into work by helping job search, lifting skill levels or assisting job mobility between sectors or regions. Programme spending appears to be relatively well balanced between active and passive support (Figure 31). A specific institute, the Institute for Employment and Vocational Training, which operates under the Ministry of Social Security and Labour, is responsible for implementing employment measures, vocational training and running public employment services. The integration of these various functions is appropriate, as is the fact that the Institute operates through a network of local public services, under the authority of regional offices.[118] It would nevertheless be desirable to carry out a systematic monitoring and evaluation of active labour market programmes in order to rationalise the numerous instruments in place.

Specific measures have been introduced over the years to improve employability of groups at risk, especially youths and the long-term unemployed. They have been relatively successful as shown by the rapid improvement in labour market outcomes (OECD 2001a). One of the latest such initiatives, the Job Offer Stimulus Programme, dating from April 2001 and in force until 2006, provides some subsidies with co-financing from European funds.[119] Maintenance of the level of jobs created over a period of at least four years is one of the main conditions of companies to benefit from financial support under this programme.

The effectiveness of the public employment service (PES) in Portugal is relatively low, mainly in terms of its job-brokerage function. The *Instituto de Emprego e Formação Profissional* (IEFP) placements as a share of total hirings in the economy are lower than in most OECD countries.[120] The pool of jobs reported to the agency by employers is typically small and contains mainly low-paying positions that are difficult to fill.[121] Measures have been taken to improve the effectiveness of PES, namely the recent creation of a self-service approach, through the use of electronically available services, as the IEFP started to publish some information on the Internet regarding job demand and supply, training actions

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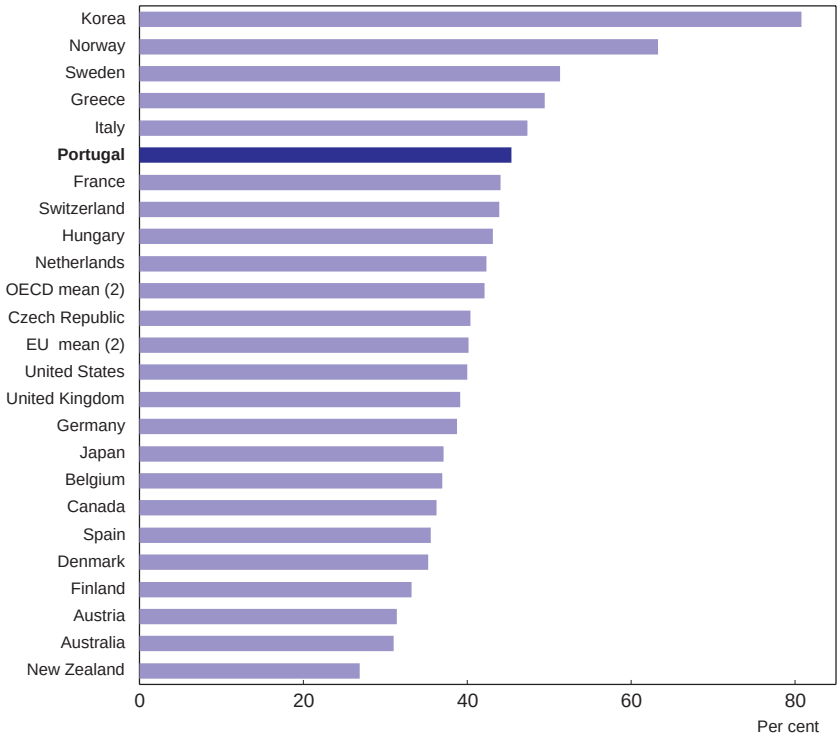
Figure 31. **Labour market policies in OECD countries**
Share of active labour market policy spending in total spending[1]



1. Average over 1999 and 2000, except for Greece (1998) and the United Kingdom (1998-1999).
2. Country mean excluding Greece and the United Kingdom.
Source: OECD, *Employment Outlook*, July 2002.

and other employment-related matters. The improvement of relations with employers, especially regarding the way they are served at job centres, the promotion of the image of the PES and the early diagnosis of employers' needs in terms of qualifications and skills should be one of the priorities. It is also necessary to implement a rapid articulation of the vocational guidance and training programs offered with the qualification and skill demands of the companies, which should be closely monitored. Additionally, further efforts should be made to enhance the exposure of the PES to market forces and the systematic and continuous evaluation of IEFP programmes should be enforced.[122]

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Increasing social insertion while preserving incentives to work

Increasingly, in OECD countries, welfare is seen as "a springboard rather than a hammock" and an objective of welfare is "making work pay". This suggests a move away from a welfare system based on passive income support and towards a more active social policy. In Portugal, the coverage of the unemployment benefits is relatively low by European standards as the eligibility conditions are relatively strict.[123] Additionally, even the minimum guaranteed income introduced in 1997 presents a number of features that minimise the risk of creating work disincentives.

The minimum guaranteed income (RMG) aims at ensuring that all individuals have the resources required for satisfying minimum needs, whilst providing for their progressive social and professional integration.[124] One of the objectives is to integrate socially-excluded people into the measures already existing under the general employment policy, rather than creating specific measures. A recent study suggests that the RMG has been particularly effective in reducing the duration and acuteness of poverty. In 2001, additional efforts were made to offer jobs or training to beneficiaries of the RMG. Since then, the government has stressed the need to tighten the operating rules of the RMG. This concern has been addressed in June 2002 in the scope of changes made to the scheme, now called social insertion income (*Rendimento Social de Inserção*, RSI).

The general social security system: the reform process has started

Portugal's general social security system, which applies to the great majority of private sector workers, is (like the civil servants' scheme) a pay-as-you-go system. The general system's main component is the contributory scheme, which provides old age, survivor and invalidity pensions as well as unemployment, sickness, occupational disease, maternity and family benefits.[125] Social security outlays have increased steadily in the last decade to around 14 per cent of GDP in 2000 (from about 10 per cent of GDP ten years earlier), of which about half are old age and survivors' pensions. This component of social security expenditures will be affected by demographic changes, as the old age dependency ratio is expected to double in the next 50 years. Alongside the public pension system, people have begun to contribute to voluntary private pension funds, encouraged by a tax credit. These funds are managed by authorised life insurance and pension fund companies. The authorities are seeking convergence of the general and civil servant regimes.

Pressures from ageing populations

Portugal is facing much the same demographic evolution as most other OECD countries in Europe.[126] Currently, there are about four persons of working age (conventionally defined as 15-64 years) for every person over the age of

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65 years. By 2050, the proportion will have fallen nearly to two-to-one.[127] This "old-age dependency ratio" has already been rising for some time because of rising life expectancy, but as from about the middle of this decade, it will start to rise much more quickly as the "baby-boom" cohort, which had comparatively few children, starts to move into retirement. The proportion of older people entitled to a public old-age pension will also rise, because of the increasing participation rates of women over the past 40 years. In the absence of further reforms,[128] expenditure on pensions for older people (old-age, survivor and disability pension) is projected to rise by about 4 percentage points of GDP between 2000 and 2040. It would subsequently fall back somewhat, on the assumptions that fertility rates rise (though not reaching levels that would stabilise the population in the long run). About half of the increase in pension spending is expected to be concentrated in this decade.[129] Since older people on average consume higher amounts of medical services than younger people, and the proportion of older people will rise, expenditure on health (and on long-term care) will also likely grow faster than GDP over future decades.[130] The Portuguese projections suggest an ageing-induced increase in health spending of 0.8 per cent of GDP by 2050, a figure at the low end of OECD country projections. No projections are available for increased expenditure on long-term care: most countries which have made such projections expect an increase of about one percentage point of GDP by 2050. In sum, age-related public expenditure in Portugal could rise by 6 percentage points of GDP over the next few decades, and after the middle of this decade, it can be expected that age-related spending will add about a quarter point of GDP to primary spending every year for 30 years or so.

To prevent such steady spending pressures being reflected in rising deficits and debt levels, deep and sustained reforms will be required, either to reduce spending on pensions, or to reduce spending in other areas, or both. The alternative of letting contribution rates to the public pension system (public and private) rise to finance the higher pension payments to retirees, can be only a small part of the solution. If this were to be the only response, contribution rates would have to rise by up to 40 per cent, driving an increasing wedge between labour costs and earnings, and lowering incentives to participate in the formal economy. It is also a solution that is undesirable on equity grounds: in effect, future generations of workers would have to pay higher contribution rates than did their parents, without acquiring rights to higher pensions. As regards reforms to non-pension spending, these have been addressed in Chapter III.

Containing the growth of pension spending will inevitably entail making pensions less generous for a given work history. Most OECD countries have already embarked on reforms of this type,[131] and Portugal has also introduced reforms (see below). The majority of countries now index pensions on prices only, or on net wages, but not gross wages. Several countries have raised, or will raise, the age at which retirement on a full pension is possible. Gradually increasing the

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importance of privately funded pensions (especially of the defined contribution type) relative to that of public pay-as-you-go pensions is also becoming common. OECD analysis strongly supports the view that the most effective reforms are those which encourage people to work longer (or the very least, remove financial incentives for workers to retire early). As well as reducing the fiscal burden of having to pay pensions to those still of working age, this has the advantage of raising the level of total output, potentially raising living standards for both retirees and those still working. The average age at retirement in Portugal (for males) is well above the OECD average and recent reforms to the Portuguese system do reduce the incentives to retire early.

Recent reforms of the social security system

In the 1990s, several measures were taken to address the expected increase in the ratio between beneficiaries and employees and enhance the financial sustainability of the system. These included an increase in retirement age for women to bring it in line with men's, the extension of the minimum contribution period, the reduction of the benefit accrual rate and the revisions of rules allowing the accumulation of pensions with labour income.[132] Portability of pensions across some private sectors remains an issue.[133] A Basic Law on Social Security, approved in July 2000, set the principles of the reform of the private sector workers' scheme.[134] While a number of reforms were included, aimed at improving the system's operation and ensuring its future finances, the basic public-managed nature of social security was not changed.[135] In November 2001, following several months of negotiations, the social partners and the government signed a social security reform agreement, introducing changes to the calculation formula of pensions and a provision allowing employees to make pension contributions, above a very high threshold, into private schemes.

In July 2002, the general principles of a revised Basic Law on Social Security were approved in Parliament and discussions with social partners have started with a view to obtaining consensus on some specific points. This new Basic Law will be enforced from 2003 onwards. The main feature of the law is the introduction of a mixed public-private system, opening up the social security system to the private sector (see Box 10). The new model adopts different thresholds allowing the combination of public and supplementary social security schemes and introduces a more equal weighting between the pay-as-you-go and the capitalisation of future benefits. The law also confirms the calculation of retirement pensions based on the full contributory history of an individual.[136] According to official estimates, this new contributory scheme will lead to an initial reduction in receipts of around 0.6 per cent, but in the long run the establishment of ceilings will lead to a decreasing public expenditure on pensions.

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Box 10. The 2002 New Basic Law on Social Security and additional measures

The new Basic Law on Social Security preserves the social functions of the state as a fundamental element of people's protection, but introduces some new features:

- The principle of social co-responsibility of the state, enterprises and families, trying to implement a progressive culture and practice of risk sharing.
- An increased balance between the pay-as-you-go and the capitalisation of future benefits, to strengthen the solidarity between the working population and pensioners.
- The principle of positive discrimination, increasing the provision of social benefits to the most disfavoured.

As regards the calculation of pensions, the law defines new rules for capping both contributions and benefits (three caps), with the objective of introducing a ceiling to public pensions (which are earnings-related) and promoting supplementary pension plans (fully funded). The thresholds would be fixed according to the beneficiaries' income, allowing them to combine the public and the supplementary social security systems, and the pay-as-you-go and capitalisation regimes. New secondary legislation will establish the caps, thresholds and contribution rates.

The supplementary (private) system is a funded one. Whatever option is chosen in the higher income groups, equality of treatment regarding taxes will be guaranteed.

Concerning the guarantee of beneficiaries' rights and the security of the assets allocated to pension plans, a pension guarantee fund will be created, functioning as a safety mechanism for insolvency situations. The introduction of certification measures and the reinforcement of mechanisms of supervision are also essential elements to guarantee the stability of the supplementary system. In order to increase mobility and competition, the portability of pension rights and financial reserves is assured.

Regarding positive discrimination, a family supplement to minimum pensions for people aged over 75 will be introduced and there will be benefit entitlement in the event of total, permanent disability for workers who have not paid the required level of contributions over their career. Some measures aimed at supporting family and maternity will also be introduced.

To facilitate the transition from work to retirement, a system of partial retirement, in combination with part-time work, will be implemented.

In the context of the new Basic Law of Social Security, the convergence of the minimum old age and disability pensions to the net minimum wage is to be achieved by 2007. The minimum pensions will be progressively increased during the next four years, in line with the contributions made, to the minimum wage level. The new pension levels will vary between 50 per cent of the minimum wage (for the social pension, *i.e.* for people who have never contributed to the system)

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and 100 per cent (for persons who have contributed for more than 30 years). In the short-run, this pension increase will benefit around 1 million people (nearly 45 per cent of total beneficiaries) and will entail a substantial increase in pension spending (estimated at around 50 million euros each year to 2007). In the future, the increase in minimum pensions will be indexed to that of minimum wages.

Improving the climate for entrepreneurship

The new government's strategy to strengthen competitiveness and efficiency in order to achieve higher long-term growth is set out in a broadly-based Productivity and Growth Programme, presented in June 2002 (see Annex IV). It aims at enhancing competition, promoting productive investment and exports, improving the business environment, including througheasier access to financing, a reduction of red tape to shorten delays in starting up new enterprises, as well as developing innovation and applied R&D. The broad thrust of the programme closely follows the policies for growth identified in the OECD's *Growth Project*. Concrete policy measures have also been announced regarding selected economic areas, with detailed schedules of implementation. A number of targeted programmes, co-financed by large EU transfers, have already been put in place to create a more favourable climate for entrepreneurship.[137] Over the last decade, significant transfers received from the EU boosted infrastructure development, but the results of co-financed investments in the domain of innovation and R&D have not been wholly satisfactory. With a view to filling this gap, one of the strategic objectives of the third CSF (2000-2006) is to promote business competitiveness and stimulate innovation.[138] The timely and accurate achievement of the new objectives is key to enhancing potential growth over the medium-term.

Promoting a business friendly environment

Evidence on firm dynamics

Recent developments of output growth and productivity (reviewed earlier in the chapter) highlight the need for Portugal to maintain higher growth on a sustained basis to allow catch-up with the European average. Growth of output and productivity takes place in individual firms. Because firms' behaviour and decisions are influenced not only by market forces but also by policy and institutions, understanding productivity dynamics at the firm level helps the identification of appropriate policies to foster growth. Evidence from firm level data available for ten OECD countries, including Portugal, provides useful insights on developments within individual firms as well as firm dynamics (entry and exit of these in markets).

In Portugal, as in other countries for which data are available, labour productivity growth in manufacturing appears to be largely explained by evolutions

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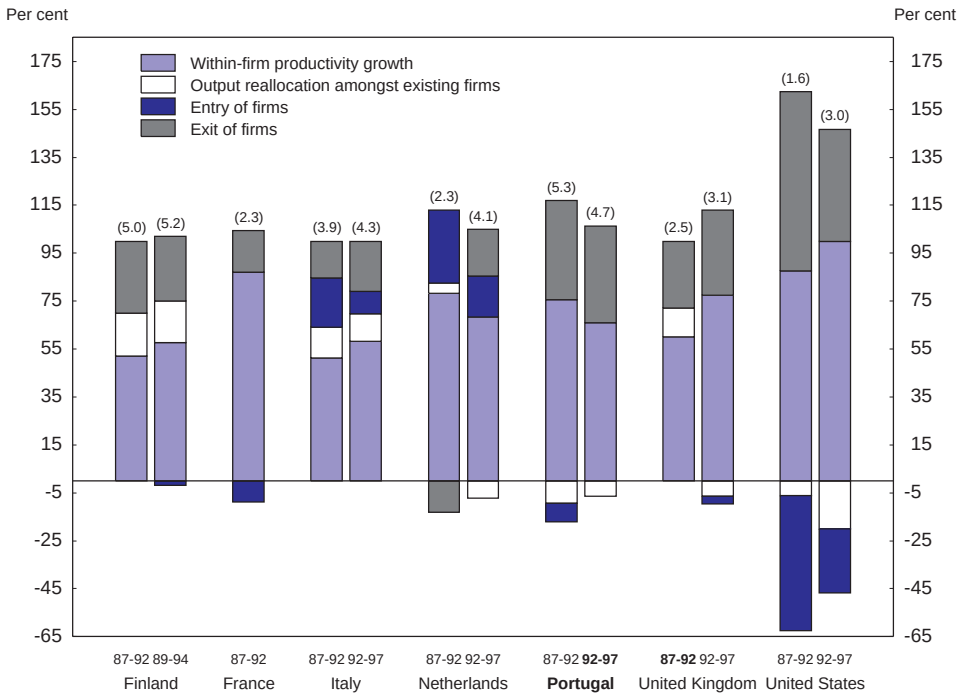
within individual firms (Figure 32). The exit of lower productivity firms also makes a substantial positive contribution to overall productivity growth, especially in the early 1990s a period of slowing activity. This process of "creative destruction" takes place on a relatively high scale in Portugal, as in the United States. The entry of new firms has a much smaller effect (negative in the late 1980s, positive in the early 1990s).[139] In service sectors, the within-firm component of productivity growth is generally large, but entry and exit of firms have variable effects according to sector and sub period.[140]

Figure 32. **Breakdown of labour productivity growth in manufacturing**[1]
Percentage share of total annual productivity growth of each component[2]



Note: Figures in brackets are overall productivity growth rates (annual percentage change).
1. For details on the method used see ECO/CPE/WP1(2002)3.
2. Components may not add up to 100 because of rounding.
Source: OECD, ECO/CPE/WP1(2002)3.

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These results have some implications for policy making and institutions. Strategies to enhance productivity growth within firms (through technology innovation, applied research and development, human and physical capital development) need to be complemented by measures to remove restrictions that slow the "exiting" of low-productivity firms or hinder the entry of others. The turnover of firms is generally higher in Portugal than in many other OECD countries, suggesting that freed resources would not remain unemployed for long. Labour market policies that facilitate the process can be combined with social programmes to cushion the transition.

Stimulating firm creation

Promoting a business friendly environment has been an important component of Portugal's strategy to strengthen competitiveness and efficiency for several years. In this context, various measures have been introduced since 1999 to cut red tape and reduce business compliance costs. Formalities for the creation of commercial business or participating in public procurement have been simplified and delays have been shortened, through the use of electronic means. The government's 2002 programme contains several measures to help companies deal with the still complex bureaucratic procedures and makes further advances in alleviating procedures, including administrative simplification and deregulation.[141] The purpose is to make it easier for firms to be created, to expand, to merge and to shut down. One important element is the planned reduction of the number of business contact points in the public administration and the increase of the effectiveness of registry and notary services, namely through privatisation of the latter while ensuring a reasonable degree of competition in the supply of notary services. The creation of a more effective bankruptcy system and the reduction of fees and fiscal burdens related to enterprise development and restructuring, merger and divestiture operations should help to improve operating conditions for companies.

The scheme to encourage investment in the tradable sector is being updated. An *Investment Code* will be created, encompassing all forms of investment incentives of a financial, fiscal or contractual nature, to simplify and systematise the legal framework of productive investment. The government is also proceeding to the elimination of multiple interlocutors needed for an investment operation that add to time delays and costs.[142] The procedures for investment support under the *Operational Programme for the Economy* 2000-06 are being modified. There will be mixed financing from the public and banking sectors; the subsidy component (public lending) now becomes conditional on results of an *ex post* evaluation of the projects based on pre-defined quantitative objectives (convertible financing).[143]

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Fostering a knowledge-based economy remains a challenge

Improvements in multi-factor productivity, which plays a crucial role in the process of economic growth, are closely related to innovation and science and technology progress. Expenditure on research and development (R&D) is an investment in knowledge that should translate into new technologies and more efficient ways of using existing resources of physical and human capital. The diffusion of information and communication technology (ICT) can also help to raise economic growth, particularly when combined with effective human resource strategies involving education and training and organisational change.

Encouraging innovation and technological progress

Despite the increases recorded in recent years, Portuguese spending on R&D as a percentage of GDP is less than half the OECD average, as is the proportion of researchers in the active population (Figure 33). As in other less advanced OECD countries, R&D activities are carried out not so much by the business sector, but rather by the higher education and government sectors, these sectors accounting for almost two-thirds of total R&D expenditure in Portugal, against less than 30 per cent on average in the OECD.[144] The shift of Portuguese industrial production and exports towards higher-technology goods is less advanced than in most other OECD countries although the rate of growth of this sector has increased substantially in the last years. The traditional sectors, which still account for the larger share of Portugal's exports, face a higher price elasticity than R&D-intensive sectors, and are exposed to intense competition from low-cost producers.

The science and technology policy agenda considers two related challenges. *First*, overcoming the Portuguese structural gap by sustaining basic research in all R&D domains. *Second*, promoting knowledge absorption and innovative procedures, in order to generate spillovers into the productive sector. The *Operational Programme for Science, Technology and Innovation* 2000-06, prepared under the aegis of the EC, addresses these two issues in a complementary way. The goal is to develop the general science and technology base through funding programmes for advanced training in R&D and to support R&D institutions and research projects subject to quality evaluation by international peer review. In addition, a national agency (*Ciência Viva*) conducts programmes to promote science and technology culture and public understanding of science.

In 2001, measures were taken to spur innovation in enterprises and R&D activities, including the introduction of a more favourable R&D tax regime.[145] The fiscal framework for venture capital was recently revised and a legal framework concerning venture capital has been implemented since 1998, creating a more favourable background to the development of this financial instrument. However, most of venture capital funds in Portugal have been used to finance expansion

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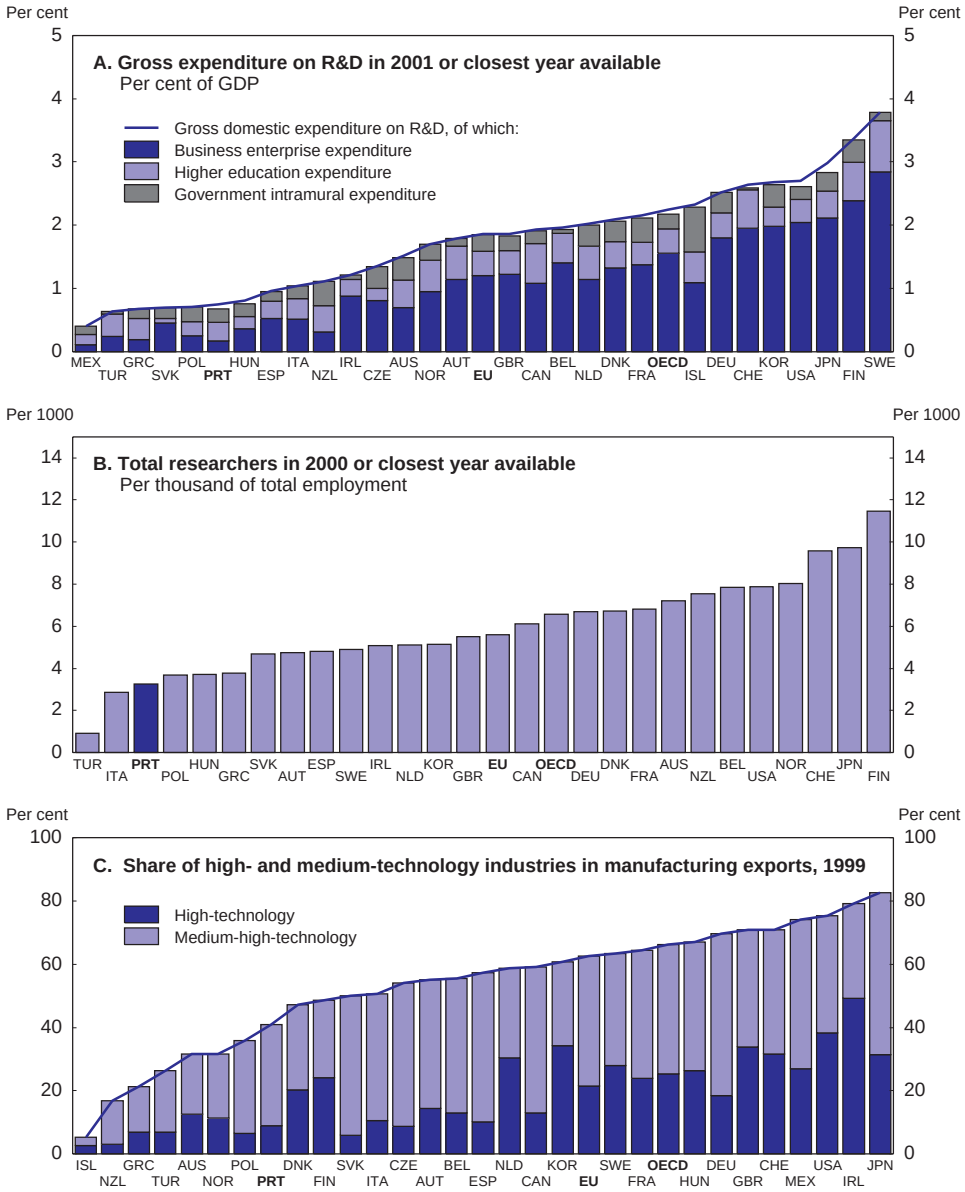
Figure 33. **Science and technology indicators**



Source: OECD, *Main Science and Technology Indicators* (2002/1); OECD, *Science, Technology and Industry: Scoreboard of Indicators* (2001).

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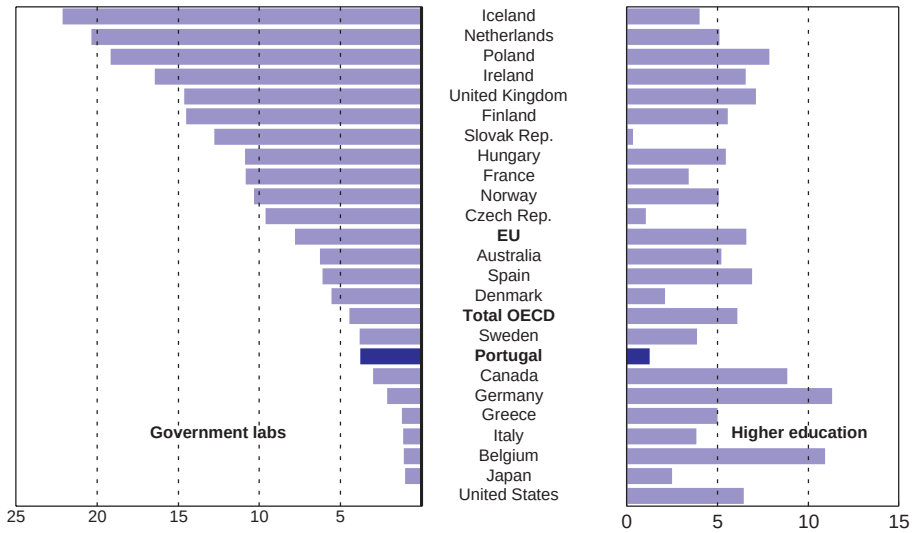
Figure 33. **Science and technology indicators** *(cont.)*

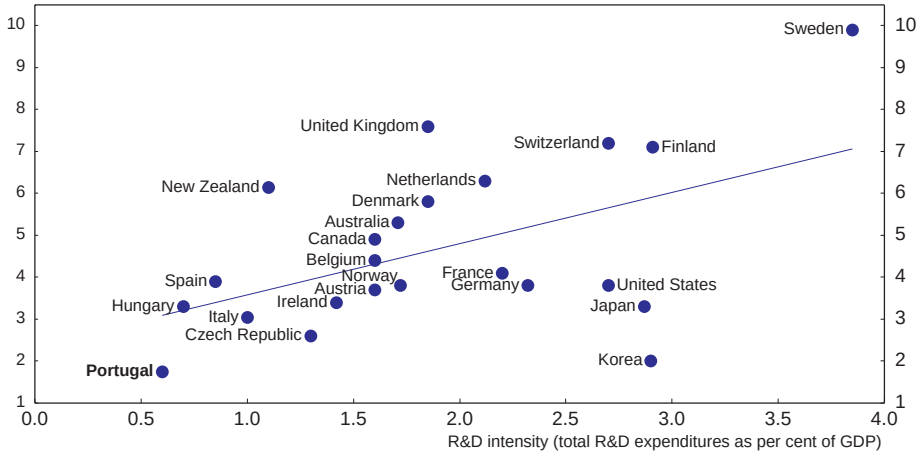
D. Few links between firms and government labs

Government and university research funded by business, per cent (1)



E. Low research productivity

Publications in the 19 most industry-relevant scientific disciplines (2)



1. 1999 or 2000 for most countries; 1999 for Portugal, 2001 for Germany and Canada.
2. World share of publications divided by GDP in US$. 1998 or latest year available.

Source: OECD, *Main Science and Technology Indicators* (2002/1); OECD, *Science, Technology and Industry Outlook* (2001).

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projects instead of start-ups. The creation of a public venture capital syndication fund should help the development of R&D activities.

Seizing the benefits of ICT *diffusion*

ICT contributed significantly to aggregate growth in several OECD countries in the past decade or so. Capturing the benefits of ICT cannot be treated as given but will depend on complementary investments and actions by private and public actors. In spite of steady progress, businesses, government and citizens have not yet embraced new technologies, the Internet and electronic commerce as readily as in other European countries. In recent years, Portugal has registered a rapid growth in the usage of the Internet, but it is still lagging behind the EU in the most common indicators, such as the use of electronic mail or the number of Internet hosts and secure servers (Figure 34). In 2001, Internet access by Portuguese enterprises was still the lowest in the EU and the share of Portuguese households with Internet access at home was far below the EU average. By contrast, a case of success in the ICT sector is the widespread use of mobile telephones in Portugal. The number of mobile phone users more than doubled during the 1990s, as deregulation and increased competition in the sector drove prices down. By 2000, Portugal had moved to just above the EU average, with about two thirds of the population using a mobile phone.

In line with EC orientations, incentives have been introduced, in the scope of the *Operational Programme for Information Society* 2000-2006 to spread ICT use among households and companies, schools and other public areas.[146] The "Internet Initiative" sets out a number of quantitative targets to be achieved in different sectors.[147] Special attention has been paid to promote the general use of ICT within the public administration, especially for procurement purposes. All new public information and invoices now have to be published on the Internet. Concerning the use of ICT in public services, Portugal ranks 5th in the benchmarking of the European Commission eEurope Programme, after Ireland, Finland, Sweden, and Denmark. In addition, the current strategy for human capital development has an explicit focus on developing competencies in ICT at various levels (see above). In order to acclimatise citizens with the Internet, public access points started to be placed in municipalities in 2001, and a tax incentive has been provided to households since 1998 for the acquisition of IT equipment.[148] The costs of Internet access for consumers, along with factors such as quality and availability, are among the key factors influencing the rate of Internet development. In the OECD area, the level of prices differs considerably from country to country, and countries with lower access costs typically have a greater number of Internet hosts. In Portugal, Internet access tariffs have been steadily declining and additional cuts are expected, given the increased liberalisation of the telecommunications market.

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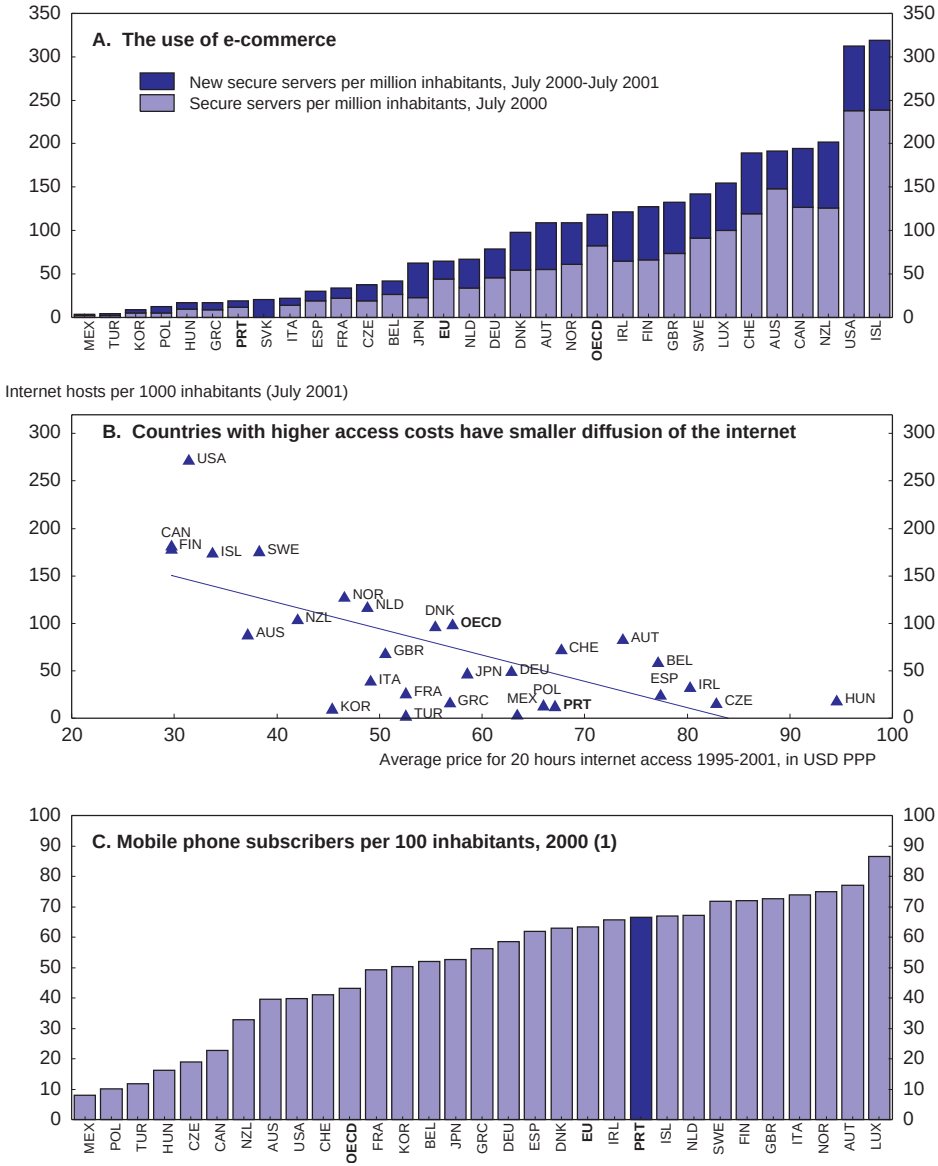
Figure 34. **ICT diffusion indicators**



1. Or closest year available.
Source: OECD, Telecommunication database; EUROSTAT.

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EU *funds supported the upgrading of physical infrastructures*

Portugal has received sizeable transfers from the EU, in the context of successive Community Support Frameworks (CSF), including both structural funding and cohesion funding.[149] The CSF funds have played an important role in the modernisation and restructuring of the Portuguese economy. According to estimates of the (now extinct) Ministry for Planning, which take into account only the demand-side effects, EU funds helped to raise the level of GDP by 3.3 per cent over the period 1994-2000, with positive effects on employment, disposable income and private consumption. Their cumulative impact on the level of investment was almost 12 per cent, and was especially strong in construction.

EU transfers also played a notable role in growth through the supply-side effects of transfers used to finance (human and capital) investment. They have allowed major improvements in transport infrastructures. Thousands of miles of new roads and motorways, major bridges, the electrification of railways, the expansion of metro lines, and the modernisation of Portuguese ports and airports were co-financed by EU structural funds. Because of Portugal's geographical position on the periphery of the EU, interconnection with the Trans-European networks remains an important objective and arrangements combining the public and private sectors are expected to continue. As EU transfers are set to decline over the longer term, the convergence process must progress further by then so that investment needs become more comparable with those of the more advanced EU countries.

Competition policy and regulatory reform

Efforts to liberalise the economy have followed two main lines of action: *i*) reducing the role of government through a broad privatisation programme, to improve efficiency and contribute to debt reduction; and *ii*) opening up network industries and developing the regulatory framework needed to strengthen competition. The institutional arrangements for a competition policy were in dire need of updating, and an independent body, the Competition Authority, has recently been created.

T*he institutional design*

The Competition Authority

The past institutional arrangements for competition, which split responsibilities and decision making between a Directorate General in the Ministry of Economy and a Competition Council, with members directly appointed by the prime minister, were not adequate to achieve a sufficient degree of transparency and to take and implement decisions efficiently (OECD, 2001a). The newly created

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competition policy authority interacts with existing regulatory bodies. It has an independent status, and concentrates powers to ensure fair competition, including the approval of mergers and the prevention and sanction of predatory and anti-competitive practices. A new competition Law will shortly be submitted to Parliament for approval. These changes, which anticipate the modernisation and decentralisation foreseen in recently approved EC rules are appropriate. They should contribute to improving policy credibility and promoting a better diffusion of the competition culture.

Experience in other OECD countries points to some desirable requirements: the Competition Authority should be technically and operationally autonomous and free to issue its decisions. The independence would be reinforced, for instance, if the president of the Commission and its members were assigned for a fixed term, longer than the government term. The enforcement of competition is also related to the resources that are committed to it. It is thus essential that an adequate number of qualified staff and financial resources be devoted to competition enforcement. Procedures of competition policy should contain adequate provisions for transparency, predictability, non-discrimination, accountability and speediness. The Authority should be subject to strict time constraints to issue its decisions. For transparency, an annual report is appropriate, including summaries of illustrative cases reviewed under the Competition Law. Decisions taken to enforce the law should be published periodically.

Independence of regulators

Portugal, like many OECD countries, has sought to make the regulators' status independent of political power, to limit the influence of private lobbies on regulatory decisions. In practice, this independence is very difficult to establish.[150] Some of the requirements are that regulators: *i*) should be separate and autonomous from the government, their appointment being made preferably through a multi-party process, involving executive and legislative bodies; *ii*) should be protected from arbitrary removal, for instance through the use of fixed term appointments; *iii*) should have adequate and reliable source of funding, such as industry fees, rather than budget allocation. The case of Portugal's telecommunications regulator, the National Authority of Communications (ANACOM), is illustrative in this context. The head of the Authority is appointed by the Council of Ministers, and it reports to the Ministry of Economy. After a recent change, the appointment is now for 5 years (as in many other OECD countries, including the United States, the United Kingdom and Spain). It is financed through industry fees, as in several other countries, which is appropriate.

The general pattern is to have several sector-specific regulators, as is the case in most European countries, including Portugal, and in the United States at the Federal level. This pattern can increase the total amount of sector-specific

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information available by reducing the control span of the regulator. It can also provide the elements needed for establishing a yardstick of competition through comparison of the various regulators behaviour. One important drawback, however, is that sector-specific regulators can be more easily captured by the regulated industries. The new government in Portugal has announced that the independence of regulatory authorities at the sectoral level would be strengthened.[151] The decision was taken to create a single regulatory body for the energy sector and in early 2002 responsibilities of the electricity regulator (ERSE) were already extended to cover natural gas.

Progress in privatisation and sectoral reform

Portugal has advanced rapidly in the past decade in privatising the large number of state-owned companies. There has also been some opening up of network industries but the results are patchy. The bulk of the privatisation programme was achieved by 1999, covering the financial sector (banking, insurance), industrial firms (cement, brewery) and network industries (electricity, telecommunications). Since then the process slowed, because of changes in the government priorities, adverse market conditions (in air transport for instance) and time required for the restructuring of some companies for privatisation. In 2000-01, no new enterprise of significant size was added to the privatisation list, the main operations consisting of further rounds of share sales in Portugal Telecom, EDP (electricity), Brisa (motorways) and GALP (oil and gas). Privatisation proceeds amounted to nearly 3 billion euro over the past two years, to be added to the 15 billion euro raised over the previous decade. As the privatisation programmes mature, public and private partnerships should grow in importance, as a means of improving efficiency and as an alternative approach to privatisation.[152]

Companies on the privatisation agenda of the new government include the national airways (TAP) and airport management (ANA), as well as paper companies (Portucel) (Table 18). In addition, the final rounds of privatisation operations already engaged are to be completed by 2004. Although recognising the need for competitive solutions in privatised sectors, the authorities seek to maintain Portuguese ownership in "strategic" sectors, and revenue raising appears to be a major objective of privatisation. The sectors that are considered as strategic are energy, transportation, natural resources (pulp and paper) water companies and radio-TV. In several enterprises in these sectors (Portugal Telecom, EDP and GALP), the state has retained, or intends to retain, special voting rights (golden shares) within the ambit permitted by the EU.[153] In addition, the Portuguese privatisation framework law prohibits the acquisition by foreign investors of more than a given number of shares in Portuguese undertakings, according to the activity sector, and it requires prior authorisation for the acquisition of more than 10 per cent in companies being privatised. This legislation is considered as a restriction on free movements of capital within the EU and should be modified

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Table 18. **The current phase of privatisation**
1999-2004

Company	Sector	% capital privatised	Proceeds (million euros)
1999			
Tabaqueira (2nd phase)	Tobacco	25	70.0
Portugal Telecom (4th phase)	Telecommunications	13	947.7
BRISA (3rd phase)	Motorways	19	498.1
Fapajal	Paper	99	4.1
Papel do Prado	Paper	95	30.7
Carbolis	Chemical (industrial gas)	100	3.0
Soporcel	Paper	10	55.0
TOTAL			**1 608.6**
2000			
Dragapor	Construction (dredging)	100	3.1
Tabaqueira (3rd phase)	Tobacco	10	27.7
GALP (2nd phase)	Gas	15	433.6
EDP (4th phase)	Electricity	19	1 779.0
Portugal Telecom (5th phase)	Telecommunications	9	960.8
TOTAL			**3 204.2**
2001			
BRISA (4th phase)	Motorways	5	137.1
CIMPOR (4th phase)	Cement	10	410.1
TOTAL			**547.2**
2002-2004[1]			
ANA	Airport management		
BRISA (5th phase)	Motorways		
EDP	Electricity		
GALP (3rd phase)	Gas		
Portucel (2nd phase)	Paper		
Telecom infrastructure	Fixed network		
REN	Electric Network		
TAP	Air transportation		

1. According to official estimates, privatisation proceeds are expected to reach about 5 billion euros, total, over the 2002-2004 period.
Source: Ministry of Finance.

(June 2002 European Court of Justice decision). In the financial sector, the State's position (it owns the *Caixa Geral de Depósitos* group) remains unchanged. Nevertheless, the privatisation of the public banking group should not be ruled out.

Although indicators of regulatory reform on product markets point to a gradual easing during the 1980s and 1990s, the process was slower than in most other OECD countries and the degree of stringency of product market regulations still prevailing in 1998 was relatively high.[154] Portugal has been making good progress in implementing the EU 2001 product market recommendations and the transposition of internal market directives into national legislation improved markedly.[155] The

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transposition rate stood at 96.1 per cent in November 2002, below the 98.5 per cent threshold established in the European Council of Stockholm and the 97.9 per cent EU average. For instance, in the road transportation sector, the EC directive for the opening of road transportation to competition was translated into national legislation in two steps, in 1999 for merchandise freight, in 2001 for passengers. In Portugal, as in other EU countries, attention is now shifting from transposition to effective implementation. Liberalisation of network industries such as electricity and telecommunications has also advanced, but it is essential to increase competition, as incumbents have retained strong market positions and some prices still remain above EU averages.

Opening up the electricity sector: Portugal remains a laggard in the implementation of EC *directives*

The liberalisation of the electricity sector has evolved differently across countries. Portugal has been one of the countries where the process has lagged. By 1996, having undergone no major liberalisation and no privatisation, it was in the same group as France, Italy, the Netherlands, Belgium, Ireland, Spain, Greece, and Canada, while other OECD countries were more advanced in the process. As the very first step in the liberalisation process, the state-owned electricity company (EDP) was restructured in 1994 and the regulatory agency ERSE became operational in 1997.[156] This is also when the first phase of the EDP privatisation was undertaken. Privatisation has been carried out over several years, with the last phase planned to be completed by 2006. Since 2002, 45 per cent of the market has been open to competition for all consumers in medium and high voltage, with full liberalisation envisaged in 2004.

By 1998, accounting separation between electricity generation and transmission had been achieved. Then, following EC directives, the legal separation was finally achieved at the end of 2000, with the detachment of the transmission grid manager from the EDP group.[157] Legal and accounting separation increased transparency. But to prevent the occurrence of discriminatory practices and enhance competition, further market opening is needed. Following the separation, the contractual relations between the company responsible for the grid and the EDP Group companies (*i.e.* power purchase agreements with EDP generation companies and supply contracts with EDP distribution) were not significantly affected.

According to the EC Directive (1996), member states are required to introduce third-party access, *i.e.* have the legal obligation to provide access to the grid under non-discriminatory conditions. This is essential to allow effective entry of new generators, hence to provide real choice to consumers; it also allows large consumers to contract directly with independent generators, by-passing distribution services. EU countries must also offer choice of supplier to consumers, starting with the "large" ones and moving in several steps to smaller ones.The EC is discussing proposals to carry out further separation in the energy sector, by imposing

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as a minimum requirement functional separation of the transmission system operator and legal separation of the distribution operators.[158]

Currently, the national electricity system is made of a Public Electricity System co-existing with an Independent Electricity System.[159] Although large customers can purchase electricity outside the public system,[160] several factors still hinder competition among electricity suppliers in either the public or the independent electricity markets. Purchasing power agreements (PPA) in the public electric system provide a guarantee on the remuneration of the investments, as they foresee a payment based on a fixed return on net asset plus depreciation. Cost fluctuations, reflecting the behaviour of fuel costs and the monetary variables relevant to PPA's, are paid according to market values. It was recently announced that PPA's will soon be renegotiated or extinguished as a move to enhance competition within the market. Since generators still benefit from such protection in the public system, there is little incentive to produce electricity in the independent system.[161] As generators are allowed to partly pass along their costs, end-user electricity prices are still rather high. There has been a rapid decline in prices over recent years, in line with the regulator's plan to bring them down to the EU average. By now industrial users prices are close to EU levels. But Portuguese electricity prices rank among the highest in the OECD for household consumers (Figure 35) Following a decision to create an Iberian electricity market, Portugal and Spain decided to strengthen the two national grids, between 2003 and 2006. The Portuguese government has further committed itself to abolish a significant number of the existing purchase power agreements with EDP generation companies, by the second half of 2003. REN will be confined to operating the national transmission grid which will be open to other players. The provision of retail services, will be left to the market, either through the negotiation of supply contracts in an "energy stock exchange" or through free bilateral contracts. The legal status of retail operators will be created and their operation liberalised, thereby allowing for a legal and functional separation of the transmission system, distribution operators and the retail operators. Several issues will need to be addressed however before the project can be carried out, to preserve competition because of differences between the two countries regarding the set of regulations prevailing.

Telecommunications: the incumbent has retained a strong market position

At the end of the 1990s, state control through share ownership was still important in many OECD countries. Incumbent public operators were still dominant in many segments and price structures were heavily distorted. Countries that were the first movers in a genuine liberalisation process were the first ones to witness changes in the market structure (number of license holders, market share of the incumbent and of the second largest operator…). By 1998, entry conditions in trunk (domestic long-distance fixed), international and mobile services had been relaxed in many OECD countries. In Portugal, mobile services were liberalised starting

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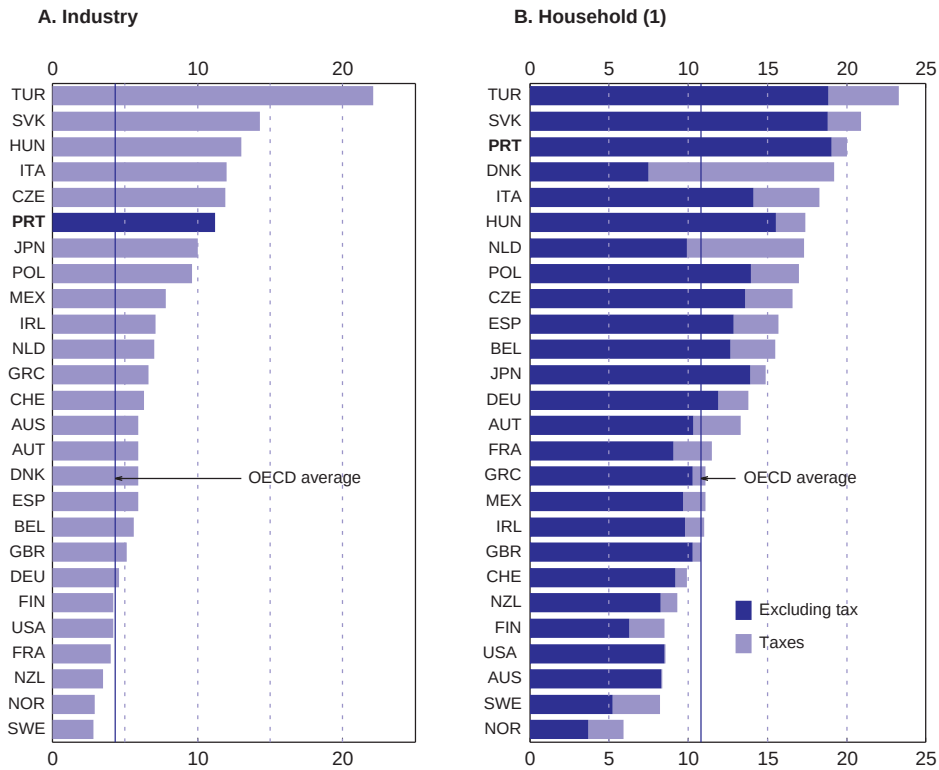
Figure 35. **Electricity prices in selected OECD countries**
In US dollars per 100 kWh (using PPPs) in 2001 or closest year available



1. Price excluding tax for Australia and the United States.
Source: IEA, *Energy Prices and Taxes*, Third Quarter 2002.

in 1991, and by 1998, as in about half OECD countries, three or more competitors were operating in this segment. However, in trunk and international services, Portugal maintained monopoly conditions. As of 1998, only five other OECD countries (Greece, Turkey and 3 Central European countries) were in that situation.[162] In 2000, competition began for all types of fixed telephone services through direct access. Indirect access was applicable on national and international calls and soon after, in January 2001, local and regional calls became eligible. Prices started falling towards the OECD average (according to ANACOM, prices of fixed telephone calls decreased by 26.1 per cent in real terms, between January 1998 and December 2001), but they were still above the OECD average (Figure 36). The privatised incumbent operates

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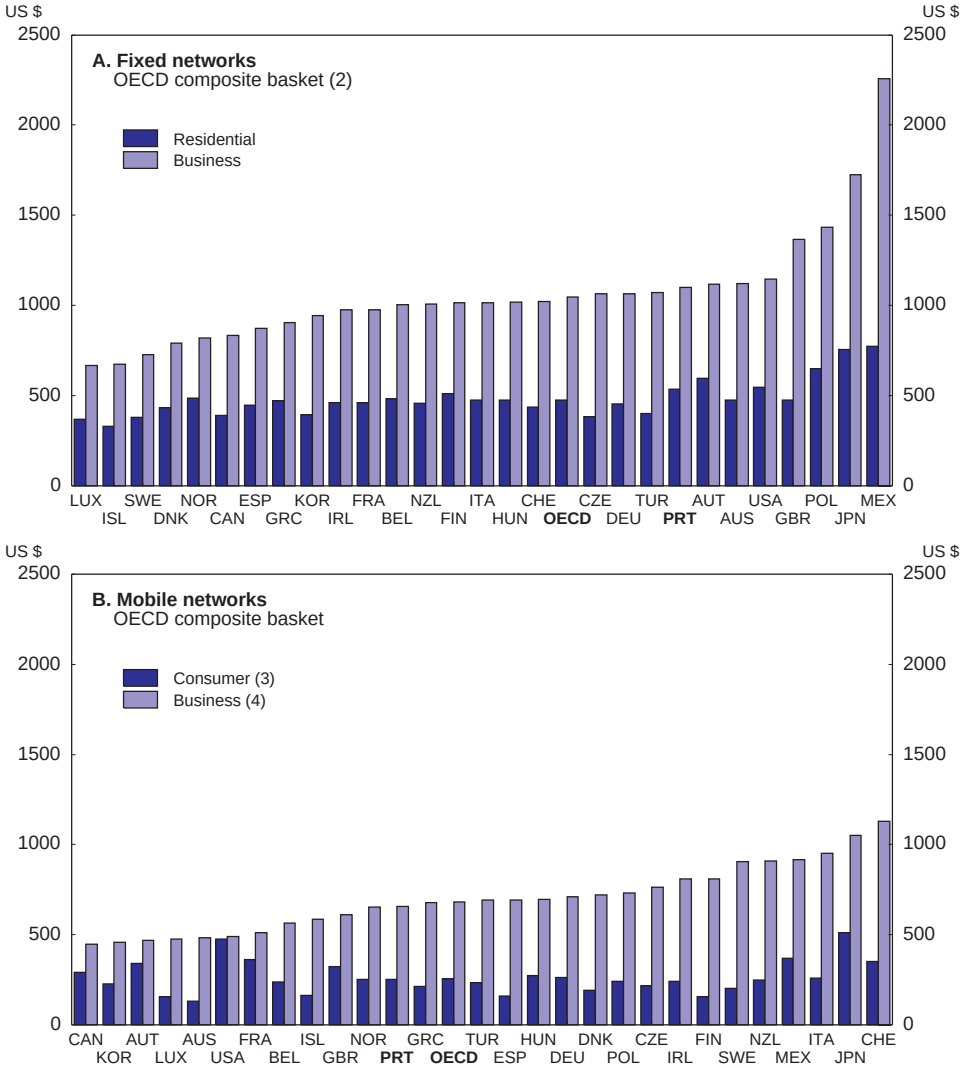
Figure 36. **Telephone charges in selected OECD countries**[1]
May 2002, in US$



1. Fixed and usage charges.
2. Domestic and international services plus calls from fixed to mobile communication networks.
3. The basket includes 50 minutes per month and excludes international calls.
4. The basket includes 300 minutes of which 60 are international.

Source: OECD, Communications database.

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local loops in the framework of a concession contract, until 2025. Although PT has a privileged knowledge of costs and demand patterns, the new operators are also starting to have those competencies.[163] The incumbent also owns the major cable TV network, but operates it through an independent company. In this context, it should be noted that PT Comunicações was declared, for two consecutive years, as an entity with significant market power (SMP) in the fixed telephony networks and/or fixed telephony services and therefore was obliged to present a reference offer for the access to the local loop as from January 2001. As a consequence, as from September 2002, that offer was available on a large scale.[164]

Concluding remarks

To sum up, the structure of the Portuguese economy has several features that should aid convergence towards EU-average productivity and living standards. Despite strict EPL, the labour market is comparatively flexible, as wages reflect more closely sectoral and skill realities than they do in most European countries. As a result, unemployment is low and employment rates are high. But the workforce is lacking in education and training, and closing the productivity gap will require sustained efforts to remedy this, including making the public education system more effective. Product markets in the private sector appear to function relatively well, with competition from and with other EU countries playing a major role. Recent reforms to encourage the creation of new firms and facilitate the exit of failing firms are a further step forward. But Portugal's comparative advantage in the production and export of low technology goods made by low-skilled and low-paid labour is not a lasting strength, as competition from developing countries becomes stronger all the time. Policies to encourage the diffusion and implementation of new technologies and production processes are required. The privatisation programme has recorded steady if unspectacular progress, and it is of paramount importance that the European Court of Justice (ECJ) ruling on Portuguese control in privatised companies be addressed and high priority be given to ensuring a competitive environment. The obligation to also maximise revenue, as set in the Portuguese legal framework, should not be allowed to conflict with this objective. Implementation of competition policy needs to be strengthened, as more competitive markets, especially in network industries, are essential to increase productivity and put downward pressure on prices. Several other OECD countries, and in particular some of Portugal's main trading partners, are moving ahead in structural reform at a rapid pace. Finally, the enlargement of the EU to new entrants will probably lead to a phasing out of transfers received, while intensifying competitive pressures. These factors make it all the more important to push ahead in policy reform.

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Box 11. Implementing structural reform: an overview

Proposal*	Action taken	OECD Assessment/recommendations
I. Improve labour force skills and competencies		
• Raise the quality of formal education, enhancing vocational and technical programmes	Broader access to pre-school education. Revision and modernisation of compulsory education curriculum	Continue with implementation. Promote the reunification of the three basic education cycles. Extend compulsory education to 12 years of schooling
	Some progress towards the development of ICT in schooling	Increased vocational and technical orientation at upper secondary level needed to reduce dropout rates
		Increase the quality and quantity of computer equipment and services
• Improve the incentives faced by teachers and school managers, as well as their accountability	Legislation to carry out comprehensive evaluation of all schools	Implement legislation and monitor the results, rewarding best performers
• Facilitate transition from school to work and support lifelong learning	Legislation to ensure that minors aged 16 and above are provided with training when employed	Continue to encourage training and re-training of those already in the labour force
	Strategy for the development of adult education. Consolidation of national certification system. Specific objectives set for firms' supply of training	Continue with implementation. Consolidate the strategy of lifelong learning. Step up progress towards combating "digital illiteracy"
• Improve quality of tertiary education, maintaining a even-handed treatment of public and private institutions	Legislation for the quality of tertiary education, including the creation of a systematic and independent system of evaluation and certification of courses/ universities	Implement legislation. Close or merge courses with very few students
		Establish a minimum positive mark in upper-secondary graduation to access to tertiary education
		Re-evaluate the operating and financing conditions of tertiary education institutions
II. Promote labour-market adaptability and mobility		
• Ease employment protection legislation	New *Labour Code* proposal includes some easing of the conditions required for fair dismissals	Continue and broaden labour market reforms. Ensure that employment protection is not an obstacle to hiring

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Box 11. Implementing structural reform: an overview (*cont.*)

Proposal*	Action taken	OECD Assessment/recommendations
• Reduce obstacles to "atypical" work contracts	Rules governing temporary employment were tightened	Review the rules. Promote easier access to temporary contracts and part-time work
• Phase out housing market restrictions		Design global plan to develop an efficient house-renting market
III. Make active labour-market policies more efficient		
• Improve the job-matching efficiency of the PES	Early intervention of public employment services, through a network of local public services. Creation of a self-service approach, through the use of electronically available services	Continue with implementation. Improve the relations with employers, monitoring closely their needs. Promote a rapid articulation of the training programs offered with the qualification and skill demands of companies. Enhance the exposure of the PES to market forces
• Rationalise the numerous instruments in place		Carry out a systematic monitoring and evaluation of all active labour market programmes
IV. Promote a business friendly environment		
• Reduce red tape and administrative burden on business	*Productivity and Growth Programme* makes further advances in alleviating procedures, including administrative simplification and deregulation	Promote the timely implementation of all measures included in the *Productivity and Growth Programme*
	Reduction of the number of business contact points in the public administration. Reform of the bankruptcy law	Implement legislation. Further increase the efficiency of registry and notary services and related judiciary processes
• Stimulate firm creation and encourage productive investment	Simplification of administrative procedures of investment projects. Revision of the legal framework for venture capital	Implement legislation and complete the *Investment Code*. Apply the rules of convertible financing to co-funded projects. Foster venture capital market
V. Promote technology and innovation		
• Increase R&D spending and rebalance toward business	Steps taken to promote science and technology culture and public understanding of science. EU-funded programmes for advanced training in R&D and for R&D research projects. Introduction of a more favourable R&D tax regime	Increase R&D spending. Continue to encourage research collaboration between universities and enterprises. Timely implementation of the *Operational Programme for Science, Technology and Innovation*

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Box 11. **Implementing structural reform: an overview** (*cont.*)

Proposal*	Action taken	OECD Assessment/recommendations
• Seize the benefits of ICT diffusion	Steps taken to spread out ICT among households and companies, schools and other public areas	Speed up the implementation of the *Internet Initiative*. Push Internet access tariffs down by increasing liberalisation of the telecommunications market
VI. Increase product-market competition		
• Reform institutional arrangements for competition policy	Creation of a independent Competition Authority and a new Competition Law	Implement legislation. Clarify the Competition Authority's interactions with sectoral regulators
	Progress in the transposition of EC directives to national legislation	Promote the effective implementation of EC directives
• Remove entry barriers and increase competition in network industries	Liberalisation process of the electricity sector has lagged. Decision to create an Iberian electric market	Apply rigorously EC directives concerning the energy market. Proceed with the implementation of the Iberian electric market
	Despite some liberalisation in telecommunications, incumbent has retained a strong market position	Enhance competition, in particular in the fixed line sub-sector. Implement the unbundling of the local loop
• Continue privatisation process	High priority was given to preserve Portuguese control and to maximise revenue for the government	High priority should be given to ensure competitive bidding and coherence of the privatisation process with competition policy. Abolish privatisation limits for all public enterprises

* Proposals are based on detailed analysis provided in previous and current *Economic Surveys*.
Source: OECD Secretariat.

The broad policy framework, articulated around three main pillars – first, macro stabilisation and budgetary reform; second, social policy and labour market reforms; and third, microeconomic policies aimed at improving the business environment – is in accordance with OECD policy recommendations. Continued investments in human capital development and advances in the employment strategy, including the labour market reform proposed and the new framework law on social security, are welcome. The 2002 Productivity and Growth Programme is appropriately aimed at fostering innovation and facilitating business, with a renewed focus on stepping up product market competition. Timely implementation of the government's reform plans is key to raising Portugal's growth potential. An overview of the main OECD recommendations for structural reform is presented in Box 11 above. Reform of budgetary processes and public administration

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are discussed in the previous Chapter and issues relating to environmental dimensions of sustainable development are examined in the next section.

Issues relating to sustainable development

There is growing concern over the problems that the pursuit of development may pose in the longer term, if appropriate measures are not taken to ensure a proper balance between economic, social and environmental performance. This section focuses on three issues relating to the interface between the economic, environmental and, to some extent, social dimensions of sustainable development that are of particular importance in Portugal. For each of them, indicators are used to examine performance and to identify any worrisome trends, and the corresponding public policies are evaluated. This section also looks at institutional mechanisms that could be used to integrate different policies affecting sustainable development (Box 12). This section deals first with air pollution. It goes on to examine policies for reducing water pollution, as well as the efforts that have

Box 12. Integrating policies relating to the different dimensions of sustainable development

A national strategy for sustainable development has been prepared in Portugal. This initiative that involved all the public institutions concerned in a forward planning exercise aimed at contributing to better policy integration. The subject has been the topic of broad public debate at several regional conferences. The strategy represents a tool for mobilising public opinion and for guidance of decision makers in formulating and implementing policies. The strategy is to result in a programme that will set target dates for the various activities listed. The measures will first need to be scoped in terms of their cost, so that the necessary trade-offs can be made among the different priorities.

In terms of evaluating public decisions concerning investment, environmental impact assessment (EIA) are prepared for most public works projects which have a significant impact on environment, and for those that are financed by Community funds, including the projects under the Operational Programme for the Economy. These EIA will have to be extended by 2004 to cover all public policies, pursuant to Community directives on plans and programmes. Cost-effectiveness analyses are done for certain major policies such as the national action plan to combat climate change, the national emission ceilings programme and the strategic plan for water supply and waste water treatment. There is still no cost-benefit analysis performed in the preparation of environmental policies. However, cost-effectiveness analyses are done for certain major policies, such as the national action plan to combat climate change. With the growing importance of environmental policies, it would be highly desirable to give more emphasis to cost-benefit analysis, as an essential tool for ensuring consistency among the efforts undertaken in different areas and, to focus on cost-effective solutions to achieve a given target.

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been made to manage natural resources on a sustainable basis, particularly with respect to fisheries and water supply.

Air pollution

Principal issues

The release of pollutants into the air raises problems of public health that, in the case of Portugal, relate primarily to the presence of suspended fine particles in the major cities. On the other hand, given its geographic situation, Portugal has one of the highest atmospheric pollutant "net exports", relative to emission levels, of any country in continental Europe, both for sulphur dioxide (65 per cent) and for nitrogen dioxide (50 per cent). About half of these net exports are deposited in the sea, with no great environmental damage. The other half is transported inland. The main challenge is to correct local pollution problems and to meet the emission targets set forth in European legislation, without imposing excessive costs on the economy.

Performance

Portugal has been one of the slowest among OECD countries in reducing the pollution intensity of its economic activity, and indeed that intensity has worsened in the case of volatile organic compounds (VOCs), with pollution intensities well above the average for the OECD area. The increase in emissions has, nonetheless, been modest. Only in the case of carbon monoxide emissions has there been any clear downward trend, and this only after substantial increases in the early 1990s (Figure 37). The growth in Portugal's electric generating capacity has not been accompanied by as marked a reduction in emissions per unit of output as in most other OECD countries (Table 19). On the other hand, transportation-generated SO_2 emissions have declined by one-third.

Such air quality problems as exist relate less to traditional pollutants than to particulate matter. As most conventional pollution emissions are transported, they have only a limited impact on air quality in Portuguese cities, with SO_2 concentrations that are among the lowest in the OECD (Figure 38). In addition, the ozone alert threshold has not been exceeded in any of Portugal's major cities since 1995. The public information threshold, which is lower, has however been briefly exceeded on occasion[165] (Figure 37). On the other hand, there is concern over the levels of fine particulate concentrations, which are carcinogenic and are harmful to the respiratory system. Concentrations of particulates with a diameter of less than ten micrometers (PM10) currently exceed the limits, on an annual average, at each of the 14 measurement stations in Lisbon and Porto. In terms of daily peaks, values were exceeded at ten of these 14 stations. The sources of particulate pollution have yet to be properly documented, but it would seem that combustion facilities and vehicles are the main culprits.[166]

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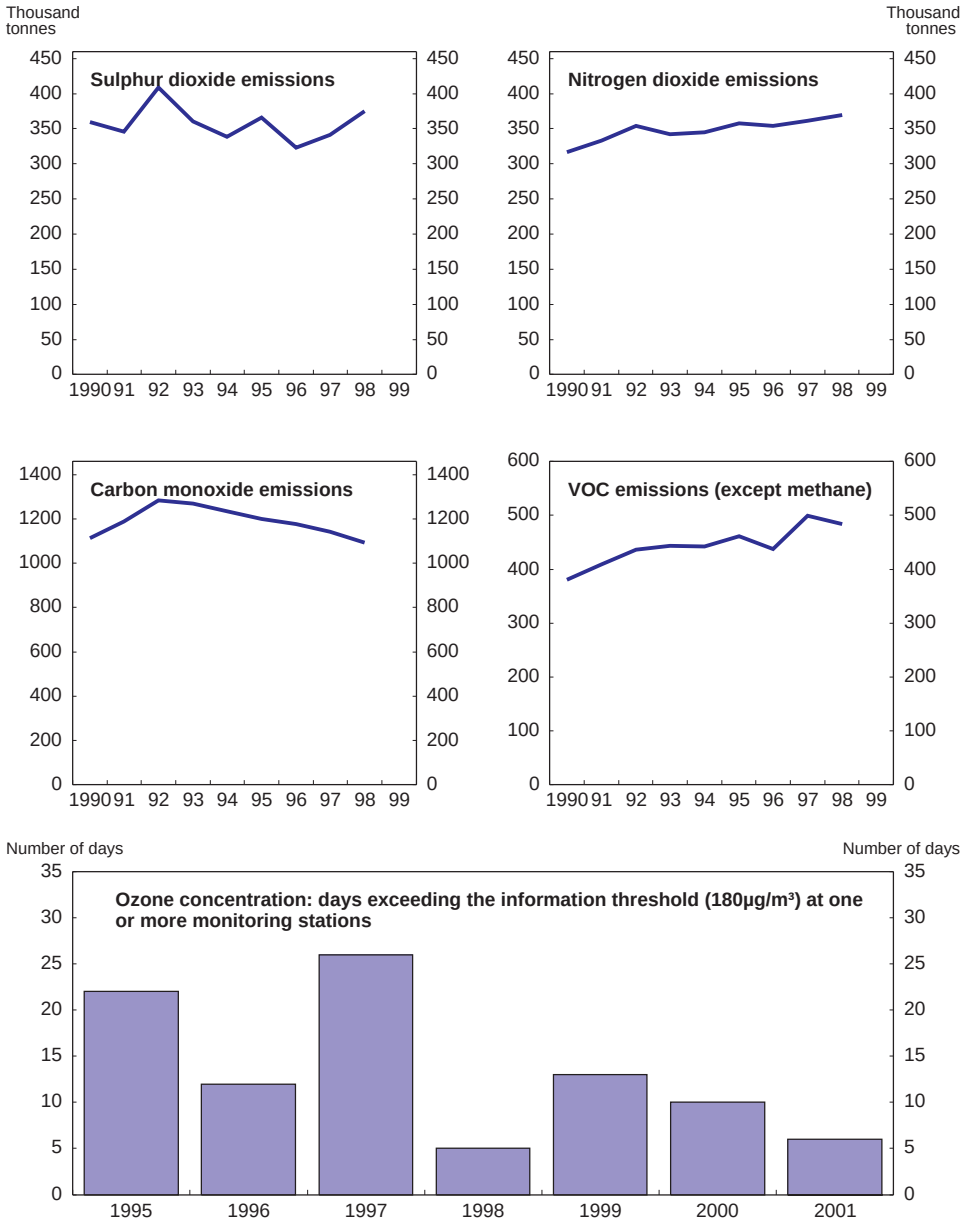
Figure 37. **Performance indicators: air pollution**



Source: EMEP; Ministry for Towns, Territorial Planning and Environment.

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Table 19. **Atmospheric pollution performance indicators**

	Change in emissions per unit of GDP, 1990-99[1]			Level of emissions , 1999[2]			Improvement in productive efficiency, 1990-1999	
	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide per unit of electricity output	Nitrogen dioxide per vehicle
	Per cent per year			Grams per dollar of GDP (1995 US$ PPP)				
Australia	3.1	–1.9	–2.9	3.9	5.5	4.1	n.a.	n.a.
Austria	–10.2	–3.5	–6.5	0.2	0.9	1.2	–77.2	–32.5
Belgium	–9.2	–3.6	–4.8	0.8	1.2	1.1	–64.4	–22.6
Canada	–5.3	–2.8	–2.9	3.2	2.6	3.5	n.a.	n.a.
Czech Republic	–19.1	–6.6	–5.7	2.1	3.0	1.9	n.a.	n.a.
Denmark	–14.2	–4.9	–5.2	0.4	1.6	1.0	–79.8	–38.9
Finland	–13.0	–3.8	–4.1	0.7	2.1	1.4	–65.3	–28.7
France	–8.3	–3.8	–5.1	0.5	1.2	1.4	–52.0	–37.8
Germany	–20.1	–7.1	–8.8	0.4	0.9	0.9	–85.2	–43.7
Greece	–1.4	–0.3	–0.2	3.6	2.5	2.6	–4.6	–22.3
Hungary	–6.1	–1.1	–3.8	5.5	2.1	1.4	n.a.	n.a.
Iceland	–1.1	–1.7	–4.9	3.7	3.9	1.4	n.a.	n.a.
Ireland	–8.1	–6.3	–7.9	1.7	1.3	1.0	–35.8	–19.0
Italy	–7.6	–4.3	–4.4	0.8	1.2	1.4	–52.8	–31.3
Japan	–1.7	–1.3	–2.4	0.3	0.5	0.6	n.a.	n.a.
Korea	–8.9	–3.0	n.a.	2.1	2.0	n.a.	n.a.	n.a.
Luxembourg	–18.7	–9.1	–7.9	0.2	0.9	0.9	n.a.	n.a.
Mexico	–0.4	–0.9	–9.3	1.6	1.6	1.4	n.a.	n.a.
Netherlands	–10.1	–6.5	–8.8	0.3	1.1	0.7	–50.2	
New Zealand	–0.4	–0.5	–1.6	0.7	3.0	2.6	n.a.	n.a.
Norway	–9.7	–2.8	–1.8	0.2	2.0	3.0	n.a.	n.a.
Poland	–10.0	–6.6	–4.8	5.1	2.8	2.2	n.a.	n.a.
Portugal	**–2.1**	**–0.9**	**0.1**	**2.4**	**2.3**	**3.1**	**–17.7**	**–18.3**[3]
Slovakia	–13.7	–7.4	–5.6	3.3	2.4	2.0	n.a.	n.a.
Spain	–5.8	–2.1	–3.5	2.2	1.7	3.6	–39.4	–14.4
Sweden	–8.3	–4.3	–3.9	0.3	1.3	2.1	–26.0	
Switzerland	–6.0	–5.4	–6.3	0.1	0.5	0.9	n.a.	n.a.
Turkey	3.2	1.0	0.0	3.4	2.3	1.6	n.a.	n.a.
United Kingdom	–13.9	–7.9	–6.6	1.0	1.3	1.4	–72.8	–51.0
United States	–5.5	–2.5	–4.7	2.0	2.7	1.9	n.a.	n.a.

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Table 19. **Atmospheric pollution performance indicators** (*cont.*)

	Change in emissions per unit of GDP, 1990-99[1]			Level of emissions , 1999[2]			Improvement in productive efficiency, 1990-1999	
	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide	Nitrogen dioxide	VOC	Sulphur dioxide per unit of electricity output	Nitrogen dioxide per vehicle
	Per cent per year			Grams per dollar of GDP (1995 US$ PPP)				
European Union	−11.1	−5.0	−5.3	0.8	1.2	1.5		
OECD Europe	−10.1	−4.8	−5.0	1.2	1.4	1.5		
OECD	−6.7	−2.9	−4.4	1.5	1.9	1.6		

Note: For the OECD average data for countries with missing data for either 1990 or 1999, data for the latest year has been substituted. Estimated data for 1999 represents about 5 per cent of the area total.

1. Australia: 1995-99 for sulphur dioxide; Korea and New Zealand: 1990-98; Mexico: 1994-98 for sulphur dioxide and VOC and 1990-98 for nitrogen dioxide; Slovakia: 1990-98 for sulphur dioxide and nitrogen dioxide and 1990-97 for VOC.
2. 1998 for Korea, Mexico, New Zealand and sulphur dioxide and nitrogen dioxide in Slovakia; 1997 for VOC in Slovakia.
3. Between 1990 and 1997 for Portugal.

Source: Cooperative Programme for Monitoring and Evaluating of Long-Range Transmission of Air Pollutants in Europe (EMEP); World Health Organisation; OECD (2002a).

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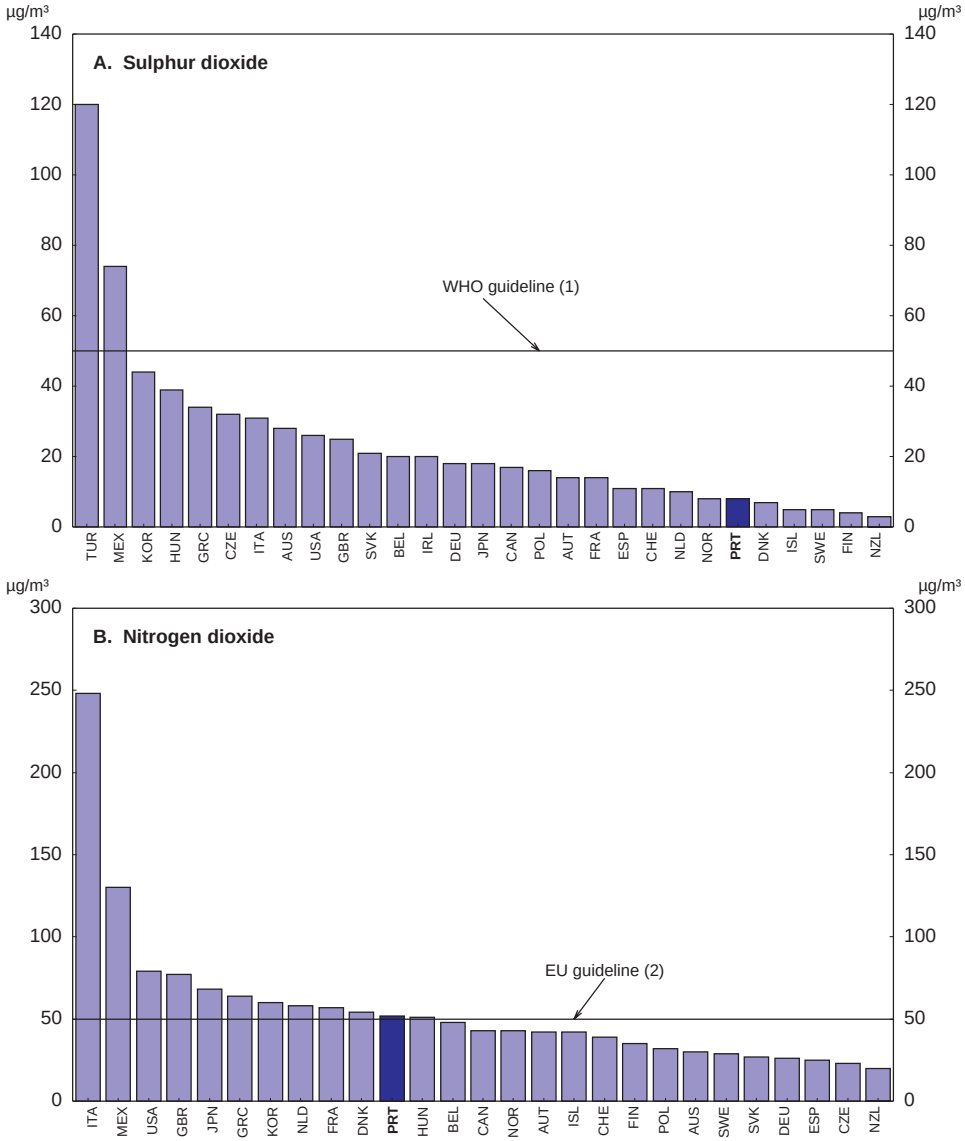
Figure 38. **Urban concentrations of atmospheric pollutants in OECD countries 1995**



1. Mean value for one year.
2. Median of annual data of one hour values for concentrations.

Source: World Bank.

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Policies

Following accession to the EU, Portugal adopted strict internationally-based targets for limiting emissions, in compliance with air pollution policies which are determined primarily by Community legislation.[167] Consistent with the long-standing approach in community environmental law, Portugal currently uses regulation as its main air policy instrument (OECD, 2001j). In 2000, Portuguese firms spent nearly € 130 million, or about 0.1 per cent of GDP, to reduce their emissions of atmospheric pollutants (INE, 2002).

If it is to fulfil its international commitments under the EU, Portugal will have not only to end the steady but slow rise in emissions but to make considerable reductions (Table 20). This is particularly the case for SO_2, NO_x and VOCs, where current emissions are more than twice the ceilings set for 2010. If they continue to be

Table 20. **Selected commitments to air quality**

Concentrations

Pollutant	Measuring period	Limit value	Date
SO_2	1 hour	350 µg/m^3: not to be exceeded more than 24 times each year	01/01/2005
	24 hours	125 µg/m^3: maximum three times each year	01/01/2005
	Calendar year	20 µg/m^3	19/07/2001
NO_2	1 hour	200 µg/m^3: maximum 18 times each year	01/01/2010
	Calendar year	40 µg/m^3	01/01/2010
NO_x	Calendar year	30 µg/m^3	19/07/2001
PM_{10}	**Phase 1**		
	24 hours	50 µg/m^3: maximum 35 times each year	01/01/2005
	Calendar year	40 µg/m^3	01/01/2005
	Phase 2		
	24 hours	50 µg/m^3: maximum 7 times each year	01/01/2010
	Calendar year	20 µg/m^3	01/01/2010
Lead	Calendar year	0.5 µg/m^3	01/01/2005
CO	8 hours	10 mg/m^3	01/01/2005
Benzene	Calendar year	5 µg/m^3	01/01/2010

Emissions

Pollutant	Emission ceiling to be met in 2010 (kilotonnes)	Emissions 1990	Emissions 1999
SO_2	160	338	380
NO_2	250	282	385
VOC	180	381	494

Source: EU Directives 1999/30/EC, 2000/69/EC and 2001/81/EC; National inventory greenhouse gases, Submission 2002.

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based on the conventional regulatory approach, measures to ensure compliance with emission ceilings could cost Portugal close to € 1.5 billion a year, or 1.3 per cent of its GDP (European Commission, 1999). The current market price of fuels, nevertheless, should help to reduce emissions as older generating plants are replaced resulting in a switch to the least-cost solution, natural gas, which is less polluting than coal or fuel oil.[168] Yet this impact will not be sufficient, and the potential size of costs is such that it will be important to seek the most economical reduction strategies possible.

Meeting national emission ceilings will require efforts on the part of industry. The action plan now in preparation includes a wide and public consultations phase, at the end of which the government will decide how the burden of the depollution effort is to be distributed. The authorities could develop plans to award emission permits as a function not only of the cost to each sector, but also in light of feasibility and equity criteria, as in the case of the allowance trading regime currently in preparation at EU level. With a regulatory device of this nature, the introduction of criteria other than marginal cost equalisation may make the overall cost of depollution greater. Sectoral targets are unlikely to result in abatement cost equalisation. Even if marginal cost equalisation were the only criterion for the initial allocation, perhaps unlikely, granting businesses emission permits that are fixed once for all could increase costs, for two reasons. F*irst*, the authorities distributing the permits cannot know the exact values of the marginal depollution costs borne by the businesses. S*econdly*, costs evolve over time, as levels of activity vary and new technologies appear. Allowing businesses to trade these quotas among themselves would, on the other hand, make it possible to equalise marginal costs and minimise the overall cost, especially if international trading were allowed to offset the narrowness of local markets.[169]

Because it is growing so strongly, the transportation sector has aroused concern over its impact on air quality mostly because of its emissions of fine particulates, concentrations of which are particularly high in the country's major cities. Diesel vehicles are today responsible for most of these emissions, and despite the reductions expected from the current timetable for stiffening standards, they will continue to emit greater volumes of particulates than gasoline-powered vehicles (Table 21). In this regard, fuel taxation provides a perverse incentive: the excise tax on a litre of diesel is 15 per cent less than that on a litre of gasoline, whereas if the harmful effects of diesel combustion were taken into account, the tax on diesel should be higher than that on gasoline (Proost *et al.*, 2001). A supplementary approach, which could be pursued only on a Europe-wide basis and over the longer term, would be to strengthen fine-particulate emission standards for diesel vehicles.[170] Ideally, sources of mobile pollution should be integrated with non-mobile sources in a pollution trading system. Such an approach might be possible for fleet operators of diesel vehicles, when emission limits are fixed for fine particle emissions. In other areas, total emissions from road transport eventually

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should decline drastically, as a result of stiffer standards for fuels and vehicles despite the expected increase in the length of road trips in coming decades (Table 21). In practice, however, very high vehicle license fees (about 40 per cent of the price of a 2000 cc vehicle) could undermine the pace at which the positive impact expected from stiffening vehicle standards occurs, by providing an incentive

Table 21. **Road transport: emission standards and emissions**

A. Emission standards for vehicles registered at different dates in EU countries

By fuel and emission standard on an urban test cycle, per vehicle-kilometre
Index Pre-Euro I gasoline car = 100, except for particulates[1]

Type of vehicle	Pre-Euro I	Euro I	Euro II	Euro III	Euro IV	Percentage change
	1990	1993	1997	2001	2006	
Petrol car						
Carbon monoxide	100	15	10	7	4	–96
Hydrocarbons	100	9	4	3	2	–98
Oxides of nitrogen	100	19	9	6	3	–97
Particulates	5	2	2	2	2	–55
Diesel car						
Carbon monoxide	7	4	3	2	2	–71
Hydrocarbons	10	4	3	2	1	–90
Oxides of nitrogen	43	29	21	13	7	–84
Particulates	100	55	31	20	10	–90
Semi-trailer truck						
Carbon monoxide	44	22	18	9	7	–84
Hydrocarbons	183	87	78	47	33	–82
Oxides of nitrogen	1 704	893	650	461	325	–81
Particulates	700	482	185	124	24	–97
Bus						
Carbon monoxide	63	28	22	11	8	–87
Hydrocarbons	83	90	84	50	35	–58
Oxides of nitrogen	795	859	614	436	307	–61
Particulates	458	304	187	125	24	–95

B. Total road transport emissions in Portugal

	Thousand tonnes, 1995	Index, 1995 = 100 per cent					
		1995	2000	2005	2010	2015	2020
CO	499	100	79	54	38	28	23
NO_X	104	100	79	55	33	24	21
VOC	101	100	70	40	18	13	12
Benzene	4	100	52	29	14	10	8
PM-diesel	3	100	79	56	32	20	16

1. No legal standard exists for gasoline cars. The base for particulates is pre-1993 diesel car emissions.

Source: Panel A: UK Department for Transport, Local Government and the Regions, *Transport Statistics Great Britain* 2001.
Panel B: European Commission, Standard and Poor's DRI, KULeuven, August 1999, Draft Final Report.

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to keep older, more polluting vehicles in operation.[171] That incentive is exacerbated by the considerable reductions in municipal road user fees (from 65 to 80 per cent) that are allowed for vehicles more than six years old.

Conclusions

In order to moderate the costs inherent in the command and control regulatory approach, a free trading market mechanism could be set on the basis of the initial allocation of emission permits to point sources, as it is currently in preparation for in-house greenhouse gases emissions at the EU level. The tradable permits market should cover not only industrial polluters but other major point sources of emissions as well, such as electric power plants, in order to reduce costs further. Vehicle taxation should be revised to encourage fleet renewal. This could be done by eliminating vehicle license fees. At the same time, the annual road-use tax could be increased and should vary in the light of a vehicle's pollution characteristics.[172] At a minimum, older vehicles, which are also the most polluting ones, should no longer benefit from tax reductions. The effort to combat particulate pollution in the cities could start with an increase in diesel taxes, to bring them up to par with gasoline taxes. If the impact of such taxation proved to be regressive, it could be offset by an appropriate change in the overall social distribution mechanism. Over the longer term, it would be desirable to strengthen Europe-wide rules on fine particulate emissions by diesel vehicles.

Reducing water pollution

Principal issues

Drinking water is an essential good, the supply of which depends in large part on the quality of both surface and ground waters. The availability of high-quality water has additional advantages as the basis for many economic activities, particularly of the leisure kind, and also contributes to the value of the landscape. The main challenge here is to consolidate the progress made in recent years, and to maintain and improve water quality at the lowest possible cost, recognising that the hitherto generous European investment subsidies, that often cover 60 per cent of capital costs, are likely to end.

Performance

Portugal has had a poor record for water quality but this situation has evolved rapidly. A start was made in reducing the pollution of watercourses during the 1990s, through the construction of purification plants that reduced the biochemical demand for oxygen (Table 22).[173] During the past 15 years, the population connected to waste water purification plants has risen from 21 per cent of all households to 55 per cent. From 1990 to 2000, a total of € 1.3 billion was spent on

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Table 22. **Performance indicators: water pollution**
Selected rivers

	Biochemical oxygen demand	Nitrates	Total phosphorus	Nitrogen balance on agricultural land
	Percentage changes, 1980-85 to last three years			1985-87 to 1995-97, annual average change
Australia				0.8
Austria	..	20.1	–42.8	–2.4
Belgium	–51.8	2.7	55.9	–0.5
Canada	..	–2.1	–19.3	8.4
Czech Republic	–45.2	–20.9	..	–5.8
Denmark	–46.9	–33.1	–49.0	–2.7
Finland	..	37.3	–27.6	–1.9
France	–38.0	10.3	–3.0	–0.9
Germany	–19.3	2.4	–64.2	–3.6
Greece	..	..	..	–5.3
Hungary	–32.2	–25.0	–27.5	n.a.
Iceland				–0.2
Ireland	..	..	..	2.5
Italy	..	30.3	–33.3	–3.7
Japan	–7.4	..	..	–0.7
Korea	..	..	..	3.9
Luxembourg	–8.9	–5.0	–2.6	
Mexico	74.1	–73.0	28.9	–2.3
Netherlands	–23.2	–13.9	–54.9	–1.8
New Zealand				3.3
Norway	..	–1.0	–21.6	0.1
Poland	–12.5	4.8	–26.1	–4.9
Portugal	**–20.4**	**19.2**	**87.3**	**3.8**
Spain	104.7	10.4	–13.1	1.0
Sweden	..	6.1	–31.6	–3.2
Switzerland	..	16.0	–27.5	–2.7
Turkey	122.0	–8.4	8.5	–3.4
United Kingdom	–16.5	7.8	16.9	–2.1
United States	–3.7	–3.1	6.4	2.2

Source: OECD, *Environmental Data*, Compendium 1999.

sewer systems and purification plants serving municipalities and industries. The percentage of public bathing sites that meet quality standards has risen considerably (Table 23). Yet despite this progress, water quality problems remain widespread: nearly two-thirds of watercourses are still polluted to a degree considered critical, serious or excessive according to national norms. The national ceilings set for concentrations of toxic substances are exceeded for nine pollutants.[174]

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Table 23. **Water quality in Portugal**

Quality of surface waters,[1] nationally (percentage of measuring stations)[2]

Quality category[3]	1990	1993	1996	1999
No pollution	0	0	0	0
Moderate pollution	20	23	22	35
Critical pollution	26	29	42	38
Severe pollution	44	40	33	25
Excessive pollution	10	8	3	2
Total	100	200	100	100

Bathing water quality
(portion of bathing areas where water quality meets the limit values of Directive 76/160)[4]

	1993		1997		2001	
	Per cent	Number	Per cent	Number	Per cent	Number
Coastal waters	58	(312)	90	(336)	97	(366)
Inland waters	17	(24)	25	(24)	84	(38)

1. Based on mandatory limit values for all water uses (human consumption, aquatic life, irrigation, recreational activities).
2. Year-over-year figures for 60 identical measuring in pollution-prone areas.
3. Quality categories using 16 chemical parameters (organic matter, suspended matter, micro-organisms, nutrient elements, heavy metals).
4. Parentheses indicate the total number of bathing places on which the percentage is calculated.

Source: OECD; European Commission.

The major causes of pollution currently are untreated municipal effluents, industrial discharges and agricultural wastes. Only five of the 27 cities located in sensitive zones have purification plants that meet existing standards (European Commission, 2002b). More than two-thirds (68 per cent) of industrial effluents are discharged without treatment. The toxic substances now polluting the water come primarily from the leather and textile industries. Agricultural nitrate runoff is growing steadily, and is the highest among the European OECD countries (Table 22).

Policies

Expanding waste water purification is the major focus of efforts to counter water pollution in Portugal, and has accounted for substantial investment during the last 15 years. Additional investments will be required, however. Portugal has set itself the objective of connecting 90 per cent of the population to purification plants by 2006. Only a small fraction of waste water collection and treatment costs is recovered from the polluters concerned – households and businesses – through a specific municipal tax. Cost-recovery rates stand at 20 per cent for households and businesses, and these are enough to cover operating expenses, but the eventual replacement of the plants is being neglected. Without metering, consumers

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are under no incentive to restrain their use of water. Metering should be introduced in order to relate the municipal tax to water consumption, so as to ensure correct appreciation of marginal costs by users and so help stabilise the use of natural resources (see below).

Industry has been very slow in treating its effluents. A 1994 decree-law called for imposing taxes on polluting industrial effluents. That legislation has never been enforced, however, and industrial wastes are still governed by an administrative penalties system that is largely ineffective, because the levels of fines are not sufficiently dissuasive. The only effective way to improve water quality is to make sure that a similar level of effort is imposed on all industrial polluters. Monitoring of industrial effluents is possible and so, in contrast to the household sector, a policy based on the taxation of pollution is possible.

Farming activities exert significant pressure on water quality. Crops account for 89 per cent of agricultural emissions of nitrates and 79 per cent of those of phosphates, with the remainder attributable to swine operations. This pressure could even intensify with the rapid growth of irrigation, which encourages a shift to more intensive and more polluting modes of agricultural production. The major problem lies in the lack of any effective incentives for reducing pollution. The best solution to this problem would be the taxation of nutrient balances, provided that reasonable estimates are available of the economic benefits that would flow from lower nitrate run-offs. A code of good agricultural practices has been introduced, and failure to observe it makes farmers liable, in principle, for financial penalties in areas identified as vulnerable to nitrate pollution. Yet not a single fine of this kind has been levied to date. Moreover, the common agricultural policy encourages excessive resort to fertilisers and pesticides, through price supports for agricultural products, and in this way contributes further to pollution. Making assistance conditional on sound environmental practices would reduce the scale of this damaging incentive.

Conclusions

Considerable investments have been made to collect and process urban waste water. The sustainability of these efforts could be enhanced by strengthening their economic and financial underpinnings, through charging more realistic prices for water and allowing municipalities to recover their costs. For households the current municipal tax could be replaced by a sewage charge based on households' gross water consumption, as a second-best solution to charging households for their actual waste production. For industrial enterprises connected to the municipal sewage networks their output of pollutants to the municipal sewer system should be monitored and taxed.[175] Untreated industrial wastes dumped into watercourses should be taxed at the cost of the pollution, as stipulated in 1994 decree-law. Since farming activities are increasingly responsible for water pollution,

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it would be desirable to change the current situation, where farmers are under no real incentive to control pollution. The penalties now on the books should be enforced. Reducing farm-sourced pollution will also require reforms to the support provided under the common agricultural policy. If water protection policy is to be as effective as possible, there should be a shift toward a common mechanism that embraces all sectors and that is based on market forces. The most efficient approach would involve trading permits between industrial, agricultural and municipal sources of nutrient pollution, within the same river basin or around principal lakes and reservoirs.

Working towards the sustainable use of natural resources

Principal issues

The sustainable use of natural resources is an issue that arises in particular for fisheries, water supply and soil erosion. For fisheries, the difficulties lie in the overall size and allocation of quotas. For water supply, the main issue is the lack of adequate user prices. When it comes to soil erosion, the problem lies primarily with the failure of owners to take proper thought for the future.

Performance

Fish stocks in the exclusive economic zone, as in the rest of the world's oceans, continue to decline (Table 24), under the impact of overfishing. The dwindling resource is reflected in catches by Portuguese vessels that are 40 per cent below what they were in 1990. On the other hand, fish consumption has more than doubled since 1980; as a result imports increased.

The rapid spread of irrigated farming constitutes the principal threat to water resources (Table 25). The amount of water used for irrigation rose by 72 per cent between 1990 and 1998 even though the surface irrigated fell by 10 per cent, a trend that is expected to continue, given the many projects now underway (Alqueva, Odeleite-Beliche and Odelouca-Funcho). This quantitative growth is of particular concern since irrigation is shifting from the areas that are best endowed with water to those that are least so (Table 25).

Policies

Fishing activity is limited by the application of the European common fishing policy. In order to take full advantage of these quotas, there are significant financial transfers to the industry that amount to more than 20 per cent of the value of the catch. The sardine catch, which represents half of the total, provide an indication of the policies adopted in response to overexploitation. In this case, the principal measures are a prohibition on fishing during the spawning months,

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Table 24. **Key indicators of sustainable use of natural resources**

	Fish catch	Fishing fleet	Water withdrawals	Transfers to the fishing industry
	Per cent change	Per cent change (tonnage)	Per cent of available resources	Per cent of landed value
	1990-2000	1985-1997		1997
Australia	1	78.1	4.3	9
Austria	61	..	2.7	..
Belgium	–28	1.6	42.5	5
Canada	–39	..	1.7L	25
Czech Republic	..	..	15.6	..
Denmark	4	–29.9	15.7	16
Finland	15	140.4	2.2	90
France	–4	4.2	23.9	18
Germany	–37	–86.0	24.4	32
Greece	–25	–38.7	12.1	13
Hungary	–56	..	5	..
Iceland	32	62.7	0.1	4
Ireland	27	..	2.6	47
Italy	–19	–12.3	32.2	5
Japan	–48	–42.1	20.8	21
Korea	–26	15.6	35.6	7
Luxembourg	..	..	3.4	..
Mexico	–3	..	17.4	1
New-Zealand	63	..	0.6	4
Norway	69	–3.8	0.7	12
Netherlands	22	20.0	4.9	8
Poland	–51	..	18.7	4
Portugal	**–41**	**–39.3**	**15**	**21**
Slovak Republic	..	..	1.4	..
Sweden	35	..	1.5	42
Switzerland	–47	..	4.9	..
Turkey	33	..	15.2	13
United Kingdom	–2	..	14.6	13
United States	–15	..	19.9	24

Source: FAO; OECD and World Bank.

and an annual ceiling set as a share of the allowable Europe-wide catch. This quota has been divided among the seven organisations representing the ship owners. It is left to the ship owners themselves to distribute licenses within their organisation: no domestic market for these quotas has appeared, and the authorities are not inclined to introduce one. These measures do appear to have achieved their objective of stabilising the stock of sardines and the Iberian fish stocks. The only case of individual quotas involves fishing on the high seas,

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Table 25. **Irrigation: scope and trend**

A. Withdrawals of freshwater by major use

	Public supply	Irrigation	Industry	Cooling of thermal power plants
	Percentage			
1991 breakdown	5	59	13	23
1998 breakdown	7	79	3	11
1991-1998 change	+86	+72	–65	–62

B. Irrigated area

Region (from north to south)		Number of hectares		Change (percentage)
		1989	1999	
North	Between Douro and Minho	225 510	148 300	–34
	Trás-os-Montes	87 750	93 100	6
	Beira Interior	112 730	92 710	–18
	Beira Litoral	155 930	104 620	–33
South	Ribatejo and Oeste	144 700	154 520	7
	Alentejo	110 750	163 990	48
	Algarve	34 220	30 010	–12
Total		871 590	787 250	–10

Source: OECD; Instituto da Agua.

beyond Portugal's exclusive economic zone. Yet the government sees to it that these quotas cannot be commercially traded among holders. The introduction of transferable individual quotas would constitute an effective means of rationalising fisheries to enable the activity to take place at lower cost and has indeed led to such results where they have been introduced (OECD, 2001j). Such a policy would tend to lead to concentration of the fishing industry. Fishermen that held saleable quota rights would be compensated. Employees, however, would have to bear adjustment costs that could be met through normal labour-market and social policies.

Water management suffers from an inadequate institutional structure that prevents the scarcity of the resource from being reflected in its price. In particular, users are not required to pay the opportunity cost of that scarcity. The 1994 decree-law imposing taxes on water use has never come into force. In any case, it provided an exemption for irrigation until 2009. That exemption would have led to even greater inefficiency, since irrigation consumes more than three-quarters of water withdrawals. The extraction and use of water is already subject to the grant as permits by the government. Either these permits should become the property

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of the current users and be transferable or they should be auctioned. In either case such a procedure should ensure that water is used in the area where it has the highest value.

Moreover, while households and industries pay a portion of the cost of treating and distributing water, this is not true in the case of agriculture. Beneficiary farmers will not be charged for any portion of the government's capital expenditures under the current programme, which runs to 2006, and they cover only a small portion of maintenance costs. In total, farmers are billed for € 8.5 million a year, while annual expenditure on supplying them with water amounts to € 310 million (Instituto da Agua, 2001) or nearly three times as much as the operating costs of sanitation systems. A clear policy of charging for irrigation water needs to be introduced, by establishing marketable rights and not underpricing government projects. As well, the common agricultural policy exacerbates these incentives for the overuse of water in two ways, first through the specific subsidies that are reserved for irrigated farming, and secondly through general farm price supports. These community subsidies have also favoured the erosion of farmlands in the southern part of the country, by making it profitable to apply intensive cultivation methods to many erosion-prone lands that were formerly devoted to livestock grazing, an activity that is much less damaging to the soil.

Conclusions

In order to move towards sustainable use of natural resources, there will have to be an end to situations where open access and subsidisation leads to depletion of natural resources. The use of water merits particular attention. Households are charged for their use of water but not at the levels that adequately reflect costs. Farmers continue to benefit from irrigation projects that have been subsidised nationally and supranationally. The trading of water extraction permits should be allowed in order to ensure that scarce water is used in its most productive use, though, from an economic point of view market prices in agriculture will not necessarily give such signals due to trade restrictions. In the area of fishing, access to stocks has been limited by the EU Common Fisheries Policy. However, the existing structure of the industry has been protected by not allowing market-based transfer of fishing rights. A more efficient industry, best able to provide a sustainable base for regional economies, would emerge from allowing a market in these quotas and through the high level of subsidies to the industry. Such a change in direction would need to be backed by policies designed to ensure the integration of people affected by restructuring into the overall labour force.

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Notes

1. The deceleration of private consumption in 2001 was amplified by the change of the automobile tax. The automobile tax on off-the-road vehicles was increased in early 2001 to equalise the tax applicable on other light passenger cars. Sales of this type of vehicle thus increased by approximately 30 per cent in 2000 – partly reflecting anticipated purchases – but then fell by nearly 80 per cent in 2001. The reduction of purchases of passenger cars accounted for around half of the slowdown in private consumption in 2001.
2. In 2001, the deceleration of real disposable income was largely due the unexpected rise of inflation, while in 2002 it reflected mainly the lower growth of effective wages.
3. The consumer's confidence indicator calculated by the European Commission showed an unusual large decline in mid-2002, perhaps related to the restrictive fiscal measures announced, and reached in October its lowest level since the series exist (June 1986).
4. In particular, with the ageing of the population, the share of the young cohort, with a traditionally low participation rate, in the total working age population diminishes. According to some estimates, the change in the age structure of the population should continue to have positive effects in the participation rate until 2006. See Banco de Portugal, "Implications of developments in the age structure of the Portuguese population for the participation and unemployment rates", *Annual Report* 2001, pages 131-134.
5. January-October 2002 average.
6. October 2002, year-on-year rate of change.
7. The standard VAT rate was raised from 17 to 19 per cent from June 2002 onwards. The mechanical impact on the HICP was distributed between June and August, due to the fact that some of the prices are collected on a quarterly basis. At the end of this 3-month period, estimates point to around ½ percentage point direct effect on year-on-year inflation. Some impact on HICP inflation also came from the increase of the *ad valorem* component of the tobacco tax (from 23 to 32 per cent in 2002).
8. October 2002 year-on-year rate of change of the overall index excluding energy, food, alcohol and tobacco.
9. Inflation differentials may develop during periods of transition, when a less-advanced economy, like Portugal, is catching up with more advanced economies. The Balassa-Samuelson model suggests that countries which are experiencing higher productivity growth in the tradables sector than in the non-tradables sector will have higher inflation rates for non-traded goods and services. This is based on the assumption that wages set in the tradables sector will be followed by wages in the non-tradables sector, raising the latter above productivity growth. This will result in an increase in price inflation in the sheltered sector relative to the exposed sector. With overall inflation

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being the average of inflation in the two sectors, the average price level relative to other countries will increase in such a scenario. Most studies for Portugal suggest that this effect accounts for 0.5 to 1 percentage points of the inflation differential with the euro area. See Costa (2000), Swagel (2000) and Brito and Correia (2000).

10. The sharp deceleration of import volumes resulted from the slowdown of some domestic demand components with higher import content, such as the consumption of durable goods and investment in equipment.

11. In average terms, the real effective exchange rate index for Portugal based on unit labour costs in manufacturing, appreciated by 2.6 per cent in 2001 and 1.8 per cent in the first nine months of 2002 compared with the first nine months of 2001.

12. The last change of monetary policy by the European Central Bank (ECB) was on the 5th December of 2002, when it cut its key-refinancing rate by 50 basis points to 2.75 per cent more than one year after the previous cut.

13. In December 2001, credit to the non-financial private sector grew 12.6 per cent (23.1 per cent in December 2000). In September 2002, the year-on-year rate of change of this aggregate stood at 8.8 per cent.

14. According to estimates of the Banco de Portugal, the indebtedness of households stood at around 95 per cent of disposable income in 2001 (5 percentage points up from the previous year). The indebtedness of non-financial corporations reached 89 per cent of GDP at the end of 2001 (almost 10 percentage points up from 2000). As a comparison, according to ECB estimates, the average indebtedness of households was 81.3 per cent of disposable income in the euro area in 2000 and, the average indebtedness of non-financial corporations 70.8 per cent of GDP.

15. The cyclically adjusted net borrowing of general government remained almost stable from 1997 to 1999 at around 3.2 per cent of GDP and increased markedly to 4.3 per cent and 4.6 per cent of GDP in 2000 and 2001, respectively. In 2002, the fiscal policy was considerably tightened and the cyclically adjusted public deficit should decrease to around 2.9 per cent of GDP.

16. The price-setting regime of consumer fuel prices was changed in March 2002. Domestic fuel prices are now set in a monthly basis in order to reflect the evolution of oil price in international markets.

17. This measure, the object of which was to cope with the worsening government balances observed during the year, resulted in investment expenditure being cut by 10 per cent compared to the initial budget (see OECD, *Economic Survey of Portugal*, 2001a).

18. This was partly a catch-up, following the freeze on investment in 2000.

19. Data based on the old methodology.

20. Such regularisation is not systematic in the EU countries.

21. Concerning tax revenues, European regulations stipulate that receipts in respect of unpaid or bad tax and social insurance contributions should not be posted as revenue in the general government accounts. Portugal enjoyed a dispensation in this respect up until June 2002.

22. According to the authorities' estimates, the wage bill overrun was probably equivalent to 0.5 per cent of GDP, *i.e.* about half of the overshoot on current expenditure. The latter is thought to derive above all from the higher than forecast salaries paid to civil servants in the Education, Defence, Justice and Finance Ministries and the continued post regardings.

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23. By December 2001, the Stability Programme referred only to the bottom end of the growth band and forecasted a deficit of 1.8 per cent of GDP.

24. Investment spending has been frozen at the previous year's level and the new rules defined in the Framework Law are in force: each Ministry has at its disposal 85 per cent of the ceiling set for 2002, the remaining 15 per cent being released only for projects that are positively assessed. These rules, which are permanent in nature, relate only to investment that is not co-financed by the European Union.

25. The object of this Law is to deal with the control difficulties observed in the past, in 2001 in particular. The government is now able to set the ceilings on new borrowing by local authorities or by enterprises they own. If the ceilings are not complied with, transfers to these authorities will be reduced. Hitherto, local authorities did in fact have limits applying to their debt service, but there no were mechanisms for making them respect these limits.

26. The OECD projections were prepared in November for the OECD *Economic Outlook*, released on 21 November 2002 (OECD, 2002a).

27. The revision of the 2002 deficit carried out for the supplementary budget (on a public accounts cash basis) included the regularisation of debts incurred in 2001 and earlier which came amounted to the equivalent of 1.6 per cent of GDP.

28. The reduction in the structural deficit would amount to 1 percentage point of GDP, according to government estimates, which are based slightly on different growth assumptions for 2002 and 2003 and different elasticities of taxes to growth.

29. The Stability Programme lists a series of measures to be adopted in these areas by 2006.

30. In recent years there has regularly been a difference between the general government borrowing requirement and changes in the public debt with the latter often larger than the borrowing requirement. This difference, reflecting off-budget operations, privatisation, or the absorption of unexpected debts, varies from year to year between 1 and 2 GDP points and is estimated to have reached 3.4 points of GDP in 2002.

31. Equivalent to 14.9 per cent of the nominal value of the government debt existing on the same date. Most of these operations were interest rate swaps carried out with the objective of adjusting the refixing profile and the duration of the debt portfolio *versus* the benchmark portfolio.

32. Reduction in tax on non-residents' interest earnings; application of Community directives and OECD recommendations on tax competition; introduction of mechanisms to prevent operations designed to minimise tax on interest income. On this last point, the object is to restrict residents' sales of securities to non-residents not liable to tax prior to payment of the coupons or dividends, the former then being bought back at a price below the sale price.

33. The ratio of public outlays to GDP is the most commonly used indicator for expressing public sector size in an international and historical context. Such comparisons should, however, be made with care. If tax expenditures for social and economic purposes and taxes on social transfers were taken into consideration for all countries, the international picture would be quite different (Adema, 2001). Unlike many OECD countries, Portugal does not tax social transfers, and provides social and economic assistance *via* tax expenditure. When taking into account tax expenditure, Portugal total public spending is even higher.

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34. The pension benefits of workers joining before 1993 are based on the last salary, as well as on permanent remuneration granted in the last two years of service. Public employees hired after September 1993 are now subject to the same provisions as those applicable to the private sector (see Chapter IV). However, with around two thirds of public employees hired before 1993, pension expenditure is set to rise rapidly relative to GDP in the medium-run.

35. According to the European Commission, EU Structural Funds contributed about ½ per cent per year to growth between 1989 and 1999. The cumulative effect added about 10 per cent to GDP in Portugal as well as Greece and Ireland. Estimating an annual series covering investment in transportation infrastructure on GDP from 1976 to 1998, Pereira and Andraz (2002) find a similar result for Portugal.

36. According to the Ministry of Planning, the CSF III, funds are expected to raise the GDP level by 2.1 per cent and to create 81 000 jobs (1.6 per cent of the labour force), over the period to 2006.

37. For example, the proportion of electrified railway lines is still well below the EU average of 48 per cent.

38. Budget estimates of payroll costs are generally prepared a few months before wage negotiations with public sector unions are concluded, explaining the existence of discrepancies between budget and actual figures. Also, a special (contingent) budgetary endowment is available to the government in order to meet "unforeseen" spending needs. This endowment has in the recent past been largely used to finance the increase in the payroll.

39. The clauses can be invoked when unexpected circumstances threaten the achievement of the budget deficit target. The contingent measures are specified *ex ante* in the budget and are the maximum amount that can be frozen by the Executive without parliamentary approval. There are no specific trigger mechanisms governing the imposition of a budget freeze.

40. Prior to 2000, budget freezes were used as a response to the overshooting of spending by some Ministries (especially health). They often affected less than the maximum authorised amount and were then partly (or mostly) reversed closer to the year-end, as revenue windfalls often offset overspending. In this way a tendency developed for the budget deficit to turn out better than targeted during most of the 1990s, in spite of spending overruns. The freezing foreseen in the 2003 budget is meant to provide a true "margin of safety", in as much as the evolution of revenues is uncertain. It is the Government's intention to unfreeze some of the amount if the macroeconomic environment turns out to be better than expected.

41. Although, the migration to the ESA95 system is in progress, the annual budget and general accounts of the State, autonomous funds and services, as well as the social security fund, are still recorded entirely on a cash basis, implying delays in recognising on-going financial commitments.

42. In the context of the Stability and Growth Pact, national governments make budgetary commitments to the EU on behalf of the general government, so that compliance concerns all levels of public administration. Some Euroarea Member States have put in place a "Domestic Stability Pact" in order to promote fiscal discipline at subnational levels (*inter alia* Germany, Italy and Spain). In 1999 the Italian financial law, for example, ushered in the so-called *Internal Stability Pact*, setting some rules on the deficit and the debt of sub-central governments (see OECD, 2002b).

43. Portugal and Centeno (2001) using the European Union Household Panel data.

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44. Although the gap is substantially reduced when differences in education levels between the public and private sectors are taken into account, it remains much larger in Portugal than in most other EU countries

45. Local government staff is insufficiently qualified with respect to increasing technical needs including experience with contracting out of tasks to the private sector. The training quality of Staff working in local government is also poor.

46. Public sector employees made redundant as a consequence of the on-going restructuring of the public administration will be placed in an employment "pool" or "bourse", in view of their reassignment. The employment "pool", which is under the responsibility of the Ministry of Finance, will centralise information on civil servants available and on job opportunities. Employees in the pool will see their salary reduced by one sixth after three months. This reduction is expected to give them an incentive to search for a new job.

47. The system does however allow for incentives to individual or team productivity of a non-pecuniary nature (*e.g.* sabbatical periods, scholarships, promotions *in situ*).

48. See the reform proposal number 16 of the aforementioned report of the ECORDEP for further details.

49. The Portuguese private sector also uses fewer external support services (20 per cent of value added) than is the case in other OECD countries (first column in Figure 21).

50. The often-cited and controversial study by Arthur Andersen (2000) concludes that in the UK, for example, public-private partnerships projects can reduce costs by 17 per cent.

51. Using public-private partnerships projects as an opportunity for shifting public investment and borrowing off-balance sheet ignores the fact that possible macroeconomic crowding-out of market activities in principle is the same for public investment and private investment *via* public-private partnerships (Van den Noord, 2002), the more so since in both cases taxes need to be levied in order to meet future capital and operating costs.

52. However these limits do not apply to medium and long-term loans contracted for carrying out projects co-financed by European structural funds, within the scope of the Community Support Framework, loans used for the repayment of other loans, loans to meet extraordinary expenditure resulting from public calamity situations, and loans for the acquisition, construction or repair of real estate for social housing purposes.

53. Based on the recent OECD Programme for International Student Assessment (PISA) the average reading, science and mathematics performance of Portuguese 15-year olds, who are approaching the end of compulsory schooling, is significantly inferior to their counterparts abroad.

54. According to recent estimates by the Ministry of Education, over the period 1990-2000, each year 13 per cent of pupils leave before completion of primary schoolings, and more than 40 per cent of those who graduate from upper secondary education have taken at least one to two additional years before successfully graduating.

55. In 2000 the Ministry of Education planned to slightly reduce the number of public schools for primary and secondary education from 14 137 in 2000-2001 to 13 671 in 2002-2003. It also planned to merge a number of schools under one administration. This is expected to improve school management, while not necessarily to reduce the number of schools. In academic year 1999-2000 30 per cent of individual schools were assembled into 400 groups including on average 10 schools each. As restructuring continuos, the number of groups is expected to rise to 1 382 in 2005-2006. *www.dapp.min-edu.pt/est_plan/pdf/mudar.pdf*

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56. According to the OECD (2001b) PISA report, in Portugal the variation of in student performance on the PISA reading literacy scale between schools is high and above the OECD average.

57. Using the micro-data of the Portuguese Labour Force Survey of 126 233 individuals between the first quarter of 1998 and the fourth quarter of 2000, Portugal and Centeno (2001) find that a college graduate woman (man) earns 92 per cent (88 per cent) more of what a woman (man) with full secondary education earns. In most European countries the wage premium associated with college education varies from a maximum of 60 per cent for women in the UK to a minimum of 40 per cent for women in Italy (Blondal *et al.*, 2002).

58. Pereira and Martins (1999) estimate the economic returns to education in Portugal in the 1982-1995 period. They find that the return was increasingly above 9 per cent for both men and women over the entire period.

59. Blondal *et al.* (2002).

60. Health is the only sector where hiring is still permitted after the 2002 budget freeze, in services that are understaffed. However, the new hiring will be based on regular civil servant contracts because the hiring freeze on fixed term and individual contracts applies also to the health sector.

61. See St. Aubyn (2002) for a survey.

62. A voluntary experimental remuneration model for general practitioners (GPs) involving 500 doctors at the national level has been recently implemented. This project requires participating doctors to organise themselves in groups, so as to enhance peer pressure within the new remuneration model. As far as hospitals are concerned, the cardiology division of the hospital of Coimbra has recently become a pilot case, where a share of doctors' pay, as well as managers' pay, has been related to workload and performance, while a new-performance related pay system for hospital doctors and other health care professionals is under study.

63. Private fees are in principle regulated by the government, but in practice the range of minimum and maximum reference prices is set by the Medical Association (*Ordem dos Médicos*) at very high levels by international comparison.

64. Central government transfers (*i.e.* the tax system) finance 90 per cent of the National Health System (NHS) budget. The remaining 10 per cent is financed by the NHS own receipts, mostly generated by hospitals charging fees for services. The annual general budget establishes a cash limit for total National Health System (NHS) spending. However, this initial allocation is perceived to a large extent as indicative, and the need for supplementary budgets has been the norm in recent years.

65. Generics are not widely used, accounting for 1 and 2 per cent of the market. They are subject to a government-established price ceiling. This is equivalent to 80 per cent of the lowest-priced similar product. Only products with market share equal or above 10 per cent are considered for comparison purposes.

66. The concept "per adult equivalent" is based on household income, not on the income of each member of the household taken separately. For the methodology see Marlier and Cohen-Solal (2000). Marlier and Cohen-Solal (2000) analyse social benefits in the European Union. Their analysis is based on the 1996 European Community Household Panel (ECHP).

67. Tax expenditures, which are not counted in income transfer expenditures, are estimated to represent a high cost in terms of revenue foregone. In 1998, the last year for which data are available, it was estimated that only 60 per cent of the gross income

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declared by households was taxable and a large share of tax expenditure went to the highest income categories (see Bronchi and Gomes Santos, 2001).

68. The Netherlands, for example, allow to carry 1 per cent of a year's budget over the following financial year (Blondal and Kronmann Kristensen, 2002).

69. In Spain, for example, local authorities may lose a share of the EU structural funds if they do not meet the spending ceilings set by law.

70. Crude estimates by the Ministry of Finance based on 1999 data, show that income distribution could be slightly improved by removing tax credits for health and education spending (Reis, 2003).

71. For guidelines on contracting out and public-private partnerships, see OECD *Economic Survey of Mexico* (2000a) and OECD *Economic Survey of the United Kingdom* (2001c).

72. Per capita income increased by close to 3 per cent in 1980-2000, about 1 percentage point higher than the EU average. Comparative indicators are from OECD (forthcoming, 2002e).

73. Specifically, the per capita growth rates that were "explained" by the regression analysis were the annual average growth rates between the average level of output in the 1974-77 period and that in the 1994-97 period. Note also that Portuguese per capita GDP rose faster than that of labour productivity, as the employment rate increased over the period, with the increasing rate of participation of women offsetting rising periods spent in compulsory and non-compulsory education, and falling participation rates of older males.

74. The recent productivity gains in many OECD countries were associated with stable or falling employment, often related with keeping low skill/low paid workers out of the job market. In Portugal, they were associated with rising employment, including for the low skill/low paid workers. The Portugal's pattern – also observed in the Netherlands – could partly explain the slow productivity gains observed in the recent upturn. See Scarpetta *et al.* (2000).

75. See Bassanini, Scarpetta and Hemmings (2001).

76. Ministry of Finance estimates, based on the revised national accounts, put potential growth at 3 per cent in the 1978-2001 period (*Cf.* Pina, 2002). Banco de Portugal estimates that potential GDP growth was close to 3 per cent in the 1973-98 period (Banco de Portugal, 1998).

77. According to Scarpetta and Tressel (2002) estimates, Portugal's technology gap would be reduced by as much as one quarter and productivity levels increased by more than 10 per cent. In Portugal, the strong increase in the levels of total factor productivity reflects comparatively strict regulations in product markets and relatively low productivity levels *vis-à-vis* the leaders. See also OECD 2002g.

78. In 2001, more than two-thirds of the population aged 25 to 64 had attained only 6 years of schooling (against an OECD average of 15 per cent) and only 9 per cent had achieved tertiary education, the lowest proportion in the OECD.

79. See the International Adult Literacy Survey (IALS), OECD (2000b). Tests refer to prose, document and quantitative literacy. Other countries ranking as low as Portugal's are Chile, Poland and Slovenia. At the lowest level (level 1) of literacy the individual tested may, for example, be unable to determine the correct amount of medicine to give a child from information printed on the package.

80. The vast majority of children are educated at public schools, although the private sector maintains an important presence in tertiary education. For a review of the evolution of the Portuguese education system until 1995, see Pereira and Martins (1999).

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81. See Clements (1999) and St. Aubyn (2002). These aspects are discussed in more detail below.

82. See European Commission – *Structural Indicators* 2002 (Share of the population aged 18-24 with only lower secondary education and not in education or training).

83. See OECD (2001e), which covers 32 countries, including 4 non-member countries (Brazil, Latvia, Liechtenstein and Russia). These results are consistent with those of the Third International Mathematics and Science Study (TIMSS), conducted in 1994-95. The TIMSS included mathematics and science tests, but not literacy ones. Portuguese students were also among the worst performers in every dimension and grade considered.

84. See Pereira and Martins (2002) for detailed evidence on the economic returns to education in Portugal in the 1982-1998 period.

85. See survey by Pereira and Lima (1999).

86. In the scope of Broad Economic Policy Guidelines (BEPG) for 2002, adopted by the EU Council, the main priorities for Portugal concerning the labour market were identified as: improving education and training systems in the framework of a better articulated lifelong learning strategy; promoting wage developments that are consistent with the maintenance of competitiveness and price stability; and modernising labour market institutions. See a brief summary of the 2002 National Action Plan for Employment in Annex III.

87. See Inspecção-Geral da Educação (2002).

88. Following the curricular revisions of the 1st and 2nd cycles already in place, the 3rd cycle revision is starting in the school year 2002/3. Among the main measures: the same disciplines are taught in the 3 years; a second foreign language and a new discipline of "Technological Education" become compulsory; classes are organised in 90 minutes periods; a new compulsory discipline of "Introduction to ICT", and national final exams of Mathematics and Portuguese language will be introduced in the 9th grade in 2004/5.

89. Measures listed here are those which would be the most promising to enhance the qualification of human resources. Specific measures aimed at improving the efficiency of education expenditure are listed in Chapter III. See also Clements (1999), Carneiro (2000) and St. Aubyn (2002).

90. The supply of law and teaching graduates by Portuguese universities continues to rise, despite the fact that most students face increasing difficulties to get a job afterwards. By contrast, in all health-related courses, the number of places is still not sufficient to cover increasing demand for higher skills in the Portuguese health sector.

91. The Ministry of Higher Education and Science was created by the new government, merging responsibilities for science and technology (previously in the now-extinct Ministry for Science and Technology) and tertiary education (formerly in the Ministry of Education). A new law for the quality and development of tertiary education was approved in July 2002 ("*Regime Jurídico do Desenvolvimento e Qualidade do Ensino Superior*"). A proposal for changes in the financing and autonomy status of universities is under preparation.

92. In 1999, 75 per cent of students were enrolled in general upper secondary programmes, compared with less than half on average in the OECD.

93. This legislation comes within the context of the tripartite agreement signed by the government and social partners in February 2001. The new legislation establishes a

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minimum period of 40 per cent of working time to be used for training. Public financial support is provided to companies to compensate for the training costs of minors.

94. The Operational Programme for Education (PRODEP III – *Programa de Desenvolvimento Educativo para Portugal*) includes two sets of measures directed to the use of information technologies in schools. The first measure establishes goals regarding the equipment of schools (one computer per 20 students by 2003 and one computer per 10 students by 2006). The second set of measures refers to promoting the production and acquisition of multimedia contents. Also through the PRODEP III, teachers are receiving ICT training.

95. The authorities expect that around 2 million citizens will be certified with this basic skills diploma by 2006. The accreditation of the institutions that will provide the Basic Skills Diploma, and further requisites related to the diploma have already been regulated.

96. See European Commission – *Structural Indicators* 2002 (Percentage of population aged 25-64 participating in education or training in the four weeks preceding the EC labour force survey).

97. Only 20 per cent of all enterprises provided continuing vocational training in 1999, the lowest percentage in the EU. *Cf.* "Continuing vocational training in enterprises in the European Union and Norway", Eurostat, *Statistics in Focus*, Theme 3 – 03/2002.

98. For more information on adult learning in Portugal, see (OECD, 2003) or visit the OECD adult learning website (*www.oecd.org/els/education/adultlearning*).

99. The strategic objectives regarding adult learning of this tripartite agreement include the promotion of quality training, the consolidation of the national certification system, the creation of a system of adult education with key competencies and different modes of education, and the development of enterprise training. In this context, it was determined that at least 10 per cent of the workers of each enterprise must participate in continuous training actions from 2002 onwards. All workers must have a minimum of 20 hours of certified training by 2003 and of 35 hours by 2006.

100. The average level of the unemployment rate in Portugal over the 1990s was much closer to that of the flexible US labour market than of other EU countries. See Blanchard and Portugal (2001). Employment rates for Portugal and other EU countries are calculated in proportion of working-age population, defined as 15 to 64 years.

101. Following the economic downturn, the rate of youth unemployment increased to 12 per cent in the third quarter of 2002, while the share of long-term unemployment stood at 38 per cent.

102. For a detailed analysis of long-term unemployment in OECD countries, see OECD (2002f), Chapter IV.

103. Adaptability captures the ability of workers and their employers to adjust to a shock by moving to another job. Flexibility refers to the sensitivity wages and prices in adjusting to shocks. It encompasses also the facility to adjust working hours and shifts. For a discussion of these notions applied to the Portuguese labour market see Traça (2002).

104. Dismissal restrictions include various notification requirements and reinstallation with full back pay if the dismissals are considered unjustified. Mandatory severance pay is one-month pay per year of tenure, subject to a minimum of three months. More importantly, firms have to follow a sequence of time-consuming and potentially production-disruptive administrative procedures (advance notice, report on the financial and technical reasons, negotiations with the representatives of the employees…).

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105. Blanchard and Portugal (1998), looking at Portuguese and US labour markets, show how similar unemployment rates can hide extremely different labour markets. The unemployment in Portugal reflects much lower flows and much higher duration than in the US. They argue that these differences may come from high employment protection in Portugal.

106. Bover *et al.* (2000) put in evidence that different collective wage bargaining systems in Portugal and Spain have an important role in explaining different labour market outcomes.

107. In Portugal, the trade unions structure is characterised by a large number of unions acting in an uncoordinated fashion. The representation of trade unions and their ability to negotiate depends on union membership. The ambiguity in the representation of each trade union undermines the potential for co-ordination among unions, often creating a situation of competition that ensures wage flexibility. Employers' associations are able to reach agreements with the least demanding unions, while waiting for the government to extent them nation-wide. See Bover *et al.* (1998).

108. Gaspar and Luz (1997) concluded that Portuguese real wage elasticity *vis-à-vis* the unemployment rate was high, according to international standards.

109. Varejão (2002) found evidence that firms tend to train temporary workers in order to be able to select the most competent for permanent jobs. See also Portugal (1999).

110. See European industrial relations observatory on-line (2002) "*Economically dependent workers*", *employment law and industrial relations* in *www.eiro.eurofound.eu.int*/2002/05/*study*/TN0205101S.*htm*l

111. See Centeno (2000). The contracting out or outsourcing of services to "independent" workers is known as "green receipts", because the workers who are being employed fill a green-coloured receipt and submit it to the enterprises to which they provide services.

112. Other features of the law include: *i*) the termination by the employer of a contract in force over 12 months, will imply that that job position cannot be filled by another employee for the next 6 months; and *ii*) employees with a fixed-term contract have the right to receive 6 months' basic pay if they are replaced by a new recruit who is given a standard contract to perform the same functions.

113. Either consecutive or interrupted (*i.e.* in the cases where the employee is asked to take breaks between contracts).

114. The measure follows on the Job Offer Stimulus Programme introduced in early 2001 (see below).

115. See OECD (2001d).

116. The Broad Economic Policy Guidelines for 2001, adopted by the EU Council, identified reducing the rigidities of employment protection legislation as one of the main priorities for Portugal concerning the labour market.

117. The Labour Code proposal extends the circumstances considered as just cause for dismissal. It adds absence through fraudulent sick leave and more than twelve unjustified late arrivals of over 30 minutes during one year. In addition, it reduces the number of unjustified absences considered as giving just cause for dismissal from six to four.

118. The regional network includes job centres, vocational training centres, business creation support centres and vocational rehabilitation centres. Charana and Rodrigues (2001) provide a detailed description of the activities of the *Instituto de Emprego e Formação Profissional* (IEFP).

119. This programme (*Programa de Estímulo à Oferta de Emprego*) aims at co-ordinating measures for stimulating job offers and to facilitate labour market entry for disadvantaged

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groups by providing financial support to companies. It provides non-refundable subsidies for hiring young people looking for their first job, long-term unemployed persons aged over 45, beneficiaries of the minimum guaranteed income and people with disabilities. The IEFP is in charge of supervising the implementation of the projects. It will also promote their co-financing by the European Social Fund and European Regional Development Fund.

120. See OECD (1998a), and OECD (2001f).

121. Addison and Portugal (2002) conclude that despite its frequency as a search vehicle, the state employment agency has a low hit rate and generally leads to lower-paying, shorter-lasting jobs.

122. For examples of best practices in OECD countries, see Martin (2000).

123. The payment of unemployment benefits is subject to prior registration with the IEFP and continued benefit receipt requires the recipient to confirm his subsequent unemployment status with the agency. There is strong evidence that the current unemployment benefit system increases unemployment duration, especially for old workers (Addison and Portugal, 2003). Access to benefits can be denied following a refusal to undertake suitable work or training.

124. Beneficiaries of the *Rendimento Mínimo Guarantido* (RMG) must follow a programme of social integration designed to provide access to vocational training, or to find a "normal" job or be employed in some form of community work. The different integration programmes are drawn up at the local level, to ensure greater proximity to the beneficiaries. Rodrigues (2001) did a simulation study on the impact of the RMG.

125. There is also a voluntary social insurance scheme – for persons capable of work who are not covered compulsorily by any social protection scheme, as well as for national citizens who pursue their occupational activity abroad and who are not covered by an international agreement concluded by Portugal – and a non-contributory scheme that provides a minimum pension and several other allowances.

126. Ireland is the major exception, with an unusually "young" population, though one that is projected to age gradually in future decades.

127. OECD, *Economic Outlook* 69, June 2001; EPL/ECFIN/655/01-EN final 2001.

128. The formula to calculating public pensions for private-sector workers was modified in 2001 to take into account the full contribution history.

129. These projections are taken from estimates submitted to the Ageing Working Group of the EU Economic Policy Committee, and are based on a comparable set of cross-country assumptions regarding productivity growth, demographic evolution and labour-market developments (See the table on page 22 of that report for the relevant data for Portugal). Portugal also submitted projections of age-related spending to a parallel exercise carried out in the OECD. However, the Portuguese assumptions concerning productivity growth in that exercise were significantly more optimistic, leading to implausibly low projections of the growth of spending on old-age pension relative to GDP.

130. This will also depend on whether prices of medical technology rise faster or move slowly than the general price level. In practice, there is a long-established trend for them to rise faster.

131. See, for example, "Maintaining prosperity in an ageing society" (OECD 1998b), "Reforms for an ageing society" (OECD 2000c) and "Ageing and Income" (OECD 2001g).

132. See Pereira and Rodrigues (2001).

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133. Bank employees, amounting to around 55 000, remain the most important exception to the general system. Bank employees, in general, contribute to a privately run, fully-funded scheme and the portability of pension rights outside the banking sector is not assured. The fact that bank employees have a separate and non-transferable pension system represents an additional impediment to the movement of labour between sectors.

134. The 1998 White Paper on Social Security found that the social security system falls short in guaranteeing the wellbeing that society would like to see, and that there is a trend towards the development of serious financial problems in the system within the next 15 years.

135. The most relevant measures included a new financing model, the creation of a new reserve fund as well as a revision of the pension formula. The new law also introduced measures to stimulate social dialogue. In particular, it created a Social Security Council (*Conselho da Segurança Social*), on which both trade unions and employers' organisations are represented.

136. At present, only the best ten out of the last fifteen years enter the pension formula.

137. The *Operational Programme for the Economy*, 2000-06, designed under the guidance of the EC, aims at providing financial and technical support for the development of an entrepreneurial culture by supporting investment projects in ICT sectors, facilitating access to finance and to foreign markets and facilitating technology transfers, with special attention given to small and medium-size enterprises (SMEs).

138. Among the priority areas for policy action defined for the 2000-2006 period: raising the qualification level of human resources, while promoting employment and social cohesion; facilitating the adjustment of the productive structure to the challenges of the information society; favouring a more sustainable development of the country; and promoting more balanced growth across regions.

139. The fact that entry of new firms makes a negative contribution to average productivity growth in several OECD countries may reflect the time horizon over which productivity growth is measured. If new entrants undergo a process of learning and selection, a five-year period as used in the OECD study is probably too short to show a high positive impact on aggregate productivity. For details on the methodology and evidence on firm dynamics, see Scarpetta *et al.* (2002) as well as Chapters III and V in *The sources of economic growth* (OECD, forthcoming).

140. In wholesale and retail trade, restaurants and hotels services, as in manufacturing, exits involve low productivity units. Entry of new firms also makes a small positive contribution.

141. Currently, the processes of creation, obtaining a licence to operate, developing a company and even terminating a firm are still rather complex and very time-consuming. The new system of registry, transformation and licensing of societies is expected to reduce the time frame from 153 to 68 days.

142. All administrative procedures of small investment projects will be concentrated in a single institution (IAPMEI). For larger investments (national and foreign), the recently created Portuguese Investment Agency (API) has one-stop-shop status. It is responsible for detecting big investment opportunities in Portugal, for full administrative assessment of such investments (including potential applications to financial incentives), for establishment and licensing procedures, and for the negotiation of special contractual regimes.

143. The big difference with the previous arrangement is that the possibility to capitalise the loan is now conditional, whereas before subsidies were granted *ex ante*, after the

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selection process. Also, the financial evaluation of projects to be selected will be made by banks. According to its new orientations, the *Operational Programme for the Economy* 2000-06 (POE), which channels the structural funds, should prioritise support based on venture capital, seed capital or development capital and rewards granted to high value added projects, through objective measures, computed throughout the project's life (and not *ex ante*, as today). Another priority is to simplify application procedures, fully reforming the systems management institutional framework, and to establish partnerships with private financial and venture capital institutions, pre-qualified and subject to contractual obligations and service supply levels.

144. See OECD (2001h) *Science, Technology and Industry Scoreboard – Towards a knowledge-based economy*.

145. Under the new tax incentive system firms can deduct 20 per cent of eligible R&D expenditures (net from direct state funding) from their taxable income and an additional 50 per cent (up to € 500 000) on incremental expenditures above the average of the previous two years. For more information on Portugal's policy for science and technology and recent efforts to spur innovation, see "Observatório das Ciências e das Tecnologias", *wwwoces.mces.pt/documentos/index.jsp*.

146. In June 2002, the Seville European Council approved the *e-Europe* 2005 action plan, to follow *the e-Europe* 2002. The new plan follows two types of actions: *i*) it aims at stimulating services, applications and content, covering both online public services and e-business; *ii*) it addresses the underlying broadband infrastructure and security concerns. According to this plan, by 2005, all EU countries should have modern online public services (e-government, e-learning and e-health services) and a dynamic e-business environment. Additionally the widespread availability of broadband access at competitive prices and a secure information infrastructure will have to be ensured.

147. Examples of targets of the "Internet Initiative" are: to reach a penetration rate of 50 per cent of the population by 2003 and to quadruplicate the number of home computers connected to the Internet. Regarding public administration, the following general targets were set: to have all official forms available on the internet in 2002; to enable widespread electronic submission of income tax declarations in 2003; and all public services to be online on the internet by 2005.

148. Around 100 public Internet access points (PIAPs) were approved in the summer of 2001. By 2003, there should be a PIAP in each local government board (around 4 500). At present there are around 800 PIAPs in town halls, as well as in public libraries, post offices and museums. The tax incentive amounts to the deduction of 25 per cent of the purchase value of the equipment from tax liability of households (under a certain ceiling). The number of households benefiting from the incentive rose from 90 000 in 1998 and 1999 to 116 420 in 2000.

149. Structural funds have been in place since the creation of the European Community to reduce the gaps between more and less advanced regions. The Cohesion Fund was introduced in 1993 to support specifically the least prosperous member countries (Greece, Ireland, Portugal and Spain) by funding investment projects for environment and transport infrastructure.

150. Details in this section and examples of best practices are based on Gonenc, Maher, Nicoletti (2001).

151. The need to strengthen the independence of the regulators is particularly acute in water and railway transportation, while in energy and telecommunications regulators are already formally independent.

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152. Typically public and private partnerships involve the retention of state ownership of some key assets while ensuring that these assets become more productive through the use of private sector inputs. Through these partnerships, the design, planning, financing, construction and management of infrastructure projects or services are privatised. See Mahboodi (2002).

153. Golden shares allowed by the EU give veto power to the governments on some decisions, such as those related to public security requirements, for instance in the supply of energy, as illustrated by the recent European court cases.

154. The only countries with a worse ranking than Portugal are Greece and Italy. See Nicoletti *et al.* (2001).

155. See EC report on the implementation of broad economic guidelines, February 2002.

156. Since April 2002 ERSE competencies have been extended to include natural gas and the electricity of the Autonomous Regions of Madeira and the Azores where the uniformization of national tariffs is a major objective.

157. *Rede Eléctrica Nacional* (REN) is the concessionaire operating the national transmission grid. Since November 2000, the Portuguese State became the major shareholder in REN, with 70 percent, while EDP owned the remaining 30 per cent. In December 2001, *Caixa Geral de Depósitos*, the public financial holding, acquired 20 per cent of REN's equity from the State.

158. Vertically-integrated firms, such as electricity companies, have undergone some form of separation. The various separation models that exist, even when they do not go as far as ownership separation, are still useful in that they can provide information that helps detect bad practices on the part of the regulated firm, and thereby can complement access regulation. For details, see OECD (2001i).

159. The Public Electricity System is characterised by capacity planning and purchasing power agreements between generators and the grid operator. The Independent Electricity System consists of the Non-Binding System (characterised by free contracts between generators and eligible consumers) and the generators in the Special Regime, which includes co-generators and generators from renewable sources.

160. Since the beginning of 2002 ERSE has extended the eligibility criteria to the Non-Binding System to all medium, high or very high voltage consumers. A time schedule for the accession of the remaining consumers to the Non-Binding System has already been set: as from January 2004 for special low voltage consumers and as from mid 2004 for household consumers.

161. The only important generators in the independent system are the co-generators and generators from renewable sources (including those that belong to the EDP Group) which sell their electricity to the grid operator at a premium. These generators have no incentives to sell to eligible consumers.

162. In Portugal, however, by 1998, there were no longer restrictions on FDI in the telecommunications sector, while in several other countries there were some. *Cf.* O. Boylaud and G. Nicoletti (2001).

163. ANACOM has been consistently developing competencies regarding the knowledge of demand patterns, as result of its statistical data collection through the analysis of the cost accounting system results of the incumbent operator, which has been audited annually since 1996, by independent entities. ANACOM is also developing an engineering based bottom-up model, which allows to estimate the costs of an efficient network operation.

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164. 100 Main Distribution Frames were offered for local loop unbundling (corresponding to 1 647 000 lines). 14 sites were ready for co-location (corresponding to 375 000 lines) and 4 operators were asking for co-location.

165. Ozone concentrations generally do not exceed the information threshold for more than a few hours, thanks to the brisk sea breezes that sweep away pollution towards the end of the day.

166. Fine sand carried by the winds from North Africa also contributes to particulate pollution, but generally involves larger particles that are less harmful when inhaled.

167. So finally bringing Portugal into line with the targets of the Convention on Long-Range Transport of Pollutants.

168. The trend to higher emissions from electricity generation should be contained in the course of the next decade, thanks to the growing use of natural gas. Current forecasts show that the share of natural gas in electricity production will rise from 5 per cent in 1998 to 42 per cent in 2010, while that for petroleum will drop from 59 per cent to 33 per cent (IEA, 2000). Assuming a 58 per cent increase in electricity output over that period, the volume of petroleum used in power stations will decline by around 10 per cent. These developments should translate into lower emissions, since in modern power stations the burning of natural gas releases only very low quantities of local pollutants.

169. In Europe, the United Kingdom and the Netherlands are introducing such markets.

170. Such a measure should be technically feasible, given progress in filter technology. Stiffer standards would be more effective than diesel taxation alone in reducing the particulate emissions of diesel vehicles offered for sale. The regulatory tool in this case is targeted directly at emissions, something that will not be possible through taxation until such time as European registration fees are harmonised as a function of pollution characteristics.

171. Although people may be encouraged to buy less polluting vehicles because the tax rate varies with engine displacement and because clean vehicles are exempt, these taxes tend to slow renewal of the fleet. Consequently, vehicle registration fees hold back the process of reducing emissions through new standards.

172. To prevent such a change from shifting the balance of revenues among different levels of government, the annual road tax surcharge could be shared between the central government and municipalities.

173. This conclusion must be accepted with caution, in the absence of satisfactory time series for evaluating water quality trends. Although an exhaustive monitoring network is now in place, this was not the case in the early 1990s, when monitoring was restricted to the most seriously affected watercourses.

174. Those pollutants are the following: 2,4-D (dichlorophenoxyacetic); anthracene; amine salt; simazine; tributyltin; trichlorophenol; ammonia; phosphorous compounds and nitrites.

175. Waste water volumes are in fact closely related to gross water consumption, making this an ideal basis for calculating sewage and water purification rates for households, given that the nature of pollution differs little from one household to the next.

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Glossary of acronyms

ABC	Activity-based costing
ALMP	Active labour market policies
ANA	Aeroportos e Navegação Aérea
ANACOM	National Authority of Communications
ANEFA	Agência Nacional de Educação e Formação de Adultos
API	Portuguese Investment Agency
BEPG	Broad Economic Policy Guidelines
BOOT	Build, own, operate and transfer
CA	Contribução Autárquica
CGA	Caixa Geral de Aposentações
CSF	Community Support Framework
DRG	Diagnostic related groups
EC	European Commission
ECB	European Central Bank
ECHP	European Community Household Panel
ECJ	European Court of Justice
ECORDEP	Estrutura de Coordenação da Reforma da Despesa Pública
EDP	Electricidade de Portugal
EES	European Employment Strategy
EIA	Environmental impact assessment
EPL	Employment protection legislation
ERSE	Entidade Reguladora dos Serviços Energéticos
EU	European Union
FDI	Foreign direct investment
FEDER	European Regional Development Fund
GALP	Petróleos e Gás de Portugal
GDP	Gross domestic product
GFCF	Gross fixed capital formation
HICP	Harmonised Index of Consumer Prices
IA	Imposto Automóvel
IALS	International Adult Literacy Survey
IAPMEI	Instituto de Apoio às Pequenas e Médias Empresas e ao Investimento
ICT	Information and Communication Technology
IEFP	Instituto de Emprego e Formação Profissional
INA	Instituto Nacional de Administração
IRC	Corporate income tax
IRS	Personal income tax
IT	Information Technology
MFP	Multi-factor productivity

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NAPE	National Action Plan for Employment
NHS	National Health System
NSR	Nossa Senhora do Rosário
PC	Personal Computer
PES	Public employment service
PGP	Productivity and Growth Programme
PIAPs	Public Internet access points
PISA	Programme for International Student Assessment
PPA	Purchasing power agreements
PPP	Purchasing power parity
PRODEP	Programa de Desenvolvimento Educativo para Portugal
PROINOV	Programa Integrado de Apoio à Inovação
REN	Rede Eléctrica Nacional
R&D	Research and development
RMG	Rendimento Mínimo Guarantido
SMEs	Small and medium enterprises
SGP	Stability and Growth Programme
SGPS	Sociedade Gestora de Participações Sociais
SCUT	Sem cobrança aos utilizadores
TAP	Transportes Aéreos Portugueses
TIMSS	Third International Mathematics and Science Study
UMTS	Universal mobile telecommunications system (third generation mobile telephone systems)
VAT	Value added tax

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Annex I

Local government funding[1]

A slow process of devolving spending competencies to local governments has recently started and municipalities are now deciding several policies and organisation matters (OECD 2001a). A new Law replaced the 1987 Law on Local Finances in 1998. This defines in detail all the revenues to which *autarquias* (municipalities) are entitled and sets an annual ceiling on their debt servicing.

Central and regional government transfers: Local authority finances largely depend on an annual transfer from the central and regional governments. In 2000, these accounted for about 40 per cent of local government total revenues. The annual transfer is distributed to *autarquias via* three different funds defined by law: the *Fundo de Base Municipal* (FBM), the *Fundo Geral Municipal* (FGM) and the *Fundo de Coesao Municipal* (FCM) that correspond to 13.6, 62 and 16.7 per cent of the transfers respectively. The remaining 7.7 per cent is distributed directly to the *Freguesias* (parishes). The FBM grants a minimum transfer to each local authority that is apportioned in equal amounts. The calculation of FGM shares is made in two steps. The first divides the total amount into three parts: the mainland and the two autonomous regions according to number of residents and hotel bed capacity, number of municipalities and geographical area. The second divides the amount amongst the component municipalities according to number of residents and hotel bed capacity. The FCM has been created to guarantee the equitable distribution of public resources and balancing social and economic differences. Finally, 40 per cent of the total transfer received by each local authority must be reserved as funds available for investment projects and recorded as capital receipts.

The 1998 Law has clarified the rules governing transfers from the central government to local authorities and made an important change to the transfer formula. Since 1999, the transfer system has been based on a backward-looking approach. The general grant to transfer each year is linked to 33 per cent of the simple average of revenues from three central government taxes: the personal income tax (IRS), the corporate income tax (IRC) and the value added tax (VAT) accrued the year before the budget law is designed. It follows that the central government transfer in year *n* is equivalent to:

$$Transfer_n = 0.33 \times \frac{IRS_{n-2} + IRC_{n-2} + IVA_{n-2}}{3}$$

According to the 1987 Law on Local finances, the total amount of the annual central government grant was transferred according to the anticipated trend in VAT according to the following formula: +

$$Transfer_n = Transfer_{n-1} \times \frac{IVA_n}{IVA_{n-1}}$$

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Under the new Law, central government transfers are higher than under the old system when the economy slows down, because they are based on past values of tax receipts. On the other hand, they are lower when the economy picks up again.

Local taxes: In 2000, own-taxes of the sub-central level of governments were the second most important source of local finance and accounted for about 35 per cent of the total. Local taxes include, *inter alia*: the *Derrama* a surcharge on the Corporate Income Tax; the *Sisa*, a tax on property transactions and the *Contribuçao Autarquica*, a tax on the value of real estates. Local authority autonomy over these revenues is confined to setting the tax rates within strict limits fixed by the Law for local finance.[2]

EU *funds*: Municipalities also receive grants from the EU through the Structural Funds and from the European Regional Development Fund (FEDER) to co-finance investment projects. Grants from the EU in 2000 amounted to about 6 per cent of total municipal revenues. Local authorities have more power to influence the amount of EU transfers than with the previous sources of revenues because the amount of transfers depends on the quality and eligibility of the individual investment projects submitted for co-financing purposes.

Borrowing: Finally, local authorities have access to capital markets and may borrow or issue bearer bonds to finance gaps between expenditures and revenues. Short-term loans may be used to meet cash constraints, whereas medium and long-term loans, including bonds, can only be used to finance investment spending or to face situations of structural financial imbalance or financial collapse. The Law for Local Finances also establishes limits on the amounts of short-term loans and on annual debt servicing, depending on the total amount of funds transferred by the State to the local authority and past investment spending. These limits do not include, however, medium and long-term loans contracted for carrying out projects co-financed by European Structural Funds, loans used for the repayment of other loans, loans to meet extraordinary expenses due to natural calamities, and loans for the acquisition, construction or repair of council housing. Moreover, the limit on annual debt servicing has become less stringent over the years as interest rates fall. As a result, a number of municipalities are currently experiencing large deficits.

The 2002 Budgetary Stability Law allows the central government to impose indebtedness ceilings in order to control local authority spending and deficit problems that may ensue. Central government transfers can be reduced if the indebtedness limits are not respected and or/if financial information is not provided in time.

Notes

1. This annex is based on the description of local government funding in Correia da Cunha and Silva (2002).

2. For an assessment of local taxes in Portugal see Bronchi and Gomes Santos (2001).

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Annex II

Private internal rates of return on education[1]

The private internal rate of return on education is an indicator that measures the effects higher educational attainment has on the earnings of workers relative to the cost of the initial investment on education. It can be interpreted as the average increase in earnings related to the attainment of extra years of schooling beyond compulsory education.

Calculation

The private internal rate of return on education (ROR) is defined as the discount rate "δ" that equalises the real costs of education at a given level during the period of study to the real gains from education thereafter. It is calculated from the following general formula:

$$\sum_{t=a}^{a+l}(1+\delta)^{-(t-a)} \times C^{e}(t) = \sum_{t=a+l+1}^{64}(1+\delta)^{-(t-a)} \times B^{e}(t)$$

where: t is the age, a the typical age at the start of the period of education in question, l the theoretical length of the education period, C^e the costs of the education and B^e the real benefits linked to an investment in the education. The benefits are assumed to last until the age of 64. The parameters a and l depend on the particular type of education beyond the compulsory level.

According to the factors that affect the costs and the benefits, different indicators can be derived. Thus considering only the gross differential earnings and the length of education one can define a narrow rate. Takinginto account factors like the tax system, the unemployment risk and private spending on education a more comprehensive rate can be evaluated.

The costs incurred by individuals are their foregone earnings net of tax while studying adjusted for the probability of being in employment, plus any education fees minus the resources made available in the form of grants and loans. In the particular case of tertiary (university) education the formula is:

$$C^{u}(t) = \left[1-\tau\left(E^{s}(t)\right)\right] \times \left[1-ur^{s}(t)\right] \times E^{s}(t) \times (1+g)^{(t-a)} + F^{u} \times (1+g)^{(t-a)} - S(t)$$

where τ is the average tax rate for base-year earnings $E^s(t)$ of a single person with upper-secondary education at age t, ur^s is the unemployment rate of people with upper-secondary education at age t, g is the growth rate of labour productivity in the economy as a whole, F^u is the annual private cost of tertiary education in the base-year, and $S(t)$ is student grants and loans at age t. The scaling factor $(1 + g)^{(t-a)}$ allows to obtain future earnings by the scaling up of base-year earnings with the productivity growth rate for the economy as a whole. For the sake of simplicity, it is assumed that average tax rates remain at the level t throughout. The costs of upper-secondary education are similarly defined.

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The benefits are the gains in post-tax earnings adjusted for higher employment probability minus the repayment, if any, of public support during the period of study and are expressed in the case of tertiary education as:

$$B^u(t) = [1-\tau(E^u(t))] \times [1-ur^u(t)] \times E^u(t) \times (1+g)^{(t-a)} - [1-\tau(E^s(t))] \times [1-ur^s(t)] \times E^s(t) \times (1+g)^{(t-a)} - R(t)$$

where $E^u(t)$ is the base-year earnings of a single person with tertiary education and $R(t)$ the repayment of loans. The benefits of upper-secondary education are similarly defined.

The calculations assume that the student is in full-time education and has no work activity, and hence no earnings while studying. Moreover, as the probability of course dropout has not been taken into account, the internal rates are conditional on successful completion of the relevant education programmes.

The case of Portugal

In the case of Portugal, two estimations of the ROR have been calculated: the ROR based on pre-tax earnings and the length of studies (the narrow rate) and a "comprehensive rate" estimated from a simplified form of the equation, taking into account only the effects of the tax system and the unemployment risk.[2] It has been assumed that the standard length of upper-secondary is 3 years and that of tertiary education 4 years, the typical ages at the start of upper-secondary education and tertiary education have been set respectively at 15 and 18. Both rates suggest a large return, at upper-secondary as well as at tertiary levels (Table A.1). These results are in line with those found in other studies.[3]

Table A.1. **Private rates of return, 1999**

Per cent

Education attainment	Narrow rate	Comprehensive rate
Upper-secondary	11.9	11.3
Tertiary	21.8	23.9

Source: Secretariat's calculations.

According to the narrow indicator, ROR for the tertiary education is much larger than the one for upper-secondary education. Different reasons may have influenced these results. *First*, the gross wage differentials are narrower between people with secondary education and people with upper-secondary education than between people with upper-education and people with tertiary education (Figure A.1). *Second*, the relatively short education programmes have also to be considered as prime determinants.

The results are not much altered for the ROR based on net earnings; this can be explained by the relatively low progressivity of the tax system.[4] Hence in the case of upper-secondary ROR the taxes reduce the internal rate by one-third of percentage point, and for the ROR in tertiary education by 0.5 percentage point.

Unemployment risk differentials are expected to increase the internal rates of return compared with the narrow rates, reflecting a lower risk for more educated people. In the Portuguese case, however, the results for the upper-secondary education have been affected negatively by an adverse labour market in 1999, the year over which the RORs have been

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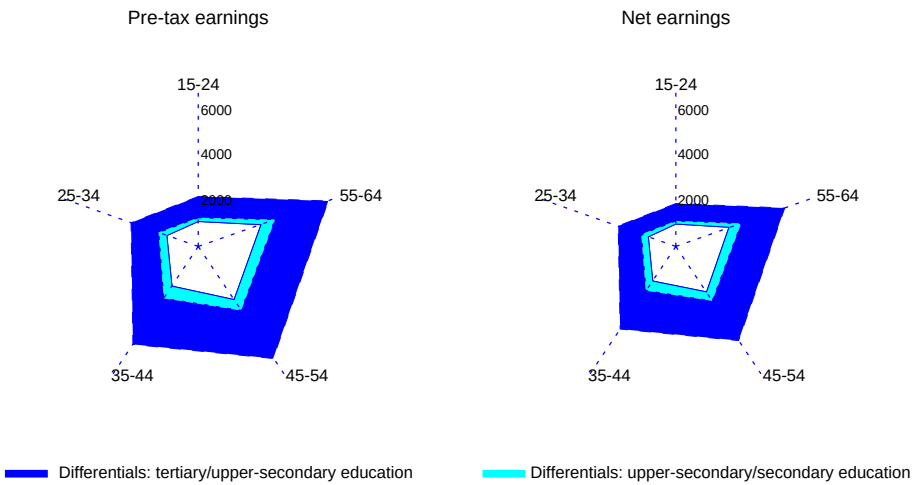
Figure A.1. **Wages differential by level of education and by age**

In escudos, men, 1999



Source: 1999, Quadros de Pessoal.

calculated, since in 1999 the global unemployment rate was 4.1 for people with upper-secondary education whereas it was 3.9 per cent for people with lower education. Conversely, for tertiary education where the unemployment rate is low at 3.1 per cent, the unemployment differential adds 2.1 percentage points to the narrow rate.

These estimations have several important limitations. They assume stability in the wage premia through the life cycle and are based on average earnings and costs. They are strongly influenced by the length of studies and in practice, there can be considerable variation in rates of return for different fields of study or particular social groups. Moreover they are likely to be biased upwards as unemployment, retirement and early retirement benefits are not taken into account. Finally the non-monetary benefits of investment in education (for instance better health) are not taken into consideration.

Data sources

The calculations have focused on ROR for men. Gross medium earnings data have been obtained from the 1999 "Quadros de Pessoal", a large data set that includes information about employees and their firms. They refer to full-time workers aged 15 to 64 and are available for ten-year age intervals and by type of education attainment. The categories of phases of education follow the ISCED convention.[5] Upper-secondary programmes are the final stage of secondary education. Entry to tertiary education requires completion of upper-secondary education.

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The income tax rates and the social contributions paid by the employers come from *Taxing Wages* 2000-2001 and refer to 1999.

The unemployment rates by education and age come from *Education at a Glance* and refer to 1999.

The labour productivity growth rate used in the calculation is set equal to the average rate in the medium-term reference scenario for 2003-07 of the OECD Economics Department.

Notes

1. For a detailed analysis of the theory and of the methodology see Blondal *et al.* (2002), "Investment in human capital through post-compulsory education and training : selected efficiency and equity aspects", OECD *Economics Department Working Paper* No. 333 available through OECD's Internet Web site *www.oecd.org/eco – http://appli1.oecd.org/olis/2002doc.nsf/linkto/eco-wkp*(2002)19/$FILE/JT00129531.PDF
2. Data on education fees, loans and grants were not strictly comparable with those of countries shown in Box 9 table.
3. Pereira and Martins (2002), "Education and Earnings in Portugal", Conference of the Bank of Portugal, 24-25 May 2002. Barceinas-Paredes *et al.* (2001), "Unemployment and returns to Education in Europe", PURE project, Final Report.
4. See OECD *Economics Survey of Portugal*, 2001.
5. For a full description see *Education at a Glance* (2002).

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Annex III

The 2002 National Action Plan for employment

The National Action Plan for Employment (NAPE) adopted in 1998, formalises the commitment assumed by Portugal, in the context of the 1997 Luxembourg Summit on Employment, to implement the employment guidelines agreed at that Summit. Such guidelines resulted in a co-ordinated European Employment Strategy (EES) based on four pillars: improving employability, developing entrepreneurship, encouraging adaptability of workers and businesses and strengthening the policies for equal opportunities. The NAPE applies the contents of the EES to the Portuguese reality with a multi-annual view (five-year period), setting objectives, quantified goals and action deadlines and defining new programmes and measures. Each year, the NAPE goes through a process of revision, in order to identify responses to new challenges. In the 2002 review, the main guidelines of previous years were maintained, although adapted to the new challenges of the EES and fine-tuned in accordance with the results of prior implementations and with the recommendations made to Portugal by the European Council. A deeper revision of the NAPE, in line with the future guidelines of the next phase of the EES, will be done in 2003.

Three essential goals were defined in the NAPE for 2002:

- Improvement of employment quality and of labour protection.
- Adequacy of labour legislation to the new needs of work organisation and to the increase of productivity and competitiveness of the Portuguese economy.
- Conciliation of the goal of a high employment level with the challenges of quality, competitiveness and technological innovation.

Taking into account these three basic targets, the following areas were identified as areas worth of particular attention:

- Qualification of human resources in order to adapt them to both workers' and enterprises' needs.
- Promotion of technological and vocational education in order to give adequate skills to the students entering the labour market after conclusion of basic education.
- Strong investment in continuing training programs for teachers, giving priority to the first cycle of basic education and to information technologies and multimedia fields.
- Reinforcement of entrepreneurship, especially in technological areas, improving the innovation and risk spirit.
- Creation of mechanisms allowing the effective development of a network for professional risks prevention.
- Launching of an integrate program against work accidents.

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- Systematisation and simplification of current labour legislation.
- Promotion of adaptability and flexibility of work organisation.
- Creation of conditions assuring social effectiveness of employment policies, facilitating entrance of young people in the active life and contributing to better workers mobility.

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Annex IV

The 2002 Productivity and Growth Programme

The sustainable increase of Portuguese productivity and competitiveness requires a deep modification of the business environment. Bearing in mind the persistent competitiveness loss observed in the last years, the government adopted a broadly based Productivity and Growth Programme (PGP) in June 2002. The PGP includes orientations valid for the duration of the legislature (four years) as well as more immediate measures, and is organised around seven key priorities.

Fostering productive investment

- Revision, simplification and systematisation of the legal framework concerning productive investment. An Investment Code will be created, encompassing all forms of investment incentives (financial, fiscal and contractual).
- Modification of the philosophy and procedures of the Operational Programme for the Economy:
  - Priority support based on venture capital, seed capital and development capital; rewards granted to high value-added projects, through quantitative measures computed throughout the project's life.
  - Simplification of application and evaluation procedures, through the reform of the programme management framework and the establishment of partnerships with private financial and venture capital institutions.
- Elimination of multiple interlocutors needed for an investment operation to reduce delays and costs:
  - Concentration in a single institution (IAPMEI) of all administrative procedures for small investment projects, creating investor cabinets that will ensure investors assistance both in what relates to new investments or expansion of existing ones.
  - Creation of the Portuguese Investment Agency (API) with one-stop-shop status for national investment projects of a higher dimension and all foreign direct investments. This agency will be responsible for full administrative assessment of such investments (including potential applications to financial incentives), for licensing and establishment procedures, and for the negotiation of special contractual regimes.
- Development of an intensive foreign direct investment attraction programme. This program will use restructured diplomatic channels and include "road shows" promoting Portugal as an investment destination in selected markets.

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- Creation of a fiscal mechanism to support investment of companies producing tradable goods and services. Companies will be allowed to retain a fraction of the corporate tax bill, provided that the fiscal reserve is used in the following two years to finance new investments through equity increases.
- Revision of legal and fiscal frameworks for venture capital societies, reducing tax levels over reinvested profits.
- Revision of labour legislation to adapt it to the new social and economic realities.
- Enlargement and dynamisation of the Business Localisation Areas (*Áreas de Localização Empresarial*), through the improvement of infrastructures and enterprise co-operation, and the simplification and speed-up of licensing procedures.
- Creation of instruments to support companies' permanent capital base (equity and long-term debt), through the reinforcement of the lending nature of the Mutual Guarantee System (*Sistema de Garantia Mútua*), and the institution of "joint" access mechanisms to capital markets for small and medium enterprises (SMEs).

Reinforcement of fair competition

- Extension of competition regulation to all economic sectors, through the concentration of the powers to ensure fair competition in a single entity with independent status (Competition Authority), including the approval of mergers and the prevention and sanction of predatory or anti-competitive practices.
- Enhancement of competition in the energy market, with a view to reducing costs supported by enterprises. Implementation of an electric Iberian market, while defending national interests and self-sufficiency in production. Further integration with Spain is foreseen with the creation of a gas Iberian market.
- Creation of a single regulatory entity for the energy sector (electricity, gas, oil).
- Continuation of telecommunications development, stimulating competition in the sector through the definition of clearer rules regarding predatory practices and market power abuse.
- Rationalisation and development of railway transport and reinforcement of port productivity and efficiency, in order to promote effective competition between alternative means of transportation.

Promotion of Portuguese companies, products and tourism

- Reformulation of the public system of external trade promotion, through the introduction of a new concept of economic diplomacy. Efforts will be concentrated in target markets and diplomatic representations will be given self-capacities for trade promotion, supported by a co-ordinating body under the joint supervision of the Ministries of Economy and Foreign Affairs.
- Public support of regular actions of trade promotion in target markets by means of contracts with major business associations.
- Concentration of tourism support and promotion in a single entity that will establish contracts for tourism promotion actions in target markets with the sectoral confederation.
- Creation of the Portuguese Brands project (*Marcas Portuguesas*) to support projects with high value-added produced and retained in Portugal, allowing the creation of effectively distinct characteristics of quality, innovation, design and services.

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– Creation of a comprehensive and updated information system – to be managed in partnership with the business associations – on international markets, business opportunities, technologies and sector-based perspectives.

Administrative simplification and deregulation

– Identification and elimination of all redundant and unnecessary regulations.
– Rationalisation and simplification of public administration structures and procedures related with the supply of services to enterprises, by adapting the structure of the Ministry of Economy to the realities and needs of the business sector.
– Settlement of minimum service levels to public organisms regarding the quality and timing of responses to companies' demands.
– Reduction of the number of business contact points in the public administration, privileging new information technologies.
– Reinforcement of the effectiveness of registry and notary services, namely through privatisation of notary services and concentration of some legal contracts' procedures.

Consolidation and revival of the business environment

– Institution of a new system of registry, transformation and licensing of societies, to reduce related time frames and bureaucracy.
– Creation of a more effective bankruptcy system.
– Elimination of fee and fiscal obstacles to enterprise development and restructuring, mergers and divestiture operations.
– Creation of mechanisms and instruments to allow a more effective and less time consuming recovery of state credits over enterprises.

Innovation and R&D

– Concentration of basic scientific research conducted by universities and research centres within the scope of the Ministry for Science and Higher Education.
– Rationalisation and reorientation of the core business of public laboratories into applied research directed to the business community, in articulation with technology centres.
– Regulation, strategic definition, information diffusion and support to private enterprises R&D projects, through public-private partnerships under the responsibility of the Ministry of Economy.
– Selective support through seed capital to young entrepreneurs to launch new businesses with proven potential and viability.
– Utilisation of structural funds in applied research projects, bringing together universities, public institutes and private enterprises, to foster patents and new products.

Privatisation

– Reduction of the public presence in the productive and infrastructure sectors, seeking to maintain the decision making centres in Portugal, while fully respecting the EC rules.
– Implementation of the privatisation process through strategies coherent with the goals of the international position of Portuguese business groups in sectors con-

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sidered of special importance, such as paper and paper pulp, forestry, energy and telecommunications.

- Privatisation of the water sector and reformulation of the legal statute and competencies of the sector regulator.
- Extension of the privatisation programme to public real estate.
- Definition and execution of restructuring programmes of public enterprises that are not subject to privatisation, with the implementation of management control systems, efficiency and productivity goals and retribution of management linked with the results achieved.

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Annex V

Chronology of main economic events

2001

January

Fuel prices raised by 3 to 4 per cent (on average).

Liberalisation of the local and regional telecommunications market.

Publication of the Stability and Growth Program for the period 2001-2004.

Re-election of Jorge Sampaio as President of the Portuguese Republic.

February

Social partners and the government sign a tripartite agreement on employment, the labour market, education and training policy, setting specific objectives for firms' supply of training.

Social partners and the government sign a tripartite agreement aimed at improving work-conditions, health and safety in the workplace and fighting against work-accidents.

April

Publication of the 2001 National Employment Action Plan (Approved in December 2001).

New Job Offer Stimulus Programme came into effect. This programme, in place until 2006, aims at co-ordinations measures for stimulating job offers and to facilitate labour market entry for disadvantaged groups through financial support to companies.

May

The ECB lowers its key-refinancing (repo) rate from 4.75 per cent to 4.50 per cent.

The government launches an Integrated Innovation Support Programme (PROINOV).

July

Publication of new legislation tightening the rules for fixed-term employment contracts (effective in August).

June

Approval of the 2001 first supplementary budget.

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August

The ECB lowers its key-refinancing rate from 4.5 per cent to 4.25 per cent.

Publication of the new Framework Law for the Budget.

September

The ECB lowers its key-refinancing rate from 4.25 per cent to 3.75 per cent.

Publication of the report on the reform of public spending (ECORDEP) by the Ministry of Finance.

October

Government presents the 2002 budget proposal (to be approved in December). The budget deficit target for 2002 is set at 1.3 per cent of GDP.

November

The ECB lowers its key refinancing rate from 3.75 per cent to 3.25 per cent.

Approval of the 2001 second supplementary budget.

Social partners and the government sign a tripartite agreement on social security reform, following the basic law approved in July 2000.

December

Centre-Right parties win the local government elections.

Resignation of socialist Prime-Minister António Guterres.

The Government submits to the European Commission the Stability and Growth Programme for 2002-05. The budget deficit target for 2002 is revised to 1.8 per cent of GDP.

2002

January

Euro banknotes and coins are introduced in Portugal.

European Commission activates an early warning proposal for Germany and Portugal.

February

National Statistical Institute sends the first Excessive Deficit Procedure notification of 2002 to the EC. The budget deficit estimate for 2001 is revised to 2.2 per cent of GDP.

European Council, considering that the Portuguese government has effectively responded to the concerns expressed in the EC recommendation, closes the early warning procedure.

March

Escudo banknotes and coins ceased to be legal tender and are no more accepted as a means of payment. Exclusive circulation of the euro in the Portugal (from 28 of February onwards).

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Eurostat, summarising the results of the February notification, announces that it cannot certify the accounts of the Portuguese general government of 2001.

Government announces a revision of the 2001 budget deficit to 2.4 per cent of GDP, based on new provisional information on local government accounts.

Centre-Right coalition wins general elections.

New legislation to encourage permanent employment by means of financial subsidies to companies that convert a fixed-term contract (on expiry) into a permanent one.

New legislation to ensure that minors aged 16 and above are provided with training when employed.

Change of the price-setting regime of consumer fuel prices. Domestic fuel prices are now set in a monthly basis in order to reflect the evolution of oil price in international markets.

April

New centre-right coalition government takes office. Durão Barroso is the new Prime Minister.

Creation of the Commission for Analysis of Public Accounts, following the non-certification of the 2001 general government accounts.

Creation of the regulatory body for the energy sector (ERSE) that covers the electricity and natural gas sectors. (ERSE already existed but was only for electricity: *Entidade Reguladora do Sector Eléctrico*).

May

Approval of the 2002 supplementary budget. The budget deficit target for 2002 is revised to 2.8 per cent of GDP and a series of emergency measures are introduced.

Publication of the 2002 National Action Plan for Employment.

June

Government presents changes to the minimum guaranteed income (RMG) scheme, now called social insertion income (*Rendimento Social de Inserção*, RSI) (approved by Parliament in October 2002 but rejected by the Constitutional Court in December).

Government presents a Productivity and Growth Programme. The programme includes orientations valid for the duration of the legislature (four years) as well as more immediate measures to increase Portuguese productivity and competitiveness.

The standard VAT rate is raised from 17 to 19 per cent.

Broad Economic Policies Guidelines for 2002 are decided in Seville European Council.

European Court of Justice rules against the Portuguese privatisation legislation that imposes restrictions on the acquisition of shares by foreign investors.

Revision of bankruptcy legislation, with a view to facilitating the exit of under-capitalised companies.

July

Publication of the report of the Commission for Analysis of Public Accounts on the 2001 fiscal situation, which concluded that the 2001 budget deficit was 4.1 per cent of GDP.

European Commission initiates the Excessive Deficit Procedure for Portugal.

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Government presents a revised Basic Law on Social Security. The main feature of the law is the introduction of a mixed public-private system, opening up the social security system to the private sector (approved by Parliament in December 2002).

Parliament approves a Law for the Development and Quality of Tertiary Education, setting the principles for the creation of a systematic and independent system of evaluation and certification of universities and courses.

Parliament approves a Law for the Evaluation System of non-Tertiary Education, creating a comprehensive system of evaluation of all schools.

Government presents the new Labour Code, revising current labour legislation in areas like fixed-term employment contracts, dismissals, collective bargaining, working-time and absence from work. The new Labour Code will bring together all labour legislation in one single document.

August

Publication of the new Law of Budgetary Stability and of the revised Framework Law for the Budget.

Government publishes the curricular revisions of the 3rd cycle of basic education (7-9 years) to start in the school-year 2002/3.

September

European Commission initiates the Excessive Deficit Procedure.

President of the Republic promulgates legislation on the reform of Public Administration, increasing the mobility of civil servants and creating an employment pool for redundant staff (*quadro de supranumerários*).

Government approves the creation of the Portuguese Investment Agency (API), with one-stop-shop status for large national investment projects and for all foreign direct investment (approved by Parliament in October).

Government approves the creation of an independent Competition Authority.

Government presents a set of decentralisation measures that transfer competencies from central to local governments in areas like education, health, culture, public works, housing and environment. These include the maintenance and building of pre-primary and basic schools and the management of non-teaching staff, and the building of health centres in partnership with the central government.

Government approves changes to the medicament policy, with a view to reduce expenditure on pharmaceuticals. These include the prescription of drugs according to the active substance so that patients can choose the least expensive medicament and establishment of ceilings (equivalent to the price of generics) on public co-participation of pharmaceuticals.

October

Government presents the 2003 budget proposal. The budget deficit target for 2003 is set at 2.4 per cent of GDP.

European Commission concludes that an excessive deficit exists in Portugal.

Approval of the Decrees-law converting 34 hospitals into public enterprises.

Iberian electric market to start until June 2003 (decision taken at the Portugal/Spain summit).

Revision of the Framework Law of Telecommunications, authorising the sale of the fixed-network.

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November

Approval of the 2003 Budget by Parliament.

European Council confirms that an excessive deficit exists in Portugal.

Parliament approves a new Framework Law on Hospital Management, establishing the transformation of some hospitals into public enterprises and new rules for their management.

December

Publication of the Stability and Growth Programme for 2002-06.

The ECB lowers its key refinancing rate from 3.25 per cent to 2.75 per cent.

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